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The Nanosyntax of the Adjective

Evidence from Czech

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Preface

This book is the result of more than ten years of joint work on various aspects of degree morphology. Our investigation was inspired by Jonathan Bobaljik's work on the decomposition of adjectival degrees. In his 2012 book, he surveyed a sample of over 300 languages and showed, based on a variety of evidence, that the positive, comparative and superlative degrees are decomposable and stand in a containment relation: the superlative is built on top of the comparative, which in turn is built on the positive degree adjective. Among the evidence adduced for this conclusion was the observation that the superlative degree often morphologically contains the comparative as a subpart, and that the comparative in turn typically morphologically contains the positive.

The starting point of this book was the observation that some languages provide a much richer morphology: we discovered cases where the comparative is derived from the positive by the addition of two or even three independent suffixes, while in other cases the comparative is not derived from the positive, but from a shorter, truncated form of the positive. Even more fascinating was the fact that some languages—Czech among these—contain both of these patterns at once: in some cases, the comparative adds three morphemes on top of the positive, in other cases, the positive is shortened and just a single morpheme appears. These intricate patterns of morphological derivation between the positive and the comparative did not at first seem to fit the general pattern and invited us to further investigate this issue. Our basic question was whether it is possible to derive these divergent patterns of morphological marking using a single functional sequence at all.

This book pulls together our ten years into this topic and argues that it is indeed possible to find a unified approach to this variation. The full morphological complexity can be derived by unifying two strands of research: on the one hand the original claim that the superlative syntactically contains the comparative, and the comparative syntactically contains the positive; and on the other hand the ingredients familiar from the semantics research distinguishing scales, standards of comparison, measure phrases, etc. We ren-

der this with a syntactic representation of two scales, with the comparative showing both scales and the positive only one, along the lines of Kennedy and Levin's 2008 work (*Measure of Change: The Adjectival Core of Degree Achievements*). Each scale is in turn represented syntactically by the same four features of standard of comparison, dimension of the scale (e.g., size or speed), direction of the scale (positive vs. negative, as in *fast* vs. *slow*) and perhaps a measure phrase (which we call 'distance'). We show that this combination yields exactly the rich morpho-syntactic patterns observed. The result of this book is thus a much more fine-grained morpho-syntactic decomposition of the adjectival degrees than previously thought.

The way we relate forms to the proposed semantic sequence relies on the theory of Nanosyntax, which originated in the work by Michal Starke in the early 2000s and has become one of the major approaches to the syntax-morphology interface. The book provides an introduction to this framework using examples of degree marking from various languages, and then zooms in on Czech, which contains fifteen different adjectival classes regarding the formation of the positive and the comparative. This covers both the truncating classes (where the comparative is formed on a truncated form of the positive) and the highly agglutinative forms in which the comparative has three extra markers added on top of the positive. We show that the cross-linguistically invariant lexicalisation algorithm of Nanosyntax allows us to capture the coexistence of all these classes within a single language.

During our work, we benefited from the constructive comments and feedback from our colleagues and the audiences at various conferences where parts of the material was presented. We also gratefully acknowledge the support of the Czech Science Foundation (grant number GA25-15601S).

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Acronyms

3	3rd person
A	Adjective (a syntactic head)
ABL	Ablative
ADJ	Adjective
ADV	Adverb
AGR	Agreement
AUG	Augment
AS	<i>as</i> , a relative clause marker in Scottish Gaelic
C _x	Comparative feature of type X
CMPR	Comparative
DAT	Dative
DEF	Definite
DEG	Degree
DIM	Dimension (a functional head)
DIR	Direction (a functional head)
DMV	Diminutive (a functional head)
DM	Distributed Morphology
F	Feature
FEM	feminine
<i>fseq</i>	Functional sequence
GEN	Genitive case
ILL	Illative case
INFL	Inflection
INS	Instrumental
LOC	Locative
MASC	Masculine
NEUT	Neuter
NOM	Nominative
Q	Quantity (a functional head)
NEG	Negation (a functional head)
PAL	Palatalisation

PL	Plural
POS	Positive degree adjective
POINT	A functional head introducing the standard
PRS	Present
REL	Relation (a functional head)
SG	Singular
SPRL	Superlative

Chapter 1

Introduction

1.1 Topic of the book

The central stage in this monograph belongs to Czech adjectival morphology. The reason for this is simple: the data found in this language sheds light on two important areas of research.

The first one is the structure of adjectival phrases in general. Specifically, the data at hand provide evidence for the existence of multiple positions of exponence (functional heads) inside the positive and the comparative degree of adjectives. We argue that inside each degree, there are at least four functional heads with two more that are optional.

The second area concerns the syntax-morphology interface in general and the theory of allomorphy in particular. Specifically, we show that the patterns of allomorphy in Czech cannot be captured by context-sensitive rules. The role of such rules in standard theories is to treat two different realisations of some feature (like plural in *fox-es* vs. *ox-en*) as conditioned by the neighbouring morphemes (e.g., realise PL as *en* when next to *ox*). Such rules are commonly used – to the extent that they are almost perceived as a-theoretical descriptive devices – but in Czech, these rules fail to provide a satisfactory analysis or description of the surface patterns (unless the patterns are treated as accidental homophony).

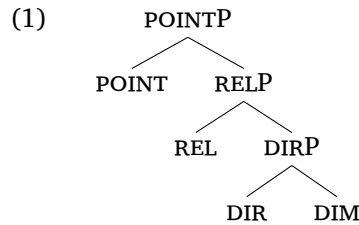
We argue that to capture the patterns, a different tool is required, namely phrasal lexicalisation as pursued within the Nanosyntax approach to grammar (Starke 2009, De Clercq et al. 2025). Using this framework, the approach to allomorphy that we develop in the monograph ultimately boils down to the claim that there is no allomorphy in the sense of one feature having multiple different realisations depending on the linear context. Instead, all variation in the form of various markers and in their distribution boils down to what features these markers actually lexicalise.

In the remainder of this first section, we elaborate on the two points that we made above regarding the structure of adjectival phrases and the syntax

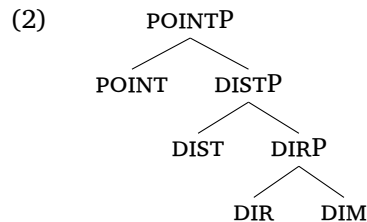
morphology interface, with the aim to give the reader an idea of how the richness of the Czech data motivate the relevant claims. At the end of this first section, we provide an overview of how the book is organised.

1.1.1 The structure of adjective phrases

In this book, we argue that the positive degree of a gradable adjectives is composed of at least four syntactically independent semantic ingredients (features), with each feature an independent functional head in the structure. Two of the ingredients represent the adjectival scale (also called a ‘measure function’); we call them DIMENSION (e.g., speed) and DIRECTION (positive direction for *fast*, negative for *slow*). The other two ingredients introduce a particular standard on the scale, which the argument must exceed (i.e., *fast* means ‘having greater speed *than some contextually salient speed standard*’). We label the projections introducing the standard REL and POINT, where the first one is responsible for RELATING the standard to the scale, and POINT is a syntactic representation of that standard, whose precise value is often determined by the context. The *fseq* of the positive degree adjective then looks as in (1):



It may not be necessary to dedicate two features to the semantic function of introducing a standard. One of the two may instead denote the distance on the scale. In the positive, this would be the distance between the standard and the argument, amounting to a measure phrase:



In this book, we focus on the fact that Czech morphology suggests four features, and not on the exact semantics of those features. The entire book can however be read with DISTANCE substituted for REL.

These four positions (however we choose to label them) can be independently motivated by the morphology of Czech adjectives. To see the contours of the argument, consider the data in (3). The table shows that Czech has (at least) four different morphological classes of adjectives, defined by the type of the marker that is found in between the root and the word-final agreement marker *-ý*. We call these different class markers *augments*. We also include a morphologically simplex adjective that has no such marker.

(3) Four classes of positive-degree adjectives in Czech

GLOSS	POSITIVE AUGMENT comparative			
‘young’	mlad-	-ý	-Ø	mlad-š-í
‘sweet’	slad-	k -ý	-k	slad-š-í
‘easy’	snad-	n -ý	-n	snaz-š-í
‘wide’	šir-	ok -ý	-ok	šir-š-í

We know that the *augments* are independent morphemes because the roots often exist without such markers in forms other than the positive. For instance, with the roots in the table, the *augments* are absent in the comparative, as shown in the final column. Their absence in the comparative makes them interestingly different from traditional derivational morphemes.¹

If this was all there was regarding the *augments*, we would have evidence for two positions: one for the root, and another one for the *augment*. However, Czech is interesting in that it has diminutive adjectives formed by a variety of suffixes, including the diminutive *-ouč*. These diminutive adjectives reveal an asymmetry among the *augments*, which is easy to capture if we assume three positions, but difficult to capture otherwise.

To see that, consider the data in Table (4). We see here that when diminutives are formed from the adjectives with the *augments* *-n* and *-ok*, these *augments* are found to the left of *-ouč* in the surface string, see the bottom two rows in Table (4). The *augment* *-k*, on the other hand, is not found to the left of the diminutive, but (presumably) to its right.

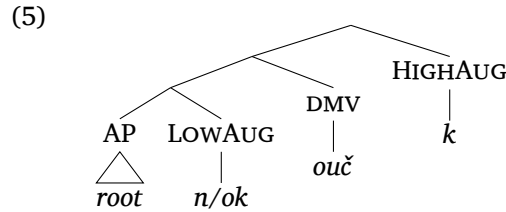
(4) Classes of adjectives in the positive degree

GLOSS	POSITIVE AUGMENT		diminutive
‘young’	mlad-	-ý	mlad’- ouč- k -ý
‘sweet’	slad-	k -ý	slad’- ouč- k -ý
‘easy’	snad-	n -ý	snad- n -ouč- k -ý
‘wide’	šir-	ok -ý	šir- ok -ouč- k -ý

While there is lots more to unpack about these forms (including the fact that all adjectives in the diminutive have the *augment* *-k* following the diminutive *-ouč*), it is clear that the data reveals an asymmetry between the *-n/-ok* *aug-*

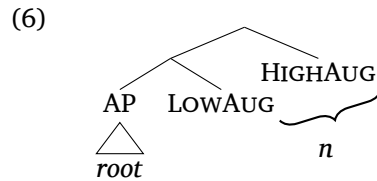
¹ This is not a phonological deletion, as we shall argue in Chapter 3. One part of the reason is that with different roots, the *augments* are preserved in the comparative.

ments on the one hand (they precede the diminutive), and the *-k* augment on the other (follows the diminutive).² This asymmetry can be captured if we posit three positions of exponents, as in (5), where different augments may occupy different positions, as revealed by the diminutive.



Given the structure in (5), and with three positions of exponence (root, low augment and high augment), it is clear why the augments *-n* and *-ok* are ordered to the left of the diminutive, and why *-k* is ordered to the right, as this corresponds to their structural positions.

However, what is not completely clear is why the low augments *-n*/*-ok* do not combine with the high augment *-k* in the absence of the diminutive. The idea that we develop in this book is that this is so because the low augments *-n* and *-ok* lexicalise not only the position described as a ‘low augment’, but, in the general case, they also lexicalise the position of the high augment, as in (6).



This is why *-n* and *-ok* generally do not combine with *-k*, unless there is a diminutive intervening between the two positions (in which case the lexicalisation depicted in (6) is no longer viable and both augments show up).

Taking stock, and assuming three positions of exponence, we can currently capture three of the four classes as depicted in (7), where the top row gives the positions in the morphological template, and the rows below depict which positions are lexicalised by which marker. Also the roots come in various sizes (they are associated to different feature sets) and, depending which features are left unlexicalised by the root (if any), one of the augments appears.

² We treat *-ouč-k* as decomposable, because there are other diminutives such as *-in-k*, or *-oun-k*, where what varies is the diminutive part, but the *-k* remains a constant.

(7) Lexicalisation table with three positions

	ROOT	LOW AUG	HIGH AUG	GLOSS
Class n	snad	n		‘easy’
Class k		slad	k	‘sweet’
Class Ø		mlad		‘young’

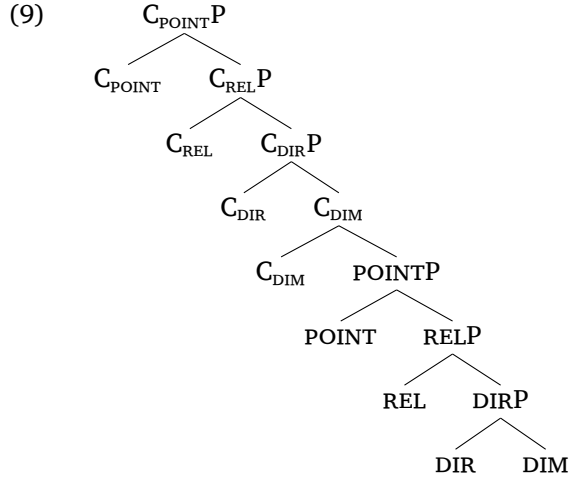
In Chapter 2, we argue that this general picture should be extended also to the augment *-ok*, as depicted in Table (8). The top row now gives the proposed ingredients of the positive, as sketched at the start of this section, and the remaining lines depict a hypothesis about how the various exponents relate to these features.

(8) Lexicalisation table with augments

	DIM	DIR	REL	POINT	GLOSS
Class I	šir		ok		‘wide’
Class II	snad		n		‘easy’
Class III		slad	k		‘sweet’
Class IV		mlad			‘young’

What evidence is there for this more elaborate picture? The reasoning starts from the fact that in the table, the augment *-ok* is the only augment that reaches down into the area where the scalar features reside (lexicalising the *DIRection* feature), while the other augments do not do so. We argue that this gives rise to another asymmetry, which is that *-ok* only gives rise to positive adjectives such as ‘tall’ (but not ‘short’), ‘deep’ (but not ‘shallow’) and so on.

Moving on to the comparative, we propose that it corresponds to the same type of scale as the positive, containing again the features *DIM*, *DIR*, *REL* and *POINT*. The full adjectival sequence is thus as given in (9). For ease of reference, and in order to be able to distinguish the positive and the comparative in the representations and the discussions in the text, we mark the comparative heads with a capital *C* and a subscript that provides their semantics, e.g. C_{DIM} , C_{DIR} , etc.



As the structure indicates, the comparative scale is derived from the positive degree. It has the same dimension as the positive-degree scale (the comparative DIM is unspecified and interpreted the same as the positive-degree DIM). However, the comparative scale is different from the positive-degree scale in that it has a different starting point, which is provided by the standard phrase. Thus, a comparative like *tall-er than John* includes two scales. One is the positive-degree scale, where the standard point is not determined by context, but specified as John's height. Then a new (comparative) scale is constructed, which has the same dimension, but starts at John's height, rather than at zero. We also note that the comparative scale may have the same or possibly a different direction than the original scale (*more* vs. *less tall*), and finally, the argument must again exceed the standard of the comparative scale (which coincides with the starting point of the comparative scale, i.e., John's height). This type of approach has been proposed in Kennedy & Levin (2008) on independent grounds (they use the term derived measure function for what we call the comparative scale).

If REL was substituted by DISTANCE, as suggested above, the two relevant distances would respectively be the amount represented by the standard phrase, and the amount by which the argument differs from the standard. In *Atka is 10cm taller than Erwin* the two relevant distances are then 'the height of Erwin' (the distance between zero and Erwin's height), and '10cm' (the distance between Atka's height and Erwin's height).

In this book, we provide evidence for the existence of such a rich comparative structure from the morphology of Czech adjectives. In particular, we observe that some adjectives require up to three new morphemes in the comparative. We show this in (10).

(10)

	POSITIVE	COMPARATIVE	GLOSS
a.	slab-ý	slab -š -í	‘weak’
b.	chab-ý	chab -ěj -š -í	‘weak’
c.	žádouc-í	žádouc -n -ěj -š -í	‘desirable’

In (10a), we see an adjective that forms its comparative by adding just a single morpheme, *-š*. In (10b), there are two morphemes, since an additional *-ěj* appears in between the root and *-š*. Finally, in (10c), we get both *-ěj* and *-š*, and a third marker (*-n*) appears in addition.

We argue that under the assumption that each morpheme is lexicalising at least one feature, three morphemes entail three extra features in the comparative, and that serves as our evidence for three out of the four heads that we are proposing.

Finally, the argument for the fourth head comes from comparative adverbs. What we observe here is that some adjectives which lack *-ěj* in the comparative adjective also lack it in the adverb (see (11a)), but some such adjectives need it in the adverb, see (11b). A third class of adjectives has *-ěj* in both forms, see (11c).

(11)

	POSITIVE	CMPR ADJ	CMPR ADV	GLOSS
a.	drah-ý	draž -š -í	dráž -e	‘expensive’
b.	slab-ý	slab -š -í	slab -ěj -i	‘weak’
c.	chab-ý	chab -ěj -š -í	chab -ěj -i	‘weak’

We argue in Chapter 3 that the ‘inconsistent’ class (11b) can only be captured if we assume that *-ěj* lexicalises a different sets of features in the adjective and in the adverb (else they would pattern the same). Now the only way they can lexicalise different sets of features is if the two sets overlap in some feature and differ in another feature (barring the possibility of accidental homophony). But if there is a feature they share, and a feature in which they differ, there must be at least two features for *-ěj* in one of the forms, which we argue to be the adjective. Combining this with the three positions of exponence arrived at earlier, this means four features in total.

It is our goal in this book to elaborate on these observations, present the data in a systematic and comprehensive fashion, rule out alternatives, and show how these proposals derive the full range of facts observed in Czech. If our approach turns out to be correct, the facts lead us to discover a rather rich set of ingredients in both positive and comparative forms. In our analysis, two of these ingredients correspond to the scale (DIMENSION and a DIREction features). The other two features, REL and POINT introduce a standard, which the argument of the adjective has to exceed. We further argue that the comparative has a recursive structure, and builds a comparative

scale (on top of the scale corresponding to the positive (similar to Kennedy & Levin’s 2008 ‘derived measure function’).

1.1.2 Allomorphy and the syntax morphology interface

The second area where the data under discussion contribute to our theoretical understanding is the domain of allomorphy and the syntax-morphology interface in general. In particular, it turns out that in Czech, it is very difficult (if possible at all) to state morpheme distributions in terms of context-sensitive rules. These difficulties reveal the need for a different approach to allomorphy, which in our case, is an account based on phrasal lexicalisation as developed within the Nanosyntax framework. In essence, our approach boils down to the proposal that there is no allomorphy in the way it is traditionally understood, in that there would be two different markers lexicalising the same feature or features, differing only in the context in which they lexicalise these features. On our approach, different markers that appear to be in an allomorphic relation never lexicalise the same features, but always lexicalise a different set of features. The context is relevant for this in as much as it determines what features the root lexicalises, and which features are left for the affixes. The division of labour between the root and the different affixes is, on our approach, algorithmically determined, relying on the so-called lexicalisation algorithm (Caha et al. 2025).

In this section, we highlight the difficulties for the traditional approach based on context-sensitive rules of exponence. However, we shall not provide a full solution to these puzzles here, as that requires the technology and elaborate representations that we develop throughout this book.

To see the issue, consider first the allomorphy of the augment. While discussing Table (3), we already mentioned that augments may disappear in the comparative, and that this depends on the root. To see the full pattern for one augment, consider four different adjectives depicted in Table (12). The adjective on the first row (12a) has *-n* both in the positive and in the comparative. The adjective on the second row only has the augment in the comparative, while the adjective on the row (12c) only has the augment in the positive. Finally, there is no augment on row (12d).

(12)

	CMPR ADJ	CMPR ADV	GLOSS
a.	jem -n -ý	jem -n -ěj-š-í	‘fine’
b.	žádouc -í	žádouc -n -ěj-š-í	‘weak’
c.	pozd -n -í	pozd -ěj-š-í	‘early’
d.	hrdý -ý	hrd -ěj-š-í	‘proud’

Focussing on the augment alone, the relevant pattern of distribution is given in (13).

(13) Four different root classes

CLASS	POS	CMPR
I	n	n
II	∅	n
III	n	∅
IV	∅	∅

Caha et al. (2024b) argue that such a distribution is impossible to capture using context sensitive rules. To see the issue, assume again that there is a morphosyntactic position, call it α , which the augment is the realisation of. If zero is the default realisation, as in (14a), we need three different rules specifying where $-n$ appears, see (14b-d). As far as we are aware, there is no way of unifying these rules into a single non-disjunctive statement that correctly predicts where $-n$ is found.

- (14) a. $\beta \Leftrightarrow -\emptyset$
 b. $\alpha \Leftrightarrow n$ in the context of Class I
 c. $\alpha \Leftrightarrow n$ in the context of Class II and CMPR
 d. $\alpha \Leftrightarrow n$ in the context of Class III and POS

Moreover, it is unclear whether the rule in (14d) can be stated at all, since, in an account where comparatives contain the positive, the positive feature (whatever feature that may be) would be also contained in the comparative, and activate the rule also there, i.e., beyond its intended domain. The same issues arise if $-n$ were chosen as the default realisation, in which case the contexts for the zero marker would have to be stipulated in very much the same way as in (14b-d).

The obvious point to make is that if multiple rules for a single marker can be multiplied like this, this amounts to accidental homophony, where each context requires a new rule. This is of course a weak theory, which makes no predictions about the possible systems of marker distribution, as any instance of a morpheme is readily accommodated by postulating a new rule.

It is our goal in this book to show that the Nanosyntax model adopted here is able to model such distributions using just a single rule. The core difference between our account and the rules as in (14) is that in Nanosyntax, zero markers arise as a result of phrasal lexicalisation, where the relevant position hosting $-n$ is lexicalised by an independently existing item. As a result, there is therefore no need to carefully orchestrate a competition between two markers, a zero and an overt marker, as the zero does not really exist in Nanosyntax.

What we further observe is that such patterns of distribution do not arise only augments, but also for the markers in the comparative forms (of which we consider the adjective and the adverb). In Table (15), we focus on the distribution of the comparative marker *-ěj*. Once again, what we observe is that this marker can be found in both the adjective and the adverb (see (15a)) or nowhere at all (see (15d)). It is also possible to see it in only one of the forms, in which case it's either in the adverb (as in (15b)) or only in the adjective (see (15c)).

(15)

	POSITIVE	CMPR ADJ	CMPR ADV	GLOSS
a.	chab-ý	chab -ěj -š -í	chab -ěj -i	'weak'
b.	slab-ý	slab -š -í	slab -ěj -i	'weak'
c.	brzk-ý	dřív -ěj -š -í	dřív -e	'early'
d.	drah-ý	draž -š -í	dráž -e	'expensive'

Summarising, the same pattern of the distribution of *-ěj* is observed, as Table (16) brings out.

(16) The distribution of *-ěj*

CLASS	CMPR ADJ	CMPR ADV
I	-ěj	-ěj
II	∅	-ěj
III	-ěj	∅
IV	∅	∅

It is easy to see that the table in (16) is, abstractly, going to lead to exactly the same issues as we observed for the analogous table (13). To show that, we again assume that there would be a single position of exponence, call it β , that shows allomorphy between *-ěj* and zero. Differently from what we tried for (28), suppose now that the overt marker *-ěj* is the default realisation, as in (17a), and that the zeroes have to be introduced by context sensitive rules. These are given in (17b-d), and, again, there is no way to write a single non-disjunctive rule that would have the same effect.

- (17) a. $\beta \Leftrightarrow -ěj$
 b. $\beta \Leftrightarrow \emptyset$ in the context of Class IV
 c. $\beta \Leftrightarrow \emptyset$ in the context of Class III and ADV
 d. $\beta \Leftrightarrow \emptyset$ in the context of Class II and ADJ

We show in Chapter 3 that the same account as proposed for the augment *-n* can be extended to the case at hand. Moreover, we show that when various patterns of augment distributions are paired with different patterns of *-ěj* distribution, we end up with fifteen different classes in total. It is our aim

to show that all these classes fall out from the Nanosyntax model that we propose here, and, moreover, we achieve this without invoking any kind of postsyntactic and purely morphological adjustments, such as Rules of Referral (Zwicky 1977; Baerman 2004) or any of the purely morphological operations assumed in Distributed Morphology (Halle & Marantz 1993), such as Impoverishment, Fusion, Fission and the like.

As mentioned above, the core of our account is based on the idea that there is no allomorphy in the traditional sense of context-sensitive rules, rather, all the patterns fall out from a model where individual roots and affixes are simply specified for the features they lexicalise, and the forms fall out from this specification.

1.1.3 The structure of the book

The book is structured as follows. In Chapter 1, we provide an overview of the framework of analysis Nanosyntax. As we do so, we draw on the data from the area of degree morphology, and, where relevant, we also point out cases where one of the degrees seems to require a more elaborate structure with multiple comparative (or superlative) features.

In Chapter 2 we discuss the positive degree in Czech, including augment distributions, diminutive adjectives and negative adjectives. In this chapter, we also provide a brief overview of earlier attempts to decompose adjectives into multiple functional heads.

In Chapter 3, we turn to the comparative, and show how the combination of augment allomorphy and comparative-marker allomorphy leads to a whole 15 classes. In this chapter, we show in detail how the Nanosyntax framework deals with these examples. In Chapter 3, we also include a brief Conclusions, formulating the general points established in the course of this book.

1.2 General architecture of the model

1.2.1 Morphemes and features

At the basis of the nanosyntactic model (Starke 2009) lie three important insights on the nature of morphemes and features. The first is fairly traditional, namely that **morphosyntactic features are more abstract than morphemes**. This is a shared trait with realisational approaches to morphology, where the abstract features are the elements that morphosyntax operates over, and they alone determine the morphosyntactic properties of various forms, independently of the morphemes that realise them.

The second insight is shared with Cartographic approaches to syntax, which propose that **morphosyntactic features are the elementary building blocks of syntactic structures**, according to the ‘One Feature, One Head’ (OFOH) principle (Cinque & Rizzi, 2010; Kayne, 2005). This means that each morphosyntactic feature is introduced as a syntactic terminal.

The third insight is specifically nanosyntactic and it starts from the traditional observation that morphemes typically realise multiple features. Since, according to the second insight, each feature is an independent syntactic terminal, this entails that such **morphemes realise multiple terminals in a fully fledged syntactic structure**. Technically, this is implemented as phrasal lexicalisation, where morphemes realise a phrasal constituent containing multiple terminals.

Let us start off by elaborating on the first issue, the abstractness of features as compared to morphemes. To appreciate this difference, consider the fact that English has (at least) three different morphemes to mark the comparative of an adjective, as shown in (18):

(18)

POS	CMPR	ALLOMORPH
tall	tall-er	(-er)
bad	worse-Ø	(-Ø/suppletion)
intelligent	more intelligent	(more)

Clearly, however, there is also a level at which these three different suffixes are different instantiations of what is underlyingly the same thing. The underlying unity manifests itself both semantically and syntactically. Semantically, there is no meaning difference between the three different ways of marking the comparative in English (abstracting away from the conceptual differences that are linked to the open-class roots). Syntactically, whatever syntactic processes are sensitive to the presence of the comparative feature make no distinctions conditioned by the particular allomorph marking the comparative (e.g., the availability of a *than*-phrase, or differential measures like *by three inches* / *by a great margin*).

The English situation is a *prima facie* case of allomorphy, in which there is an abstract feature CMPR, which may be realised by different morphemes.³ This distinction between an abstract invariant feature and its variable realisations can not only be observed within a single language, as in the English case just illustrated. It is also highly relevant when we make crosslinguistic comparisons, where the feature CMPR is present across a wide range of languages (perhaps even universally), despite there being a multitude of variable morphemes realising it.

³ We want to point out that we use the concept of a morpheme in its traditional sense (the smallest segmentable meaningful unit). This is different from the Distributed Morphology framework, where morpheme refers to a syntactic terminal (containing an abstract feature or feature bundle), whereas its realisation is referred to as an exponent.

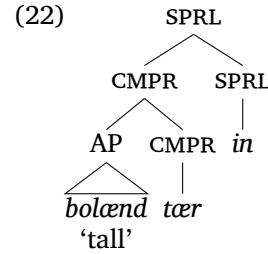
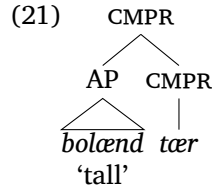
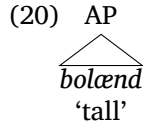
The fact that in English, there is only a loose formal resemblance between the different allomorphs means that it makes little sense to promote one of them to the underlying form, from which the others are derived by phonological rules. Rather, the underlying entity that ties the allomorphs together is devoid of a phonology: it is an abstract feature CMPR. This state of affairs also motivates late insertion: what is syntactically relevant is the feature CMPR, not any of its instantiations. This fact follows directly if what is present in the syntax is the feature and not the form. The form is inserted after the syntax has run its course, i.e. post-syntactically.

Let us now introduce the second idea, which is that individual features are organised in hierarchical structures. In order to show this, we turn our attention to the marking of the positive, comparative and superlative degree in Persian, see (19). The relevant fact is that the comparative in (19b) is based on the form of the positive (19a), and the superlative (19c) is built on top of the comparative.

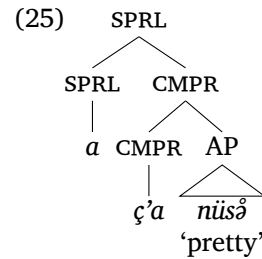
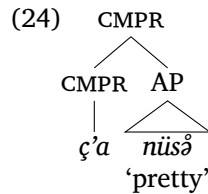
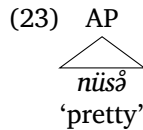
- (19) a. Reza **bolænd** = e
 Reza tall is
 ‘Reza is tall.’
- b. Reza æz Javad **bolænd -tær** = e
 Reza from Javad tall CMPR is
 ‘Reza is taller than Javad.’
- c. Reza æz hæme **bolænd -tær -in** = e
 Reza from all tall CMPR SPRL is
 ‘Reza is the tallest of all.’

In his typological work, Bobaljik (2012) found quite a number of languages like Persian, where the superlative is built on top of the comparative. At the same time, there do not seem to be any languages where the containment relations would be the other way round, with the superlative contained in the comparative.

In order to capture this generalisation (and also for other reasons), Bobaljik (2012) proposed that the comparative and superlative are not two different values of a single degree head; rather, the SPRL is an independent feature that is added on top of the comparative. The three degrees then correspond to the structures in (20)-(22), where each feature is an independent syntactic terminal. Specifically, the structure of the superlative (22) literally contains the comparative structure (21), and the comparative in turn contains the positive (20).



The structure where CMPR and SPRL are distinct syntactic terminals makes empirical predictions about the range of possible orders of CMPR and SPRL with respect to the base. In recent work on the topic of linearisation (Cinque 2005; Abels & Neeleman 2012; Medeiros 2018), it has been argued that a hierarchical structure like the one in (22) makes strongest predictions when both morphemes are found on the same side of the head. In such examples, the structures predict a strong (possibly absolute) tendency for orders where the CMPR is closer to the root than SPRL. Persian, which we saw in (22), illustrates this tendency in the position after the root. Ubykh, as shown in (23), manifests the same ordering, with CMPR closer to the root than SPRL, but in a position before the root (data from Bobaljik 2012, 31).



Caha (2017b, 889ff) discusses the languages in Bobaljik's sample from the perspective of linear ordering and argues that the attested and unattested morpheme orders are precisely as expected if Bobaljik's layered representations like the one in (25) are adopted. This provides evidence for not only postulating more features in the superlative in relationship to the comparative, but, indeed, evidence for proposing that each of the features occupies its own independent terminal, with its own position in the syntactic hierarchy.

In this book, as in Nanosyntax in general, we use this finding as a rule of thumb: whenever the empirical evidence suggests that we should split a single head (e.g., DEG) into multiple features (CMPR, SPRL), we will always understand each such feature to be an independent head. We will likewise assume that these heads are present universally, with language variation restricted to the way in which these structures are realised by language-specific lexical items. That way, we aim to reduce the variation among languages not

to the abstract features present in their structure, but rather to the component of grammar where variation is the norm, namely the lexicon (Borer 1984).

This takes us directly to the third insight of the Nanosyntax framework, which is the nature of portmanteau morphemes. Portmanteau morphemes (or cumulative morphemes) are morphemes that realise multiple grammatical features. For example, Latin is known to have portmanteau morphemes for number and case. However, in Nanosyntax, many more morphemes than traditionally thought end up classified as portmanteaus. For examples, consider the fact that in English, the marking of the superlative is monomorphemic (*-est*). However, under Bobaljik's decomposition, the superlative has at least two features. This means that *-est* (which realises these two features) is also a portmanteau.

The specific way how Nanosyntax models the fact that a single morpheme lexicalises multiple terminals is similar to other processes that affect multiple terminals. For instance, when multiple terminals move together (e.g., to the position left of the verb in V2 clauses), this is modelled as phrasal movement: the terminals move together because they are all contained in a larger unit that moves. When ellipsis targets multiple terminals (e.g., a verb and its object), this is understood as an ellipsis of a constituent that contains these terminals (i.e., a VP ellipsis). In a similar way, Nanosyntax proposes that when multiple terminals are lexicalised together, this is because lexicalisation targets a phrasal node where all the terminals are contained. As a result, the grouping of multiple terminals for the purpose of lexicalisation does not depend on some arbitrary definition, but relies on a grouping mechanism (constituency) that independently exists in grammar and, therefore, represents the zero hypothesis.

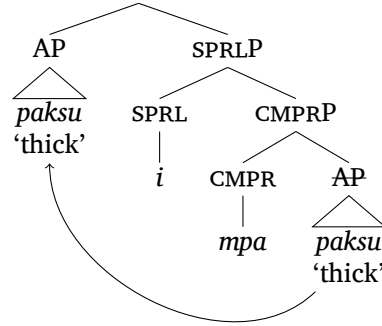
To see how this works in practice, let us return to the English *-est* in forms like *smart-est*. We have already mentioned that if we take the evidence provided by Bobaljik (2012) seriously, we are led to analyse *-est* as a portmanteau morpheme that realises a constituent containing CMPR and SPRL.

In order to see how this constituent arises, it is instructive to consider the linear arrangement of degree morphemes in Finnish. In this language, as Bobaljik (2012, 56) observes, the SPRL marker is closer to the root than CMPR. The relevant example is shown in (26c), and it features the superlative *-i* in between the root *paksu* 'thick' and CMPR *-mpa*.

- (26) a. *paksu -un*
 thick -ILL
 b. *paksu -mpa -an*
 thick -CMPR -ILL
 c. *paksu -i -mpa -an*
 thick -SPRL -CMPR -ILL

In Cinque (2005) (also in Abels & Neeleman 2012), this type of example is derived by moving the root leftwards by phrasal movement, which allows it to cross CMPR and SPRL without inverting their order, as in (27).⁴

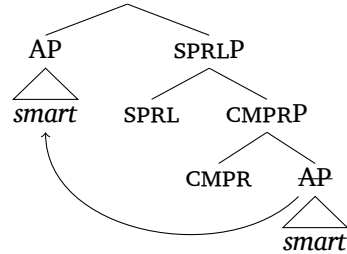
(27)



The reliance on phrasal movement is driven by theoretical simplicity: we know that the grammar contains phrasal movement, and given that phrasal movement has the right properties to generate these orders (as argued in Cinque 2005), phrasal movement seems to be the default choice.

When dealing with portmanteau morphology, we assume that an analogous movement takes place in the English superlative as in Finnish, yielding a structure where the root evacuates from inside SPRLP, see (28). This is exactly the same movement that is independently needed in (27).

(28)



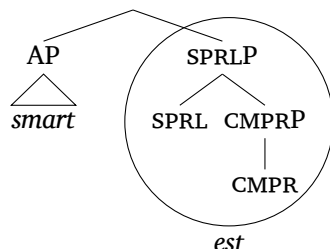
The structure resulting from the movement is in (29).⁵ With AP evacuated, CMPR and SPRL form a (remnant) constituent, SPRLP. This constituent is targeted by lexicalisation. To our mind, this idea is theoretically attractive because it leverages independently existing grammatical processes (move-

⁴ We do not label the node created by the movement of the AP, but nothing hinges on this at this point. Generally, we only label nodes that arise by merging meaningful elements and not label nodes that arise by movements which do not contribute to interpretation.

⁵ We leave out the trace of the moved AP, for reasons that we return to below.

ment and constituency) to deliver the required grouping of CMPR and SPRL for the purpose of lexicalisation.

(29)



In general, the Nanosyntactic analysis of portmanteau morphology thus tries to maximally leverage the existing tools (syntactic hierarchies, phrasal movement, constituency) and develops a model of portmanteau realisation that essentially only adds the possibility to link phrasal nodes to phonological interpretation, which is what portmanteau morphology boils down to on this approach.

Summarising, features in Nanosyntax are both smaller and more abstract than morphemes. They are the language-invariant atoms of language, and belong to a cognitive system that is independent of the realisation of these features in individual languages. The atoms compose through the application of Merge, forming larger constituents. Since in the nanosyntactic architecture, syntax is the only compositional system, anything that is internally complex has to be the product of a syntactic derivation. The visible units of language, like morphemes, words, phrases, etc. correspond to these larger constituents created by Merge, i.e. they are the products of a syntactic derivation. Since the visible units of languages are typically internally complex and consist of several features (i.e. syntactic constituents created by Merge), these visible units must correspond to phrase-level constituents. The concept of phrasal lexicalisation means that morphemes lexicalise these constituents.

1.2.2 The lexicon and language variation

In this section, we lay the ground for capturing language variation in Nanosyntax. We begin by highlighting the areas of grammar where we believe no variation exists. The first place with no variation are the features. In Nanosyntax, similarly to Cartography, the underlying features and their hierarchies are assumed to be universal, i.e., invariant across languages. Similarly to mainstream Principles & Parameters models, such as Minimalism, Nanosyntax also maintains the idea that the operations of the narrow syntax are the same across languages, and therefore, that language variation is con-

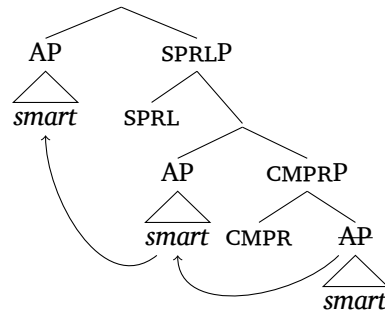
fined to the lexicon, the one place in the grammar where the evidence for variation is overwhelming. By its desire to reduce all variation to the lexicon, Nanosyntax aligns with the so-called Borer-Chomsky conjecture (Borer 1984, 2; Baker 2008, 353).

However, there are two differences in how the Borer-Chomsky conjecture is implemented in standard minimalist models and in Nanosyntax. In standard Minimalism, as well as in models such as Distributed Morphology (Halle & Marantz 1993) or the Exoskeletal model (Borer 2005), lexical items are essentially arbitrary bundles of features, and they are language-specific. Also, these language specific items are the input to Merge. As a result, languages may differ in the bundles that are the input to Merge (see e.g. Bobaljik & Thràinsson 1998 on the Icelandic-English difference in verb movement). This makes it easy for lexical items to influence how Merge operates, either directly via features related to selection, or indirectly via the checking of uninterpretable features. As a result of Merge interacting with different atoms in different languages, different structures arise.

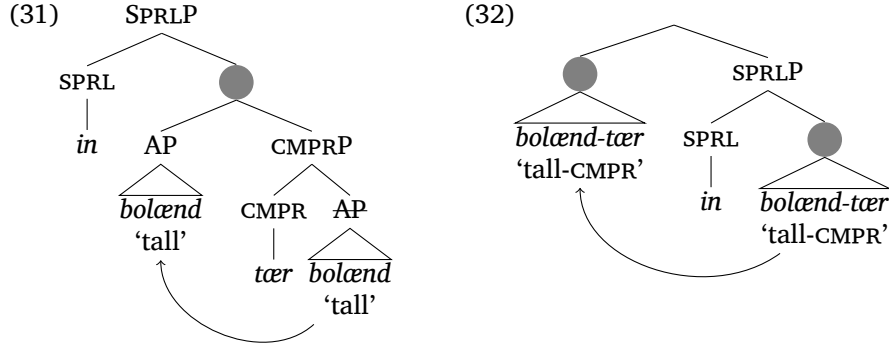
This is different from Nanosyntax. As highlighted in the previous section, one of the core claims in Nanosyntax is that the atoms of Merge are individual features (like CMPR and SPRL). Therefore, there are no language-specific feature bundles preceding syntax: all languages start from the same atoms. The only place where we find a language-specific lexicon in Nanosyntax is after syntax. This leads to an apparent paradox: on the one hand, Nanosyntax portrays the lexicon as the only source of linguistic variation; yet at the same time, the lexicon seems to come into the picture too late to influence syntactic derivations and thus be responsible for crosslinguistic variation.

To see the conundrum in concrete terms, consider again the difference between English and Persian. In English, we have portmanteau marking of the superlative (*tall-est*), while in Persian, we have agglutinative marking (*bolænd-tær-in* ‘tall-CMPR-SPRL’). The specific challenge is that in our account, the two languages must have slightly different syntactic structures, and these must somehow derive from different lexical entries. For English, we have proposed that the AP moves outside of SPRLP without any pied-piping, as in (30), where we show the idea that the AP moves cyclically, rather than in one single step.

(30)



This structure cannot be correct for Persian, because in Persian, the CMPR marker is closer to the root than the SPRL. Therefore, we need to invoke the so-called roll-up derivation, where the AP first moves across CMPR, creating a higher unlabelled node above CMPRP, as in (31), where the relevant node is circled. Subsequently, as shown in (32), movement of AP pied-pipes this unlabelled node across SPRL.



Pictured like this, the only difference between English and Persian is whether the second movement step pied-pipes the circled node along (Persian) or not (English). Pied-piping is a well-known parameter of variation (Cinque 2005), and reducing the difference between Persian and English to pied-piping is, therefore, a step forward. However, the question is how the differences between English and Persian post-syntactic lexical entries influence (or trigger) these movements, given that they only lexicalise the output of such derivations.

The solution adopted in the Nanosyntax literature (see Starke 2014, 2018) is that syntax actually produces various structures blindly, and ships them to the interface for externalisation. However, not all such representations can be successfully externalised (or ‘lexicalised’). In concrete terms, the idea is that the Persian lexical entries are stored in such a format that they would not be able to lexicalise CMPR and SPRL features in any other structure than that shown in (31) and (32). Similarly, the English lexical entries are such that they could not lexicalise the Persian derivation or any other derivation but the one appropriate for English. That is why the derivations proceed differently in different languages.

Thinking further along these lines, it seems that for each feature we add (e.g., CMPR), there are the following options. Either the lexical head of the extended projection (i.e., the root) moves across CMPR (as in Persian or English) or it does not (as in Ubykh). When movement happens, it is either just the root that moves (as in English, where only the AP moves across SPRL), or pied-piping takes place (as in Persian). Since these three options must be contemplated for every feature, the number of potential derivations

becomes large (if F is the number of features, the number of derivations is F^3).

In order to cut down the exponential growth of derivational options, the Nanosyntax model always works only with a single derivational path, which is cyclically checked against the lexical resources of particular languages. Always when a new feature F is introduced, there are three derivational options (stay, move, move + pied-pipe) that are tried in that order, and after each step, the derivation is checked against the lexical resources as to whether it can be externalised. If matching succeeds, this means that the structure can be externalised, and a new feature may be introduced. If the structure cannot be externalised, a different option is tried. This sequence of steps is embodied in the so-called lexicalisation algorithm.

- (33) Lexicalisation Algorithm (based on De Clercq et al. 2025)
- a. Merge F and lexicalise FP
 - b. If (a) fails, evacuate the closest labelled node containing the lowest feature of the *fseq*
 - c. If (b) fails, evacuate the immediately dominating constituent and lexicalise (recursive)

Notice how clause (b) directly corresponds to the type of movement that we discussed for English above, while clause (c) formalises the pied piping or roll-up type of movement that we needed for Persian. In English, what is evacuated out of SPRLP in (30) is the closest constituent to SPRLP that contains the lexical head, i.e., the AP, which represents the lowest feature in the functional sequence. After this evacuation, SPRL and CMPR form a constituent, and *-est* will be inserted. This step, however, fails in the case of Persian, since Persian has no such portmanteau. Therefore, a higher node is pied-piped in (32). We shall illustrate the working of this algorithm when discussing concrete derivations below.

Phrasing this in terms of cycles, we can say that nanosyntax adopts a version of a cyclic lexicalisation theory, in which every Merge step is a cycle in the sense that the syntactic derivation interfaces with the lexicon after each Merge step. Every time Merge adds a feature, that procedure makes sure that the feature can be ‘externalised,’ i.e., mapped onto PF by a matching lexical item. If a match is found, the derivation proceeds to the next Merge step. If not, the lexicon sends the structure back to the syntax for further manipulation, i.e., movements. These interactions are schematised in Figure 1.1.

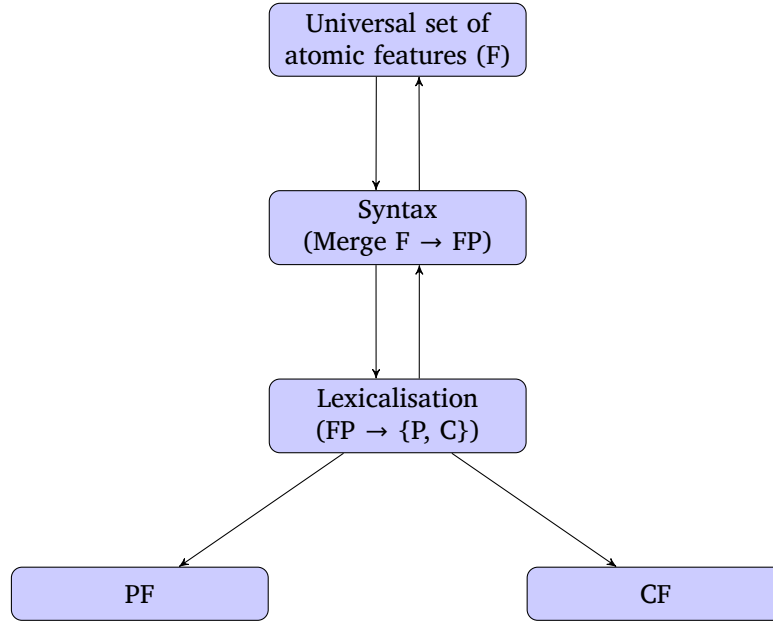


Fig. 1.1 Nanosyntactic model of grammar

We can see here that the model starts from single features. These are the input to the syntax, i.e. Merge. The first operation of Merge takes two features F_0 and F_1 and creates a syntactic phrase F_1P . This constituent is then checked against the lexicon for a lexical item that can lexicalise it. The postsyntactic lexicon is represented in the diagram by the formula $FP \rightarrow \{P, C\}$, since it links syntactic constituents (FPs) to phonology (P) and/or concepts (C). This allows FPs to be mapped onto phonological form (PF) and conceptual form (CF).

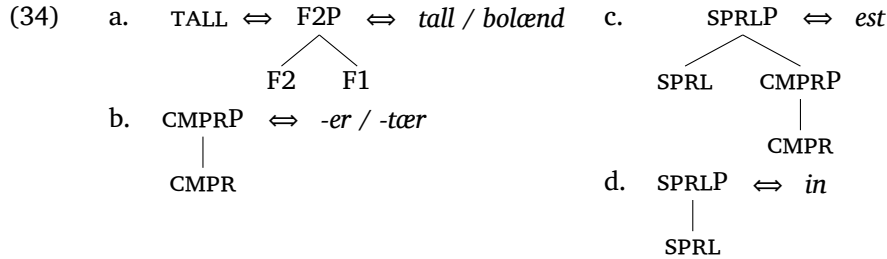
If lexicalisation of F_1P is successful, syntax selects the next feature F_2 from the list of atomic features, merging it with the existing F_1P to create F_2P , which is again checked against the lexicon. If lexicalisation is not successful, i.e. if the lexicon does not contain a lexical item matching the structure created by the syntax, FP gets sent back to the syntax. In syntax, the phrase marker undergoes a predetermined and universal set of rescue operations (evacuation movements out of FP), specified in the Lexicalisation Algorithm (33). These modify the syntactic structure by applying a particular movement operation, changing the non-externalisable phrase marker. These syntactic operations have the potential to create a structure that is externalisable, i.e., one where FP (after evacuation) is matched by some item in the lexicon. Therefore, after each such rescue operation, the lexicon is again consulted, until FP is successfully lexicalised.

The bidirectional arrows in the diagram connecting the features, syntax and the lexicon represent the cyclical nature of the model, where after features are selected for Merge, Merge applies, and the lexicon is consulted. Then, depending on the outcome, a syntactic node FP can be sent back to the syntax multiple times within the same cycle. Finally, after each successfully completed cycle, the syntax has renewed access to the list of atomic features.

1.3 Deriving agglutination and portmanteau realisation

In this section, we illustrate the functioning of the system by showing how the empirical difference between the Persian and the English superlative are derived from a universal set of features, a universal mode of interaction between the syntax and the lexicon, and with the difference between the two languages being entirely situated in their different postsyntactic lexicons. Readers who are well-versed in nanosyntactic derivations may well read this section diagonally, or skip it altogether.

In (34), we provide the set of lexical entries: (34a) gives the lexical entry for the roots *tall* and *bolænd* ‘tall.’ We show these in a single representation because the difference between English and Persian only concerns the different phonological form; the lexical trees of the roots are identical across the two languages.

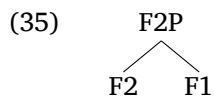


Since both of them represent the positive degree, we attribute to them the same features. The specific features are labelled as F1 and F2, which is a stand-in for specific features that we introduce in Section 1.9. Both roots are also linked to the concept of tallness (or ‘length on the vertical plane’), marked by small caps (TALL), but have a different phonology.

In both languages, the comparative is formed by a morpheme that is suffixed to the root. Because of the parallel behaviour, the entry for the comparative morpheme (*-er/-tær*) is the same in both languages (34b). The entry lexicalises a phrasal node (CMPRP) with a single daughter CMPR. The only difference between the languages rests in the entry for the superlative. The English superlative is a portmanteau (realising both CMPR and SPRL) (34c).

The Persian superlative *-in* is an agglutinative marker, realising only SPRL (34d).

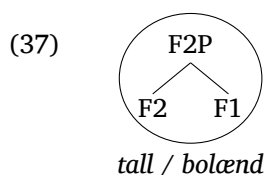
The derivation starts by merging F1 and F2, producing F2P, see (35). Following the lexicalisation algorithm (33), we now have to check the lexicon for a lexical item with a matching constituent.



In its simplest form, matching involves identity between a syntactic structure and a lexical tree.

- (36) *Condition on matching* (De Clercq et al., 2025)
 A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

Whenever the lexical entry contains a constituent that is identical to the one produced by syntax, matching obtains. This identity requirement is met in both languages by the lexical tree for the roots: *tall* in English and *bolænd* in Persian (recall (34a)). We indicate successful matching with a circle around the relevant constituent, as follows:



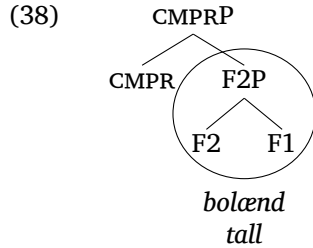
It will become relevant that the identity requirement is satisfied whenever the lexical entry contains a constituent identical to the syntactic structure, even when the lexical entry contains additional structure. The reasoning that leads to that conclusion is the following. Suppose that XP is a constituent stored in the lexicon. Suppose further that XP contains YP. It follows that YP is a constituent stored in the lexicon. Therefore, identity between syntactic and lexical constituents obtains even when the relevant lexical constituent is embedded within a larger structure.⁶

⁶ Because of this, the Condition on Matching used to be called the Superset Principle in earlier works on Nanosyntax, see (i).

- (i) *The Superset Principle* (Starke, 2009)
 A lexically stored constituent L matches a syntactic phrase S iff L contains the syntactic phrase dominated by S as a subtree.

However, is not a separate principle from the Condition on Matching in (36), but rather a different way of encoding it. Since identity may be considered a more fundamental relation than containment, we will use (36) henceforth.

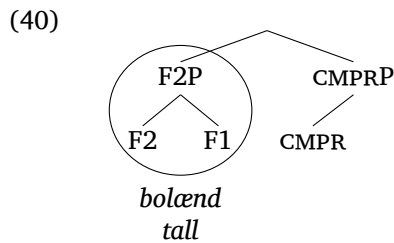
Let us now return to the derivation in (36). Since the lexicon contains a matching item, the derivation continues by merging the feature CMPR on top of (36), producing (38), and the lexicon is consulted again.



At this point CMPRP no longer finds a match in the lexical item (34a). Since there is no matching lexical item, the object constructed by syntax cannot be mapped onto PF and ‘crashes’ at the interface. Therefore, the derivation is sent back to the syntax, which will modify the structure by evacuating various constituents out of CMPRP, hoping that after material is out of it, CMPRP will find a match and it will be possible to map it onto PF. The evacuation movements apply in the order as described in the lexicalisation algorithm (33), repeated for convenience in (39).

- (39) Lexicalisation Algorithm
- Merge F and lexicalise FP
 - If (a) fails, move the closest labelled node containing the lowest feature of the *fseq*
 - If (b) fails, move the immediately dominating constituent and lexicalise (recursive)

Since the first option (i.e., non-movement given in (39a)) failed, we apply the second option, i.e. we move the closest labelled constituent that contains the lowest feature of the *fseq* (this being the feature F1). The relevant constituent closest to the root node is F2P, and its movement creates (40).

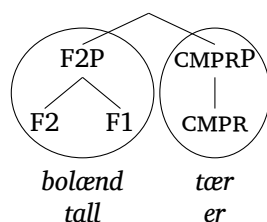


Before moving on, we note two technical characteristics of the trees created by this type of evacuation movement in nanosyntax. The first is that the trace of F2P in the original position is eliminated, and so is the node dominating the trace. This convention makes it easier to evaluate matching. The second

is that the root node of the tree after movement is left unlabelled. Such unlabelled nodes can by definition never be targeted by step (40b) of the Lexicalisation Algorithm, but can only be moved as a result of pied piping (40c). The relevance of this will become clear as we proceed.

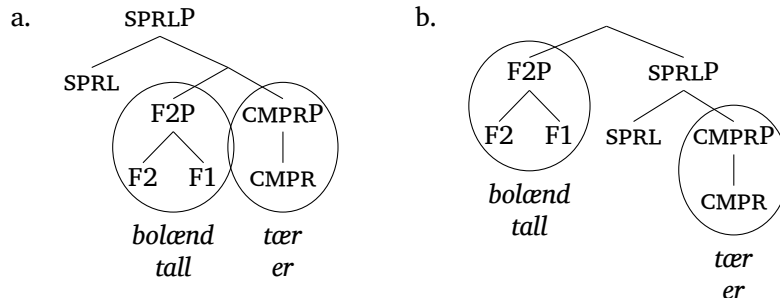
Once (40) is created, the lexicon is consulted again. Now the right hand part of the tree finds a match in the comparative markers *-er/-tær* (recall (34b)), so the CMPRP is successfully lexicalised:⁷

(41)



Since the CMPRP in (41) finds a match, there is a way to map the derivation onto PF, and the derivation may continue by merging the next feature in the *fseq*. The next feature to merge is SPRL, which yields (42a), but no lexical item in the lexicon provides a match for SPRLP. We therefore resort to (39b), which instructs us to move the closest labelled node containing the lowest feature. This is again the node F2P: since the sister node of SPRL is unlabelled, it is not the target of movement at this stage. Moving F2P across SPRL results in (42b).

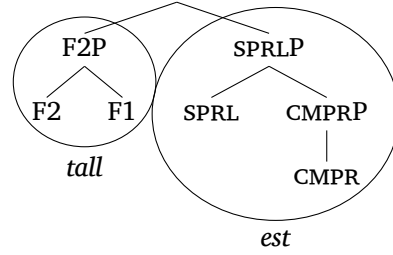
(42)



Now the SPRLP on the right in (42b) is a match for the English superlative marker *-est* (recall (34c) above).

⁷ The CMPRP in (40) is also contained in the entry for *-est* given in (34c). Therefore, *-est* is a match here. The morpheme *-est* does not appear because it loses the competition against *-er*, which is a better match. We come back to this in the following section.

(43)



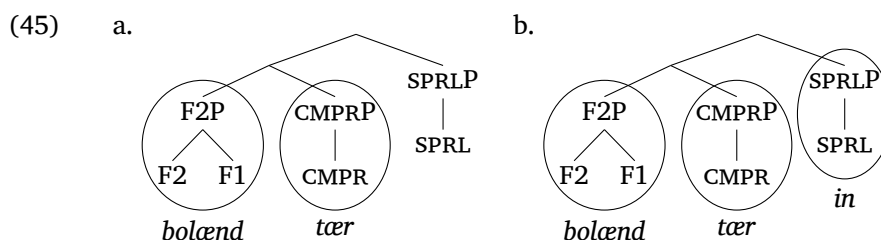
Observe that in (43), the lexicalisation of SPRLP replaces the earlier lexicalisation at the level of CMPRP, which we have indicated in the tree in (43) by removing the circle around CMPRP. Another case where this happens is where a suppletive root (like *worse*) replaces an earlier lexicalisation (*bad*) (see Sect. 1.6.1 and 1.7 for more discussion of suppletion). This replacement is referred to as Cyclic Override in the Nanosyntax literature:

(44) Cyclic Override

A lexicalisation of a higher node overrides the lexicalisation of all nodes contained in it.

Cyclic override is not an independent principle, but a consequence of the lexicalisation algorithm. As an effect of the algorithm, matching always targets higher and higher FPs as they get created, overriding previous matches. To understand this principle, it is important to keep in mind that matching does not lead to an immediate pronunciation; matching only serves the purpose of filtering out derivations that cannot be externalised. The actual pronunciation only happens at the very end of the derivation. Also, the principle does not imply that lexicalisations at earlier cycles get discarded. In fact, as we shall see below, in cases where the derivation returns to earlier cycles, these earlier lexicalisations may become relevant again (see Sect. 1.8 for discussion of backtracking derivations). The principle should rather be understood as reflecting what is the lexicalisation at the current cycle (the one matching the highest FP).

At this point, we have successfully derived the English superlative. Let us now turn to what happens in Persian. In this language, the lexicalisation of the SPRLP in (43) fails, since the Persian lexicon (unlike the English one) contains no matching lexical entry for the SPRLP in (43). In order to lexicalise SPRL in Persian, we have to proceed to step (39c) of the Lexicalisation Algorithm. Therefore, in Persian, the movement step given in (43) is undone, and the derivation goes back to (42a). Taking this structure as the starting point, syntax tries pied piping the immediately dominating node, leading to a rollup movement. In this step, syntax thus moves the complement of SPRL, yielding the structure in (45a). Matching is successful, as shown in (45b), since the SPRLP is identical to the lexical entry of *in*, recall (34d).



It is convenient to depict the results of such derivations in the form of a lexicalisation table, such as the one given in (46). In the table, yellow marks suffixes that are specified for just SPRL, cyan marks suffixes specified for just CMPR, and red codes suffixes that lexicalise both CMPR + SPRL. The roots are in grey.

(46)

LANGUAGE		DEGREE	F1	F2	CMPR	SPRL
ENGLISH	POS		<i>tall</i>			
	CMPR		<i>tall</i>		<i>er</i>	
	SPRL		<i>tall</i>			<i>-est</i>
PERSIAN	POS		<i>bolænd</i>			
	CMPR		<i>bolænd</i>		<i>tær</i>	
	SPRL		<i>bolænd</i>		<i>tær</i>	<i>in</i>

The top row gives all the relevant features, and each row then gives a particular form, starting with the English triplet positive-comparative-superlative, and the Persian one below it. The absence of a feature in a given form is marked by a black cell (e.g., the positive lacks both CMPR and SPRL). When a morpheme lexicalises a particular set of features, it is placed below those features in the table.

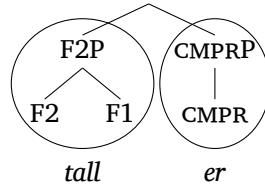
To summarise, this section provided a concrete example of how postsyntactic lexical entries influence derivations. The idea is that syntax generates structures and ships them cyclically to the interface. When the structure can be lexicalised (externalised, mapped onto PF), external merge may continue adding features. However, if there is no matching lexical item, rescue movements must take place. Cross-linguistic differences arise as a result of lexical differences: one language may simply lack the resources to lexicalise a structure which the other language successfully lexicalises.

1.4 Syncretism

In this section, we investigate a case of formal identity between the realisation of distinct meanings, standardly referred to as syncretism. The empirical discussion uses examples from Scottish Gaelic and Italian.

In order to see how syncretism arises in Nanosyntax, let us come back to the structure of the English comparative, repeated for convenience in (47) (recall (41)).

(47)



The new aspect of this derivation that we want to highlight is that there are actually two matching items in the English lexicon for the realisation of CMPRP: namely, both *-er* and *-est* are a match. To see that, consider again their entries in (48a,b), repeated from (34b) and (34c) respectively. The fact that both match the CMPRP in (47) is highlighted in the entries by bold red: the bold red constituent is identical to the CMPRP in (47).

(48) a. **CMPRP** $\Leftrightarrow$ *er*
 |
 CMPR

b. SPRLP $\Leftrightarrow$ *est*
 |
 SPRL **CMPRP**
 |
 CMPR

As a consequence, the CMPRP in (47) can be lexicalised by either (48a) or (48b), since both match. Obviously, in English, we want to avoid it being lexicalised by the superlative suffix (48b). This is the work done by the Elsewhere Principle, which designates a winner in cases where two (or more) lexical items compete with each other for the lexicalisation of a syntactic phrase. A general formulation of the *Elsewhere Principle* is given by Kiparsky (1973):

(49) *The Elsewhere Principle*

In case two rules, R_1 and R_2 , can apply in an environment E , R_1 takes precedence over R_2 if it applies in a proper subset of environments compared to R_2 .

Applied to the case at hand, bigger lexical items apply in more environments than smaller ones, given that they contain more subconstituents. This means that the smaller, better fitting, lexical entry will win against the bigger one,

which has more ‘superfluous’ features. In this way, the Elsewhere Principle ensures that in the English comparative, the suffix *er* wins against *est*, both being candidates in virtue of the Superset Principle.

However, this reasoning opens the possibility that a language lacks a dedicated comparative suffix like the one in (48a), and it only has an entry like the English (48b). In such a language, the derivation would proceed as in English, and the only difference would be that the comparative would be marked the same as the superlative. We believe that such a pattern is found in a number of languages discussed by Bobaljik (2012, 53-55), including Scottish Gaelic, Italian, Gulf Arabic, Greek, Maltese, Livonian, Irish, Manx and others. The example in (50) illustrates this pattern with an example from Scottish Gaelic.

- (50) a. an gille òg
the boy young
‘the young boy’
b. an gille as òig-e
the boy AS young-CMPR
‘the younger/youngest boy’ (Scottish Gaelic, ?, ex. 14, 50)

The ambiguous comparative/superlative form in (50b) is formed by the addition of schwa (written as *-e*) to the positive, compare (50a). ? notes that the addition of the schwa triggers consonant mutations at the right edge of the base as well as certain vocalic changes inside it, which we do not discuss here for reasons of space.⁸

In some languages, it is possible to distinguish the comparative and the superlative due to the syntactic context, notably due to the presence of the definite article in the superlative, as illustrated in (51c) for French (Kayne 2008, 297,315).

- (51) a. le livre court
the book short
‘the short book’
b. le seul livre plus court que ...
the only book more short than ...
‘the only book shorter than ...’
c. le livre *(le) plus court
the book the most short
‘the shortest book’

An interesting observation is that the definite determiner in (52c) cannot be absent for the superlative reading to arise; its absence yields the comparative, see (52b). One could therefore speculate that French may not exhibit a true syncretism between the comparative and the superlative, and that the pre-adjectival definite article takes part in realising SPRL.

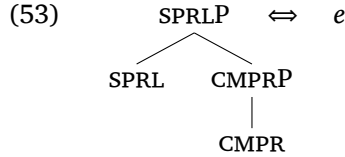
⁸ The morpheme *as*, which precedes the adjective, is a reduced-relative-clause marker.

However, this does not seem to be the case in all languages, such as Scottish Gaelic in (50), or in the closely related Italian. As Kayne (2008) notes, Italian actually disallows the definite article before the adjective, see (52a). In this language, the form *più* thus appears to be genuinely ambiguous, as demonstrated in (52b), which features a clear comparative reading due to the presence of the comparative standard.

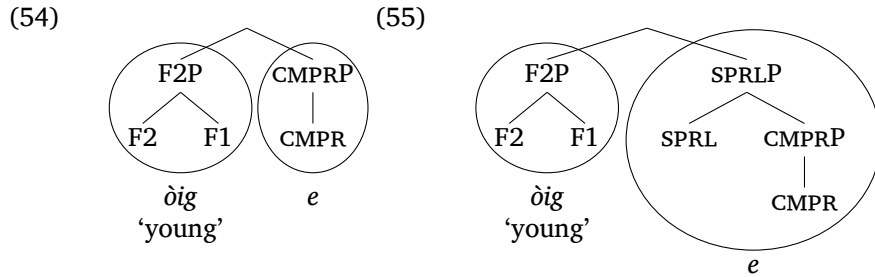
- (52) a. il libro (*il) *più* corto
 the book the more short
 ‘the shortest book’ (Italian, Kayne 2008, 301)
 b. la ragazza *più* bella di Maria
 the girl more beautiful than Maria
 ‘the girl more beautiful than Maria’ (Loccioni 2018, 100)

Summarising, Scottish Gaelic and Italian seem to instantiate a pattern where the comparative and superlative are marked by one and the same marker, namely by *-e* and *più*, respectively.

As explained above, we can capture the difference between English and Gaelic-type languages by the inventory of lexical items in the postsyntactic lexicon. Whereas English has separate markers for the comparative and the superlative suffix (recall (48a) and (48b)), Scottish Gaelic or Italian lack a dedicated marker for the comparative analogous to *-er*. The lexical entry for the Scottish Gaelic *-e* is in (53).



This affix will combine with the root in the same manner as explained above for English, and it will realise both the comparative and superlative structures as given in (54) and (55) respectively.



As a way of summarising the discussion, we depict the Scottish Gaelic pattern in the lexicalisation table (56) for comparison.

(56)

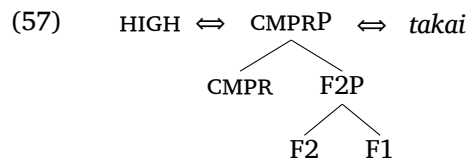
LANGUAGE	DEGREE	F1	F2	CMPR	SPRL	
ENGLISH	POS	<i>tall</i>				
	CMPR	<i>tall</i>		<i>er</i>		
	SPRL	<i>tall</i>			<i>est</i>	
PERSIAN	POS	<i>bolænd</i>				'tall'
	CMPR	<i>bolænd</i>		<i>tær</i>		
	SPRL	<i>bolænd</i>		<i>tær</i>	<i>in</i>	
GAELIC	POS	<i>òg</i>				'young'
	CMPR	<i>òig</i>		<i>e</i>		
	SPRL	<i>òig</i>		<i>e</i>		

The syncretic pattern of Gaelic thus follows from the basic principles of Nanosyntax as laid out in Sect. 1.2, in particular the Superset Principle, as well as the structure of the Gaelic lexicon, which, unlike the English one, lacks a competing lexical item that only lexicalises the CMPR head. In English, the presence of *-er* blocks the superlative suffix from lexicalising the comparative. In Gaelic, since there is no competitor, the syncretism between the comparative and superlative suffixes arises.

1.5 Root size

This section introduces the idea that languages may not only vary in the lexical entries associated to affixes, but also regarding the constituents that are associated to root morphemes. The section considers the expression of the positive, comparative and superlative degrees in Japanese and Armenian, arguing that the patterns of marking can be captured under the proposal that root morphemes may realise the CMPR and SPRL features.

The core idea is illustrated in (57), using the Japanese adjective *takai* 'high' as an example. According to this idea, the root morpheme *takai* may lexicalise the whole CMPRP.



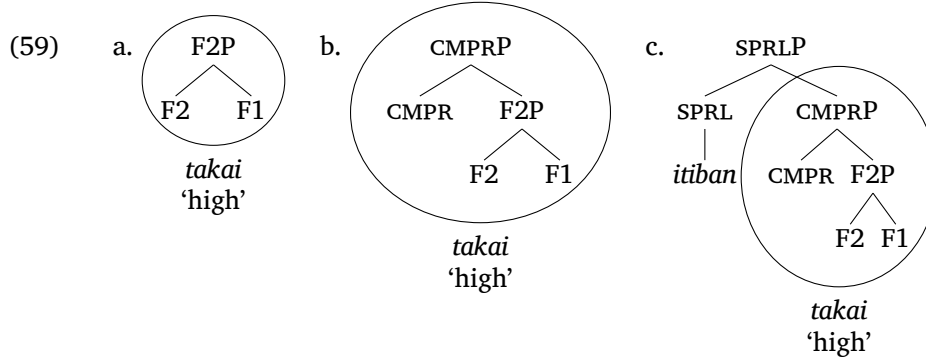
If a root morpheme has such an entry, the Superset Principle allows the root alone to lexicalise both the positive (F2P) and the comparative (CMPRP). We therefore expect that such roots have no additional degree marker in the positive and in the comparative. However, the root cannot lexicalise the

superlative. We therefore expect that the root must be accompanied by a degree marker in the superlative.

This is indeed the situation in Japanese, as shown in (58) (Arii 2011, 104, cf. Bobaljik 2012, 21-22).

- (58) a. Kono yama-wa takai
 this mountain-TOP high
 ‘This mountain is high.’
 b. Kono yama-wa ano yama -yori takai
 this mountain-TOP that mountain -than high
 ‘This mountain is higher than that mountain.’
 c. John-ga itiban takai yama-ni nobot-ta
 John-NOM most high mountain-to climb-PAST
 ‘John climbed the tallest mountain.’

The sentences (58a) and (58b) show that both the positive and the comparative degree in Japanese correspond to the bare root.⁹ The superlative in (58c) must include an additional marker *itiban* (literally ‘number one’).¹⁰ This pattern follows if Japanese adjectives are of size CMPRP, as proposed in (58). The lexicalisations are then as shown in (59).



The lexicalisations of the positive and the comparative are trivial (see the discussion of syncretism above). The superlative must realise the SPRL feature, which cannot be lexicalised by the root. That is why *itiban* appears.¹¹

⁹ The comparative has an optional *motto* ‘more.’

¹⁰ An alternative marker *mottomo* ‘most’ is also possible instead of *itiban*.

¹¹ There is much to be said about prefixal realisation of functional heads (see Starke 2018), but we ignore this issue at this point because it would lead us too far afield. In general, we assume that prefixal markers correspond to complex constituents created in a separate workspace, utilising a clause of the lexicalisation algorithm that we have not introduced yet. In a nutshell, prefixes are currently assumed to be formed by merging F with an empty set, forming a constituent lexicalised by the pre-marker *itiban*. The full structure would then be as depicted in (i).

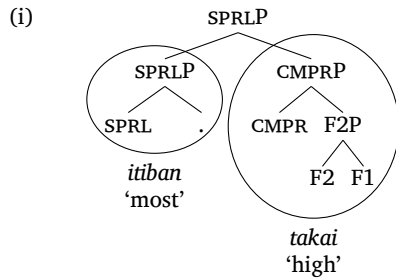
There are reasons to think that there are also languages with roots that can extend all the way up to SPRLP. In these languages, the adjectival root receives no additional marking in any of the degrees. An example of this type is provided from (Western) Armenian in (60) (Bobaljik 2012, 88; 91).

- (60) a. Artak-ə **partsrahasag e.**
 Artak-DEF tall be.3SG.PRS
 ‘Artak is tall.’
 b. Artak-ə Bagrat-e-n **partsrahasag e.**
 Artak-DEF Bagrat-ABL-DEF tall be.3SG.PRS
 ‘Artak is taller than Bagrat.’
 c. Artak-ə amen-e-n **partsrahasag e.**
 Artak-DEF all-ABL-DEF tall be.3SG.PRS
 ‘Artak is the tallest.’ (lit. ‘Artak is taller than all.’)

Comparing (60a) with (60b), we can see that Armenian signals the comparative reading only by including in the sentence the standard of comparison (‘than Bagrat’), which is marked with ablative case. The adjective is unchanged compared to the positive. In the superlative, the *than*-phrase includes a universal quantifier, which signals the superlative reading.

In our analysis, we follow the intuition that the presence of a (universally quantified) *than*-phrase in (60b-c) is not a trigger of the comparative degree reading, but rather a consequence of it. To highlight this intuition, we compare the example (60) to the Persian example (61) (repeated from (19) for convenience). Our point is that since the Persian examples in (61b-c) do not differ from the Armenian ones in (60b-c) in terms of the presence or absence of a *than*-phrase, this is not what we should focus on. Rather, the difference is only in the form of the adjective.

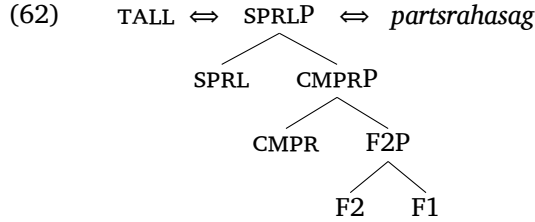
- (61) a. Reza **bolænd** = e
 Reza tall is
 ‘Reza is tall.’
 b. Reza æz Javad **bolænd -tær** = e
 Reza from Javad tall CMPR is



¹¹ The comparative has an optional *aveli* ‘more.’ Bobaljik (2012, 89) suggests that this may be a case of reinforcement.

- ‘Reza is taller than Javad.’
 c. Reza æz hæme bolænd -tær -in =e
 Reza from all tall CMPR SPRL is
 ‘Reza is the tallest of all.’

Following this intuition, we portray the difference between languages like Persian and Armenian in terms of root size. While Armenian roots can realise the whole SPRLP (and all its subconstituents), Persian roots cannot do so. This analysis is depicted in (62) for the Armenian adjective ‘tall.’ With such an entry, the adjective will appear bare in all three degrees.



The alternative analysis would be one where the Armenian degree constructions represent a form of implicit comparison, which relies on ‘the inherent vagueness of gradable adjectives to achieve comparative-like readings in certain contexts’ (Bobaljik, 2012, 91). Bobaljik argues against such an analysis for Armenian, on the basis of two pieces of evidence: the absence of an entailment to the positive (Kennedy, 2007), and the availability of measure phrases specifying the difference between the two individuals being compared (see also Beck et al. 2009). The first property concerns the fact that a sentence like (61b) does not entail the positive degree, i.e. it does not entail that Artak is tall (for instance if both Artak and Bagrat are short). This would be expected if (61b) were to involve some vague comparative-like use of the positive. Similarly, measure phrases that occur with the adjective in (61b) are interpreted as specifying the difference between the height of the two individuals being compared, and not the absolute height of the subject of the adjective, which they do in the positive degree.

- (63) Artak-ə Bagrat-e-n jergu santim partsrahasag e.
 Artak-DEF Bagrat-ABL-DEF two cm tall be.3SG.PRS
 ‘Artak is 2 cm taller than Bagrat.’

As pointed out in Bobaljik (2012, 20), more than half of the world’s languages belong to the two types discussed in this section, which means that the syncretism of the adjectival root between the positive and the comparative is common (see also Stassen 2013, §3).

Table (64) adds the two new language types discussed in this section into our lexicalisation table.

(64)

LANGUAGE	DEGREE	F1	F2	CMPR	SPRL	
ENGLISH	POS	<i>tall</i>				
	CMPR	<i>tall</i>		<i>er</i>		
	SPRL	<i>tall</i>		<i>est</i>		
PERSIAN	POS	<i>bolænd</i>				'tall'
	CMPR	<i>bolænd</i>		<i>tær</i>		
	SPRL	<i>bolænd</i>		<i>tær</i>	<i>in</i>	
GAELIC	POS	<i>òg</i>				'young'
	CMPR	<i>òig</i>		<i>e</i>		
	SPRL	<i>òig</i>		<i>e</i>		
JAPANESE	POS	<i>takai</i>				'high'
	CMPR	<i>takai</i>				
	SPRL	<i>takai</i>			<i>itiban</i>	
ARMENIAN	POS	<i>partsrahasag</i>				'tall'
	CMPR	<i>partsrahasag</i>				
	SPRL	<i>partsrahasag</i>				

1.6 Root-driven allomorphy of affixes

We now turn to the treatment of allomorphy in Nanosyntax. Rather than proposing a new mechanism (e.g., context-sensitive rules), the account leverages the tools that we already have, namely the idea of root size and a cyclic derivation. In this book, it is our goal to show that these tools (as described above) are fully capable of deriving all the patterns of allomorph distribution in Czech adjectives, without the need to invoke context-sensitive rules.

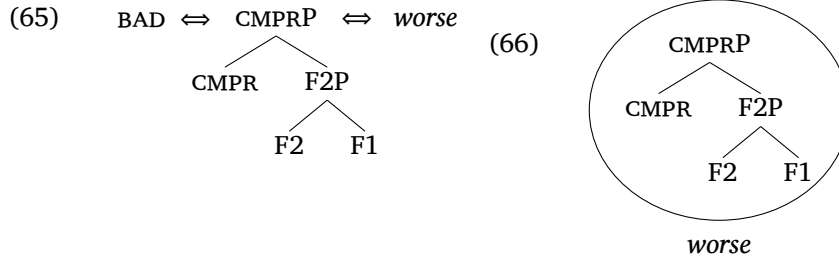
Purely descriptively, we distinguish here two different types of allomorphy depending on its directionality. One type of allomorphy extends from the affix to the root (or, more generally, from the outside of the word towards the root). This type of allomorphy will be discussed in Section 1.7. Another type of allomorphy extends from the root out. This type of allomorphy will be discussed in this section.

The section is structured as follows. In Section 1.6.1, we introduce root size as a factor that not only differentiates among languages (as we have just shown), but also as a tool that may be used to determine the allomorphy of suffixes language-internally. In Section 1.6.2, we introduce root shape as a relevant factor for determining allomorphy. Section 1.6.3 provides more evidence for the relevance of root size and shape from the domain of Latin degree marking. In Section 1.6.4, we introduce a theoretical device referred to as a pointer.

1.6.1 Allomorphy as an effect of root size

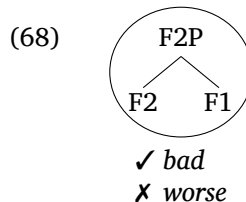
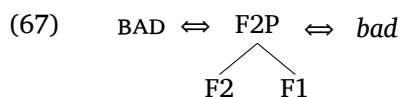
As a starting point, consider again Table (64). With the exception of English, we have so far discussed languages where we find just a single pattern of comparative marking. However, this is not a logical necessity. There is no reason why, in a single language, we could not find multiple different roots: some realising only the features of the positive (as in Persian), others realising the features of the comparative (as in Japanese), and yet other roots realising the features of the superlative (as in Armenian). In such a language, we would get multiple ways of marking the comparative and superlative.

As a case in point, consider two of the three different comparative allomorphs in English discussed above (see (18)), namely *worse-Ø* and *tall-er*.¹² As a starting point, recall that we have already discussed the derivation of *tall-er* in Section 1.3. The idea was that *tall* lexicalises F1 + F2, and *-er* lexicalises CMPR. Given that *worse* has no affix in the comparative, it is tempting to analyse it as realising the whole CMPRP, as in (65). Given this specification, *worse* is capable of realising all the features of the comparative without the need for a suffix, see (66). This lexicalisation pattern is analogous to the comparative structure provided for Japanese, recall (59).



In Japanese, the same root as in the comparative also appears in the positive, in other words, there is syncretism. This is not the case for English *worse*, which has the positive *bad*. However, this is an orthogonal difference relating to syncretism, which has to do with the presence/absence of competitors, as we saw above (Sect. 1.4). The difference between Japanese and English can be captured by proposing a dedicated root for the positive meaning in English, namely *bad*, as in (67).

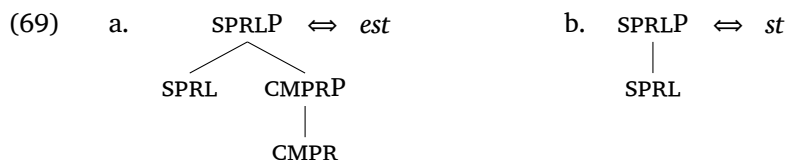
¹² We discuss the treatment of the English allomorphy in more detail in Section 1.6.2, and the current case only serves the purpose of an illustration.



In the positive degree in (68), both lexical items are candidates for lexicalisation, but *bad* wins the competition, as indicated by the checkmark.

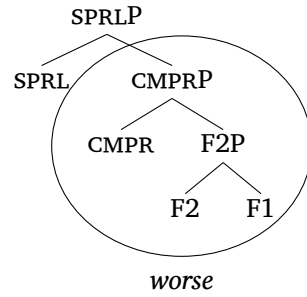
It is interesting to note that ‘tall’ and ‘bad’ also differ in the marking of the superlative, *tall-est* vs. *wors-(s)t* (cf. **tall-st*, **wors-est*). This difference is even more interesting, because it is not a difference between a zero and an overt marker. It is a difference between two different overt markers, *-est* and *-st*, with the latter contained in the former. This situation can also be captured in terms of root size.

To see how, recall first from Section 1.3 that the English *-est* is the realisation of CMPR + SPRL, as shown in (69a), repeated from (34c). However, given that *worse* already lexicalises CMPR inside the root, *-est* may not be the best item to use with this root. In particular, one could propose that *-st* only realises SPRL, see (70).

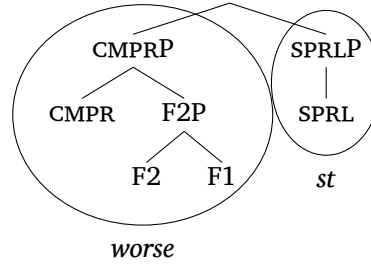


This specification would lead to a derivation where we first add the SPRL on top of the comparative, as in (70). According to the lexicalisation algorithm, SPRLP must now be matched against a lexical entry. However, *worse* is not a match because its entry in (65) does not contain the node. Therefore, the closest labelled node containing the bottom of the fseq must be moved across SPRL: this is CMPRP, and it results in (71). Here the remnant SPRLP can be realised by the *-st* allomorph, see (71).

(70)



(71)



The lexicalisation table in (72) summarises the discussion. The main idea is that proposing different root sizes within a single language allows us to capture root-driven allomorphy. For instance, we can now capture the fact that the superlative marker is different for the adjectives ‘tall’ and ‘bad.’

(72)

LANGUAGE		DEGREE	F1	F2	CMPR	SPRL
ENGLISH	POS		<i>tall</i>			
	CMPR		<i>tall</i>	<i>er</i>		
	SPRL		<i>tall</i>		<i>-est</i>	
ENGLISH	POS		<i>bad</i>			
	CMPR		<i>worse</i>			
	SPRL		<i>worse</i>			<i>st</i>

Summarising the results of this section, we have shown how root size can be used as a tool to explain root-driven allomorphy, i.e. the selection of different affixes by different classes of roots. Roots can differ arbitrarily in their size, but since this is a type of variability that is needed independently, the phenomenon of root-driven allomorphy does not require the introduction of new machinery in the model. Affixes in turn can vary in size as well, so as to match the roots they appear with, i.e. they continue the lexicalisation of the functional sequence exactly at the point at which the root achieves its maximal lexicalisation potential. These combined size differences account for the correct distribution of allomorphs of both the root and the affixes.

1.6.2 Allomorphy as a result of root shape

The idea to use arbitrary root size as a tool for modelling root-driven allomorphy has proven extremely productive in the Nanosyntax literature, and it has been applied to a number of empirical domains (see, e.g., Caha 2021 for using root size to model declensions). However, as a result of this research, it has been discovered that root size alone is not enough, and that

additional parameters of variation must be invoked, namely root shape (Blix 2022, Cortiula 2025).

In order to explain why an additional parameter is needed to capture root-driven affixation, we will now adopt the proposal in De Clercq & Vanden Wyngaerd (2017), which argues that the projection of the superlative should be split into two projections, SPRL1 and SPRL2. After we explain the reasons for the split proposal, we show why root shape is needed.

De Clercq & Vanden Wyngaerd (2017) argued for the split superlative proposal on the basis of Latin. However, similar reasons can also be found in English, when we consider additional details of superlative marking in this language. In particular, we would like to address two questions that the proposal presented in the previous section leaves open.

The first question has to do with the issue of irregular roots other than *worse*. If *-st* simply attaches on top of a comparative like *worse*, yielding *wors-(s)t*, why doesn't this generalise across all similar cases? For instance, why isn't the superlative of *less* **less-(s)t*, but *leas-(s)t*, with a different root allomorph? In Nanosyntax, one would like to attribute to the two root allomorphs *less* and *leas* different sizes (just like in the case of *bad–worse*), but this seems impossible based on Table (72). This is depicted in the middle segment of Table (73), which shows that when we segment *leas-st* into only two morphemes, we cannot distinguish between *less* and *leas* based on root size.

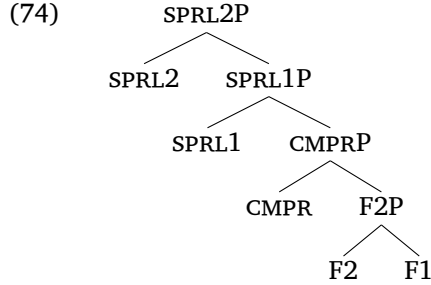
(73)

LANGUAGE		DEGREE	F1	F2	CMPR	SPRL
ENGLISH	POS		<i>bad</i>			
	CMPR		<i>worse</i>			
	SPRL		<i>worse</i>			<i>st</i>
ENGLISH	POS		<i>little</i>			
	CMPR		<i>less</i>			
	SPRL		<i>leas</i>			<i>st</i>
ENGLISH	POS		<i>tall</i>			
	CMPR		<i>tall</i>	<i>er</i>		
	SPRL		<i>tall</i>	<i>/ɪ/</i>		<i>st</i>

The second question ultimately boils down to the first one, and it concerns the two superlative markers, *-est* and *-st*. Clearly, the markers are not phonologically unrelated. Specifically, the allomorph *-est* (/ɪst/) contains the allomorph *-st*. Now if we try to decompose *-est* into *-e* and *-st* (as shown on the last line of (73)), then we get a similar conundrum: why does *-st* not attach to the comparative *-er*, resulting in *tall-er-st*, similar to what happens in Per-

sian?¹³ Why must *-e* appear as the realisation of CMPR in the superlative, but *-er* in the comparative? Both issues suggest that additional structure is involved in the functional sequence of the superlative, beyond the two heads CMPR and SPRL.

It turns out that both conundrums can be solved if we adopt the proposal by De Clercq & Vanden Wyngaerd (2017) that the superlative is formed by adding two additional heads on top of the comparative, rather than just one. The proposal is depicted in (74).



Once the superlative is decomposed, we can treat *-st* as an independent morpheme realising SPRL2, and the rest of the facts easily falls into place. The analysis is illustrated schematically in Table (75).

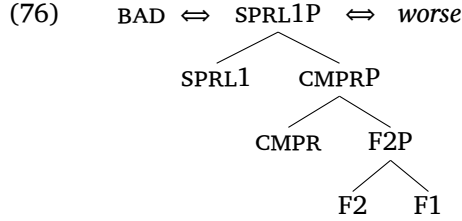
(75)

LANGUAGE	DEGREE	F1	F2	CMPR	SPRL1	SPRL2
ENGLISH	POS	<i>tall</i>				
	CMPR	<i>tall</i>	<i>er</i>			
	SPRL	<i>tall</i>	/ɪ/			<i>st</i>
ENGLISH	POS	<i>bad</i>				
	CMPR	<i>worse</i>				
	SPRL	<i>worse</i>				<i>st</i>
ENGLISH	POS	<i>little</i>				
	CMPR	<i>less</i>				
	SPRL	<i>leas</i>				<i>st</i>

Due to the presence of the additional SPRL1, we can make a size difference between *less* and *leas-*, with *leas-* being bigger in that it also lexicalises SPRL1, in contrast to *less*. We can also understand why we do not get *tall-er-st*: this is because *-er* does not lexicalise SPRL1, while /ɪ/ does, lexicalising CMPR + SPRL1.

¹³ Or similarly to the Cimbrian adjective ‘pretty:’ *šua—šuan-ar—šuan-ar-ste* (Bobaljik 2012, 31).

Worse now becomes a special case, whose lexical tree is of the size SPRL1P, which, due to the Superset Principle, also allows the lexicalisation of CMPRP, contained inside the entry.



This entry also explains why we only get *-st* in the superlative of *worse* (and not *-est*): since the root extends up to SPRL1P, only *-st* is needed to lexicalise the one remaining feature of the superlative, SPRL2. The same is true for the root *leas-*, which is likewise of size SPRL1P. The only difference with *worse* is that *leas-* has a competitor that is a closer match in the comparative (*less*), and therefore *leas-* is blocked from lexicalising the comparative, whereas *worse* is not.

While this analysis derives the facts of (75), a number of issues arise when we consider an additional triplet *good-better-best*. One of these issues will motivate the introduction of root shape as a parameter on which lexical entries can vary. Following Bobaljik (2012), we assume that *good* has the same suppletive root both in the comparative and in the superlative, namely *be(tt)*. This root has a floating *t* at the end, which is pronounced only if a vowel follows (similarly to the floating *n* of the indefinite article *a(n)*, which is only pronounced in front of a vowel). Given this decomposition, and keeping the same features as in (75), the lexicalisation table for *good* would look as in (77):

(77)

LANGUAGE		DEGREE	F1	F2	CMRP	SPRL1	SPRL2
ENGLISH	POS		<i>good</i>				
	CMRP		<i>be(tt)</i>	<i>er</i>			
	SPRL			<i>be(tt)</i>			<i>st</i>

There are, however, two problems with this setup. The first is that the root *good* lexicalises the same features as *be(tt)* in the comparative (which is analogous to the problem of *less* and *leas-* above). As we observed above, it is impossible that the exact same heads get spelled out by different morphemes. The second issue arises with the superlative *be(tt)-st*. Given that *be(tt)* combines with *-st* (yielding *best* because the floating consonant drops), we apparently must attribute to the root *be(tt)*- the same entry as to *worse*, namely as lexicalising SPRL1P. However, we just saw that this entry predicts (due to the Superset Principle) that *be(tt)* on its own should be able to lexicalise the CMPRP, which is not the case; we get *bett-er* instead.

The solution to the first problem is the same as the one for *less* and *leas*-above, where we decomposed the superlative head. We shall also decompose CMPR into two different heads, CMPR1 and CMPR2 (see also Caha, 2017a; Caha et al., 2019). Once we do so, we can update our lexicalisation table as in (78), which shows the suppletive *good* alongside the regular adjective *tall*.

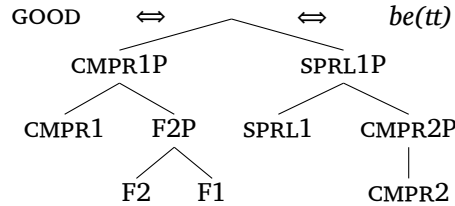
(78)

LANGUAGE		DEGREE	F1	F2	CMPR1	CMPR2	SPRL1	SPRL2
ENGLISH	POS	<i>good</i>						
	CMPR	<i>be(tt)</i>				<i>er</i>		
	SPRL				<i>be(tt)</i>			<i>st</i>
ENGLISH	POS	<i>tall</i>						
	CMPR	<i>tall</i>				<i>er</i>		
	SPRL	<i>tall</i>				/ɪ/		<i>st</i>

After the decomposition, we now have a size difference between *good* in the positive and *be(tt)* in the comparative; they no longer lexicalise the same features. Note that the proposal is also compatible with regular adjectives such as *tall*, which we now specify as lexicalizing CMPR1P; their use in the positive is a regular superset effect.¹⁴

Let us now turn to the second problem, which is that if *be(tt)* lexicalises features up to SPRL1, we expect it to be able to lexicalise the whole comparative on its own, without the need for *-er* (in the same manner as *worse*). The solution arises from the proposal that in addition to root size, there is another parameter that distinguishes among different roots, namely root shape. The gist of the proposal is shown in (79) (inspired by the approach in Blix 2022). We refer to trees like in (79) as Complex Left-Branch trees (CLB trees) because they have a complex phrasal node on the left branch (not just simple features). In (79), CMPR1P is a complex left branch.

(79)



The lexical entry (79) associates the phonology *be(tt)*- to a tree that is also of the size SPRL1P, i.e., containing all the features from F1 to SPRL1. How-

¹⁴ Assuming that *-er* only lexicalises CMPR2, we can also capture the difference between analytic and synthetic comparatives in English in terms of root size. Caha (2017a) and Caha et al. (2019) propose that roots which fail to lexicalise CMPR1 must combine with *more*, because only *more* (and not *-er*) lexicalises CMPR1 + CMPR2.

ever, the features are organised differently: the tree depicts a scenario where CMPR1P cyclically evacuates, and lands outside of SPRL1P. Since lexical entries in Nanosyntax link well-formed syntactic structures to phonology, such a scenario can be implemented without any changes to the structure of the theory: it represents a logical option that is already a part of the system.

The way this lexical entry helps us solve the issue of the comparative is that in this entry, the features of the comparative (F1, F2, CMPR1 and CMPR2) do not form a constituent to the exclusion of SPRL1; as a result, they can never be targeted by this entry in the comparative, where SPRL1 is absent. On the other hand, the features F1, F2 and CMPR1 do form a constituent to the exclusion of other features, so it is possible for *be(tt)* to lexicalise these features. This is how the lexicalisation in table (78) arises: F1, F2 and CMPR1 are lexicalised by *be(tt)*, but CMPR2 is lexicalised separately as *-er*.

While items like the one in (79) have been heavily exploited in the Nanosyntax literature (see Caha et al. 2024b, Caha 2025, Cortiula 2025), there is a question linked to their acquisition. Namely: why do learners postulate such entries? In this book, we observe that lexical items with complex left branches generally correspond to an independently needed step in a derivation of a regular class. In what follows, we exemplify this on the entry for *be(tt)* and show that the entry (79) represents a stage in the derivation of a ‘regular’ root like *tall*.

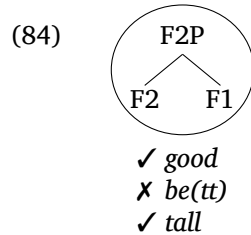
To show that, we first provide the lexical trees for the roots *tall* and *be(tt)*, as well as for the suffixes */ɪ/* and *-st*.



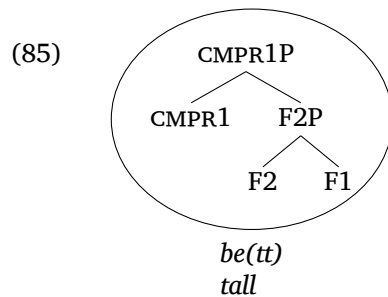
Once we compare these trees with the tree for *be(tt)* in (79), we can see that the tree for *be(tt)* as if combines the lexical entry for *tall* on the left branch, and the one for */ɪ/* on the right branch. Thus, the lexical tree for *be(tt)* in (79) is in fact a stage in the derivation of *tall* corresponding to an intermediate stage *tall* + *ɪ*.

In order to demonstrate this, we run the derivation for *tall* and *be(tt)* in parallel. In this derivation, the roots *be(tt)* and *good* compete with each other, because they are linked to the same concept. However, the root *tall* does not compete with them, it is in free variation. The choice between *tall* on the one hand and *good/be(tt)* on the other depends on what concept we want to talk about.

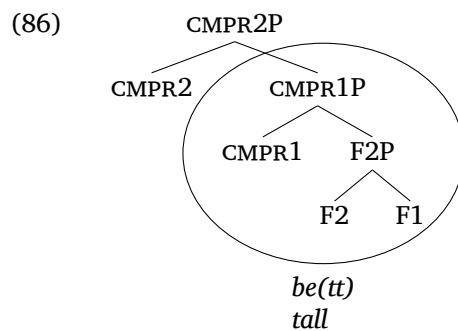
We start at the derivation at the stage where we merge F1 and F2 together. At this point, both *good* and *be(tt)* will be candidates, but *good* will win the competition. The root *tall* is also a candidate here, if TALL is the intended concept.



Next, when CMPR1 is merged, *good* will no longer be able to lexicalise the structure, and only *be(tt)* will, because its lexical entry contains CMPR1. The root *tall* can also be inserted (depending on which concept we want to talk about).

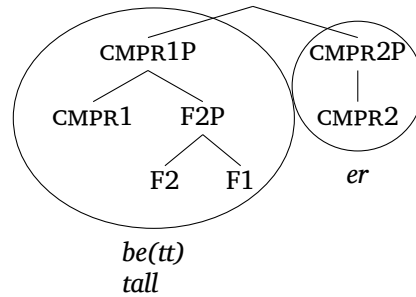


After merging CMPR2 on top of CMPR1P, the tree in (86) is not contained in either lexical entry. This is trivial for *tall*, which completely lacks the feature CMPR2. The lexical entry (79) contains this feature, but it does not contain the relevant constituent.



What we need to do, therefore, is apply the second step of the Lexicalisation Algorithm (Evacuate Closest). Starting from (86) we move CMPR1P, which is the closest labelled node containing the lowest feature of the fseq. In the resulting structure, the nodes must be lexicalised separately. As before, the CMPR1P on the left branch in (87) is contained in the lexical entry of both *be(tt)* and *tall*, and may be therefore lexicalised by either (depending on the concept). The remnant CMPR2P is lexicalised by *er*:

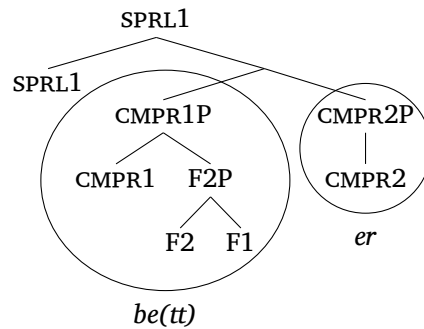
(87)



In this manner, we derive the fact that, even though the lexical entry of *be(tt)*- contains all the features of the comparative (and even one more), it still needs the suffix to lexicalise CMPR2P. The shape of its lexical entry is such that it does not contain the constituent that we built in (86), triggering evacuation movement and suffix insertion.

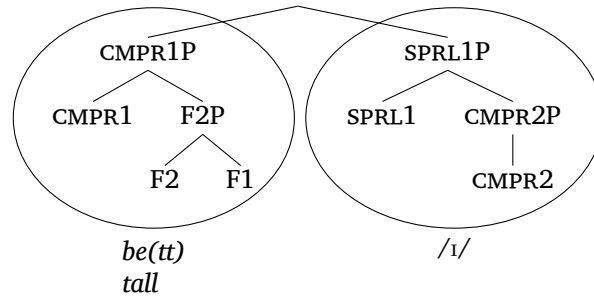
Let us next see what happens if we continue the derivation to build a superlative. When SPRL1 is added to (87) (as in (88)), we find no match.

(88)



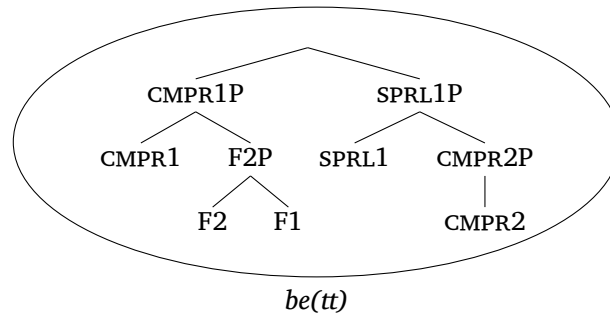
Therefore, we move the closest labelled node containing the bottom of the fseq (CMPR1P), and achieve the structure (89).

(89)



In (89), the left and the right branches can be lexicalised separately, by the root *be(tt)/tall* and the suffix */I/*, respectively. However, for *bett*, there is also a match at the level of the top node with the entire lexical entry of *be(tt)*-. This lexicalisation overrides the ones at the lower nodes, as shown in (90).

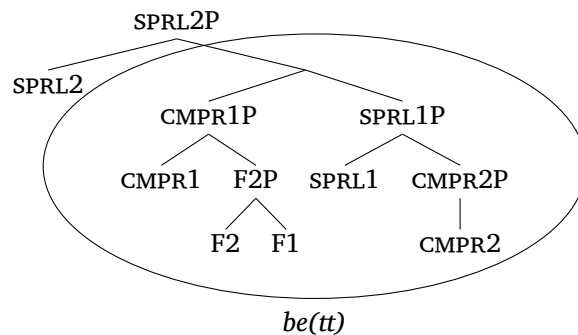
(90)



The lexicalisation in (90) (and its relation to a regular adjective in (89)) is what we wanted to show: namely that a lexical item with a complex left branch (like *be(tt)*) does not arise out of nowhere, but corresponds to a stage in the derivation of a regular adjective. (89) and (90) are identical structures, just lexicalised differently.

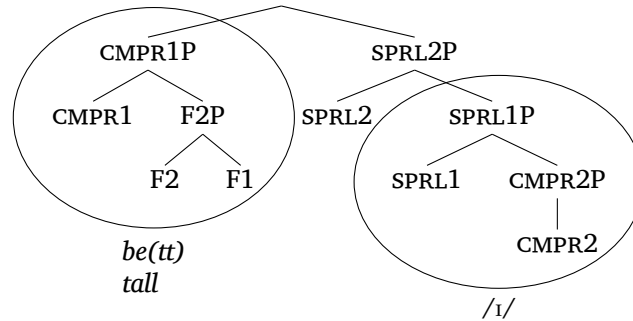
In the final step of the derivation of the superlative, **SPRL2** is merged, but it will not be directly lexicalisable (which is true for both roots).

(91)



Movement of the closest labelled node containing the lowest feature of the fseq, CMPR1P, results in (92).

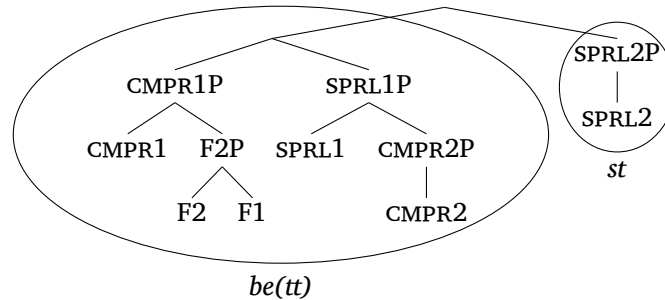
(92)



Observe how this movement destroys the constituent that matched the lexical entry of the root *bett-*, since the right branch is no longer a match. This means that we revert to the earlier lexicalisation of the respective left and right branches at the previous cycle of the derivation, as shown by the circles in (92).

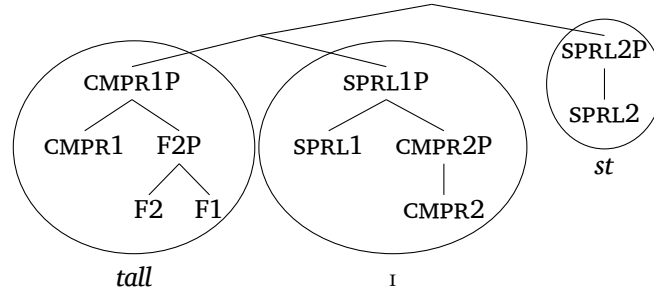
Since lexicalisation fails in (92), we undo the movement, and starting from (91), we pied-pipe the (unlabelled) node immediately dominating CMPR1P, in effect applying rollup movement. This results in (93), where the left branch continues to be a match for the root *be(tt)*, while the right branch can now be lexicalised as *-st*. We derive the correct lexicalisation *be(tt)-st*, which is pronounced as *best*.

(93)



The exact same structure is also derived for a regular comparative like *tallest*, with a simple difference that lexicalisation targets lower nodes:

(94)



For completeness, we show the lexicalisation table for four types of English adjectives in (95):

(95)

LANGUAGE		DEGREE	F1	F2	CMPR1	CMPR2	SPRL1	SPRL2
ENGLISH	POS	<i>tall</i>						
	CMPR	<i>tall</i>			<i>er</i>			
	SPRL	<i>tall</i>			<i>I</i>			<i>st</i>
ENGLISH	POS	<i>good</i>						
	CMPR	<i>be(tt)</i>			<i>er</i>			
	SPRL	<i>be(tt)</i>						<i>st</i>
ENGLISH	POS	<i>bad</i>						
	CMPR	<i>worse</i>						
	SPRL	<i>worse</i>						<i>st</i>
ENGLISH	POS	<i>little</i>						
	CMPR	<i>less</i>						
	SPRL	<i>leas</i>						<i>st</i>

Summarising, at this point we have introduced two tools that Nanosyntax uses to model allomorphy going from the root out. One of them is root size, which determines the maximum number of features that a root may lexicalise. Affixes typically start lexicalising features directly above this threshold. Since the threshold is different for different roots, affixes start lexicalising at different places, and differ as a consequence. For example, the root *worse* is stored as lexicalising SPRL1P, and it therefore needs no affix in the comparative, and it only needs *-st* in the superlative. The root *tall* only lexicalises CMPR1P, and it needs *-er* in the comparative, and *-e-st* in the superlative.

Root shape is another parameter along which roots may differ. For instance, *be(tt)* and *worse* are both of size SPRL1P, combining directly with *-st* in the superlative, with no /t/ in between the root and the SPRL2 marker. However, they differ in the comparative: while *worse* can lexicalise all the features of the comparative, *be(tt)* cannot, and combines with *-er*. This dif-

ference is coded as the difference in the shape of the root *be(tt)*. This will give rise to the pattern shown in (96), where *be(tt)* is shown to belong to an intermediate class, patterning with regular adjectives (class I) in the comparative, but with the suppletive ones of class III in the superlative.

(96)

	POS	CMPR	SPRL
I	tall	tall-er	tall-est
II	good	bett-er	be(tt)-st
III	bad	worse	worse-st
	little	less	leas-st

Such intermediate classes can be captured by Complex Left-Branch trees. As a result of its absolute size, *be(tt)* patterns like *worse* in the superlative, and they both take just ST, since they are both SPRL1P roots. However, as a result of its shape, it does not contain the full comparative, and lexicalises only CMPR1P like the regular Class I adjectives.

1.6.3 More root size and shape: Latin

In order to show the explanatory power of the root size and shape paradigm, we examine the case of Latin comparatives and superlatives, as described in De Clercq & Vanden Wyngaerd (2017). The relevant fragment is shown in (97), and it contains three different root classes, which differ in the allomorphy of their comparative and/or superlative. In particular, we can see that the two classes on top have a comparative in *-i-or* (highlighted by shading), while the lowermost class has just *-or* (apparently, with a missing *-i-*). In the superlative, the topmost class has *-i-ss-im-us*, while the two lower classes have just *-im-us* (apparently, with a missing *-i-ss-*). ‘Missing morphemes’ are marked by gaps.

(97) Latin (De Clercq & Vanden Wyngaerd 2017)

	POS	CMPR	SPRL	GLOSS
I	alt-us	alt-i-or	alt-i-ss-im-us	‘tall’
	stabil-is	stabil-i-or	stabil-i-ss-im-us	‘firm’
II	facil-is	facil-i-or	facill -im-us	‘easy’
	acer	acr-i-or	acerr -im-us	‘sharp’
III	parv-us	min -or	min -im-us	‘small’

The reason why we chose this paradigm is because it allows us to see that root size is not enough to explain the pattern. The reason is simple: if al-

lomorphism was only determined by root size, then for the comparative, we would have to conclude that Class II has the same root size as Class I. However, for the superlative, we would have to conclude that Class II has the same size as Class III. But this is impossible; Class II cannot be the same as both Class I and Class III, since Class I and III must have a different size (as they take different allomorphs).

The solution is going to be that there are two effects underlying this pattern. Class I and Class III differ in root size, which is why they always have different endings, and why Class III always has fewer endings. The ‘alternating’ Class II in the middle has a lexical entry with a Complex Left Branch. The total size of Class II roots will be like the size of Class III, which is why they have the same superlative. However, because of the constituency issue, Class II will not be able to lexicalise the comparative in the same way as Class III (i.e., Class II is comparable to *be(tt)* discussed above).

We start the analysis by introducing the functional morphemes in Table (97). In the comparative, we follow De Clercq & Vanden Wyngaerd (2017) and analyse *-i* as a comparative marker, and *-or* as an agreement marker. This is due to the fact that forms like *alt-i-or* ‘taller’ are masculine/feminine nominative singulars; the nominative neuter is *alt-i-us*. We therefore adopt the idea that *-i* is the comparative marker, while *-or* is a part of the declensional system (gender/number/case). As such, *-or* will be disregarded from the analysis, which only considers the actual degree markers.

Regular superlative forms like *alt-i-ss-im-us* ‘tallest, MASC.SG,’ stack a number of morphemes after the root. The final morpheme *-us* is again case/number/gender inflection, leaving us with the string *-i-ss-im*. We again follow De Clercq & Vanden Wyngaerd (2017) in treating the initial *-i* as the comparative marker, to which the superlative adds two markers, namely *-s* (for SPRL1) and *-im* (for SPRL2). We further assume that the marker *-im* triggers the gemination of the preceding consonant. We therefore mark *-im* as *-Gim*, for geminating *-im*.

We show the proposed decomposition and the assumed lexicalisation in (98). Class I is the class with most suffixes, all the other classes arise by reducing the number of morphemes from the root out, which we analyse as phrasal lexicalisation.

(98)

	DEGREE	F1	F2	CMPR1	CMPR2	SPRL1	SPRL2
I	POS	<i>alt</i>					
	CMPR		<i>alt</i>	<i>i</i>			
	SPRL		<i>alt</i>	<i>i</i>	<i>s</i>		<i>-Gim</i>
II	POS	<i>facil</i>					
	CMPR		<i>facil</i>	<i>-i</i>			
	SPRL			<i>facil</i>			<i>-Gim</i>
III	POS	<i>parv</i>					
	CMPR		<i>min</i>				
	SPRL			<i>min</i>			<i>-Gim</i>

The reason for proposing the geminating *-im* is the following. If we look at Table (97), we can see that in the classes II and III, the superlative lacks *-i-* and *-s*, and therefore, *-Gim* attaches directly to the base. In class II, this allows us to see that indeed, *-Gim* triggers (expectedly) the gemination of the preceding root consonant (e.g., *facill-im-us* ‘easiest’ or *acerr-im-us* ‘sharpest’). In forms such as *min-im-us* on the bottom row, we assume that *-n* is immune to gemination for phonological reasons.¹⁵

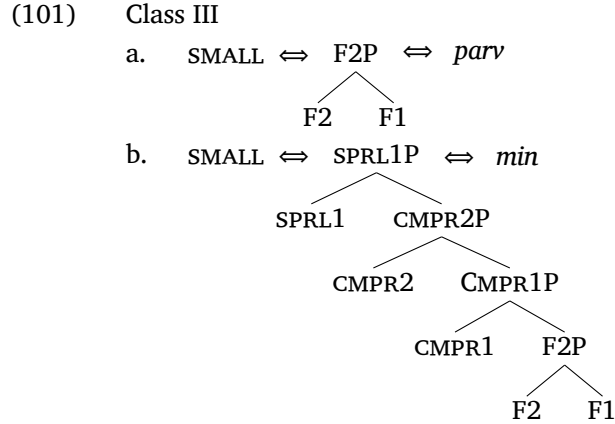
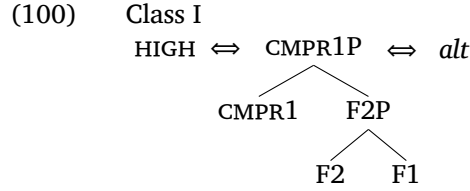
The table already clearly shows the problem that Class II would encounter if we could only rely on root size: the root *facil* is of size SPRL1P, given the absence of *-s* in the superlative, but then we cannot explain why the root cannot lexicalise the comparative and needs a suffix.

We are now going to show how the lexicalisation table is derived. The lexical entries for the suffixes are provided in (99). The comparative marker *-i* is given in (99a). The superlative markers are in (99b,c).

(99)	a.	CMPR2P	⇔	<i>-i</i>	c.	SPRL2P	⇔	<i>-Gim</i>
		CMPR2				SPRL2		
	b.	SPRL1P	⇔	<i>-s</i>				
		SPRL1						

With the functional morphemes in place, let us turn to the roots. They are listed below for Class I (100) and Class III (101) roots. We shall show how these work first since they have no CLB.

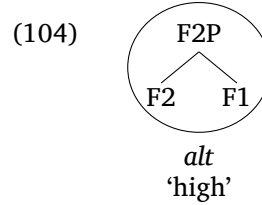
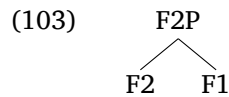
¹⁵ Alternatively, we could treat the geminator G as an independent morpheme, which is missing in the bottommost class along with other suffixes. However, since that would lead to a further decomposition, we adopt the slightly more conservative position that the absence of gemination in *min-im-us* is not enough to motivate the decomposition.



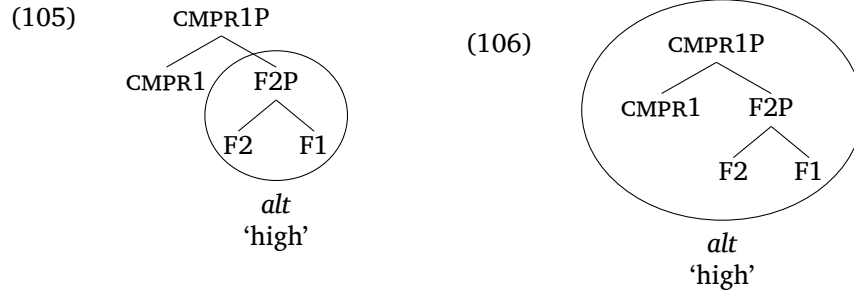
Before we run the derivations, we repeat here for convenience the lexicalisation algorithm in (102).

- (102) Lexicalisation Algorithm (based on De Clercq et al. 2025)
- Merge F and lexicalise FP
 - If (a) fails, move the closest labelled node containing the lowest feature of the *fseq*
 - If (b) fails, move the immediately dominating constituent and lexicalise (recursive)

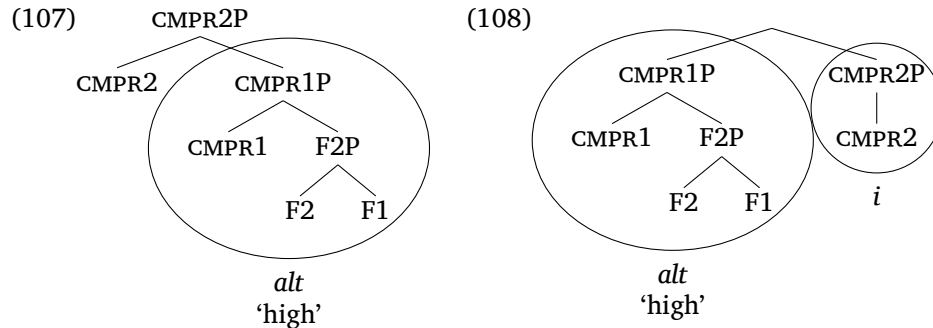
We begin with Class I, instantiated by the triplet *altus-altior-altissimus* ‘high-higher-highest’. We assume that this class has a root of the size CMPR1P (see (100)). The derivation begins by assembling F1 and F2 (103). We get a match, since F2P is contained in (100). The correct positive form *alt-* is thus derived (ignoring agreement) (104).



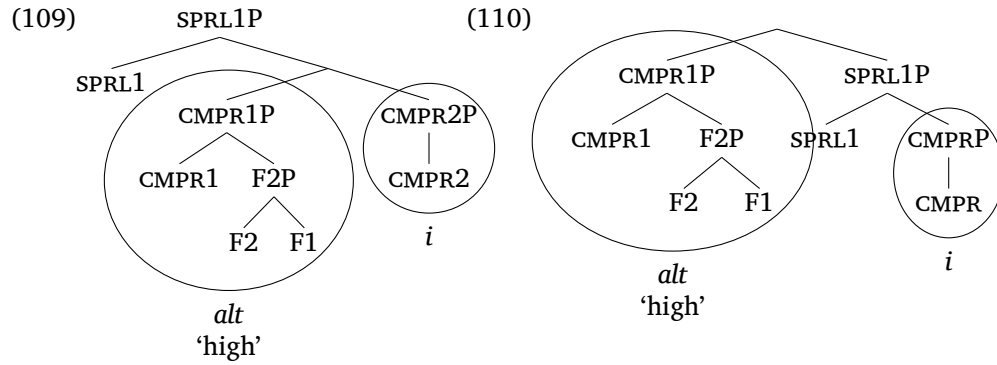
To derive the comparative, the derivation then proceeds to merge CMPR1 (105), which again finds a match in (100), so that lexicalisation can proceed as in (106).



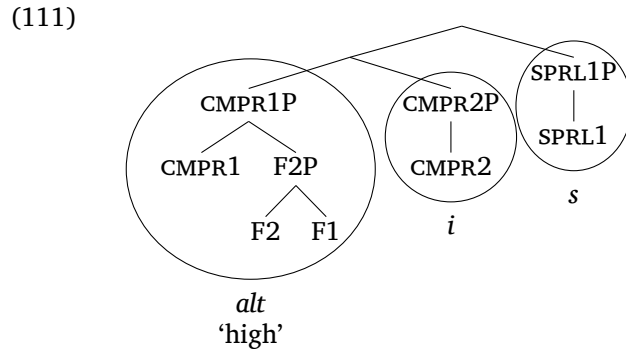
Next, CMPR2 is merged, creating (107). This structure is not contained in the lexical entry for the root *alt*, so that lexicalisation movements are triggered. We move the closest labelled node containing the bottom of the fseq, which is CMPR1P, creating (108). The remnant CMPRP on the right branch is matched by *-i*, recall (99a). As a result, the structure leads to the correct comparative form *alt-i* (again ignoring the agreement).



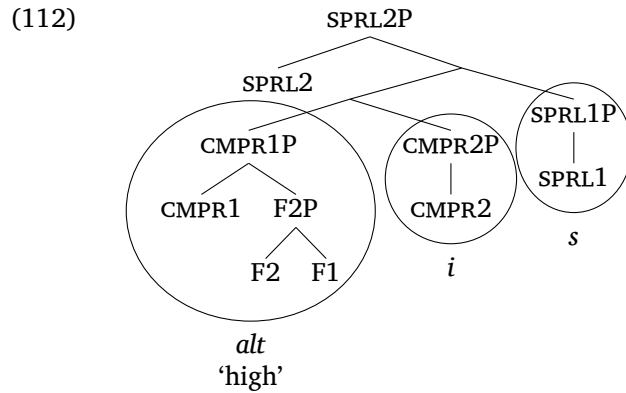
To derive the superlative, the next step is to merge SPRL1 (109). Lexicalisation without movement is impossible, so the closest labelled node containing the bottom of the fseq is moved, yielding (110). We note here that this independently needed step of the derivation is identical to the lexical item we shall propose for the intermediate Class II.



However, there is no match for SPRL1P in (110), so we go back to (109), and we pied-pipe the (unlabelled) node immediately above CMPR1P, yielding (111). Here SPRL1P is matched by -s, which just lexicalises the SPRL1 feature (recall (99b)).

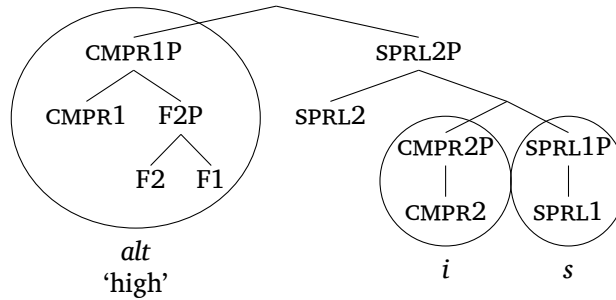


As the next step, we merge SPRL2, yielding (112).



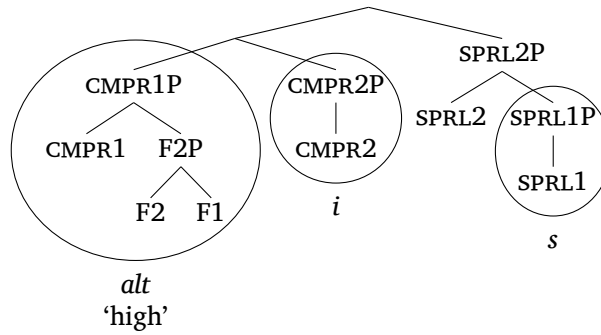
There is no match, so we move the closest labelled node containing the bottom of the fseq (CMPR1P), with the result in (113).

(113)



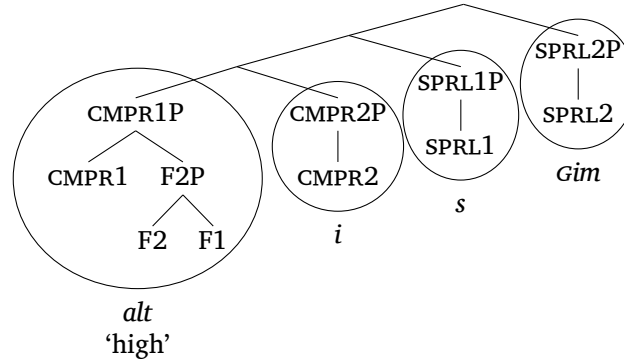
There is no match, so we undo this movement and revert to (112). Starting from that structure, we pied-pipe the node immediately above CMPR1P, creating (114).

(114)



However, there is still no match for the SPRL2P on the right branch. In this case, we revert to (112), and we again pied-pipe, since this is a step in the algorithm that is applied recursively. That is, we pied-pipe a higher node than the one we pied-piped in the previous evacuation movement step. This movement in fact amounts to a roll-up movement of the entire complement of SPRL2, and it leads to the structure in (115).

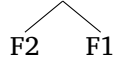
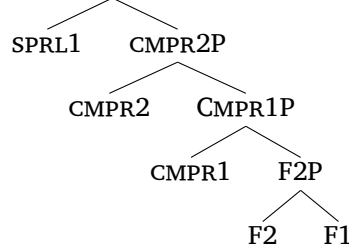
(115)



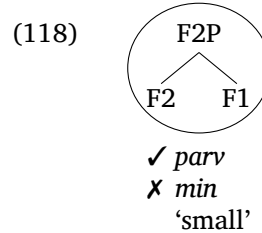
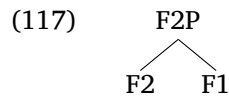
In this structure, the remnant SPRL2P on the right branch finds a match in *-Gim* (recall (99c)). This correctly produces the superlative *alt-i-ss-im-(us)*.

We now turn to Class III, with the triplet *parvus-minor-minimus* ‘small-smaller-smallest’. For ease of reference, we repeat the lexical entries for the roots in this class below. Since the root is suppletive, we postulate two entries.

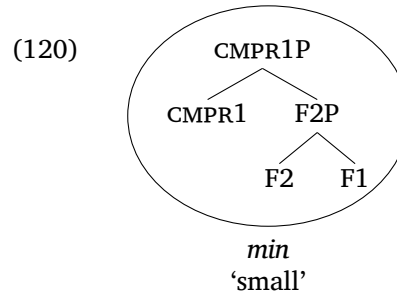
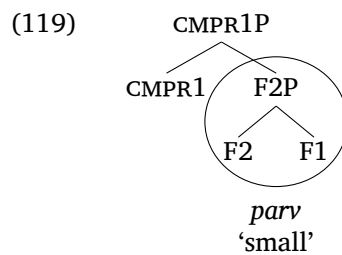
(116) Class III

a. SMALL $\Leftrightarrow$ F2P $\Leftrightarrow$ *parv*b. SMALL $\Leftrightarrow$ SPRL1P $\Leftrightarrow$ *min*

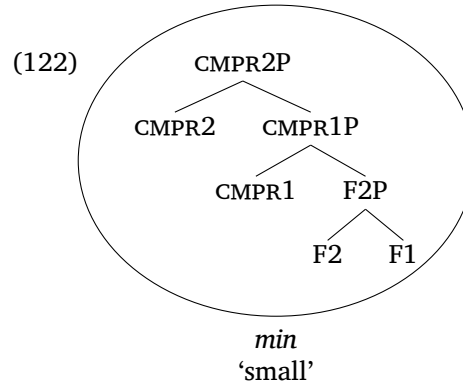
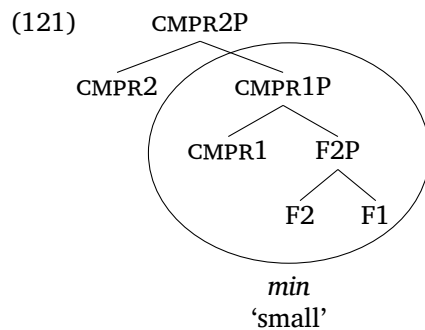
The derivation again starts by assembling F1 and F2 (117). These are realised as *parv-* (118), which is the correct form of the positive (disregarding agreement). The entry *min-* is also a match for (118), but it loses the competition because it is a better match.



The derivation then merges CMPR1 (119). This structure is contained in the root *min*, so lexicalisation succeeds without the need for any movement (120).

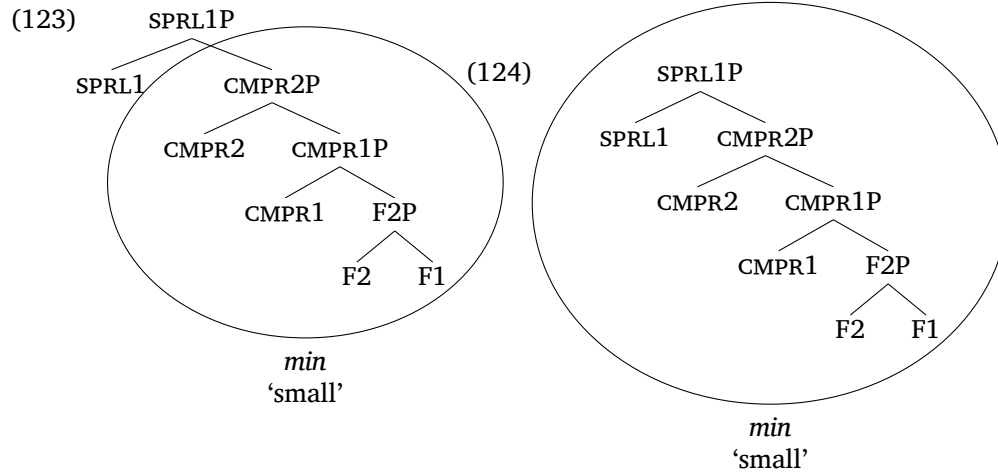


The next step is to merge CMPR2, which is again lexicalised by the root without the need for movement. This is the correct form of the comparative (disregarding agreement).

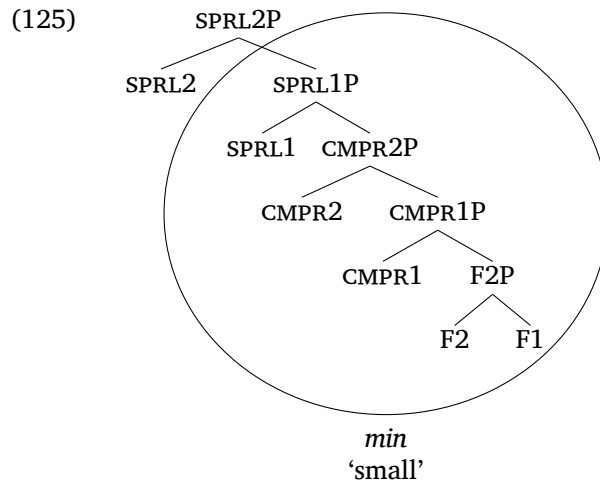


This is how root size explains the presence/absence of the comparative marker: since the root *min* 'small' is able to pronounce CMPR2, -i does not surface. In contrast, the Class I root *alt* 'high' cannot lexicalise CMPR2, and therefore, CMPR2 requires movement and -i.

The next step is to merge SPRL1, after which lexicalisation again succeeds without any movement.

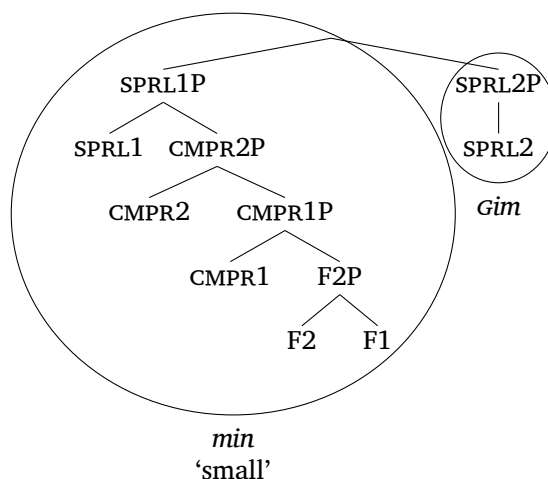


When SPRL2 is added, we get the structure (125).



However, there is no match, so evacuation movements are triggered. The first option is to move the closest labelled node containing the lowest feature of the fseq, which is SPRL1P, yielding (127). Here, *-Gim* is a match, and the correct superlative *min-im* is derived.

(126)



The result is shown in the lexicalisation table in (127) (repeated from above). We can see that the differences between the roots of Class I and Class III can be explained by root size.

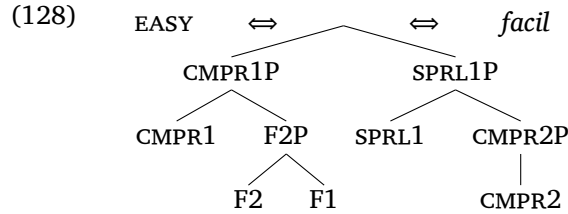
(127)

	DEGREE	F1	F2	CMPR1	CMPR2	SPRL1	SPRL2
I	POS	<i>alt</i>					
	CMPR		<i>alt</i>		<i>i</i>		
	SPRL		<i>alt</i>		<i>i</i>	<i>s</i>	<i>Gim</i>
II	POS	<i>facil</i>					
	CMPR		<i>facil</i>		<i>i</i>		
	SPRL			<i>facil</i>			<i>Gim</i>
III	POS	<i>parv</i>					
	CMPR			<i>min</i>			
	SPRL			<i>min</i>			<i>Gim</i>

However, if we now move to the Class II root *facil-is* 'easy,' we run up against the limits of root size as the only tool for the modelling of allomorphy. This is because in order to model the comparative *facil-i-or*, we must maintain that *facil* does not lexicalise CMPR2 (see the second row of Class II in Table (127)). At the same time, to model the superlative *facill-im-us* (which lacks the comparative *-i*), we must maintain that the root does lexicalise CMPR2 (as well as SPRL1), which would be a contradiction if lexical entries did not contain Complex Left Branches. The superlative is depicted on the final row of Class II of Table (127).

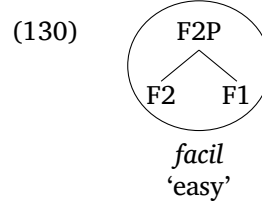
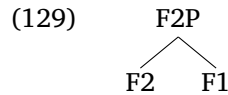
The pattern that we see in a lexicalisation table such as (127) is of course reminiscent of the one we saw before for the English triplet *good-better-best*, which is an intermediate class that patterns with a smaller root size in one column, and with a bigger one in another (table (97) represents this visu-

ally). This type of pattern can be modelled if we assume that the lexical entry for the root *facil*- ‘easy’ contains a Complex Left Branch. The entry is given in (128), and it corresponds to the stage in the derivation of a regular adjective that we saw in (110).

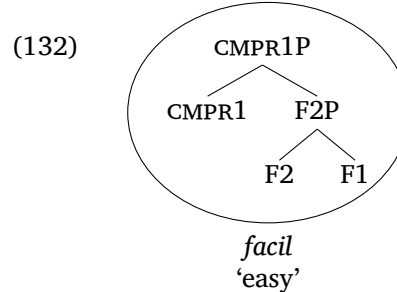
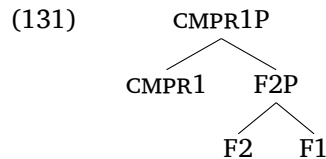


The main property of the entry is that the features F1, F2, CMPR1 (on the left branch), and CMPR2 (on the right branch) do not form a constituent, which is why the root cannot lexicalise the four features of the comparative (see the middle row of class II in Table (127)).

The actual derivation that produces the lexicalisation in Table (127) starts again by merging F1 and F2 (129). We get a match, since F2P is contained in (128), and we therefore lexicalise as in (130).

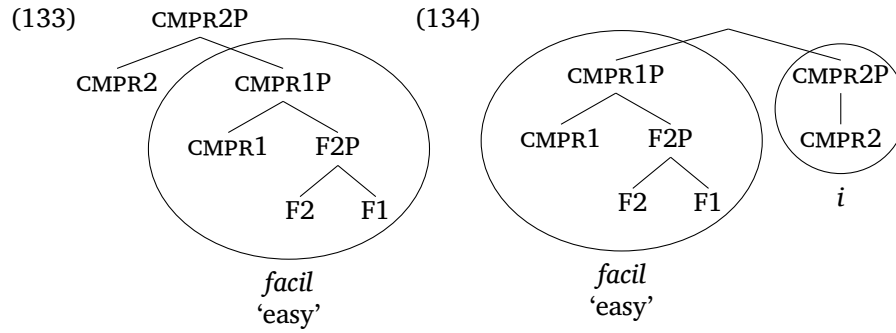


The next step is to merge CMPR1, which can again be lexicalised by the root, as CMPR1P is identical to the left branch of the lexical entry.

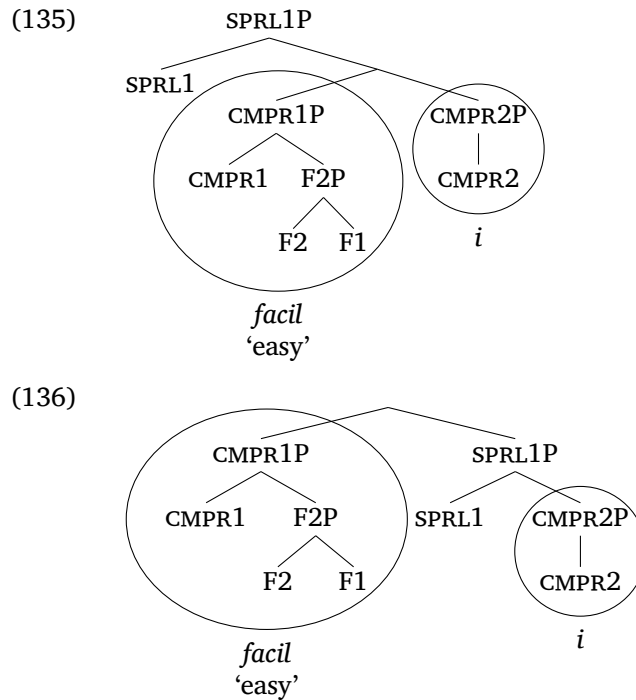


We next merge CMPR2, which produces (133). This structure is not contained in the lexical entry for the root *facil*, so lexicalisation triggers movements. This will move the sister of CMPR2, since this is the closest labelled node containing the bottom of the fseq. This creates a constituent on the right that is matched by the suffix *-i*, which leads to the correct comparative form

facil-i (ignoring agreement) (134). Note that *facil-* in (128) cannot lexicalise the entire structure of the comparative, since it does not contain either (133) or (134).¹⁶ So far, the form of the comparative is thus the same as with *alt-i-(or)* ‘higher.’



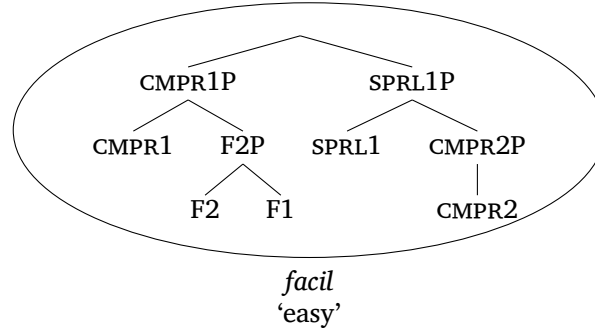
The next step is to merge SPRL1 (135). Lexicalisation without movement is impossible, so we move the closest labelled node containing the bottom of the fseq, CMPR1P, yielding (136).



¹⁶ Also note that *facil-* in (128) is a match for the right branch of (ii), but it will lose against *-i*.

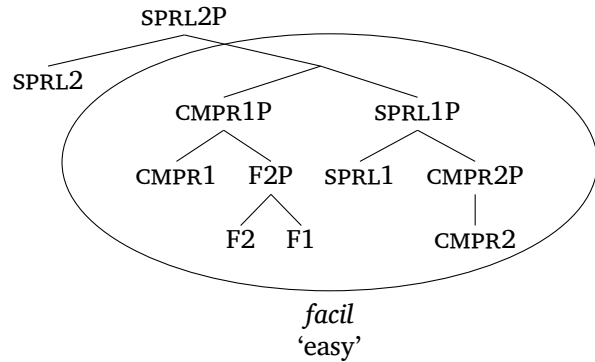
This is where *altus* ‘high’ and *facilis* ‘easy’ diverge. With *alt-* ‘high’, there was no match for a structure like this. However, *facil-* ‘easy’ can actually lexicalise the whole structure (136) because it contains it (in fact, it is identical to it). This is depicted in (137).

(137)



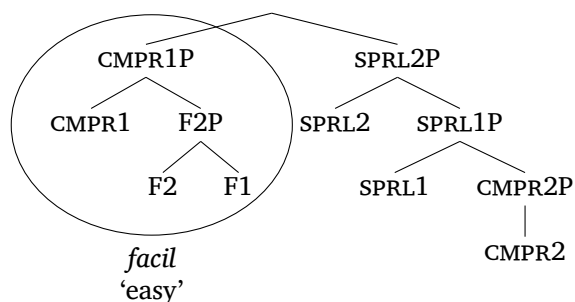
By virtue of lexicalising all the features of SPRL1P, the allomorph of the superlative is going to be the same as for the Class III roots like *min-i-im-us* ‘smallest,’ i.e., *facill-im-us* ‘easiest’ (disregarding the gemination). The structure is derived as follows. First, we merge SPRL2.

(138)



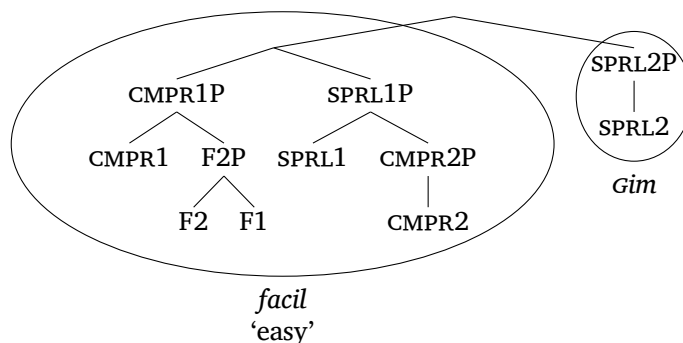
However, SPRL2P in (138) finds no match, so we move the closest labelled constituent containing the bottom of the fseq, CMPR1P, yielding (139).

(139)



However, there is still no match for the **SPRL2** on the right branch. Therefore, we pied-pipe, moving the whole complement of **SPRL2** in (138), yielding (140), which leads to the correct form of the superlative (*facill-im-(us)*).

(140)



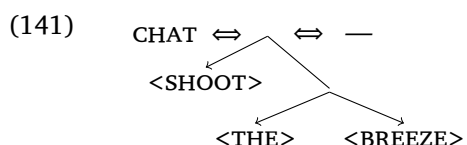
Summarising, the core idea of phrasal lexicalisation—that lexical items pair well-formed trees with phonology—allows us to model selection between a root and its allomorphs by pairing different roots with different structures. The two parameters along which these structures vary are the size of the structure, and also its shape. We illustrated this with the example of Latin, where the roots *alt-* 'high' (class I) and *min-* 'small' (class III) differ in their size: the former lexicalises features up to **CMPR1**, the latter up to **SPRL1**. On the other hand, the roots *facil* 'easy' (class II) and *min-* 'small' (class III) have the same size, but differ in shape. They can both lexicalise features up to **SPRL1** (given that both have the superlative in *-im-us*), but *facil-* has a complex left branch, which prevents it from realising **CMPR2** in the comparative structure.

1.6.4 Two routes to root suppletion

In this section, we introduce pointers. Pointers are a theoretical device that is needed for a variety of reasons, and it will be used here to model some instances of suppletion. In the previous section, we handled suppletion without pointers, which means that pointers are strictly speaking not needed to handle the phenomenon. However, modelling suppletion by pointers has slightly different outcomes than when we don't use them, and this is also why we introduce this device here. In Chapter 3, pointers will allow us to capture certain patterns that we would not be able to capture otherwise.

Since both pointers and constituent matching are needed independently of suppletion, we do not see the dual route to suppletion as an issue. In fact, this section considers data from Ukrainian suggesting that both routes may actually be needed.

A pointer is a device that allows for a lexicalisation at a higher node to make reference to specific lexical items at lower nodes. For instance, an idiom like *shoot the breeze* is a phrasal lexical item that has to be stored with a non-compositional meaning 'chat.' However, it does not lexicalise specific features, but a constituent that contains specific lexical items. The device that allows the idiom to make reference to these lexical items is called a pointer, and it is depicted in (141) as an arrow leading to the lexical item pointed to (see, e.g., De Clercq & Vanden Wyngaerd 2019b).

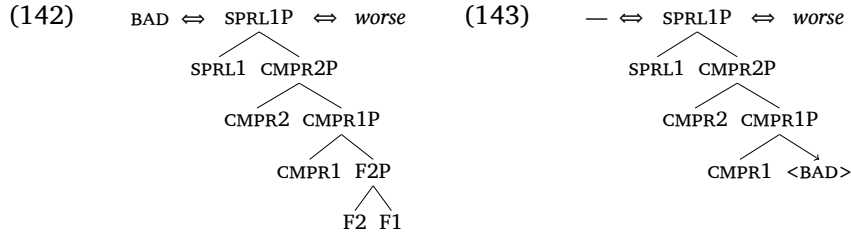


As usual, the lexical item in (141) matches a syntactic node if it is identical to it. When a node contains a pointer, identity is determined by reference to the relevant lexical item. Thus, the lexical item in (141) matches a node when the relevant branches are matched by the lexical items pointed to, i.e., by <SHOOT>, <THE> and <BREEZE>. Note that under this conception, the pointer does not point to the phonology of the lexical item or to its concept, it points to the entry as such, i.e., as a unit that links together sound, meaning and syntax.

As the next point, note that the lexical item in (141) is only associated to a concept: thus, when it matches a structure, the previous concepts are overridden. However, since the item lacks phonology, the phonology derived at previous cycles remains intact.

Vanden Wyngaerd et al. (2021) propose an approach to suppletion along similar lines. For them, suppletion is a 'phonological idiom,' i.e., a lexical item similar to the one in (141) but one that only overrides phonology, and leaves the concept intact. For instance, Vanden Wyngaerd et al. (2021) propose that *worse* does not need to lexicalise all the functional projections of

the comparative as we suggested in (76) (repeated in (142) for convenience). Instead, *worse* only lexicalises the comparative (or superlative) projections and contains a pointer to <BAD>, as in (143).



Note that the pointer-containing entry for *worse* in (143) only contains a phonological information but no concept. Thus, it only overrides the phonology *bad*, but not the concept BAD, so whatever concept has been introduced by the lexical item <BAD> remains active (it is not overridden).

To some extent, (142) and (143) may seem as two equivalent ways of encoding suppletion. However, there are slight differences between them that will lead us to use pointer-containing entries in Chapter 3. Let us comment on the difference that is going to be relevant for us.

The difference concerns the structures that the respective items match. For instance, the lexical item in (142) matches F2P because it contains it. (It loses in competition to <BAD>, but that is a different story.) On the other hand, the lexical item in (143) does not match F2P because it does not contain a constituent identical to F2P (it only has a pointer to an entry that matches such a constituent). It is only when the CMPR head is merged, projecting CMPR1P, that both versions of the suppletive entry represent a match. This difference is going to be relevant for us later on.¹⁷

Let us now reflect over what may seem to be an unwanted situation where we have two possible routes to suppletion. To be able to refer to these, let us call the first route to suppletion ‘concept sharing,’ since here we have two independent items of different sizes sharing a concept. The second route to suppletion could be called ‘pointer suppletion,’ as we have one item pointing to the other.

We do not think of this as theoretical redundancy, because the tools that provide multiple routes to suppletion are independently motivated, and they have independent non-overlapping uses. Moreover, we suspect that multiple routes to suppletion may be justified on empirical grounds, leading to slightly different cases of suppletion. A situation that seems to require ‘concept sharing’ are cases of the so-called ‘disuppletion’ (Arregi & Nevins 2014).

¹⁷ Another difference between entries with and without pointers concerns ‘gaps.’ Thus, imagine that F1P was a full phrase that would be lexicalised by *bad*. Imagine now that we merged CMPR on top of F1P, skipping F2. In such case, the entry with a pointer would match, but the one without it would not. Pointers were used to handle gaps in, e.g., Caha & Pantcheva (2012) as well as Vanden Wyngaerd et al. (2021).

The authors use this term to describe a situation where an adjective has both a regular comparative and a suppletive one. In such cases, the suppletive form and the regular form diverge in meaning. It seems advantageous to treat this instance of suppletion as ‘concept sharing,’ because concept sharing may not be perfect (allowing for only partial overlap of the senses). In other words, we find a suppletive form in the shared sense and a regular form in another sense.

On the other hand, pointers do not allow for such flexibility, and are not expected to give rise to disuppletion: the suppletive form is expected to always override the regular form, inheriting its concept (whatever that may be).

A particularly complex case along these lines is found in Ukrainian, as discussed by Vyshnevskya (2025a, 56-7). Vyshnevskya points out that Ukrainian has three adjectives meaning ‘good’ (with various nuances). In the comparative, we have two suppletive forms for ‘better’: *lip-š-yj* and *kraš-č-yj*, as shown in (144).

(144) Three positive-degree adjectives and two suppletive comparatives

GLOSS POS	POS	CMPR	GLOSS CMPR
‘good, kind’	<i>dobr-yj</i>	<i>kraš-č-yj</i>	‘better’
‘good, pretty’	<i>xoroš-yj</i>	<i>lip-š-yj</i>	‘better’
‘good, pretty’	<i>xarn-yj</i>		

In addition, two of the positive-degree adjectives have a regular comparative when the meanings diverge from the meaning ‘better.’ However, one of the adjectives (*xoroš-yj*) never shows a regular comparative form, see (145).

(145) The regular forms of the same adjectives

GLOSS POS	POS	CMPR	GLOSS CMPR
‘good, kind’	<i>dobr-yj</i>	<i>dobr-iš-yj</i>	‘kinder’
‘good, pretty’	<i>xoroš-yj</i>	—	
‘good, pretty’	<i>xarn-yj</i>	<i>xarn-iš-yj</i>	‘prettier’

Let us start by considering the asymmetry observed in (145) (some roots have a regular comparative, some don’t). Based on the fact that *xoroš-yj* ‘good, pretty’ has no regular comparative (**xoroš-iš-yj*), we hypothesise that this root is pointed to by one of the suppletive roots, which therefore overtakes the concept and prevents this root from forming a regular comparative. The proposal for the adjectives *dobr-yj* and *xarn-yj* is then the opposite, namely that they are not pointed to by any of the suppletive roots. Therefore, their suppletive patterning in (144) must be captured by ‘concept sharing.’

The specific account says that when these adjectives are used in the meaning ‘good’, a suppletive form with the same concept is preferred because it has a bigger root, and allows for a direct lexicalisation without evacuation

movements at the relevant cycle. However, when the intended meanings diverge, the regular forms *dobr-iš-yj* and *xarn-iš-yj* are derived, as in (145). This explains why some roots are dissuppletive (their suppletion is based on concept sharing) but other roots are not (they get always overridden).

Let us now turn to the issue of which of the suppletive roots includes a pointer to *xoroš*, and which one of them is just a large root meaning ‘good.’ The expectation is that the root which contains a pointer can never be used to lexicalise anything else than a comparative, as it does not match F2P, it only points to it. This description matches the root *lip-š-yj* ‘better,’ which, as far as we are aware, has no other use outside of the comparative. On the other hand, we propose that the comparative *kraš-č-yj* contains a large root without a pointer. We think of the root *kras-* as representing an instance of ‘concept sharing’ because it also has other uses. For instance, with a derivational morpheme/augment *-iv*, it means ‘beautiful’ (*kras-iv-yj*).¹⁸

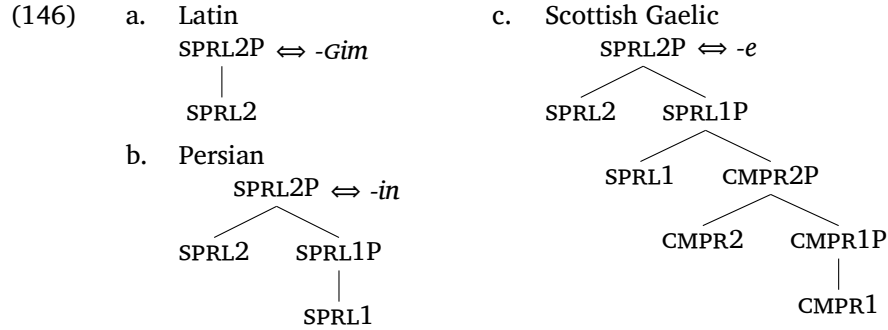
Regardless of how this issue is resolved, we conclude this section by repeating the main message, which is that besides ‘concept sharing,’ root suppletion can also be implemented via pointers. Because of the differences between the two implementations (as highlighted above), we will use pointers again in Chapter 3 to deal with root suppletion in Czech. In this chapter, we shall continue discussing suppletion in terms of ‘concept sharing,’ though nothing hinges on this decision.

1.7 Affix-driven allomorphy of roots

This section turns to allomorphy that is going from affixes to roots. This section first discusses, in abstract terms, the theoretical aspects of the lexicalisation procedure that will be used for modelling this kind of allomorphy. We then illustrate the technology using a case study from Egyptian Arabic.

The mechanism through which affixes select a particular base is, once again, a consequence of the format of lexical items (pairing well-formed syntactic structures with phonological representations), plus the idea that matching requires identity between syntax and the lexical entry. To see the idea, compare the three markers for SPRL2P in (146). They are all markers that we have already encountered.

¹⁸ We think the root is *kras-* because, as Vyshnevskaya (2025a, 49) points out, *s*-final roots in Ukrainian, when followed by the comparative *-š*, give rise to *šč*. For instance, the adjective *vys-ok-yj* ‘tall’ forms the comparative by attaching *-š* to the root, yielding the abstract form *vys-š-yj* ‘taller.’ On the surface, this is realised as *vysč-č-yj*, and that is why we interpret the form *kraš-č-yj* ‘better’ as arising from *kras-š-yj*.



In (146a), we see the Latin marker -*Gim*. In (146b), we see an updated version of the Persian superlative marker -*in*: Before splitting the SPRL marker, we specified -*in* for SPRL, so after the splitting, we simply update the lexical entry accordingly. Finally, (146c) is a similarly updated entry of the Scottish Gaelic -*e*, which spells out both the comparative and the superlative (recall that it is ambiguous between the two readings). (147) shows this in the format of a lexicalisation table.

(147)

LANGUAGE	DEGREE	F1	F2	CMPR1	CMPR2	SPRL1	SPRL2
Latin	SPRL			<i>min</i>			- <i>Gim</i> ‘small’
Persian	SPRL	<i>bolænd</i>	<i>tær</i>			<i>in</i>	‘tall’
Gaelic	SPRL	<i>og</i>		<i>e</i>			‘young’

The point we want to make with these entries is that they all have something in common: they all lexicalise SPRL2. Also, in this capacity, they are unrivalled: they are the only markers in their respective languages capable of lexicalising SPRL2. This means that the only way to lexicalise SPRL2 in these languages is to use these markers.

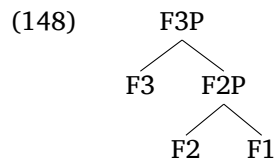
However, these markers differ in how many additional features they have. This has different consequences for the base in different languages. To see why, consider the fact that lexical items cannot lexicalise terminals, given that the lexicalisation algorithm only targets phrasal nodes through its “Lexicalise FP”. Therefore, in order to lexicalise SPRL2, these markers must lexicalise the whole SPRL2P.

This entails that, for instance, the Scottish Gaelic marker -*e* has to lexicalise all the features from CMPR1 up to SPRL2. By doing so, it prevents any other marker from realising these features. As a result, -*e* can only combine with the positive degree corresponding to F2P. By analogy, in order for the Persian marker to lexicalise SPRL2, it must also lexicalise SPRL1. As a result, this marker always attaches to the comparative base, rather than to the positive base (i.e., differently from the Scottish Gaelic marker). In Latin,

the SPRL2 marker attaches to a ‘half superlative,’ i.e., to a base that already lexicalises SPRL1.

Generalising, let us call the lowermost feature of a suffix ‘the foot’ of that suffix. The observation that we just made is that the foot of the suffix determines the features that the base lexicalises, and thereby also the form of the base.

In order to fully appreciate the power of this base-selecting property of suffixes, let us turn to Egyptian Arabic. During the discussion, we will need to add a feature to the positive, so instead of two features, we will now operate with three features. This is shown in (148). This decomposition goes in the direction of the one proposed in Section 1.1, which we further elaborate on in Section 1.9.

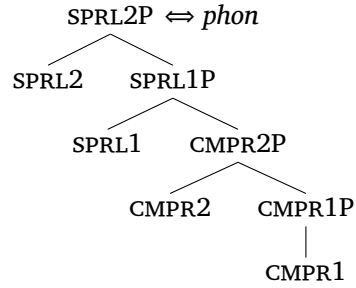


With the assumptions about the foot of suffixes and about the extra feature in the positive in place, let us turn to the data. The basic triplet of a positive, comparative and a superlative is provided in (149a-c) respectively. We can see that the comparative and the superlative are syncretic.

- (149) Egyptian Arabic
- a. Ahmad Tawiil.
Ahmad tall
‘Ahmad is tall.’ (Grano & Davis 2018, 137)
 - b. Ahmad aTwal min Basem.
Ahmad taller from Basem
‘Ahmad is taller than Basem.’ (Grano & Davis 2018, 138)
 - c. Da aTwal walad
This tallest boy
‘This is the tallest boy’ (Grano & Davis 2018, 26)

To account for this, we could treat Egyptian Arabic on a par with Scottish Gaelic, and assume that its ambiguous comparative/superlative entry is as given in (150). In (150), there is some phonology *phon* associated to the exact same constituent as in Gaelic.

(150) Egyptian Arabic comparative/superlative



As the next step, let us now determine the value of *phon*. Since comparative formation in Egyptian Arabic is templatic, we need to look in more detail at the root and pattern morphology of Arabic, see (151).

(151) Egyptian Arabic (Davis 2016, 10)

GLOSS	POS	CMPR	PATTERN
'long'	Tawiil	aTwal	aCCaC
'big'	kibiir	akbar	aCCaC
'clever'	šaaTir	ašTar	aCCaC
'bad'	wihiš	awhaš	aCCaC
'wide'	waasi?	awsa?	aCCaC
'narrow'	dayya?	adya?	aCCaC
'fat'	tixiin	atxan	aCCaC
'difficult'	Saʔb	aSʔab	aCCaC
'poor'	faʔiir	afʔar	aCCaC
'far'	biʔiid	abʔad	aCCaC
'pleasant'	laTiif	alTaf	aCCaC
'unpleasant'	baayix	abwax	aCCaC
'beautiful'	gamiil	agmal	aCCaC
'cheap'	rixiiS	arxaS	aCCaC
'broad'	?ariiD	aʔraD	aCCaC
'quick'	sarii?	asra?	aCCaC
'old'	?adiim	aʔdam	aCCaC
'easy'	sahl	ashal	aCCaC
'clean'	niDiif	anDaf	aCCaC

The table shows that the pattern of the comparative adjective is always aC-CaC, where C stands for the consonants of the triliteral root, i.e., the same consonants as in the positive (e.g. *Twl* for 'tall'). The vowels are in blue to make it easier to recognise the pattern in the comparative.

Having determined the abstract comparative template, the following question arises: what is the base that the templatic marker applies to? As Davis (2016) notes, there are two logical options. One option is that the

comparative is formed on the basis of the positive, whose consonants are fitted into the comparative template (vowels discarded). The second option is that the comparative is derived from something more abstract, namely an abstract root consisting of the three consonants, i.e., those marked by the symbol C in the template.

The lexical entry that we originally suggested in (150) makes a prediction that the comparative is derived from the positive. This is because the lowest feature of the affix (i.e., its foot) determines the size of the base with which the affix combines. Since the entry (150) has its foot at CMPR1, it predicts that the comparative template applies to the positive degree (which is the complement of CMPR1).

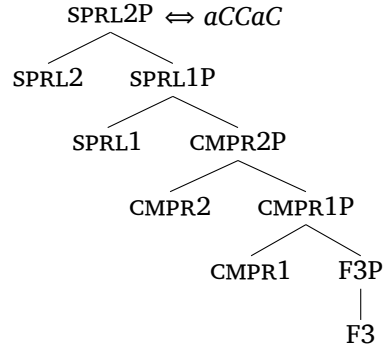
However, as Davis (2016) argues, this is empirically wrong. This can be seen when we look at the data in (152). One of the clearest reasons for the conclusion that the comparative template does not apply to the positive is that the positive degree sometimes includes consonant-containing prefixes (as on the top row) or suffixes (on the remaining rows). These extra consonants are marked in bold in (152).

(152) Egyptian Arabic (Davis 2016, 13)

GLOSS	ROOT	POS	CMPR
‘big’	nsb	mu-naasib	ans ab
‘bad’	ksl	kaslaan	aks a al
‘wide’	tʔb	taʔbaan	atʔ a b
‘narrow’	rfʔ	rufayyaʔ	ar f aʔ
‘fat’	ʔSr	ʔuSayyar	aʔ S ar

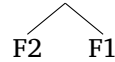
These additional consonants are, however, always ignored by the comparative template: the only consonants that ever play any role are those of the tri-consonantal root. Davis (2016) therefore concludes that morphologically speaking, the comparative is not derived from the positive in Egyptian Arabic, but from a more abstract base (the tri-consonantal root). We propose that this can be captured if the lexical entry of the comparative template *aCCaC* is not the one we gave above, but should rather look as in (153). The main update compared to the original entry in (150) is that the lexical tree includes the projection F3 below CMPR.

(153) Egyptian Arabic comparative/superlative



Recall that in order to lexicalise the comparative, the marker (153) will have to lexicalise all the features contained in the CMPRP, i.e., including F3. This way, the comparative marker prevents the base from lexicalising F3, which leads one to conclude that the base to which the *aCCaC* attaches lexicalises just F2P (a constituent different from the positive). This naturally leads to an analysis where the tri-consonantal root realises just F2P, as suggested in (154):

(154) F2P $\Leftrightarrow$ *triliteral root*



This leads to the result that morphologically speaking, the templatic marker applies to the tri-consonantal root. The two entries in (153) and (154) thus encode the observation that the comparative suffix in Egyptian Arabic does not combine with the positive form, but with a smaller constituent, which in this case we have identified with F2P, which is lexicalised by the triliteral the root. In (155) we show this in the form of a lexicalisation table

(155)

LANGUAGE	DEGREE	F1	F2	F3	CMPR1	CMPR2	SPRL1	SPRL2
Egyptian Arabic	POS	Twl	?					'long'
	CMPR	Twl		<i>aCCaC</i>				
	SPRL	Twl				<i>aCCaC</i>		

We return to the shape of the morpheme that lexicalises F3 in the positive in the following section (Sect. 1.8).

Summarising, affixes realising particular features (such as SPRL2 or CMPR) may differ in their lowest feature (their foot). The foot deserves special attention, because it determines the morphological form of the base to which the markers attach. We illustrated this on the basis of Egyptian Arabic, where postulating a syncretic CMPR/SPRL lexical entry with CMPR as a foot leads to a wrong result. However, by manipulating the foot (i.e., by placing it lower,

namely at F3), we were able to capture the fact that the templatic marker does not attach to the positive, but to a smaller constituent, F2P, which is in turn lexicalised by the tri-consonantal root.

1.8 Backtracking

In this section, we continue looking at Egyptian Arabic and we use it to illustrate an additional feature of the lexicalisation algorithm, namely Backtracking. Backtracking is introduced in (156d) as a new step (d) in the algorithm, following the movement steps.¹⁹

- (156) Lexicalisation Algorithm (based on De Clercq et al. 2025)
- a. Merge F and lexicalise FP
 - b. If (a) fails, move the closest labelled node containing the lowest feature of the *fseq*
 - c. If (b) fails, move the immediately dominating constituent and lexicalise (recursive)
 - d. If (c) fails, go back to the previous cycle and try the next option for that cycle

Backtracking is used when we add a feature F to the derivation, we try all the options under (a-c), and still cannot lexicalise FP, i.e. the derivation reaches a dead end. In such cases, backtracking tells us to remove the feature F, and go back to the previous cycle where lexicalisation succeeded. Let us call the cycle to which we backtrack F-1. The observation is that despite a successful lexicalisation at the F-1P level, we added a new feature F, and the derivation reached a dead end. So the idea is that we change something at the F-1 cycle, so that when we add F again, we add it to a different structure, and this may lead to a successful lexicalisation of FP.

To achieve this, what backtracking does at the F-1 level is that it basically tells us to forget about the successful lexicalisation at the cycle F-1, and act as if there was no success. As a result, the derivation automatically tries the next option of the algorithm. For instance, if at the cycle F-1 lexicalisation succeeded without movement, then the Evacuate-Closest option will be tried. If the Evacuate-Closest option succeeds, the structure of F-1P is different from the previous successful lexicalisation, which required no movement. So when we again add F on top of F-1P, lexicalisation may succeed, because we are adding F to a different structure.

¹⁹ Notice that in all derivations so far, we never needed any option other than those given under (a-c) of the algorithm (156). Therefore, even if we ran all our derivations with this fuller version, this would not have changed anything, since this last option would never have been activated.

In order to see an empirical case where backtracking is needed, consider the question of how exactly the positive derives from the tri-consonantal root in Egyptian Arabic. Logically, there are three options. Option one is that the positive is decomposed into a trilateral root and an affix, a templatic marker, with entries like those in (157). This templatic marker would combine with the tri-consonantal root (lexicalising F2P) and provide a fixed arrangement of the consonants and also provide fixed vocalic slots. The second option is that the positive root is not decomposed, as shown in (156), which means that it would be only loosely formally related to the tri-consonantal roots.

(157) OPTION 1
Decomposed positive

a. F2P $\Leftrightarrow$ *trilateral root*

F2 F1

b. F3P $\Leftrightarrow$ *template*

F3

(158) OPTION 2
Nondecomposed positive

F3P $\Leftrightarrow$ *phon*

F3 F2P
F2 F1

Finally, both options may be attested in the grammar, depending on the root: roots that would have an irregular entry would follow the irregular pattern, due to the preference for non-movement lexicalisation, which the entry (158) provides. Roots lacking an entry for F3P would make use of a fixed templatic shape to form their positive via the entry (157).

When we consider the actual data, it turns out that there are five different shapes that the positive can have. They are listed in (159) (from Grano & Davis 2018).

(159) The shapes of the positive in Egyptian Arabic (Grano & Davis 2018, 139)

- a. CiCiC
- b. CaaCiC
- c. CiCiC
- d. CaCiC
- e. CaCC

Regarding their distribution, Grano & Davis (2018, 138) note that “the distribution of the shapes in [(159)] across tri-consonantal roots does not appear to be predictable along any dimension, phonological or otherwise. The distribution of shapes thus bears the hallmark of a lexically idiosyncratic strategy for fitting a tri-consonantal root into a phonotactically licit shape rather than a form that could be identified as a contentful morpheme.”

While random, the different shapes of the positive are not equally attested in the sample offered by Davis (2016). For instance, the shape (159e) is attested only for 2 adjectives out of 27 given in Davis (2016). On the other hand, the positive of the shape (159d) is found in 11 out of those 27 cases.

So what we suggest is that a natural analysis would understand the most frequent pattern as a templatic realisation of the positive, along the lines of the entry (157), while treating the most infrequent shapes as irregular, along the lines of (158).²⁰

Now the derivation of the frequent cases like *Tawiil-aTwal* ‘long(er)’ will not lead to any complications. We provide relevant lexical entries for the root (160) and the suffix (161) below.

$$(160) \quad \text{LONG} \Leftrightarrow \text{F2P} \Leftrightarrow \text{Twl} \quad (161) \quad \text{F3P} \Leftrightarrow \text{CaCiiC}$$

$\swarrow \quad \searrow$
 F2 F1

$\downarrow$
 F3

In the lexicalisation table we gave in the previous section, we are now able to fill in the shape of the morpheme that lexicalises the F3 feature.

(162)

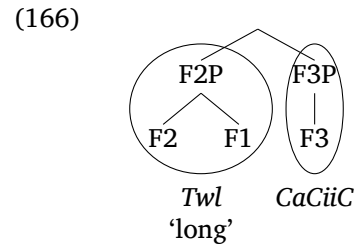
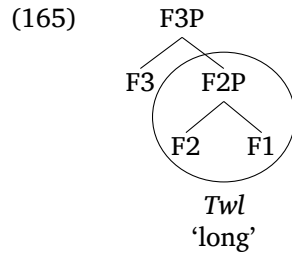
LANGUAGE	DEGREE	F1	F2	F3	CMPR1	CMPR2	SPRL1	SPRL2
Egyptian	POS		Twl	CaCiiC				‘long’
Arabic	CMPR		Twl	aCCaC				
	SPRL		Twl	aCCaC				

The derivation first constructs F2P, as in (163), and lexicalises it via the tri-consonantal root (164).



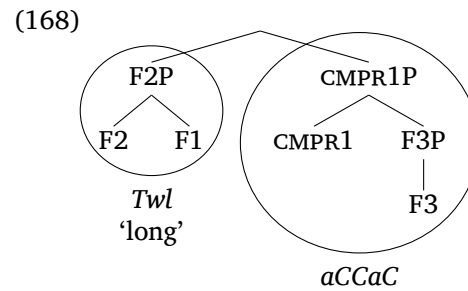
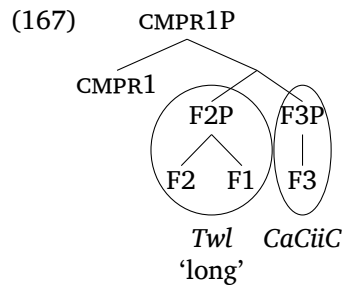
When F3 is merged, the ‘regular’ roots cannot lexicalise F3P (by definition, regular roots lack a non-compositional F3P). Therefore, they must undergo movement of the closest labelled node containing the bottom of the fseq, as in (166), and the F3P so created is lexicalised by the templatic marker *CaCiiC*.

²⁰ Obviously, the actual distribution of patterns in Arabic might be different from the one in the small sample of Davis (2016). This does not affect the logic of the argument, however, which only needs one pattern to be the most frequent one.

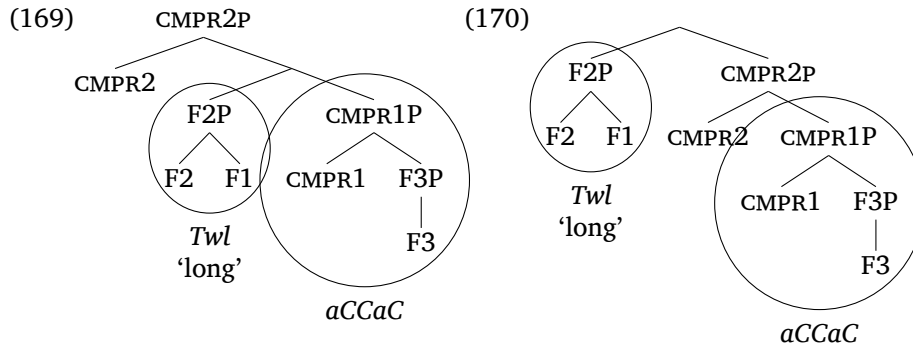


In (166), the consonants are fitted into the template and the form *Tawiil* 'long' arises.

Suppose we now want to derive the comparative, we start by adding the CMPR1 feature as in (167). There is no match for this structure, and the closest labelled node containing the bottom of the fseq, F2P, is moved, yielding (168).



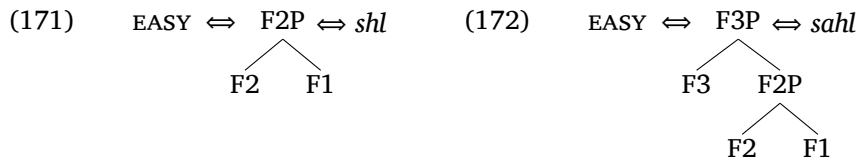
In (168), the remnant CMPR1P on the right is a match for the lexical entry of the comparative template in (153) above. This template is therefore inserted as the realisation of the CMPR1P, overriding the template found in the positive degree. To derive the comparative form, we need to merge CMPR2 (169). This structure cannot be lexicalised directly, and movement will apply to the closest labelled node with the lowest feature, i.e., F2P. After evacuation, we derive (169).



The CMPR2P on the right in (170) is now a match for the comparative marker, and is lexicalised by it, producing the correct comparative *aTwal* 'longer.'

However, the derivation looks differently for irregular roots like *sahl* 'easy.' The positive degree is not regular, and we shall treat it in terms of a suppletive root, which is not to be decomposed. On the other hand, the comparative *ashal* is compositional: it consists of the triliteral root *shl* and the regular comparative affix *CaCiiC*.

The derivation of the comparative crucially requires backtracking. Let us show how this works. We start by giving the lexical entries for the root *shl* in (171), as well as the nondecomposed (irregular, suppletive) positive *sahl* 'easy' (172).

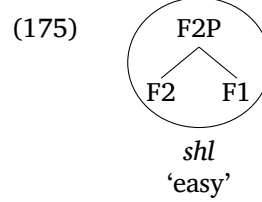
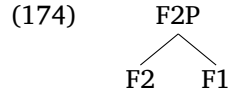


The corresponding lexicalisation table is given in (173). What we see here is that the root *sahl* and the templatic suffix can both lexicalise F3. Since both cannot do it, one of them will have to 'shrink' and use only a subpart of its entry for matching. Since the comparative suffix cannot give up on its lowest feature, it is the root that has to 'shrink' by giving up on its topmost feature. This leads to the shift from *sahl* to *shl*.

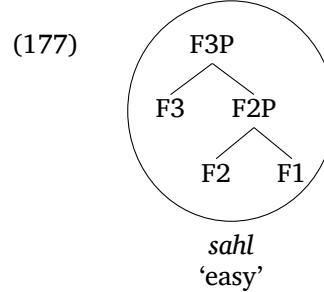
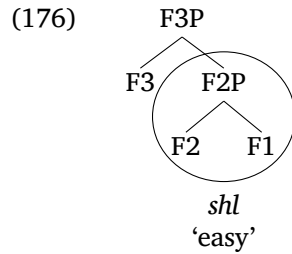
(173)

LANGUAGE	DEGREE	F1	F2	F3	CMPR1	CMPR2	SPRL1	SPRL2
Egyptian	POS		<i>sahl</i>					'easy'
Arabic	CMPR		<i>shl</i>		<i>aCCaC</i>			
	SPRL		<i>shl</i>			<i>aCCaC</i>		

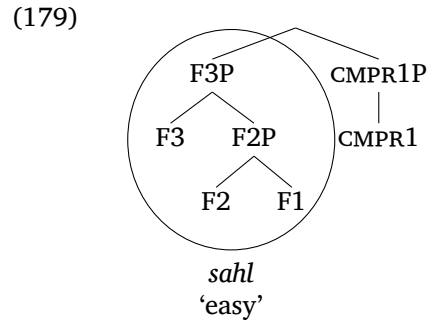
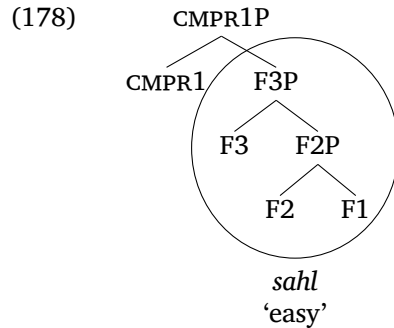
Let us now go through the derivation. We again start by merging F1 and F2, finding a match in (172), and realising the structure with the tri-consonantal root *shl* (175).



When F3 is merged to an irregular root, we lexicalise F3P without movement, since there is a match in (172), a lexical entry with the same concept as our earlier lexicalisation at F2P. This will allow it to grow the root to size F3P (see also (158)).



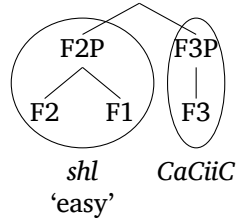
When CMPR1 is merged on top of (177), we reach a dead end. Non-movement lexicalisation does not work, since the CMPR1P in (178) is not contained in any entry. Movement of the closest labelled node containing the root provides a structure that cannot be lexicalised either, see (179).



The problem with (179) is that the comparative marker in (153) is not a match, since the comparative marker has a foot in F3, while the remnant CMPR1P on the right branch of (179) does not contain F3.

At this point, we have exhausted all options and we therefore have to backtrack. This means that we must go back to the previous cycle, i.e., F3P. When we first lexicalised this cycle, we succeeded without any movement, recall (177). However, since that led to a dead end, we now pretend that we had no match, and we try the next option. The next option is movement of the closest labelled node containing the bottom of the fseq. This is F2P, the complement of F3, and moving it produces the structure (180). This leads to lexicalisation of the F3P as the positive degree marker *CaCiiC*.

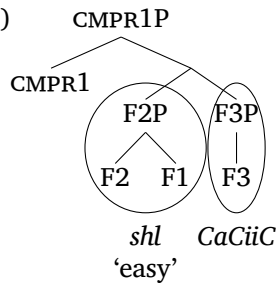
(180)



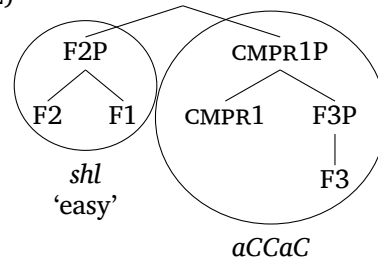
This structure corresponds to a non-existent regular positive **sahiil* for 'easy.' This regular positive, however, never surfaces, since the derivation only produces this form as an intermediate representation on its way to the comparative.

Since we now managed to lexicalise the F-1 cycle differently, we try to add F (i.e., CMPR1) again, yielding (181). Since (181) finds no match, we perform movement of F2P, the closest labelled node containing the bottom of the fseq, resulting in (182). In this structure, we can lexicalise the remnant CMPR1P on the right branch by the regular comparative template *aCCaC*.

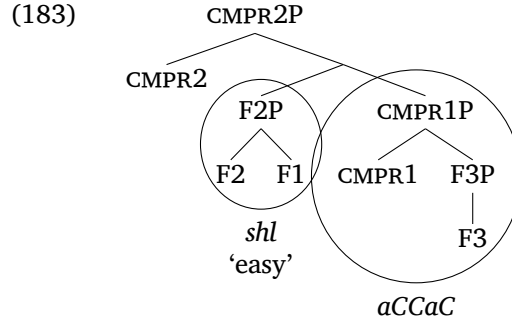
(181)



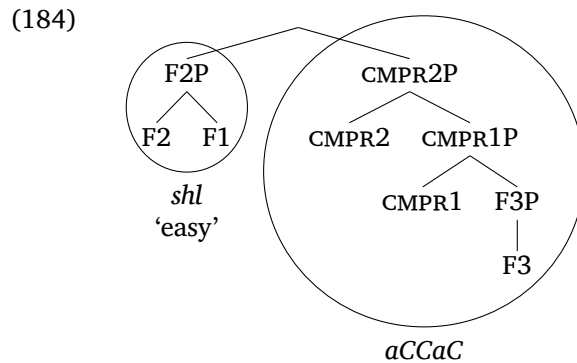
(182)



In the next step of the derivation we merge CMPR2, see (183).



After Evacuate-Closest movement of F2P, we lexicalise CMPR2P as the comparative templatic marker:



The combination of the root *shl* and the templatic affix yields the correct comparative *ashal* 'easier.'

This shows how backtracking works, and illustrates a case where it is needed.

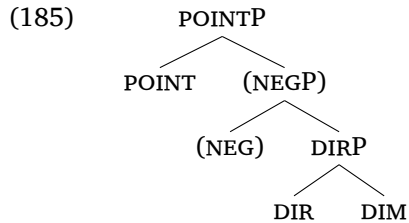
1.9 Roots

The previous sections have sketched the role and the position of features in the nanosyntactic model, and how features interact with the lexicon in derivations that are driven by the Lexicalisation Algorithm. This section turns to the features that reside at the lower part of the structure below CMPR1, which we have been for the most part conveniently labelling F1 and F2, adding an F3 later on.

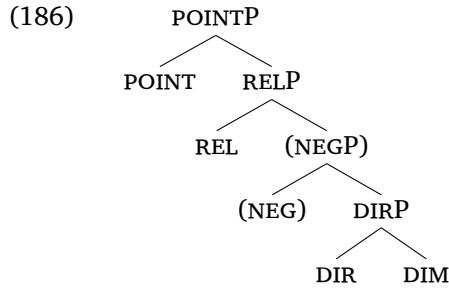
A common approach of conceptualising the lowermost positions would be to think of them as $\sqrt{}$ node and a categorising head *a* (see, e.g., Smith et al. 2019). While all we said so far is compatible with this claim, here we suggest a different interpretation of these projections. Drawing inspiration

from work by Ramchand (2008), we adopt here an approach that rejects the existence of a category-less root $\sqrt{}$, and adopts functional heads all the way down. Our specific proposal, as briefly introduced in Section 1.1, builds on the proposal by Vanden Wyngaerd et al. (2020), which we shall be using, albeit with some minor modifications. At the bottom of the hierarchy, there is a dimension, which could be size, velocity, height, colour, etc. The first feature is therefore DIM. The DIM feature has the effect of distinguishing one scale from another (Kennedy, 1999). Scalar adjectives come with an ordering on top of a dimension, which is represented by the feature DIR. Directions may be reversed by means of an optional reversal operator (called NEG in De Clercq & Vanden Wyngaerd 2019a). The presence or absence of NEG distinguishes antonymic pairs of adjectives from one another (like *tall-short*, *wet-dry*, *heavy-light*, etc.). Such pairs of adjectives thus involve the same dimension, but the direction of the scale is different: positive for *tall*, negative for *short* (see De Clercq & Vanden Wyngaerd 2018 for the syntactic relevance of this feature NEG in negative adjectives like *short*).

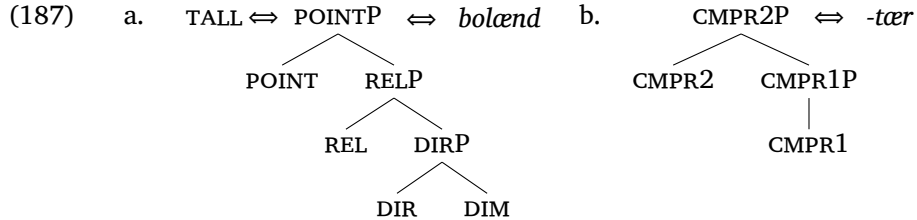
Gradable adjectives refer not just to a scale, but to a proper subpart of a scale, which can be a point or interval or area of the scale. We therefore take POINT to be the next feature in the hierarchy. This feature will divide the scale into two parts, one which exceeds this point and another one, which does not. The argument that the adjective is predicated of will have a degree of the relevant property that falls within the part of the scale that exceeds the relevant POINT (in either of two directions). Putting all of this together, we arrive at the structure in (185a) for a positive degree adjective, with the optional NEG feature in brackets.



In chapter 2, we shall elaborate this structure with the functional heads involved in the positive degree. In order to accommodate the rich morphology of these forms, it will turn out that we need to enrich this proposal by Vanden Wyngaerd et al. (2020) by an additional functional head located below POINT, which we shall call REL for ‘relational.’ The function of this head can be thought of as establishing a relationship between the scale and an individual, comparable to a measure function (type $\langle e, d \rangle$; Kennedy 1999), which maps an individual onto a degree.



As a result, there is no $\sqrt{}$ node at the bottom of the hierarchy; instead, functional heads reach all the way down. However, even in such a model, it is still possible to make a distinction between functional (closed-class) and lexical (open-class) vocabulary items. The distinction is whether or not a lexical item has a concept associated with it. We already saw this in the lexical items we gave for the Persian adjective *bolænd* ‘tall’ and the comparative marker *tær* in (34a) and (34b) above, repeated here in an updated form:



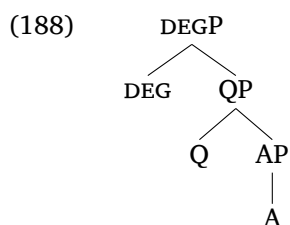
The comparative marker *tær* in (187b) is exhaustively defined in terms of its association of a phonology with a structured set of functional heads. These heads have, apart from an obvious syntactic relevance, a semantic interpretation. The lexical vocabulary item (187a) has the same, but it adds a concept, shown as TALL. This seems to be the minimal distinction that needs to be made between lexical and functional vocabulary items, and in nanosyntax it is also the maximal one.

We finish this section with a short terminological note. We shall henceforth refer to vocabulary items that serves as the base for further affixation with the term ‘root’. Adjectival roots carry a concept, but they are not to be confused with the roots familiar from DM ($\sqrt{}$), which we have discussed above.

Since roots lexicalise syntactic trees and not terminal nodes, they may vary in size, i.e. not all roots, even those belonging to the same word class, have to lexicalise the same number of features. The concept of a variable root size provides a powerful tool to account for a number of empirical phenomena, like allomorphy, suppletion, and zero marking, as we have shown above, and shall show in the following chapters.

1.10 A brief note on history

We conclude this chapter with a brief note on history. The attempts to construct an elaborate functional sequence for the adjective are found in the literature as early as Bresnan (1973), which was later developed in Corver (1997). Corver's proposal involves in total three heads in the extended projection of the adjective, DEG, Q and A, as depicted in (188).²¹



If we would compare Corver's proposal to ours, we would place the marker of the comparative (see Chapter 3) into Corver's DEG. We understand Corver's A as analogous to our scale, consisting of DIM, DIR and the optional NEG (see Chapter 2). Corver's Q corresponds to the part of our sequence which deals with setting the standard, i.e., REL and POINT. Let us now briefly review Corver's evidence.

This split degree proposal is based on certain distributional patterns of adjectival modifiers. DEG elements include the morphemes *too*, *as*, *so*, and *how*, which are in complementary distribution with each other, and therefore are taken to occupy the same position. The head Q hosts *much*, which is mostly invisible, but which surfaces in cases of so-called *much*-support, as shown in the following examples (Corver, 1997, 127).

- (189)
- a. John is fond of Mary. Maybe he is too much so.
 - b. John is fond of Mary. Maybe he is as much so as Bill.
 - c. The weather was hot in Cairo—so much so, that we stayed indoors all day.
 - d. John told me he was afraid of spiders, but I wonder how much so he really is.
 - e. John is wild about Madonna, but I am not really that much so.

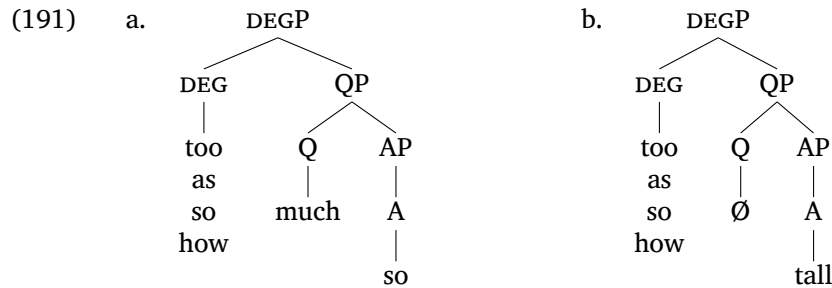
Much support occurs when the DEG head is filled, and the position of the adjective is taken by the pro-form *so*. Interestingly, *much*-support is impossible if *so* is absent, and the adjective takes its place:

- (190)
- a. John is too (*much) fond of Mary.
 - b. John is as (*much) fond of Mary as Bill.

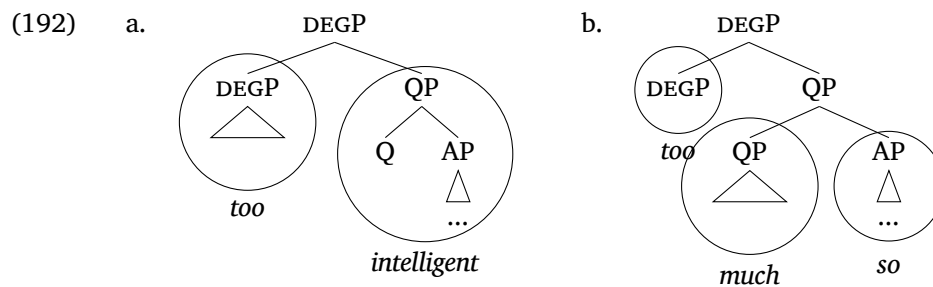
²¹ The equivalents of these heads are in fact also present in Bresnan's work, but she calls DEG Det, and for her Det is contained inside QP.

- c. The weather was so (*much) hot in Cairo that we stayed in-doors all day.
- d. I wonder how (*much) tall John is.
- e. I am not really that (*much) wild about Madonna.

The examples provided in (189) and (190) are depicted arboreally in (191a) and (191b), respectively. Notice that when the position of A is filled by *so*, we can clearly see three different positions. The reason why we do not see all three expressed with adjectives is that in these cases, Q is expressed by $\emptyset$.



While providing evidence for three positions, it is not at all obvious why the Q head would have to be filled in the case of (189) (with *so*), but has to remain empty in (191b) (with the adjective). However, adopting phrasal lexicalisation, we can easily make sense of this difference. All we have to say is that ordinary adjectives and the proform *so* differ in size: adjectives are bigger, and can lexicalise the feature(s) indicated by the head Q in Corver's system, and which are lexicalised by *much* in cases of *much*-support. The proform *so* is smaller, and needs another morpheme (namely *much*) to lexicalise Q. This is shown schematically in (192).



The obligatory absence of *much* in (192a), as well as its obligatory presence in (192b), follows automatically: in (192a) the root lexicalises the features

marked by the head Q in the tree, whereas in (192b) the root cannot do so, and neither can the DEG morphemes, so that *much*-support is needed.²²

There is an independent piece of evidence that supports this analysis. There exists in English a class of adjectives that does occur with *much*, such as *different*, *dependent*, *similar*, *alike*, *afraid*, *aware*, as the following examples (from the online SketchEngine corpus) illustrate:

- (193)
- a. South Vietnamese rainstorms are much different from those in the States.
 - b. The newly launched Rs 129 plan is much similar to the Vodafone prepaid plan of Rs 119.
 - c. Our militia are much afraid of being called to Norfolk at this season.

Observe that in these examples, no DEG element is present (although it can be present). In this respect, these adjectives contrast with the overwhelming majority of English adjectives, which are incompatible with *much*, also in the absence of a DEG element.

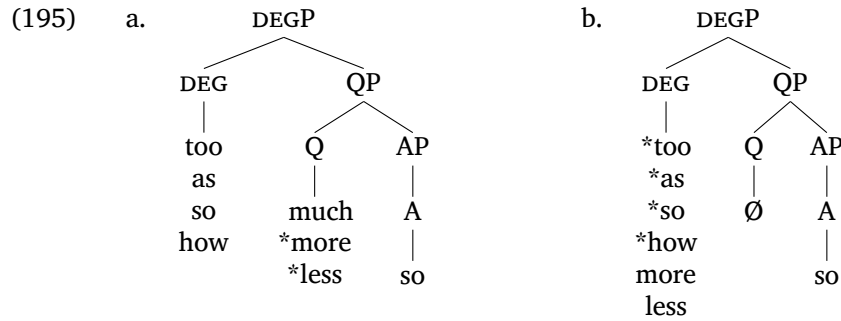
- (194)
- a. John is (*much) fond of Mary.
 - b. The weather was (*much) hot in Cairo.
 - c. Gleb is (*much) tall.
 - d. I am not really (*much) wild about Madonna.

This minimal contrast between the adjectives of (193) and those of (194) seems arbitrary, i.e. not predictable from any independent property of the adjectives in question. But it can easily be made sense of by assigning different root sizes to both classes, with those of (193) being the same size as the pro-form *so*, while those of (194) can lexicalise a bigger chunk of the adjectival functional sequence, including what Corver designates as Q.²³

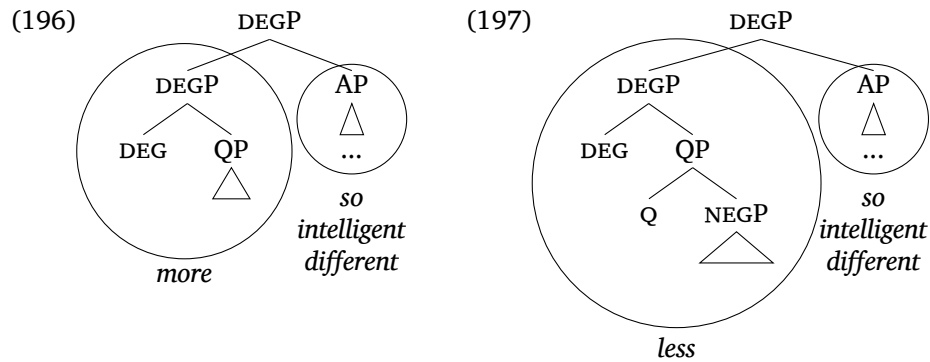
Another relevant piece of data concerns the position of *more* (which we take to be the comparative of *much*), as well as its polar opposite, *less*. In Corver's split degree system, these elements are hard to give a place. If we put *more* and *less* in Q, we expect them to alternate with *much*, which they do not (as shown in (195a)). If we put them in DEG, we expect them, incorrectly again, to be unable to occur with *so* while Q remains empty. This is shown in (195b).

²² For the sake of simplicity, we show *too* and *much* as lexicalising the heads DEG and Q, respectively, of Corver's split degree system. This is a simplification, in that it is likely that *too* and *much* lexicalise more features, but since this is independent of the point we wish to make, we will not go into this further here.

²³ A question that arises is why adjectives like *different*, *similar*, etc. can appear without *much*. We believe that this is a nongradable use, which we take to require a smaller amount of structure.



These distributional facts can directly be accounted for by assuming (much in the spirit of Bresnan 1973), that *more* and *less* are portmanteaus, which lexicalise the union of the sets of features indicated by DEG and Q, as shown in (196) for *more*. We take *less* to add a feature NEG, as in (197).



The fact that *more* and *less* lexicalise both DEG and Q explains why they are incompatible with other DEG elements (see (195a)), and why they can occur with *so* if there are no other DEG elements, as shown in (195b).

Observe further that the distinction between the two root types in English is neutralised with *more* and *less*, which can occur with the small roots like *so* and *different*, as well as with larger roots like *intelligent*. This is because the Q feature is lexicalised by *more/less*, causing the root to shrink.

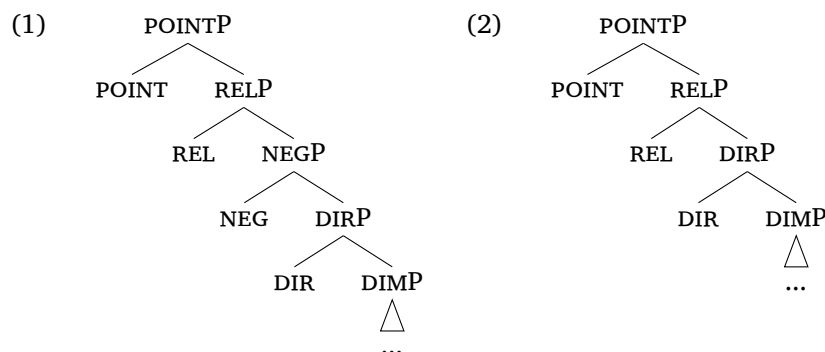
While this brief section mainly wanted to couch the new proposal in the generative tradition which it builds on, it also brings the discussion naturally to the Czech evidence for a decomposed adjectival structure, which we will zoom in on in great detail in the next chapters.

Chapter 2

The positive

2.1 Introduction

This chapter undertakes an in-depth investigation of positive degree adjectives in Czech. Our goal is to present morphological evidence for the existence of at least four functional projections in the structure of the positive, with an optional fifth feature for negation. The maximal structure with negation is given in (1), the smaller structure without negation in (2).



As explained in Section 1.9, the labels of the projections refer to a DIM(ension) (e.g., speed), with a particular DIR(ection). When NEG is absent, as in (2), this direction is positive (*fast*). When NEG is present, as in (1), the direction is reversed (*slow*). The combination of DIM, DIR and NEG determines a particular scale.

The REL head then introduces a relation between the scale and a point, whose specific value is specified by the projection POINT. The argument of the adjective must have a degree of the property that exceeds this point. When the scale is open, the value of POINT is provided by context, which accounts for context sensitivity of positive degree adjectives (*a big fly* refers to a different point on the scale than *a big elephant*). When the scale is (par-

tially) closed, the POINT coincides with the minimal or maximal point, e.g., *wet/dry* (Kennedy & McNally 2005).

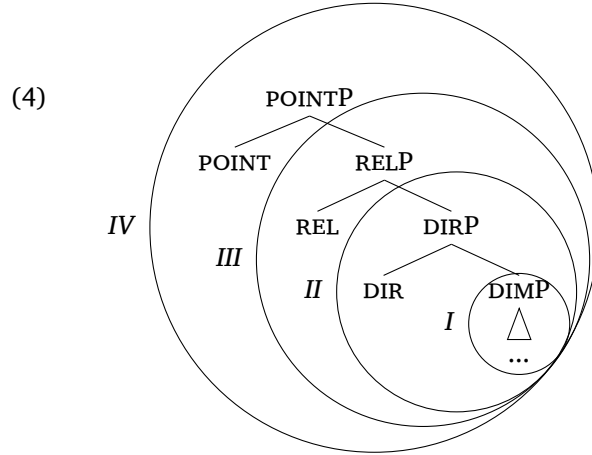
In the trees in (1) and (2), we draw a triangle below DIMP, which is important in that lexicalisation will be able to target DIMP on its own, i.e. it can mean that DIMP contains multiple features. However, it is also possible that the derivation just starts with DIM. We remain agnostic about this and use the triangle to leave this issue undecided.

Let us now turn to the evidence for the decomposition in (1) and (2), which comes from the morphology of positive-degree adjectives in Czech. The main observation is that in Czech, many adjectives are morphologically complex, and exhibit a segmentable morpheme in between the root and the agreement ending, as shown in (3). We refer to such morphemes as AUGMENTS.

- (3) a. pěk-**n**-ý ‘nice’
 b. hez-**k**-ý ‘pretty’
 c. div-**ok**-ý ‘wild’

Augments are generally obligatory: if a root has an augment in the positive, the adjective cannot be used without the augment. Augments are also not interchangeable: it is generally the case that a particular root requires one specific augment, and rejects all others. The selection between a particular root and a specific augment appears to be arbitrary, i.e., not determined by some independent property of the root like phonology or meaning. The root and the augment are often (though not always) inseparable.

Our analysis of augments will be couched in terms of (arbitrary) root size (see Section 1.6; see also (Taraldsen, 2010, 2025; Márkus, 2015; Caha et al., 2019; Cortiula, 2023, 2025; Caha & Taraldsen Medová, 2025; Bergsma et al., 2025; Vyshnevskaya, 2025b) amongst others). The idea is that different adjectival roots are lexically associated to different sizes of the adjectival structure, with the left-over features left for the augments to lexicalise. We depict this schematically in (4). Given the decomposition proposed in (2), we can define four different root classes differing in size, each class marked by a roman number (I-IV).

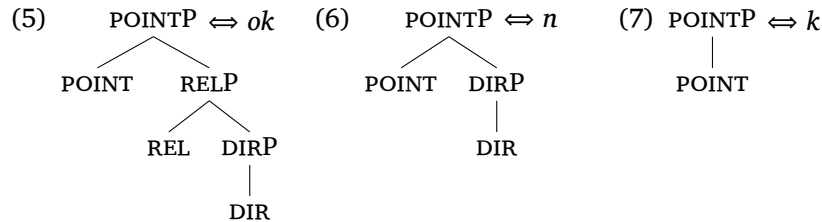


In this chapter, we argue that the four different root classes depicted in (4) map onto the three different augment classes listed in (3), plus a $\emptyset$ class (when the adjectival root lexicalises all the features). This is depicted in Table 2.1.

Table 2.1 Synoptic table of the augments

	ROOT SIZE	AUGMENT	AUGMENT FEATURES	EXAMPLE	GLOSS
I	DIMP	<i>-ok</i>	POINT, REL, DIR	div-ok-ý	‘wild’
II	DIRP	<i>-n</i>	POINT, REL	pěk-n-ý	‘nice’
III	RELP	<i>-k</i>	POINT	hez-k-ý	‘pretty’
IV	POINTP	$\emptyset$		mil- $\emptyset$ -ý	‘kind’

In structural terms, the entries of the augments look as follows:¹



Relying on the approach highlighted in the introduction, the pairing between roots and augments then follows from the lexicalisation algorithm. According to this approach, each root lexicalises as many functional heads

¹ We will modify these lexical entries in view of their interaction with comparative morphology in the following chapter, but these changes do not affect the derivation of the positive.

as allowed by its entry due to the preference for non-movement lexicalisation. When the root no longer matches the entire structure, complement movement is triggered and the augment is introduced. Since different roots have different sizes, different augments are needed. The following lexicalisation table summarises how individual roots, by virtue of the size of their entry, control the augment that they require.

(8) Lexicalisation table with augments

	DIMP	DIR	REL	POINT	GLOSS
Class I	div		ok		‘wild’
Class II	pěk		n		‘nice’
Class III	hez		k		‘pretty’
Class IV		mil			‘kind’

The chapter is structured as follows. Sect. 2.2 introduces the different augments. Sect. 2.3 considers diminutive adjectives. Sect. 2.4 looks at negative adjectives, and how negation interacts with the augments. Sect. 1.10 makes a historical note.

2.2 Introducing augments

In this section, we provide the necessary background on the Czech augments. We start by providing a sample of adjectives that belong in the individual classes in Table 2.2.

Going left to right, the adjectives in leftmost column have no augment, those in the second column have the augment *-n*, then *-k*, and finally *-ok*. We can see that these are relatively sizeable classes, with the exception of *-ok*, which is limited to the four adjectives in the final column. The *-k* adjectives listed in the table come close to an exhaustive list, while the first two classes are much larger than what is shown.

As a first rough approximation, the augments can be likened to the English adjectival suffix *-y*. Sometimes, *-y* is used as a derivational suffix (*slim-y*, *greed-y*, *speed-y*, *cheek-y*, etc.). However, with other roots, *-y* looks like a purely functional ‘augment’ that attaches to roots that do not independently exist (*prett-y*, *happ-y*, *cos-y*). Similarly, in Czech, the augments sometimes attach to nominal roots, but other times, they combine with bound roots that do not exist independently.

A couple of examples are provided in (9) for the augment *-n*. In (9a), we show examples where the adjective may appear noun-derived. In (9b), we show cases where no such noun exists. The cases where *-n* is used as

Table 2.2 Illustrating the four classes of Czech adjectives

POS	GLOSS	POS	GLOSS	POS	GLOSS	POS	GLOSS
blb-ø-ý	'stupid'	čer-n-ý	'black'	blíz-k-ý	'close'	div-ok-ý	'wild'
čil-ø-ý	'lively'	drs-n-ý	'rough'	brz-k-ý	'early'	hlub-ok-ý	'deep'
čir-ø-ý	'pure'	hluč-n-ý	'noisy'	břit-k-ý	'sharp'	šir-ok-ý	'wide'
čist-ø-ý	'clean'	hod-n-ý	'kind'	heb-k-ý	'smooth'	vys-ok-ý	'tall'
dlouh-ø-ý	'long'	hroz-n-ý	'terrible'	hez-k-ý	'pretty'		
dobr-ø-ý	'good'	jas-n-ý	'clear'	hlad-k-ý	'smooth'		
drah-ø-ý	'expensive'	jem-n-ý	'fine'	hoř-k-ý	'bitter'		
drz-ø-ý	'cheeky'	krás-n-ý	'beautiful'	krát-k-ý	'short'		
hloup-ø-ý	'stupid'	lev-n-ý	'cheap'	kluz-k-ý	'slippery'		
hol-ø-ý	'bare'	mast-n-ý	'fatty'	krot-k-ý	'tame'		
hust-ø-ý	'dense'	mír-n-ý	'mild'	křeh-k-ý	'fragile'		
chud-ø-ý	'poor'	něž-n-ý	'tender'	leh-k-ý	'easy'		
jist-ø-ý	'secure'	nut-n-ý	'necessary'	měk-k-ý	'soft'		
krut-ø-ý	'cruel'	pěk-n-ý	'pretty'	měl-k-ý	'shallow'		
mal-ø-ý	'small'	pev-n-ý	'firm'	mrz-k-ý	'meager'		
mil-ø-ý	'lovely'	pl-n-ý	'full'	níz-k-ý	'low'		
mlad-ø-ý	'young'	prázd-n-ý	'empty'	plyt-k-ý	'shallow'		
nah-ø-ý	'naked'	rov-n-ý	'straight'	prud-k-ý	'steep'		
plach-ø-ý	'timid'	sil-n-ý	'strong'	říd-k-ý	'thin'		
ploch-ø-ý	'flat'	sla-n-ý	'salty'	slad-k-ý	'sweet'		
slab-ø-ý	'weak'	slav-n-ý	'famous'	sliz-k-ý	'slimy'		
slep-ø-ý	'blind'	sluš-n-ý	'kind'	ten-k-ý	'thin'		
star-ø-ý	'old'	skrom-n-ý	'modest'	těž-k-ý	'heavy'		
such-ø-ý	'dry'	smut-n-ý	'sad'	trp-k-ý	'bitter'		
tich-ø-ý	'quiet'	snad-n-ý	'easy'	úz-k-ý	'narrow'		
tup-ø-ý	'blunt'	šťast-n-ý	'happy'	vel-k-ý	'big'		
tvrđ-ø-ý	'hard'	tuč-n-ý	'fat'	vlh-k-ý	'wet'		
zl-ø-ý	'evil'	vol-n-ý	'free'	vrat-k-ý	'unstable'		

a derivational morpheme provide evidence for its status as an independent morpheme.²

² The morphemes that follow the nominal roots are inflectional endings (marking gender, number, case).

(9)

	NOUN		ADJECTIVE	
a.	čest	‘honour’	čest-n-ý	‘honest’
	síl-a	‘strength’	sil-n-ý	‘strong’
	stříbr-o	‘silver’	stříbr-n-ý	‘silver’
b.			skrom-n-ý	‘modest’
			čer-n-ý	‘black’
			pěk-n-ý	‘pretty’

Table (10) shows the same thing for the augment *-k*.

(10)

	NOUN		ADJECTIVE	
a.	sliz	‘slime’	sliz-k-ý	‘slimy’
	slad	‘malt’	slad-k-ý	‘sweet’
	břit	‘blade’	břit-k-ý	‘sharp’
b.			krát-k-ý	‘short’
			leh-k-ý	‘easy’
			hez-k-ý	‘pretty’

Finally, the same can be observed also for the few instances of the augment *-ok*, as shown in (11).

(11)

	NOUN		ADJECTIVE	
a.	výš-e	‘height’	vys-ok-ý	‘high’
	šíř-e	‘width’	šir-ok-ý	‘wide’
b.			hlub-ok-ý	‘deep’ ³
			div-ok-ý	‘wild’

Summarising, there are examples where augments attach to nominal roots and turn them into adjectives. This shows that augments are independent morphemes, and they help ‘non-adjectival’ roots become adjectives. This supports the idea that they lexicalise adjectival functional heads.

At the same time, some of the roots do not exist independently as nouns or other parts of speech. In our analysis, this is reflected by the fact that the roots also lexicalise a part of the adjectival functional structure (but not the whole of it), so the root and the augment need to complement each other to form an adjective. This analysis is enabled by the decomposed structure pro-

³ The noun *hloub-k-a* ‘depth’ exists, but notice that this also has a suffix, so the root is more abstract than in the case of the first two adjectives, where the root alone can act as a noun. There is also the noun *div*, meaning ‘(a) wonder.’ We don’t see any link between the meanings, so we consider these unrelated roots.

posed in (2), where the property of ‘being a gradable adjective’ is distributed across multiple projections.

2.2.1 The absence of augments in the comparative

There are, in Czech, more adjectivising morphemes than the three we investigate here in detail. The main reason why we focus on the three augments in Table 2.2 as a special group is because they interact with degree morphology in a way that is *a priori* unexpected for derivational morphemes. Specifically, with some roots, they are only present in the positive, but absent in the comparative. In (12a), we illustrate this property for the augment *-n*.

(12)

	ROOT	AUG	AGR	ROOT	AUG	CMPR	AGR	
a.	snad-	n-	ý	snaz-		š-	í	‘easy(er)’
b.	hod-	n-	ý	hod-	n-	ějš-	í	‘kind(er)’

However, the pattern in (12a) (where the augment disappears in the comparative) is only found with a subset of adjectives with the *-n* augment. With other roots, such as in (12b), the augment must be preserved.⁴

Whether the augment is preserved in the comparative or not is dependent on the root: with *hod-* ‘kind’ given in (12b), the augment must be present, with *snad-* ‘easy’ in (12a), it is absent. This does not correlate with the ‘derivational’ status of the augment: both adjectives in (12) are of the *prett-y* type, i.e., with bound roots.⁵

In (13), we illustrate the same thing for the augment *-ok*. (13a) shows the pattern of the root *hlub-ok-ý* ‘deep,’ where the augment is dropped in the comparative, even though it is obligatory in the positive. (13b) shows the adjective *div-ok-ý* ‘wild.’ With this adjective, the augment is obligatory both in the positive and in the comparative.⁶

⁴ We can further observe that the presence/absence of the augment is reflected in the allomorphy of the comparative marker (*-š* vs. *-ějš*). We investigate comparative marking in Chapter 3.

⁵ For completeness, let us mention that *snad* in Czech exists as an adverb meaning ‘hopefully.’ There are no reasons to think that the adjective *snad-n-ý* ‘easy’ is related to it.

⁶ The form of the augment in the comparative, namely *-oč*, is due to a regular palatalisation process, this time marked on the consonant rather than on the vowel as in (12).

(13)

	ROOT AUG AGR			ROOT AUG CMPR AGR				
a.	hlub-	ok-	ý	hlub-	š-	í	‘deep(er)’	
b.	div-	ok-	ý	div-	oč-	ejš-	í	‘wild(er)’

To complete the overview, (14) shows that the augment *-k* may also be missing in the comparative, as in (14a), or it may be present, as in (14b).⁷

(14)

	ROOT AUG AGR			ROOT AUG CMPR AGR			
a.	blíz-	k-	ý	bliž-	š-	í	‘close(r)’
b.	hez-	k-	ý	hez-	č-	(š) í	‘pretty(er)’

Other derivational markers never drop in the comparative. We illustrate this in (15) for another productive derivational morpheme *-ov*. This morpheme also derives adjectives from nouns, like the noun *hlad* ‘hunger’ (15). However, unlike the other augments, it is always found both in the positive as well as in the comparative.

(15)

ROOT AUG AGR			ROOT AUG CMPR AGR				
hlad-	ov-	ý	hlad-	ov-	ějš-	í	‘hungry(er)’

Another difference between *-ov* and the three augments we focus on (*-n*, *-ok* and *-k*) is that *-n*, *-ok* and *-k* are in complementary distribution with each other. However, *-ov*, does not have this property. For instance, there are roots that may combine either with *-ov* or *-n*, as in (16).

(16)

- a. výživ-a
nutrition-NOM.SG.FEM
- b. výživ -n -é jídlo
nutrition -AUG -AGR food
‘nutritious food’
- c. výživ -ov -ý specialista
nutrition -AUG -AGR specialist
‘nutrition specialist’

We can see a clear meaning difference here, where *-n* derives a gradable adjective, and *-ov* a non-gradable one.

Some roots accept either *-ov* or *-k*, as in (17). Again, *-k* yields a gradable adjective, *-ov* a non-gradable one.

⁷ In the form ‘prettier,’ the augment *-k* is palatalised to *č*, and we put the comparative *-š* in the parentheses. We do so because it is not reflected in the orthography: the standard spelling of the form is *hez-č-í*. However, we suggest in Chapter 3 an underlying form *hez-k-š-í*, with the cluster *č-š* simplified to *č*.

- (17) a. sliz
slime.NOM.SG.MASC
b. sliz -k -ý povrch
slim -AUG -AGR surface
'slimy surface'
c. sliz -ov -á vrstva
slime -AUG -AGR layer
'slime layer'

There seem to be no analogous minimal pairs for the augment *-ok*, since there are only four adjectives in this class. Despite the absence of minimal pairs, similar contrast obtains. To see that, consider first two nominals based on the root *výš-*, meaning literally 'height'. The one in (18a) has no nominalizer, only an inflectional marker. The one in (18b) has a suffix *-k*, a common nominalizer in Czech (homophonous with one of the augments).

- (18) a. výš -e platu
high -INFL salary.GEN
'salary level'
b. výš -k -a platu
high -NOMINAL -INFL salary.GEN
'salary level '

The examples in (18) show that both nouns can be used as a label for the actual dimension (i.e., 'height' or 'size/level (on a vertical plane)') without implying the positive degree. In examples like (18), the two nouns are synonymous, though the noun in (18b) would be preferred when referring to physical height (e.g., the height of a person).

In (19), we compare two adjectives derived from these nouns. In (19a), there is an *-ok* adjective derived from the bare root. In (19b), there is an *-ov* adjective derived from the suffixed noun.

- (19) a. vys -ok -á budova
high -AUG -AGR building
'high building'
b. výš -k -ov -á budova
high -NOMINAL -AUG -AGR building
'high-rise building'

Again, the contrast goes in the same direction: the adjective with *-ok* in (19a) is a run-of-the-mill gradable adjective. The one with *-ov* in (19b) is a non-gradable adjective.

In cases like the ones discussed here, the distribution of *-ov* vs. the other augments cannot be described as allomorphy, because they give rise to different kinds of adjectives. While *-n*, *-k* and *-ok* yield gradable quality adjectives, *-ov* appears in non-gradable 'relational' adjectives.

Summarising, the three augments that we focus on (namely *-n*, *-k* and *-ok*) differ from other augments in being in complementary distribution with respect to each other, and in that they may be missing in the comparative. In such cases the root appears without any augment, directly followed by the comparative marker. We have also seen that other augments (like *-ov*) are less strongly associated to gradability than the ones we analyse here. This fact leads us to two conclusions.

The first one is that characterising the augments as simple category-changing morphemes is not completely accurate. For instance, they do not only turn nominals into adjectives, but also appear with bound roots. In addition, one would not expect derivational morphemes to disappear in the comparative. Similarly, the all-encompassing label “category-changing morpheme” does not capture contrasts such as the one in (19). In our approach, these facts are to be fitted into the general logic of ‘division of labour’ between the root and the suffix in the lexicalisation of functional heads of gradable adjectives, as depicted in the lexicalisation table (8). The specific details of how this division of labour plays out in the positive and the comparative is the main topic of this book.

2.2.2 How augments are selected

In this subsection, we establish the fact that for each adjectival root, it is an arbitrary property of that root which augment (if any) it combines with. By the word ‘arbitrary’ we mean that the pairing between roots and augments cannot be deduced from some independent property of the root (like its phonology or semantics). We therefore conclude that which augment a root combines with must be somehow independently coded in the lexicon. As highlighted at the outset, our approach to modelling this effect is to rely on the arbitrary lexical size of each root.

Let us start by considering the augment *-n*.⁸ The example (20) shows that the presence vs. absence of this augment cannot be determined by the phonology of the root. To show this, the example features two homophonous roots, namely *lev-*. In (20a), the root means ‘left’ and in this meaning, it requires no augment. In (20b), the very same root means ‘cheap,’ and it requires the augment *-n*.

- (20) a. lev- á noha
 left AGR leg
 ‘(the) left leg’
 b. lev- n- á noha
 cheap AUG AGR leg

⁸ Given the scarcity of the roots that take *-ok*, we skip this augment in the discussion below.

‘(the) cheap leg’

The point is that the presence vs. absence of *-n* cannot be determined by inspecting the phonology of the root. This is because the root has the same phonology in (20a) and (20b), namely *lev-*.

A similar point pertains to the meaning of the root. In Czech, there are two synonymous roots meaning ‘rough,’ namely *hrub* and *drs*. However, as the example (21) shows, one of them requires the augment *n* while the other doesn’t.

- (21) a. *hrub- á pokožka*
 rough AGR skin
 ‘a rough skin’
 b. *drs- n- á pokožka*
 rough AUG AGR skin
 ‘a rough skin’

This shows that meaning alone is insufficient to determine whether this meaning will require an augment or not. Therefore, the distribution of the augment cannot be deduced from (or reduced to) the meaning of the root.

Finally, it is also not the case that *n* can be reliably predicted from the morphosyntactic status of the base, e.g., whether it is a noun or an adjective. For instance, one could argue that the adjective ‘honest’ in (22a) has an augment because the root is a noun meaning ‘honour,’ as in (22b).

- (22) a. *čest- n- ý*
 honest AUG AGR
 ‘honest’
 b. *čest*
 ‘honour’

However, there are a plenty of bound roots like the one in (23).

- (23) a. *drs- n- ý*
 rough AUG AGR
 ‘rough’
 b. **drs*
 Int: ‘roughness’

The augment with *drs-n-ý* ‘rough’ in (23a) cannot be due to the fact that the base is a noun, since the relevant noun does not exist, see (23b).

Similarly, it is not the case that nouns always need an augment to become adjectives. We illustrate this by the contrast between (24) and (25). First, in (24), we show that the noun *stříbr-o* denoting ‘silver’ as a metal can be turned into an adjective using the augment *-n*. The adjective has both a colour and a material reading. (In the colour reading, it is gradable.)

- (24) a. *stříbr- o*
 silver NOM.NEUT.SG
 ‘silver (metal)’
 b. *stříbr- n- ý*
 silver AUG AGR
 ‘silver (colour/material)’

Compare to this the noun *zlat-o* ‘gold’ in (25a). This noun also denotes a metal and belongs in the same gender and noun class as *stříbr-o* ‘silver.’ However, it rejects the augment, see (25b).

- (25) a. *zlat- o*
 gold NOM.NEUT.SG
 ‘gold (metal)’
 b. *zlat- ý*
 gold AGR
 ‘golden (colour/material)’

The adjective *zlat-ý* ‘golden’ also has the two readings as the adjective *stříbr-n-ý* ‘silver.’ Therefore, there does not seem to be any relevant difference correlating with the presence of the augment in (24b) and its absence in (25b).

We conclude from this that neither the phonology of the root, its semantics or its morphological category can be considered an independent predictor of the appearance of the augment. Therefore, we consider the presence/absence of the augment an arbitrary property of the root: for each root, the child must learn whether it requires an augment or not.

The same conclusion generalises to the choice of a particular augment. To show this, let us now turn to the augment *-k*. In (26), we show that the appearance of the augment *-k* is difficult (if not impossible) to predict from the phonology of the root. The example (26) gives three roots of a very similar phonological shape (all ending in the sequence *lad* and differing only in the initial consonant), which each take a different augment.

- (26) a. *mlad- ý*
 young AGR
 ‘young’
 b. *slad- k- ý*
 smooth AUG AGR
 ‘sweet’
 c. *klad- n- ý*
 positive AUG AGR
 ‘positive’

This suggests that the quality of the augment (just like its presence) cannot be deduced from the phonology of the root alone.

In (27), we show that the appearance of *-k* also cannot be predicted from the meaning of the root. Specifically, both adjectives in (27) mean ‘pretty,’ but one takes *-n* and the other one takes *-k*.

- (27) a. *hez- k- á hudba*
 nice AUG AGR music
 ‘nice music’
 b. *pěk- n- á hudba*
 nice AUG AGR music
 ‘nice music’

The synonymous nature of these adjectives is supported by the following considerations. As noted in the corpus study by Křivan (2012, 42), *pěk-n-ý* and *hez-k-ý* ‘pretty’ are attested with about the same frequency in the positive degree (*pěk-n-ý* is slightly more common). However, the comparative of *hez-k-ý* is about 20 times more frequent than that of *pěk-n-ý* ‘pretty.’ Křivan explains this by relying on the observation that the roots are synonymous to a point where they are basically interchangeable in any context, and the preference for *hez-k-ý* in the comparative is driven by the fact that its comparative only has two syllables (*hez.čí* ‘prettier’), while the comparative of *pěk-n-ý* has three syllables (*pěk.něj.ší* ‘prettier’). The point is that given their synonymy, speakers do not need to consider any meaning difference and simply select the shorter form for reasons of efficiency.

This is relevant here inasmuch as it shows the high degree of the synonymy of the roots. Despite their synonymy, they combine with different augments. Therefore, the choice of the augment (*-n* vs. *-k*) cannot be determined by the meaning of the root.

Finally, the morphosyntactic category of the base is also irrelevant for the augment *-k*, which can attach to nominal roots as in (28), as well as to bound roots, as in (29).

- (28) a. *sliz- k- ý*
 slime AUG AGR
 ‘slimy’
 b. *sliz*
 slime
- (29) a. *heb- k- ý*
 smooth AUG AGR
 ‘smooth’
 b. **heb*

Summarising, we have seen that the augments under discussion are selected by specific roots, and their distribution cannot be predicted on the basis of the meaning, phonology, or some independent morphosyntactic property of the base.

2.2.3 A sample derivation

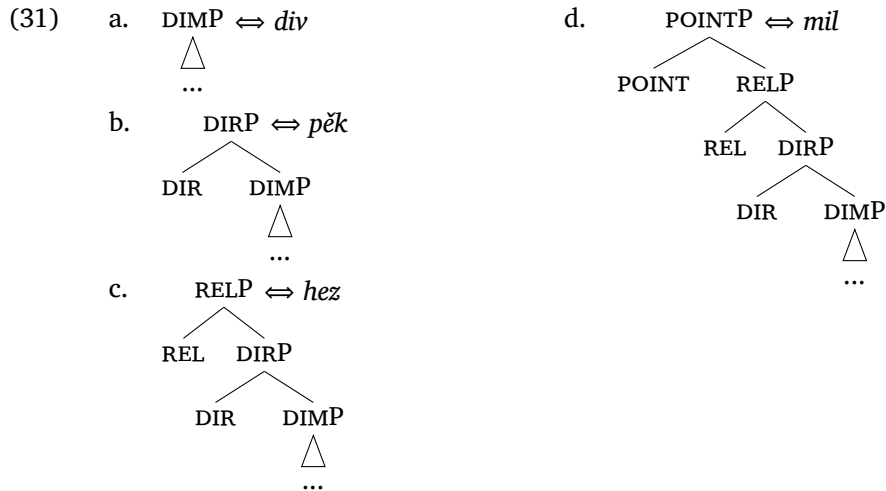
In this section, we provide a sample derivation of the *-ok* adjectives (Class I), *-n* adjectives (Class II), *-k* adjectives (Class III), and augment-less adjectives (Class IV). Our main goal is to show that the selection between the root and the suffix can be understood in terms of root size: different roots are associated to a different number of features, and this automatically leads to different augments. There is no need for context-sensitive rules of the sort used in Distributed Morphology or other theories. This is going to prove crucial when we turn to the comparative, where augment distributions are impossible to capture using such rules, recall Section 1.1.2.

We repeat the lexicalisation table with the four classes from (8) above as (30) here.

(30)

	DIMP	DIR	REL	POINT	GLOSS
Class I	div		ok		'wild'
Class II	pěk		n		'nice'
Class III	hez		k		'pretty'
Class IV		mil			'kind'

The corresponding entries for the roots are given in (31).



It is easy to see that each root in (31) lexicalises a different number of the features of the positive, and it leaves a different residue, i.e., a different number of features that it does **not** lexicalise. These unlexicalised features are left for the augment to lexicalise. In (32), we repeat the entries for the

(32) $\text{POINTP} \Leftrightarrow ok$ (33) $\text{POINTP} \Leftrightarrow n$ (34) $\text{POINTP} \Leftrightarrow k$

Diagram (32) structure:

- POINTP branches into POINT and RELP.
- RELP branches into REL and DIRP.
- DIRP branches into DIR.

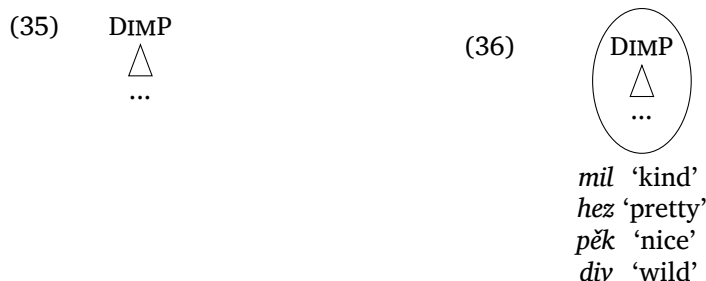
Diagram (33) structure:

- POINTP branches into POINT and RELP.
- RELP branches into REL.

Diagram (34) structure:

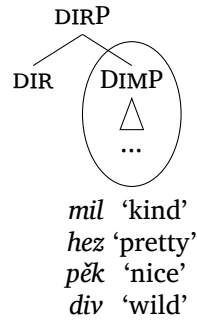
- POINTP branches into POINT.

All derivations start by assembling the dimension, i.e., DIMP. All of the roots can lexicalise this structure, as shown in (35). The choice among roots is not subject to the Elsewhere Condition, but happens on the basis of free choice, i.e. they are inserted on the basis of the relevant concept.

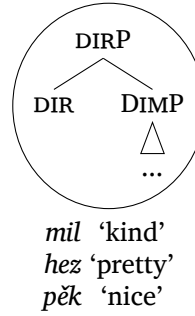


Since (36) can be lexicalised, the derivation may continue by Merge F (37). We therefore merge DIR with (36), yielding (37). At this point, the derivations diverge. The three roots that contain DIR in their lexical entry are a match, so the structure (37) is lexicalised without movement, as in (38).

(37)

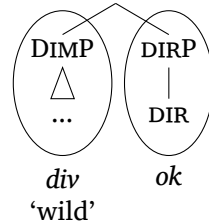


(38)



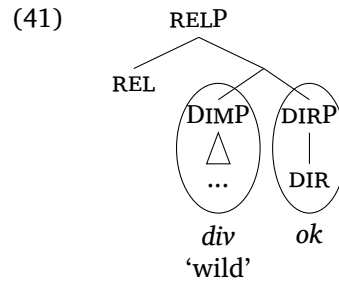
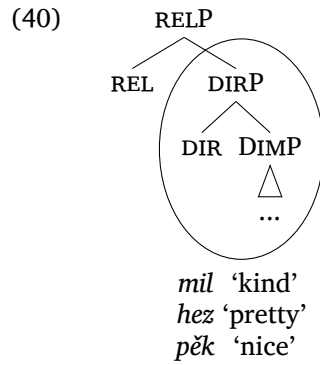
However, since the root *div* 'wild' does not have DIR in its lexical entry, evacuation movement is triggered of DIMP, which is the closest labelled node containing the bottom of the fseq.⁹ This results in (39), and the lexicalisation of DIR by *-ok*.

(39)

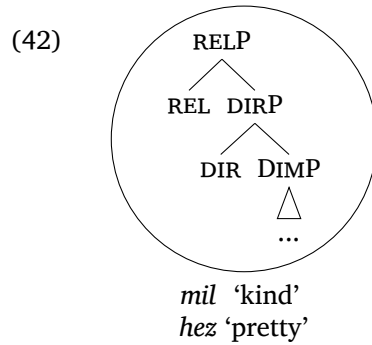


The derivation goes on by merging REL. In the roots that so far triggered no evacuation movement (Classes II-III-IV), REL will simply be merged on top of DIRP, as in (40). In the case of *-ok* roots (Class I), REL comes on top of a structure to which movement has applied in the previous cycle (41).

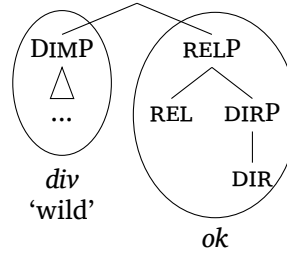
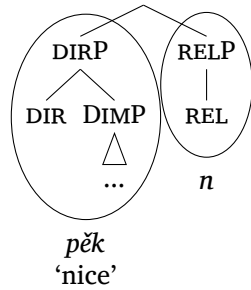
⁹ There is a potential issue here with the Elsewhere Condition, in that we do not want bigger roots to compete with smaller ones for lexicalisation. If we did, bigger roots would always override smaller ones since they do not require movement, thereby making the smaller ones unpronounceable. This is obviously an unwanted result. To avoid this, we assume that only lexical entries which the same concept compete with each other (but see Caha et al. 2019 for a different approach).



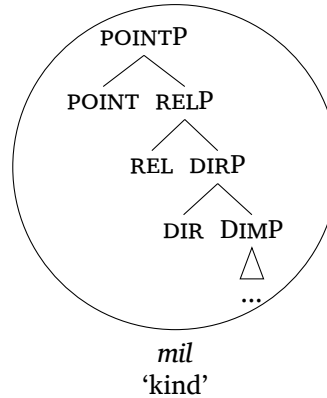
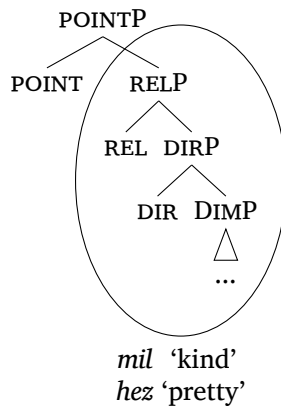
Now there are only two roots left that can lexicalise RELP directly, without movement. These are the roots that contain RELP in their entry (Class III *pěk*- 'pretty' and Class IV *mil*- 'kind'). This is shown in (42):

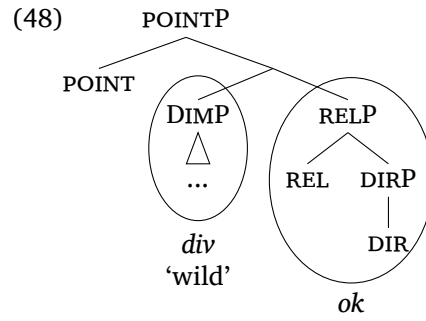
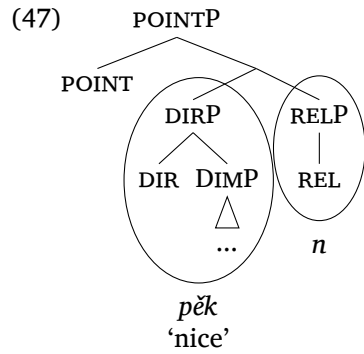


However, the Class II root *pěk* 'pretty' cannot lexicalise RELP (similar to the Class I root *div* in (41)). Since matching fails, evacuation movement is triggered, which will move the first labelled node containing the bottom of the fseq. This is the node DIRP in the case of the Class II roots, resulting in (43). In the case of the Class I root *div*, evacuation movement will target DIMP (compare (41)):

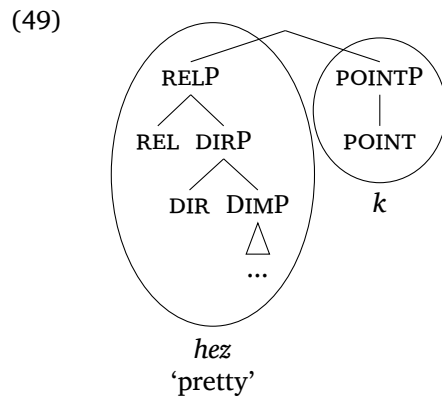


In the final step of the derivation, we merge POINT. Again this yields different outcomes depending on the type of root we are looking at. In the Class III and IV roots, POINT will come on top of a structure without movement, as shown in (45)-(46). In the Class II (47) and Class I roots (48), it comes on top of a structure to which movement has already applied. Lexicalisation will be straightforward without movement only in the case of the Class IV roots (46).



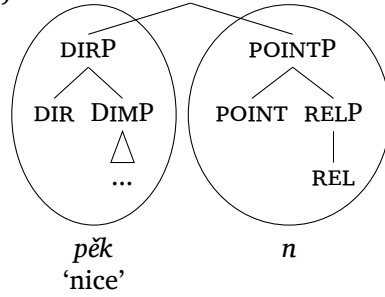


With the roots other than the Class IV roots, evacuation movement will apply, but it will target a different node in each class. In the case of the Class III roots like *hez*, the closest labelled node containing the bottom of the fseq is RELP (see (45) above), resulting in (49). Here the remnant POINTP is successfully lexicalised by *-k*.

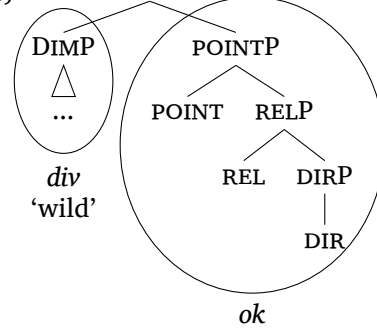


In Class II, movement targets DIRP (see (47)), yielding (50), and in Class I it targets DIMP (see (186)), resulting in (51). In either case, the augment that was already introduced in earlier cycles grows to the size of POINTP.

(50)



(51)



The lexicalisations thus derived match the lexicalisation table (8) above, repeated for convenience in (52).

(52) Lexicalisation table with augments

	DIMP	DIR	REL	POINT	GLOSS
Class I	div			ok	'wild'
Class II	pěk			n	'nice'
Class III	hez			k	'pretty'
Class IV		mil			'kind'

2.3 Diminutive adjectives

So far, we have shown that selection between roots and augments can be understood in terms of root size. Under this proposal, each augment lexicalises a different set of features. This section provides independent evidence for this proposal from diminutives, including evidence for the different relative sizes that we have assigned to the augments.

The section is structured as follows. In Section 2.3.1, we propose a specific position for the diminutive head in the sequence of projections established so far, namely between REL and POINT. In Section 2.3.2, we discuss the predictions that arise for the form of the diminutive when the specific position of the DMV head is combined with the account of augments (as depicted in (52)). In Section 2.3.3, we discuss various allomorphs of the diminutive head. In Section 2.3.4, we show how the current lexicalisation algorithm produces the correct forms of the diminutive based on different root size. In Section 2.3.5, we turn to several diminutive forms that require invoking root shape as the relevant parameter of variation.

2.3.1 A diminutive head

In Czech, both nouns and adjectives may carry a variety of diminutive morphemes. In (53), we provide an example with the diminutive *-ink*. In (53a), we show this morpheme on the noun ‘mum’ in an endearing function. On the adjective ‘small,’ it has an intensifying function.¹⁰

- (53) a. mám -a ~ mam -**ink** -a
 mum -INFL mum -DMV -INFL
 ‘mum ~ mummy’
 b. mal -ý ~ mal -**ink** -ý
 small -INFL small -DMV -INFL
 ‘small ~ very small’

The meaning of the diminutive depends in part on the adjective. With negative adjectives like ‘small,’ the meaning can be either intensifying or evaluative, indicating a positive attitude of the speaker towards something (possibly to the size of the object, the object itself, or possibly also conveying a general affection towards the addressee, e.g., a child). With positive adjectives, diminutives only have the emotive meaning, but not the intensifying one. We illustrate this in (54) with a different diminutive, *-ičk*, see (54b).

- (54) a. vod -a ~ vod -**ičk** -a
 water -INFL water -DMV -INFL
 ‘water ~ water + positive attitude’
 b. star -ý ~ stař -**ičk** -ý
 old -INFL old -DMV -INFL
 ‘old ~ old + positive attitude’

Diminutive adjectives reveal two kinds of asymmetries between the arguments we are focussing on. The first asymmetry has to do with the base to which the diminutive attaches. Specifically, the diminutive morpheme attaches to a base that includes the root and the augment in classes I and II, but it attaches to the bare root in Class III and IV, see (55).

- (55) The base of the diminutive

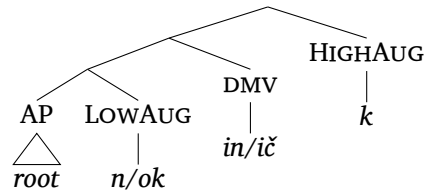
	CLASS	AUGMENT	BASE
a.	CLASS I	- <i>ok</i>	ROOT-AUG-DMV
b.	CLASS II	- <i>n</i>	ROOT-AUG-DMV
c.	CLASS III	- <i>k</i>	ROOT -DMV
d.	CLASS IV	-∅	ROOT -DMV

¹⁰ We do not decompose the diminutives *-ink* and *ičk* here, though later – in Section 2.3.3 – we argue that they are actually bi-componential, namely *-in-k* and *ič-k*.

Moreover, the different diminutives always end in *-k*. It is *a priori* unclear whether the *-k* is a part of the diminutive marker, or whether it is the *-k* augment. In our account, we shall treat the *-k* that appears after in diminutives as a separate morpheme, the same one as the augment *-k* (as we discuss in greater detail in Section 2.3.3). If this is correct, and all diminutive adjectives include the augment *-k* at their right edge, we arrive at the second asymmetry: only the augment *-k* follows the diminutive marker.

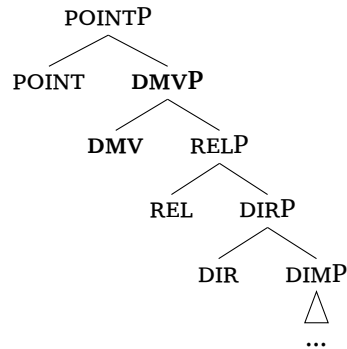
This shows that the augments are not equal, and occupy different structural positions. We depict this asymmetry in (56). The tree clearly shows the need for at least three projections in a non-diminutive adjective: going bottom up, we need one position for the root, one for the low augments *-n/-ok*, and one for the high augment *-k*.

(56)



Based on these asymmetries, we propose that the diminutive head DMV occupies a position in the tree in between POINT and RELP.

(57)



That way, the base to which the diminutive attaches will include the augments of classes I and II, since these realise heads below DMV. However, the augment *-k*, which lexicalises the POINT projection above DMV, is not found in the base to which the diminutive marker attaches. This is in line with the analysis of the augments as in (56). In what follows, we show that the two asymmetries between the augments noted above can be captured under this proposal interacting with the account of augments established so far.

2.3.2 The predictions

Let us start by presenting the lexicalisation table in (58), where we have placed a diminutive head DMV in between REL and POINT, as proposed in (57). The head DMV in the table is realised by a dedicated (agglutinative) marker *dmv*, which may take various forms (as we saw in the preceding section), and which we will discuss in greater detail in Section 2.3.3.

(58) Diminutives

	DIMP	DIR	REL	DMV	POINT	GLOSS
Class I	div	ok	<i>dmv</i>	k	‘wild’	
Class II	pěk	n	<i>dmv</i>	k	‘nice’	
Class III	hez		<i>dmv</i>	k	‘pretty’	
Class IV	mil		<i>dmv</i>	k	‘kind’	

What is important now is the position of DMV in the hierarchy. Since it comes on top of RELP, the marker that lexicalises DMV is predicted to attach to a form that lexicalises RELP. The roots in Classes III and IV are large enough to lexicalise the whole RELP; therefore, in these classes, the diminutive attaches directly to the root. In Classes I and II, the root cannot lexicalise the whole RELP, and augments are therefore needed. This way, we derive the first asymmetry.

Another interesting prediction concerns the lexicalisation of POINT in the diminutive. In particular, the DMV head separates POINT from REL, thereby affecting the lexicalisation of POINT in all classes where a portmanteau morpheme lexicalises the sequence REL and POINT.

For instance, the roots of Class IV are lexically specified as lexicalising all the heads of the positive (see the bottom row in (52)). But when DMV is added, the root can only lexicalise the sequence DIM, DIR, REL. It cannot lexicalise POINT across DMV because the lexical entry for the root does not contain DMV. As a result, it cannot lexicalise a constituent that contains DMV. Therefore, in the diminutive of Class IV, POINT remains without lexicalisation.

What is predicted to happen in this case (as depicted in Table (58)) is that POINT will have to be lexicalised separately, i.e., by an extra morpheme. Now according to our original proposal in (52), the augment *-k* is dedicated to the task of realising POINT (recall the analysis of Class III in (52)). Therefore, nothing else said, we expect that the diminutive marker in Class IV will be followed by *-k*, lexicalising POINT. We will see that this prediction is also borne out, and it represents the second asymmetry noted in the previous section.

In Classes I and II, the fate of POINT is similar. Under normal circumstances, the augment (as found in the positive) lexicalises POINT along with REL. However, due to the intervening DMV, the augments in Classes I and II

cannot lexicalise POINT across the intervening DMV. As a result, POINT must be again lexicalised by *-k*.

The discussion is summarised in (59).

(59) Diminutive adjectives

a. **The form of the base**

- (i) In Classes I and II, the diminutive attaches to the root plus the augment
- (ii) In Class III and IV, the diminutive attaches directly to the root

b. **The emergence of *-k***

In all classes, the diminutive marker is followed by *-k*

In effect, once we place the DMV head below POINT, and combine this with our account of augments, we predict exactly the two kinds of asymmetries that we have observed between the augments. To the left of the diminutive, we only find the augments *-n* and *-ok*, but not *-k*. To the right of the diminutive morpheme, we only and always find the augment *-k*.

With the expectations in place, let us turn to the full set of facts. These are depicted in Table (60). We use here yet another diminutive, namely *-ouč* and segment the *-k* as a separate marker. The data are organised according to the individual classes. We comment on the data below the table.

(60) Diminutive adjectives in Czech¹¹

	POSITIVE	DIMINUTIVE	GLOSS
Class I	šir- ok -ý div- ok -ý	šir- ok -ouč- k -ý div- ok -ouč- k -ý	'wide' 'wild'
Class II	příjem- n -ý hod- n -ý šťast- n -ý jem- n -ý	příjem- ň -ouč- k -ý hod- ň -ouč- k -ý šťast- ň -ouč- k -ý jem- ň -ouč- k -ý	'pleasant' 'kind' 'happy' 'fine'
Intermediate Class	drob- n -ý chlad- n -ý	drob- -ouč- k -ý chlad- -ouč- k -ý	'tiny' 'cold'
Class III	leh- k -ý heb- k -ý krát- k -ý hlad- k -ý	leh- -ouč- k -ý heb- -ouč- k -ý krat- -ouč- k -ý hlad- -ouč- k -ý	'light' 'smooth' 'short' 'smooth'
Class IV	blb-ý mal-ý čist-ý slab-ý	blb- -ouč- k -ý mal- -ouč- k -ý čist- -ouč- k -ý slab- -ouč- k -ý	'stupid' 'small' 'clean' 'weak'

Let us first look at the data from the perspective of the prediction (59b), which says that in all classes, the diminutive marker (-*ouč* in (60)) is always followed by the augment -*k*, i.e., even when the base adjective lacks this augment. This prediction is borne out: all the diminutive forms have the augment -*k* after the diminutive. This indicates that the augment -*k* indeed resides structurally above DMV, unlike all the other augments.¹²

The second prediction (recall (59a)) concerns the base of the diminutive, i.e., the form to which the diminutive marker attaches. Here we expect an asymmetry among the augments, namely we expect that -*n* and -*ok* should precede the diminutive marker, while -*k* never does (it follows DMV). This prediction is borne out as well. Table (60) shows that -*ok* and -*n* of Classes I and II indeed precede the DMV -*ouč*. We can also see that in Class III, -*k* is never found to the left of the diminutive marker. In this class, the diminutive always attaches to the bare root.

In the middle of the Table (60), there is a (small) class of adjectives that show intermediate behaviour: they pattern with the Class II adjectives in the positive in that they have the -*n* augment, and they pattern with Class III in not having this -*n* augment in the diminutive. These adjectives present a puzzle because they do not conform to our expectation based on how the classes should behave based on their root size. We are going to argue that these adjectives actually have the same root size as Class III adjectives (hence no augment to the left of the diminutive), but they have a different root shape, which is why they exhibit the augment -*n* in the positive. We discuss these adjectives in detail in Section 2.3.5.

As a way of summarising this section, we provide a synoptic overview of the way the positive adjectives are paired with their diminutive versions in Fig. 2.1.

The five arrows represent the five different groups found in Table (60). As we already stated in (59) above, the figure shows that, regardless of the root class, in the presence of the diminutive, the augment -*k* invariably pops up. Another generalisation is that the augment -*k* is never found to the left of the diminutive, but always to its right, while all the remaining augments (whenever present) are to the left of the diminutive.

In Section 2.3.4, we show in detail how these classes can be captured in terms of root size, ordered from smallest (top) to largest (bottom). We shall also explain why -*k* shows up in the diminutive of all root classes. In Section 2.3.5, we look at the interesting fact that the *n* class has two different

¹¹ The forms in the table are, for the most part, taken from the corpus study of Czech diminutives by Káňa (2016, 197). The forms in Class I are our own because Káňa (2016) does not mention them, presumably for the reason that they are not attested in the corpus. Nevertheless, we think that this is an accidental gap. The forms we give in Class I are attested on google and are judged as acceptable by native speakers.

¹² The only potential issue here is whether *ouč-k* should be decomposed: could it be the case that the diminutive is -*oučk* and there is no augment following it? We address this issue in Section 2.3.3 and provide reasons for the assumed segmentation, i.e., *ouč-k*.

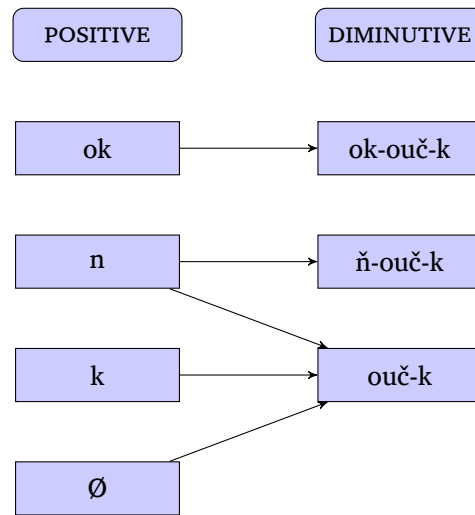


Fig. 2.1 Adjective-diminutive pairs

diminutive forms, depending on the root. This account will be couched in terms of root shape.

2.3.3 The form of the diminutive

In this section, we argue that the diminutive *-oučk* is not an indivisible morpheme, but should be segmented into the diminutive *-ouč* and the augment *-k*. The issue of segmentation (*-ouč-k*) vs. non-segmentation (*-oučk*) arises because the diminutive *-ouč* is always followed by *-k*, for reasons that, under our analysis, are linked to the need to lexicalise POINT. However, this forced co-occurrence of the morphemes makes the segmentation harder to motivate. As a result, some descriptive works, e.g., Šimandl et al. (2016), treat *-oučk* as non-segmentable (or minimally, the source does not suggest any segmentation). On the other hand, Rusínová & Nekula (2016) in their overview of diminutive morphology suggest a division into “primary, simple affixes” and “secondary, compound affixes,” treating *-ouč-k* as an instance of the latter.

Before we present our reasons for segmentation, let us point out that our analysis would also work with a non-decomposable diminutive *-oučk*. This alternative analysis is shown in (61) (where *dmv* is again a variable for the various forms that the diminutive marker may take).

(61) Non-decompositional analysis of *-oučk*

	DIMP	DIR	REL	DMV	POINT	GLOSS
Class I	div	ok		<i>dmv</i>		‘wild’
Class II	pěk	n		<i>dmv</i>		‘nice’
Class III	hez			<i>dmv</i>		‘pretty’
Class IV	mil			<i>dmv</i>		‘kind’

According to this analysis, the non-decomposable diminutive *-oučk* lexicalises both DMV and POINT. The analysis still requires an asymmetry between the augment *-k* on the one hand and the other augments *-ok/-n* on the other, since *-k* is never found to the left of *dmv*. So even under this analysis, we still gain evidence for a structural asymmetry between the augments, which in turn supports our idea that the augment *-k* occupies a high position in the functional structure of the adjective (higher than DMV). Given the possible alternative analysis in (61), the reason for defending the segmentation is not that the segmentation is necessary to prove our point about augments. The point is that there are independent reasons for assuming that *oučk-k* should be decomposed.

Our argument for separating the final *-k* from the *-ouč* part is that Czech has a large number of formally distinct diminutives. We do not fully understand their meaning differences and their internal composition, but all the different forms of diminutive adjectives have one constant, which is that they end in *-k*. This strongly suggests that the final *-k* is an independent morpheme.

The following table (62) shows several forms of the diminutive for Class II adjective *skrom-n-ý* ‘modest,’ for Class III adjective *ten-k-ý* ‘thin’ and for Class IV adjective *mal-ý* ‘small.’

(62) The richness of diminutive forms (Káňa 2016, 202-8)

BASE	<i>ten-k-ý</i> ‘thin’		<i>mal-ý</i> ‘small’		<i>skrom-n-ý</i> ‘modest’
DMV1	ten -ouč	-k-ý	mal -ouč	-k-ý	skrom-ň -ouč -k-ý
DMV2	ten -ič	-k-ý	mal -ič	-k-ý	skrom-ň -ič -k-ý
DMV3	ten -oun	-k-ý	mal -oun	-k-ý	skrom-ň -oun -k-ý
DMV4	ten -in	-k-ý	mal -in	-k-ý	
DMV5	ten -ounin	-k-ý	mal -ounin	-k-ý	

We call the different diminutive forms DMV1-DMV5, as marked in the first column of Table (62). The forms are taken from the corpus study by Káňa (2016), whose ultimate source is the Czech National Corpus.¹³

¹³ There are gaps in the paradigm of the adjective *skrom-n-ý* ‘modest’ in the last column: these forms are not attested in the corpus. The native intuitions of one of the authors (Caha) suggest that these gaps may be accidental (i.e., that the forms are grammatical even though they are unattested).

The main point is that all the diminutives end in *-k*, and it would be strange if Czech included five different, unrelated and non-segmentable diminutive suffixes, all of which end in *-k*. The segmentation makes this fact non-accidental: all the diminutives end in *-k* because they share the same morpheme, namely the augment *-k*.

Taking the idea of a segmentation for granted, let us now address the issue how one should analyse the final *-k*. This question arises in light of the fact that nominal diminutives also use *-k* to mark the diminutive, as illustrated in Table (63).

(63)	BASE	SINGLE DMV	DOUBLE DMV	GLOSS
	lod'	lod'-k-a	lod'-ič-k-a	'boat'
	mám-a	mam-k-a	mam-in-k-a	'mother'
	Michal	Miš-k-a	Miš-un-k-a	proper name

The data show that various nouns have a simple diminutive with just *-k*, or a 'double' diminutive formed by various markers in addition to and preceding *-k*. The three rows in Table (63) give three different forms of the double diminutive, which correspond to some of the adjectival diminutives seen in (62). For instance, the sequence *-ič-k* on the first row in (63) corresponds to the DMV2 form in (62). The form *-in-k* on the second row of Table (63) corresponds to DMV4.

The pattern where an adjectival augment is formally identical to a nominal diminutive (or a subpart thereof) is not one that is restricted to Czech. An example is English, where the augment-like status of the morpheme *-y* in adjectives has been mentioned earlier (*trick-y*, *happ-y*, etc.). The same morpheme also occurs in nominal diminutives:

(64)	PEOPLE	daddy, mummy, granny, nanny
	PROPER NAMES	Johnny, Lizzy, Ricky, Becky
	OBJECTS	potty, dolly, cookie
	ANIMALS	birdy, doggy, kitty, piggy, ducky, horsy, ducky, froggy, fishy
	BODY-PARTS	tummy, tootsy (= bum), fanny, willy, pussy

Another case like this is found in Turkish (Türk 2020). Consider first the fact that nominal diminutives are formed from the base by the addition of *-cık*, see (65).¹⁴

(65)	Turkish diminutives (Türk 2020)		
	BASE	DIMINUTIVE	GLOSS
	kapak	kapak-çık	'lid'
	gemi	gemi-çık	'boat'

¹⁴ *-cık* is subject to vowel harmony.

When such diminutive markers attach to adjectives, they intensify the meaning, which is why Türk (2020) calls them augmentative. We list several relevant forms in Table (66). The top part of the table (above the dividing line in the middle) provides two examples representing augmentatives of consonant-final roots (on the first two lines), and two examples of vowel-final roots (on the two lines below this). The consonant-final roots are characterised by the appearance of an epenthetic vowel between the root and the suffix.

(66) Turkish adjectival diminutives (augmentatives, Türk 2020)

BASE	GLOSS	AUGMENTATIVE	GLOSS
genç	‘young’	gence-ci-k	‘very young’
dar	‘tight’	dara-cı-k	‘very tight’
ince	‘thin’	ince-ci-k	‘very thin’
kısa	‘short’	kısa-cı-k	‘very short’
mini-k	‘tiny’	mini-ci-k	‘very tiny’
sıca-k	‘warm’	sıca-cı-k	‘very warm’
yumuşa-k	‘soft’	yumuşa-cı-k	‘very soft’
küçü-k	‘small’	küçü-cü-k	‘very small’

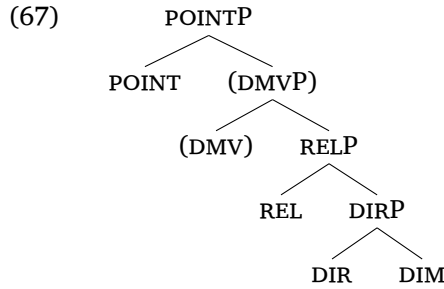
The lower part of the table shows that when the diminutive morpheme is added to a *-k* final adjective, the diminutive does not appear to the right of *-k*, but to its left, creating the impression that the root-final *-k* disappears. This does not look like a regular phonological alternation, because the noun ‘lid’ in (65) keeps the final *k* in *kapak-çık* ‘lid.’ Rather, Türk (2020) suggests that the final *-k* in Turkish positive adjectives is an independent suffix (an augment), and that the diminutive *-ci* infixes between the root and the suffix (as in Czech *-k* adjectives). In cases where the root does not feature this *-k* morpheme (the ones above the dividing line in the table), *-k* surfaces following the diminutive morpheme, also like in Czech. The augment status of the Turkish *-k* is further supported by the fact that it also drops in the causative of some *-k* final adjectives, e.g. *yukse-k* ‘high’ and *yukse-lt* ‘to ascend’.

To conclude, in this section, we defended a segmentation of diminutives in Czech into *-dmv-* and *-k*. We focused on the diminutive marker *-ouč-k* and hence argued for it to be divisible into *-ouč* and *-k*. We also addressed the observation that *-k* appears both with adjectives and nouns, thus raising the question whether the final *-k* should be analysed as an (adjectival) augment or a (nominal) diminutive morpheme. Regarding this issue, we suggested that this may turn out to be an unnecessary choice because the augment function and the diminutive function may be seen as related. Since the two functions are lexicalised the same not only in Czech, but also in other languages, this may suggest that there is something deeper to explain. We leave the investigation of this question for future research, but we shall continue our analysis under the assumption that the adjectival diminutive features

the augment *-k*. As explained at the start of the section, this is not required to demonstrate the asymmetry between *-k* and *-ok/-n*, but it is an analysis that naturally follows from our proposal.

2.3.4 Deriving the diminutive adjectives (Classes I-IV)

In this section, we provide a step-by-step derivation of the diminutive forms. As a starting point, we repeat here the structure with an optional DMV, which sits between REL and POINT:

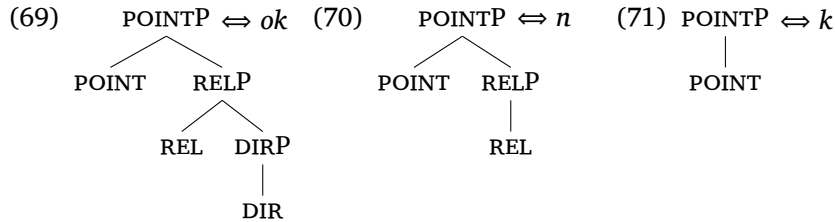


In (68) we repeat (58), which provides a synoptic lexicalisation table showing the four classes of roots, arranged by increasing root size.

(68) Diminutives (Classes I-IV)

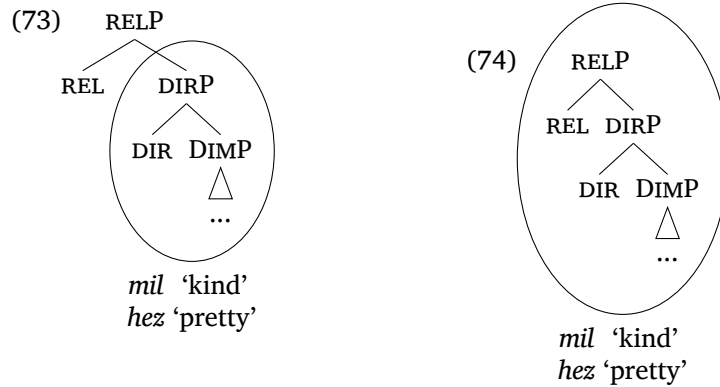
	DIMP	DIR	REL	DMV	POINT	GLOSS
Class I	div	ok	dmv	k		'wild'
Class II	pěk	n	dmv	k		'nice'
Class III	hez		dmv	k		'pretty'
Class IV	mil		dmv	k		'kind'

This section highlights the details of the derivation of Classes I-IV (leaving the Intermediate Class for Sect. 2.3.5). The lexical entries assumed for the augments in Classes I, II, and III are repeated below from (5)-(7). We also add the entry for the diminutive morpheme *ouč* in (72).



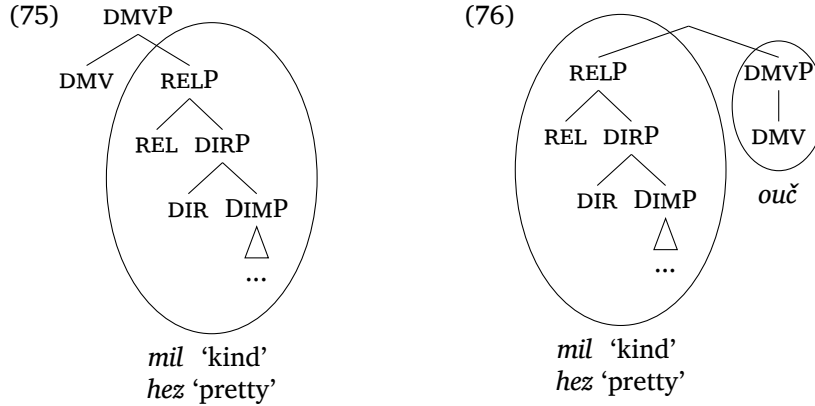
$$(72) \quad \begin{array}{c} \text{DMVP} \Leftrightarrow \text{ouč} \\ | \\ \text{DMV} \end{array}$$

Let us start with the diminutives of the large roots belonging to Class III and IV, which are characterised by the fact that the diminutive immediately follows the root. Recall from (31) that these roots are of size RELP and POINTP, respectively. This means that in the course of the derivation, when RELP is assembled, as in (73), both types of roots can lexicalise RELP without any movement, as shown in (74).¹⁵



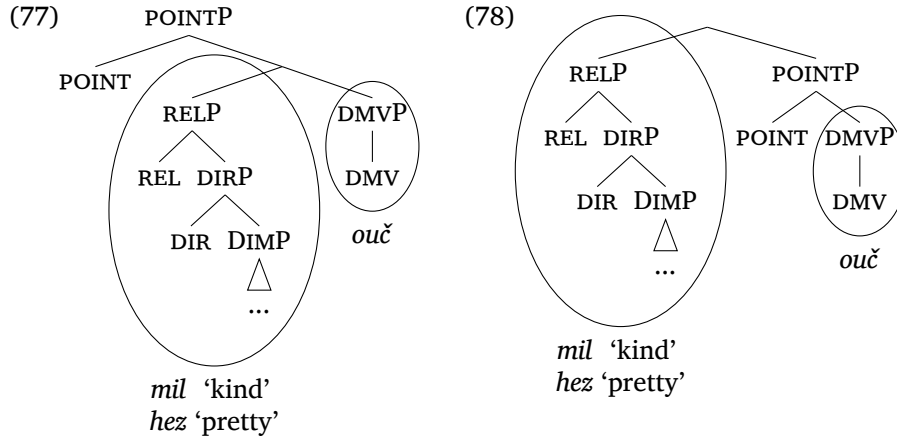
In the derivation of the positive degree, we merged POINTP on top of (74). That led to the emergence of the augment in Class III, because Class III roots cannot lexicalise POINTP (*hez-k-ý* 'pretty'), whereas in Class IV, the root could lexicalise POINT (*mil-ý* 'kind'). However, the diminutives differ in that DMV must be merged first, before any augment has appeared, as shown in (75).

¹⁵ Recall the discussion surrounding (40).



The DMVP in (75) finds no match in the lexicon, since no root contains DMV. Therefore, evacuation movement is triggered. The closest labelled node containing the bottom of the fseq is RELP, as in (76), and successfully lexicalises the remnant DMVP by *-ouč*.

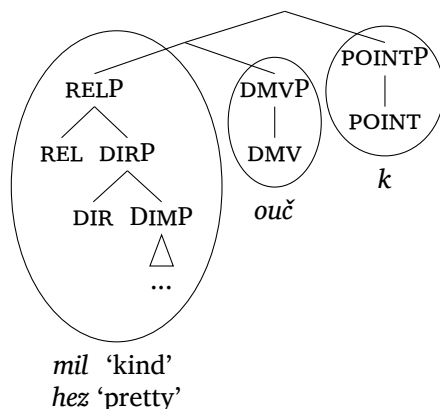
Crucially, no augment is needed for either root class as yet. The augment only appears when POINT is introduced on top of (76), see (77).



The tree in (77) finds no match in the lexicon. Evacuation movement first targets RELP, the closest labelled node containing the bottom of the fseq, resulting in (78), but there is no match here either.¹⁶ Since our first attempt at movement did not help, we go back to (77) and try moving the node immediately dominating RELP, yielding (79).

¹⁶ If we pursued a theory with non-decompositional *-oučk*, we would propose an entry for *-oučk* such that it would match the remnant POINTP in (78), leading to the lexicalisation table (61).

(79)

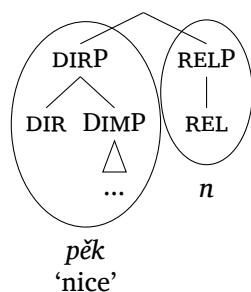


In (79), the augment *-k* matches the remnant POINTP, and lexicalisation succeeds. We have now finished the derivation of the diminutive forms in Classes III and IV.

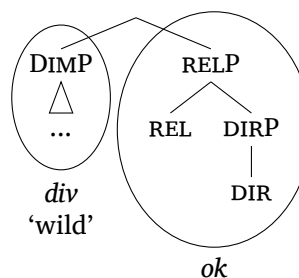
As we can see, the diminutive form requires the augment *-k* both in Class III adjectives and in Class IV adjectives. This is an interesting result, especially in Class IV adjectives, which have no augment in the simple positive form. The reason why the augment emerges in the diminutive is the structural intervention by DMV, which prevents the root from lexicalising POINT. As a result, the augment appears. Summarising, Classes III and IV have the same diminutive form (*root-ouč-k* + AGR), despite the fact that they have a different positive degree. This is because in the diminutive the roots of both classes lexicalise a constituent of the same size (REL P).

We now turn to the Class I and Class II roots, which have an augment preceding the diminutive marker. The Class I roots are of size DIMP, while the Class II roots have the size DIRP. We start off by repeating from (43) and (44) above the stage in the derivation where RELP has been merged and successfully lexicalised by the augment:

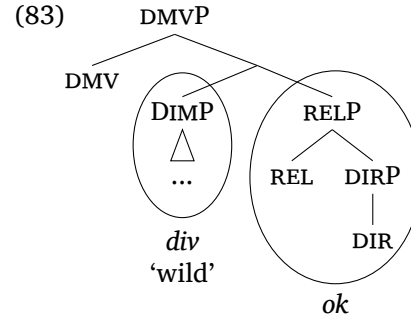
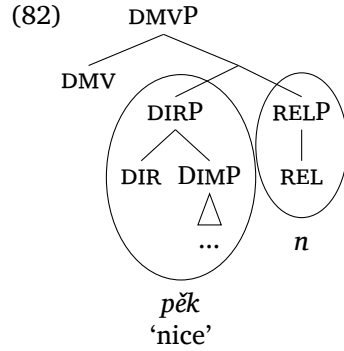
(80)



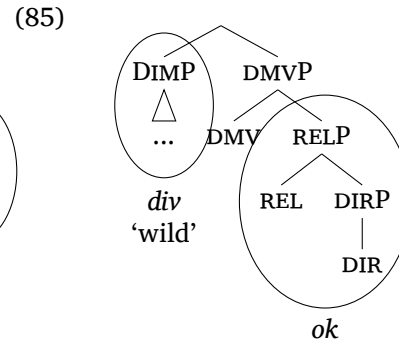
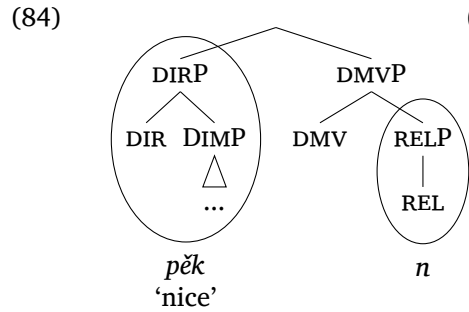
(81)



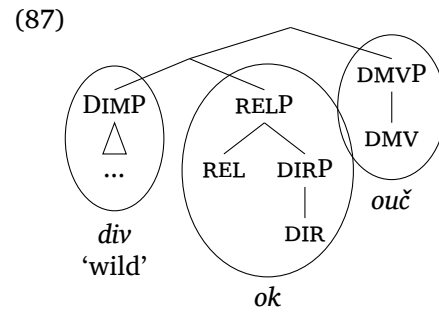
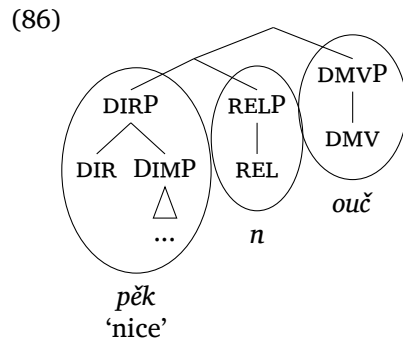
As in the previous two classes, the diminutive head is added on top of the structure that was created to lexicalise RELP:



Since neither of these trees finds a match in the lexicon, evacuation of the closest labelled constituent containing the root is tried. In the case of (82), this movement operation will target DIRP, resulting in (84), while in the case of (83), it will target DIMP and result in (85).

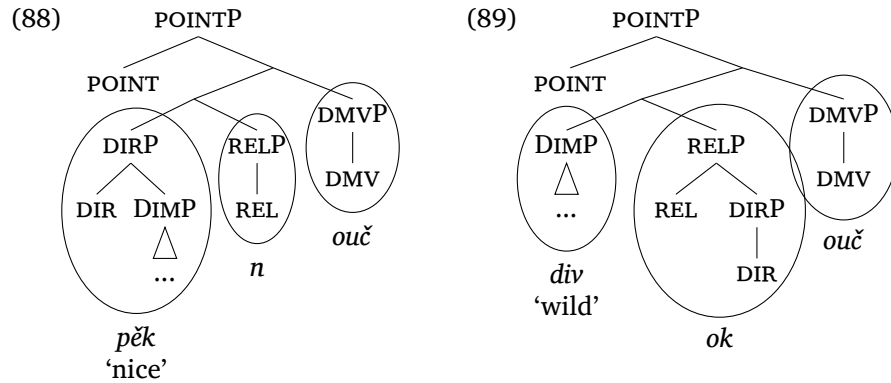


However, there is no match in either case, since neither the *-n* nor the *-ok* augment can lexicalise DMV. We therefore undo the movements, and move the immediately dominating node, yielding (86) and (87), respectively.

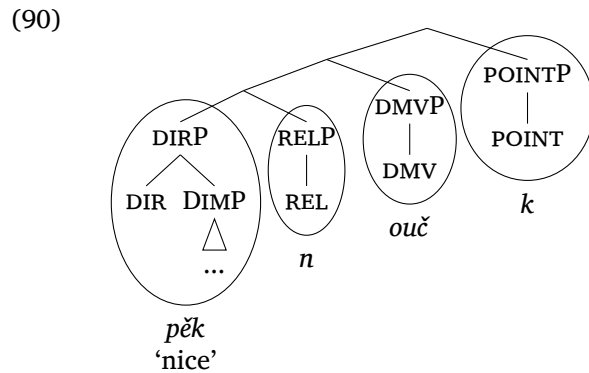


Here the remnant DMVP can be lexicalised without further ado. Note that the diminutive *-ouč* in (88) attaches to a base that contains *-n/-ok*, which is the correct result.

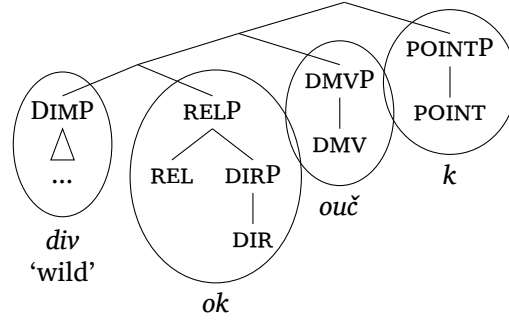
The derivation of the positive still requires the merger of POINT, as shown in (88) for the Class II root, and in (89) for the Class I root.



In neither case does the POINTP find a match. Evacuation movement targets DIRP with the Class II roots (88) and DIMP with the Class I roots (89). This will again fail to yield a lexicalisation in both cases, so that further movements will be triggered. After a series of intermediate steps, which we leave to the reader to fill in, this eventually leads to the evacuation of the node which is the sister of POINT, and successful lexicalisation of the remnant POINTP by *-k*. In (90) we see this for the Class II root, and in (91) for the Class I root.



(91)



Summarising, this section showed in detail how the diminutive forms in Table (92) arise.

(92) Lexicalisation table with augments (Classes I-IV)

	DIMP	DIR	REL	DMV	POINT	FORM	GLOSS
Class I	div	ok				<i>div-ok-(ý)</i>	'wild'
	div	ok		<i>dmv</i>	<i>k</i>	<i>div-ok-ouč-k-(ý)</i>	'wild, DMV'
Class II	pěk	n				<i>pěk-n-(ý)</i>	'nice'
	pěk	n		<i>dmv</i>	<i>k</i>	<i>pěk-ň-ouč-k-(ý)</i>	'nice, DMV'
Class III	hez				<i>k</i>	<i>hez-k-(ý)</i>	'pretty'
	hez			<i>dmv</i>	<i>k</i>	<i>hez-ouč-k-(ý)</i>	'pretty, DMV'
Class IV	mil					<i>mil-(ý)</i>	'kind'
	mil			<i>dmv</i>	<i>k</i>	<i>mil-ouč-k-(ý)</i>	'kind, DMV'

The main results are the following. One, we have successfully encoded the difference between Class I/II (diminutive *-ouč* attaches to an adjective with an augment) and Class III (diminutive *-ouč* attaches to the bare root). This comes about as a result of a structural asymmetry: only the augments *-n/-ok* lexicalise heads below DMV. Two, the diminutive is invariably followed by the augment *-k* in all classes. Descriptively, this breaks down into two special sub-cases. In Class III, the non-diminutive adjective already has the augment *-k*. In this class, the diminutive thus 'infixes' in between the root and the original augment. In Classes I, II and IV, which do not have the augment *-k* in the bare positive, the diminutive *-ouč* attaches to the positive form, and the augment *-k* appears in addition after *-ouč*. In our account, the appearance of *-k* in these classes is a consequence of structural intervention: DMV sits in between REL and POINT, making it impossible for POINT to be lexicalised along with REL (as it is in the non-diminutive adjective). Therefore, a new morpheme (i.e., *-k*) is introduced to realise POINT.

2.3.5 Diminutives: the *chladný* class as an ‘intermediate’ class

In this section we return to the interesting case discussed in Section 2.2, which is the case of adjectives like *chlad-n-ý* ‘cold.’ The regular and the diminutive forms of this class are given in Table (93) (repeated from (60) above), under the label ‘intermediate Class’.

(93) Diminutive adjectives in Czech

	POSITIVE	DIMINUTIVE	GLOSS
Class I	šir- ok -ý div- ok -ý	šir- ok -ouč-k-ý div- ok -ouč-k-ý	‘wide’ ‘wild’
Class II	příjem- n -ý hod- n -ý šťast- n -ý jem- n -ý	příjem- ň -ouč-k-ý hod- ň -ouč-k-ý šťast- ň -ouč-k-ý jem- ň -ouč-k-ý	‘pleasant’ ‘kind’ ‘happy’ ‘smooth’
Intermediate Class	drob- n -ý chlad- n -ý	drob- -ouč-k-ý chlad- -ouč-k-ý	‘tiny’ ‘cold’
Class III	leh-k-ý heb-k-ý krát-k-ý hlad-k-ý	leh- -ouč-k-ý heb- -ouč-k-ý krát- -ouč-k-ý hlad- -ouč-k-ý	‘light’ ‘smooth’ ‘short’ ‘smooth’
Class IV	blb-ý mal-ý čist-ý slab-ý	blb- -ouč-k-ý mal- -ouč-k-ý čist- -ouč-k-ý slab- -ouč-k-ý	‘stupid’ ‘small’ ‘clean’ ‘weak’

The puzzle presented by the *chlad-n-ý* class is that these adjectives have the augment *-n* in the positive degree, but lose it in the diminutive (e.g., *chlad'-ouč-k-ý* ‘cold, DIM’). If we only used root size as a way of distinguishing among various classes, we would get a paradox. On the one hand, the presence of *-n* in the positive indicates a root size of DIRP, as in Class II (recall Table (92)). On the other hand, roots of size DIRP are predicted to require *-n* also in the diminutive (see again Table (92)). There is, in other words, no explanation in terms of root size alone for the special behaviour of the adjectives in the intermediate Class, like *chlad-n-ý* ‘cold.’

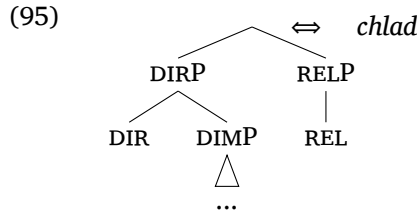
The puzzle can be resolved by relying on root shape. This solution rests on the observation that the puzzling class can be thought of as a class that alternates between Class II and IV, as depicted in (94). In the positive, the

puzzling adjectives pattern with Class II and have *-n*. In the diminutive, they lack *-n* and pattern with Class III.¹⁷

(94) *Chladn-n-ý* ‘cold’ as an intermediate class

	POS	DMV	GLOSS
Class II	pěk- n-ý	pěk-ň-ouč-k-ý	‘nice’
	jem- n-ý	jem-ň-ouč-k-ý	‘fine’
Intermediate Class	chlad- n-ý	chlad’ -ouč-k-ý	‘cold’
	drob- n-ý	drob -ouč-k-ý	‘tiny’
Class III	leh-k-ý	leh -ouč-k-ý	‘light’
	heb-k-ý	heb -ouč-k-ý	‘smooth’

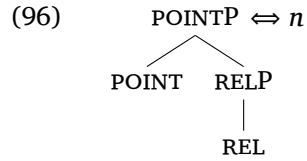
We have already encountered such an intermediate class in the discussion of Latin superlatives (recall (97) in Chapt. 1, Sect. 1.6.3), and we will encounter more such intermediate classes as we proceed. The solution we proposed in Chapt. 1 exploited the possibility of assigning to lexical entries a particular root shape. In particular, all the facts fall in place when we assign to the roots like *chlad* ‘cold’ the lexical entry in (95), which is the same size as Class III roots, RELP, but with a Complex Left Branch DIRP.



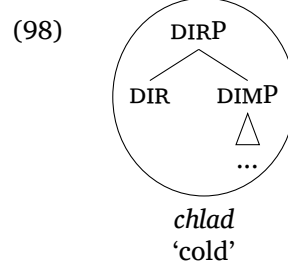
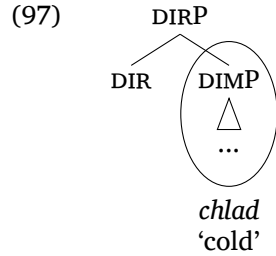
This entry captures the intermediate status of *chlad*: the left branch has the size DIRP, like all roots that take the augment *-n* in the positive. However, since the whole entry has all the features of a root with size RELP (but arranged in a different shape), it patterns with Class III in the diminutive, i.e. it is able to lexicalise all the features in the complement of DMV.

To show how exactly this solution works, we will now run the whole derivation. For convenience, we repeat the entry of the augment in (96).

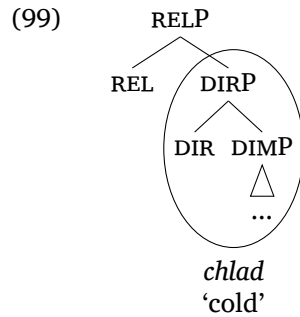
¹⁷ Since in the diminutives in Classes III and IV are indistinguishable, the intermediate Class could also be thought of as intermediate between Class II and Class IV. However, in terms of the size of its roots, it is identical only to Class III roots, as we shall explain below.



The derivation starts by constructing DIMP, which, being a subtree of (95), can be lexicalised as *chlad*. The syntax then merges DIR, as shown in (97), and lexicalisation succeeds again.

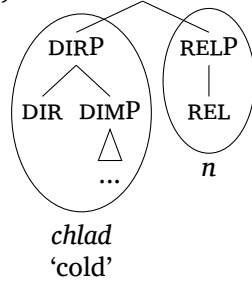


As the next step, REL is merged, yielding (99).

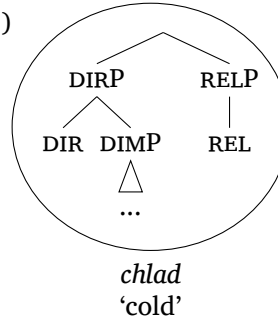


The root can now no longer lexicalise the structure. Even though the lexical entry for *chlad*- contains the feature REL (recall (95)), it does not contain a constituent that is identical to (99). Evacuation movement applies, targeting DIRP, and leading to the structure in (100)/(101). In (100), the remnant RELP can be lexicalised by the augment *-n*. However, we have another option for a match: the root node of the tree is now exact match for the lexical entry in (95), leading to the lexicalisation of this node by the root, and overriding the lower lexicalisations, as depicted in (101).

(100)



(101)

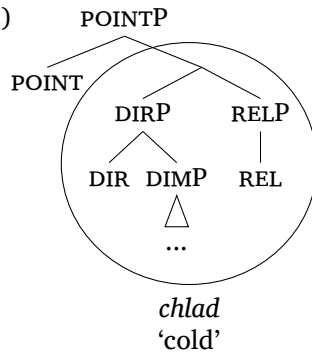


The fact that the root can lexicalise a constituent of size RELP explains the absence of the augment in the diminutive, as we shall proceed to show next.

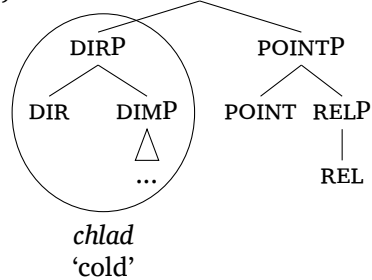
Starting from (101), the subsequent derivation faces a choice: if we want to merge a diminutive adjective, the feature DMV is next up to be merged. The augment *-n* does not appear in this case. The other option is to produce a regular positive, directly merging POINT. In this case *-n* does appear. We now show this in turn.

Consider the derivation of the non-diminutive adjective first. The structure resulting from merging POINT is given in (102). Since (102) cannot be lexicalised directly, evacuation movement targets DIRP, creating the configuration in (103).

(102)



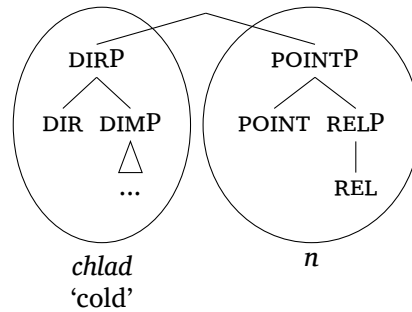
(103)



Now notice that after this movement, DIRP and REL no longer form a constituent, as they are separated by POINT. As a result, REL can no longer be lexicalised by the root along with DIRP (as it did in (102)). As a result, the root now only lexicalises DIRP after it has moved across POINT. This is another instance of something we observed earlier, namely that lexicalisations of earlier cycles (in this case DIRP) are not deleted under cyclic override, but survive and can be reverted to if the configuration changes.

The feature that is new in the current cycle, POINT, can be straightforwardly lexicalised by *-n* as a part of the remnant POINTP, as shown in (104):

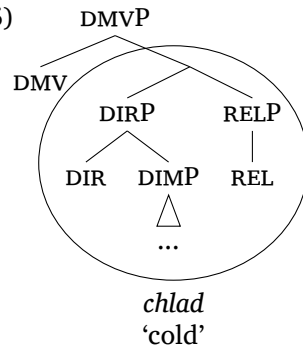
(104)



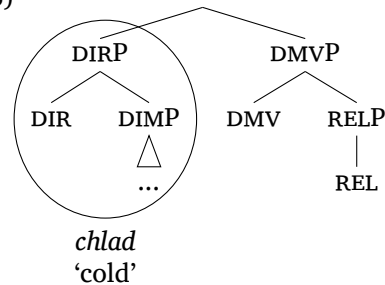
This explains why we get the *-n* augment in the positive, despite the fact that the *chladný* 'cold' roots are of size RELP, and would, on the face of it, be expected to occur with the *-k* augment. The reason for this is that the lexical entry with the CLB triggers movement of DIRP in an early stage of the derivation, creating a constituent on the right-hand side of the tree that has REL at its bottom, and it is this constituent that grows to become the *-n*-augment in the (non-diminutive) positive.

It remains to explain the derivation of the diminutive adjective, starting from the stage in the derivation depicted in (101). After this, DMV is merged, yielding (105). Direct lexicalisation is not possible, so evacuation movement applies, targeting DIRP, and resulting in (106). In this tree, the root again 'shrinks' and reverts to lexicalising merely DIRP, since REL no longer forms a constituent with DIRP.

(105)

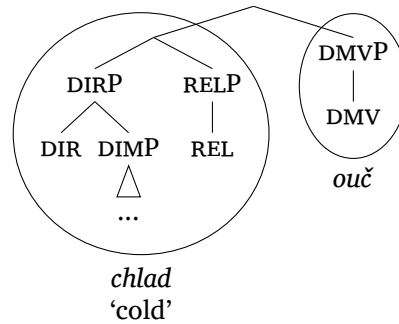


(106)



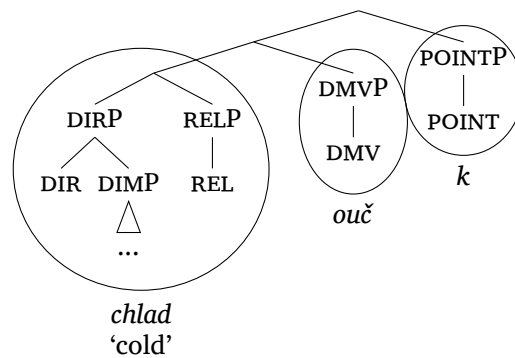
However, unlike what happened in the derivation of the positive, the remnant constituent [DMV [REL]] finds no match in the lexicon. Therefore, the derivation reverts to (105) and moves the node dominating DIRP, producing (107). In this tree, the configuration where the root lexicalised a structure with a CLB is preserved, so that the root continues to lexicalise RELP (along with DIRP), without the need for an augment. After movement, the diminutive marker *-ouč* matches the remnant DMVP.

(107)



The remainder of the derivation is identical to that of all the other diminutive adjectives that we have discussed so far: after the merger of POINT, evacuation movement will apply recursively to the closest labelled node containing the bottom of the fseq and dominating nodes, until the entire complement of POINT is evacuated, and POINTP is lexicalised by *-k*. The final result of the derivation of the diminutive adjective is shown in (108):

(108)



The lexicalisation table (109) compares the lexicalisation in the intermediate class (in the middle) with the four major classes we discussed earlier (repeated from (92)).

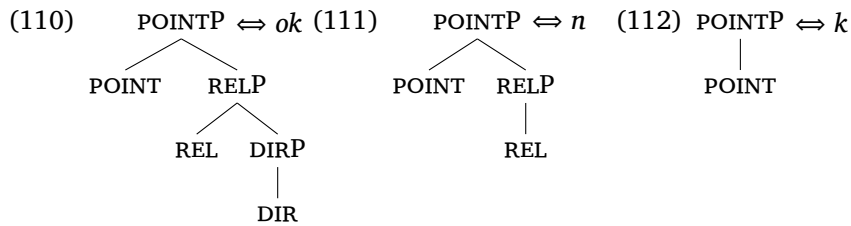
(109) Lexicalisation table with augments

	DIMP	DIR	REL	DMV	POINT	FORM	GLOSS
Class I	div	ok				<i>div-ok-(ý)</i>	‘wild’
	div	ok		<i>dmv</i>	<i>k</i>	<i>div-ok-ouč-k-(ý)</i>	‘wild, DMV’
Class II	pěk	n				<i>pěk-n-(ý)</i>	‘nice’
	pěk	n		<i>dmv</i>	<i>k</i>	<i>pěk-ň-ouč-k-(ý)</i>	‘nice, DMV’
Intermediate Class	chlad	n				<i>chlad-n-(ý)</i>	‘cold’
	chlad			<i>dmv</i>	<i>k</i>	<i>chlad'-ouč-k-(ý)</i>	‘cold, DMV’
Class III	hez				<i>k</i>	<i>hez-k-(ý)</i>	‘pretty’
	hez			<i>dmv</i>	<i>k</i>	<i>hez-ouč-k-(ý)</i>	‘pretty, DMV’
Class IV	mil					<i>mil-(ý)</i>	‘kind’
	mil			<i>dmv</i>	<i>k</i>	<i>mil-ouč-k-(ý)</i>	‘kind, DMV’

The main observation is that in the positive, the intermediate class has the same lexicalisation pattern as Class II, lexicalising only DIRP (corresponding to the complex left branch). In the diminutive, it has the same pattern as Class III (or IV), lexicalising RELP, which is the total size of the entry. As to which of these options is chosen depends on how the derivation proceeds following the invariant lexicalisation algorithm.

2.3.6 Summary

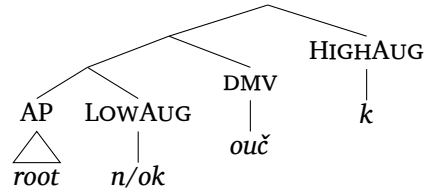
At this point, we have finished a discussion of diminutive adjectives. The goal of the discussion was to provide independent evidence for our analysis of augment allomorphy, such that different augments realise different functional heads. According to this analysis, the augment *-k* only lexicalises functional projections above the diminutive, while the other two augments also lexicalise functional projections below the diminutive. This follows from the lexical entries in (110), repeated from (5)-(7).



As we saw earlier, in the presence of the diminutive, the augments part ways: *-n/-ok* are low augments appearing below the diminutive, and *-k* is

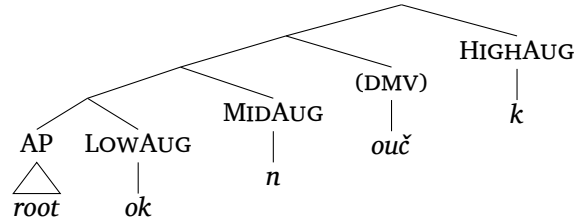
a high augment appearing above the diminutive, as shown in (113), where the heads (like LOWAUG etc.) show the position of the lowest feature of the augments that they dominate.

(113)



However, we have so far not discussed any evidence for a structural distinction between the low augments *-n* and *-ok*. Because of their overlap in feature specifications, they are in complementary distribution. However, in the next section, we discuss reasons why we think that the lowest feature of *-ok* (its foot) is lower than the foot of *-n*, as depicted in (114).

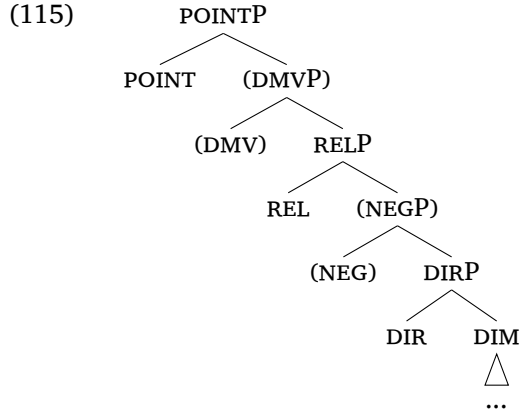
(114)



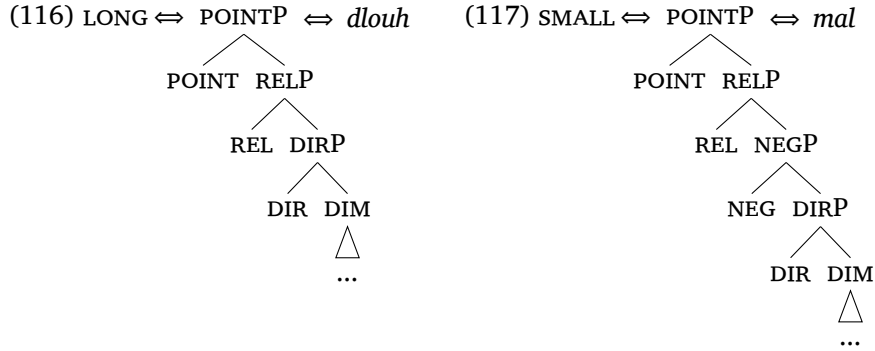
2.4 Negative adjectives as evidence for the relative position of *-n* and *-ok*

Our argumentation concerning the relative sizes of the augments *-n* and *-ok* rests on the interaction of negation with these augments. Specifically, we argue that the lowest feature of *-n* is above the functional projection NEG, while *-ok* reaches below NEG. If correct, this provides an independent evidence for the tentative proposal made in the previous sections, namely that *-ok* lexicalises projections lower down than *-n*.

In order to develop the argument, we first need to discuss the position of negation in our functional sequence, and the consequences that has on the lexicalisation of roots and augments. In particular, recall that in negative adjectives, an optional NEG head appears in between DIR and REL:



Semantically, NEG acts as a scale reverser (Ladusaw, 1980), which reverses the scale (or direction) of the gradable adjective. As such, it comes on top of DIR, but below REL in the functional sequence, as shown in (115). This head is what differentiates positive from negative adjectives, in such antonymic pairs as *tall* and *short*. As argued in De Clercq & Vanden Wyngaerd (2019a), the semantic distinction between these adjectives manifests itself syntactically, in a way that follows from the presence of the NEG head in negative adjectives. To give a concrete example from Czech, in (116) and (117) we give the lexical entries for the positive adjective *dlouh-ý* ‘long’ and *mal-ý* ‘small’, respectively. As the entries show, negative adjectives lexicalise an extra NEG head.



We shall not provide derivations for these adjectives, as they are fairly trivial, with the root growing and lexicalising all the features of the positive degree.

At this point, we want to motivate the position of NEG in the functional sequence of the gradable adjective, which is relatively low. In doing so, we shall also develop an argument showing that the foot of the augment *-ok* is below that of *-n*. The argument will proceed in two stages. First, we shall argue that NEG must be lower than POINT. Next, we shall show that NEG is lower than REL as well.

We start out by presenting an argument to suggest NEG has to sit lower than POINT. As we have observed above, the interpretation of positive degree gradable adjectives involves a standard of comparison. This contextual standard is associated with the feature POINT. As observed in De Clercq & Vanden Wyngaerd (2018, 2019a), positive and negative adjectives behave symmetrically with respect to the standard. This is illustrated in (118).

- (118) a. Linnea is tall
 > ‘The degree to which Linnea is tall exceeds some contextual standard’
 b. Linnea is short
 > ‘The degree to which Linnea is short exceeds some contextual standard’

Crucially, the meaning of (118b) does not arise as a consequence of sticking a negation on top of (118a). This would result in (119):

- (119) ‘It is not the case that the degree to which Linnea is tall exceeds some contextual standard’

Such a semantics would be compatible with a situation where Linnea is neither tall nor short, i.e. a situation where Linnea’s height falls below the standard for counting as tall, but also below that for counting as short. This is a situation that can be described by (119), but it is not a meaning that (118b) can have.

Instead, positive and negative adjectives have symmetrical semantics with respect to the standard, as informally represented in (118). This can only arise if the standard (represented by POINT in our approach) sits above the scale reverser NEG, the feature that distinguishes positive from negative adjectives, as in (120a):

- (120) a. POINT > NEG > ...
 b. NEG > POINT > ...

If NEG were higher than point, as in (120b), we would derive an asymmetry between positive and negative adjectives, with negative adjectives having the interpretation in (119), contrary to fact.

A second argument suggesting that POINT is higher than NEG is based on the polarity of the roots with the *-k* augment. The empirical observation is that *-k* comes on top of both positive and negative roots. Relevant examples are given in Table 2.3. The line separates one antonymic pair (above the line) from nonantonymic ones, below the line.

The facts in the table suggest that the roots that combine with *-k* can either contain or fail to contain NEG. This is only possible if the polarity feature NEG is below the projection lexicalised by *-k*, which is, by hypothesis, POINT: the complement of POINT can then either be a RELP without NEG (positive root) or a RELP containing NEGP, for a negative adjective. This is illustrated in

Table 2.3 Positive and negative adjectives in Czech with the *-k*-augment

POSITIVE A		NEGATIVE A	
těž-k-ý	'heavy'	leh-k-ý	'light'
vel-k-ý	'big, large'	krát-k-ý	'short'
hez-k-ý	'pretty'	níz-k-ý	'low'
heb-k-ý	'smooth'	měl-k-ý	'shallow'
slad-k-ý	'sweet'	břit-k-ý	'ugly'
blíz-k-ý	'close'	křeh-k-ý	'fragile'

the lexicalisation table in (121), which shows a positive adjective in (121a), and a negative one in (121b).

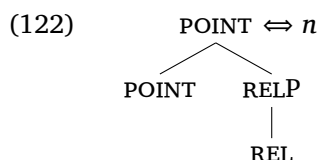
(121)

	DIM	DIR	NEG	REL	POINT	GLOSS
a.		leh			k	'light'
b.		těž			k	'heavy'

The positive adjective *leh-k-ý* 'light' lacks NEG, which is marked in black. Its root extends up to REL, and it needs *-k* to lexicalise POINT. The negative adjective *těž-k-ý* 'heavy' is identical, except that it has a NEG feature in its root.

In contrast, if NEG were higher than point, *-k*-adjectives should all be of the same polarity (depending on whether *-k* does or does not lexicalise NEG). Now one might object at this point that the matching condition does not make this conclusion compelling, since a lexical tree [_{NEGP} NEG [_{POINTP} POINT]] could lexicalise only POINTP (for a positive adjective), since POINTP is contained in the lexical entry, or NEGP (for a negative adjective). However, as we shall see in chapt. 3, the augment *-k* in fact also lexicalises certain higher features that belong to the fseq of the comparative. That makes NEG an intermediate feature in its lexical entry, and comparative adjectives with *-k* would be predicted to all be of the same polarity, contrary to fact.

We now move on to argue in support of our second claim, which is that NEG sits below REL. The argument is based on the polarity of the roots that combine with the *-n* augment. Given that *-n* (like *-ok*) precedes the diminutive morpheme, while *-k* follows it, we concluded above that *-n* has a foot in REL. We repeat the lexical entry for *-n* in (122):



If this is indeed correct, we expect *-n* to combine with roots of different polarities: positive roots if NEG is absent, and the root lexicalises DIRP, negative ones if the root lexicalises NEGP. This expectation is indeed borne out. Table 2.4 lists a whole series of both positive and negative adjectives with the *-n*-augment.

Table 2.4 Positive and negative adjectives in Czech with the *-n*-augment

POSITIVE A		NEGATIVE A	
jas- <i>n-ý</i>	‘bright’	tem- <i>n-ý</i>	‘dark’
šťast- <i>n-ý</i>	‘happy’	smut- <i>n-ý</i>	‘sad’
rov- <i>n-ý</i>	‘straight’	špat- <i>n-ý</i>	‘bad’
snad- <i>n-ý</i>	‘easy’	vad- <i>n-ý</i>	‘faulty’
sil- <i>n-ý</i>	‘strong’	hnus- <i>n-ý</i>	‘dirty’
pěk- <i>n-ý</i>	‘pretty’	hroz- <i>n-ý</i>	‘horrible’
sluš- <i>n-ý</i>	‘decent’	chlad- <i>n-ý</i>	‘cold’
plat- <i>n-ý</i>	‘valid’	drob- <i>n-ý</i>	‘tiny’
hod- <i>n-ý</i>	‘kind’	bíd- <i>n-ý</i>	‘miserable’
krás- <i>n-ý</i>	‘beautiful’	smut- <i>n-ý</i>	‘sad’
klid- <i>n-ý</i>	‘calm’	lev- <i>n-ý</i>	‘cheap’

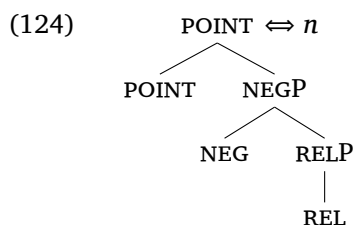
These facts require that we put the NEG head below REL, as shown in the lexicalisation table in (123). The position of the diminutive head DMV is indicated as a reminder for the reason why *-k* sits higher in the structure (although no diminutive morpheme is present in any of the examples we shall discuss in this section).

(123)

	DIM	DIR	NEG	REL	DMV	POINT	GLOSS
a.	leh					k	‘light’
b.		těž				k	‘heavy’
c.	jas			n			‘bright’
d.	tem			n			‘dark’

In (123ab) we repeat the two polar opposite adjectives with the augment *-k* from (121). In (123cd) we provide another such pair of polar opposites, but with the augment *-n*. The positive root of *jas-n-ý* ‘bright’ lexicalises DIM and DIR, and has no NEG. The negative root of *tem-n-ý* ‘dark’ extends up to NEG. Both roots require the augment *-n* to lexicalise the remaining features of the positive, up to POINT. The presence of a DMV morpheme in the structure will interrupt the growth of *-n* but not that of *-k*, as explained in detail in sect. 2.3.

If NEG were higher than REL (and lower than POINT), it would be inside the sequence of features lexicalised by *-n*, and *-n* could only lexicalise positive adjectives (with a lexical entry as in (122)), or only negative adjectives (if it were as in (124)).



We thus conclude that REL is above NEG, which also makes sense semantically, in that we first construct the scale (with a positive or negative direction) and then we merge REL, to introduce the POINT.

Though the augment *-ok* is far less widespread, and so generalisations about it are based on a small sample, it seems that *-ok* only gives rise to positive adjectives, e.g. *šir-ok-ý* ‘wide’, *hlub-ok-ý* ‘deep’, *vys-ok-ý* ‘tall’.

Table 2.5 Positive adjectives with *-ok* have negative counterparts with *-k*

POSITIVE A	NEGATIVE A
vys-ok-ý ‘high’	níz-k-ý ‘low’
šir-ok-ý ‘wide’	úz-k-ý ‘narrow’
hlub-ok-ý ‘deep’	měl-k-ý ‘shallow’
div-ok-ý ‘wild’	krot-k-ý ‘tame’

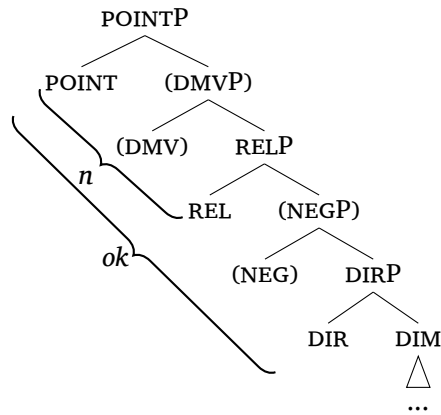
The uniform polarity of adjectives with *-ok* follows if *-ok* has a foot below REL, and NEG is right below REL. If *-ok* lexicalises the sequence DIR-REL-POINT then all such adjectives can only be positive, since there is no way of lexicalising NEG by the root. On the other hand, if *-ok* lexicalised a sequence including NEG, we would expect such adjectives to only be negative. As Table 2.5 shows, the former assumption is the correct one. We thus assign the features DIR-REL-POINT to *-ok*, as on line (e) in the lexicalisation table (125), and this makes it incompatible with negative adjectives, see line (f).

(125)

	DIM	DIR	NEG	REL	DMV	POINT	GLOSS
a.	leh					k	'light'
b.		těz				k	'heavy'
c.	jas			n			'bright'
d.	tem			n			'dark'
e.	vys	ok					'tall'
f.	root		*ok				

We can summarise the situation depicted in the table arboreally as in (126). The braces indicate the features lexicalised by the augments *-ok* and *-n*, respectively (since *-k* only lexicalises POINT, we leave it to the reader to mentally supply this third brace).

(126)



To conclude: the optional DMV head allows us to make a distinction between the augments that follow the diminutive morpheme (*-k*), and those that precede it (*-n* and *-ok*), but it did not allow us to determine which of *-ok* and *-n* lexicalises which projections. In this section, we investigated the optional NEG head, which sits low enough in the structure to allow us to distinguish between *-ok* and *-n*. The high position of *-k* predicts that roots of both polarities can occur with it, and the same holds for *-n*, which leads us to propose that *-n* (just like *-k*) has a foot that is higher than NEG. In this analysis, the *-ok* augment reaches lower into the structure than NEG, making it only able to derive positive adjectives, since roots that combine with it cannot lexicalise NEG.

Chapter 3

The comparative

Abstract This chapter proposes a decomposition of the comparative adjective into 4 heads. These four heads jointly define a ‘comparative scale’ that has the same dimension and direction as the positive scale (which also features four heads). We provide a motivation for the linguistic relevance of these heads by the morphology of Czech comparative adjectives, which exhibit up to three extra morphemes in the comparative (compared to the positive). Another piece of evidence comes from allomorphy. We show that Czech exhibits 15 different morphological classes of adjectives, considering their positive, comparative and comparative-adverb forms. We reduce this variation down to two parameters, namely whether the augment is present or absent, and whether one of the comparative markers is present or absent. We argue that the specific type of allomorphy exhibited by these markers cannot be captured using context-sensitive rules. Against this background, we show how these 15 classes are derived using the Nanosyntactic lexicalisation algorithm, providing evidence both for the rich structure, and for the general approach to the syntax-morphology interface adopted in this book.

3.1 The functional sequence of the comparative

In this chapter, we provide morphological evidence for the existence of four heads in the functional structure of the comparative. The evidence comes from the extremely rich morphology (and allomorphy) of the Czech comparative. The most striking fact concerns the existence of adjectives that add three morphemes to the positive root to mark the comparative: the augment *-n*, and two comparative morphemes *-ěj* and *-š*. Some of the relevant examples are given in (1).¹

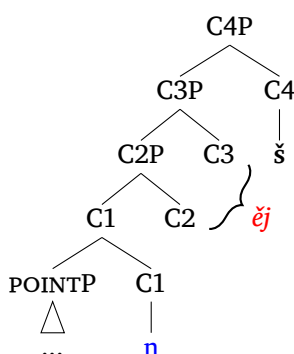
¹ The roots of the adjectives in (1) all end in *c*, suggesting this may be in independent morpheme. This does not affect our argument, however.

(1)

POS ADJ	CMPR ADJ	GLOSS
žádouc-í	žádouc- n-ěj-š -í	‘desirable’
ne-mohouc-í	ne-mohouc- n-ěj-š -í	‘incapable’
strhujíc-í	strhujíc- n-ěj-š -í	‘breathtaking’
vše-mohouc-í	vše-mohouc- n-ěj-š -í	‘almighty’

We shall provide an analysis for this class of adjectives such that the augment *-n* appears in the comparative of this class to lexicalise the lowest comparative head, call it C1.² The comparative marker *-ěj* lexicalises the two higher ones, call them C2 and C3. Finally, the marker *-š* lexicalises the topmost comparative head, C4. We depict this schematically in (2).

(2)



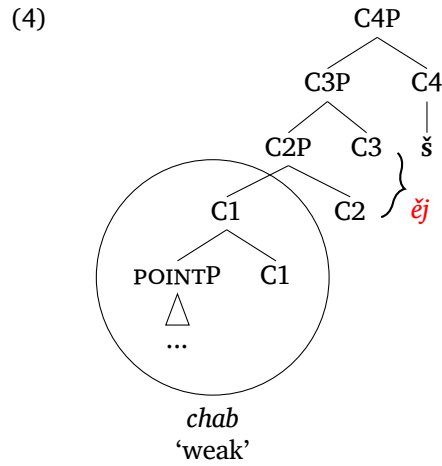
The proposal needs no elaboration where markers only lexicalize a single projection. Under the assumption that each marker occupies its own head, it is relatively easy to see how adding three markers requires at least three projections. However, it is not immediately clear why *-ěj* should lexicalise two heads. The argument for this comes from comparative adverbs, more specifically from the distribution of *-ěj* in the comparative forms. The relevant pattern is shown in (3).

(3) Comparative adverbs in augmentless adjectives

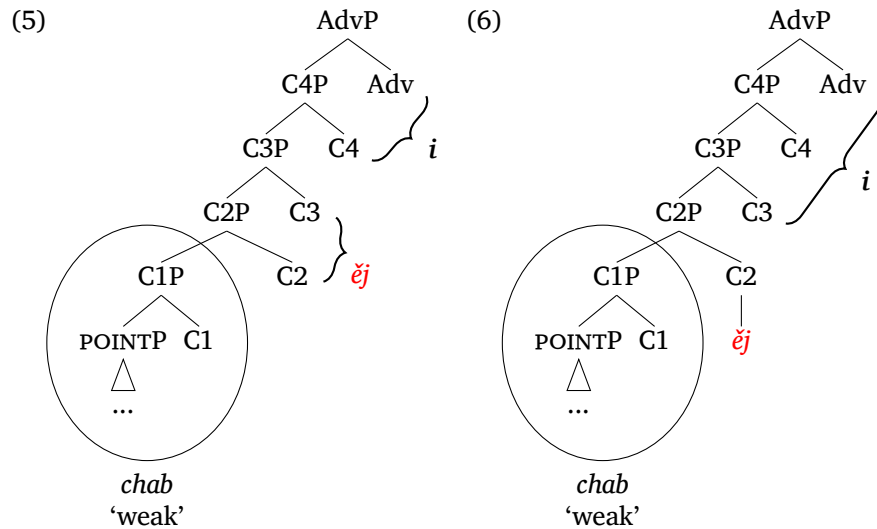
POS ADJ	CMPR ADJ	CMPR ADV	GLOSS
chab-ý	chab- ěj-š -í	chab- ěj -i	‘weak’
slab-ý	slab- -š -í	slab- ěj -i	‘weak’

Let us start to unpack this table from the adjective *chab-ý* ‘weak’ on the first line. This adjective only needs two morphemes in the comparative. We shall analyse this as other cases where augments are missing, namely as a case where the adjective root lexicalises projections up to C1, see (4).

² We note, though, that in the final analysis, the augment *-n* will lexicalize not only C1, but also the features inside the positive, as argued in the previous chapter.



The second thing to note is that in the comparative adverb *chab-ěj-i*, the comparative-adjective marker *-š* is replaced by the comparative-adverb marker *-i*. One possible analysis would be that the adverbial *-i* lexicalises not only the adverbial head, but also the topmost comparative head (i.e., the one lexicalised by *-š*). This is shown in (5).³



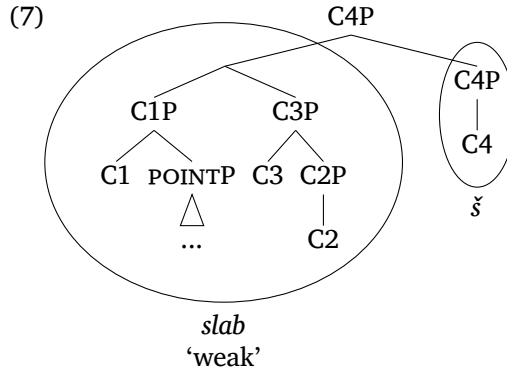
However, spreading *-ěj* across two heads allows for another possible analysis of the adverb, which is depicted in (6). Here the adverbial *-i* lexicalizes

³ In the positive degree, the adverbial marker is generally *-ě*, which replaces the agreement marker, e.g., *stroh-ý* 'strict' ~ *stroz-e* 'strictly.' It is a different marker from the *-i* that we find in the comparative, which is therefore consistent with the idea that the comparative-adverb marker lexicalises some comparative heads.

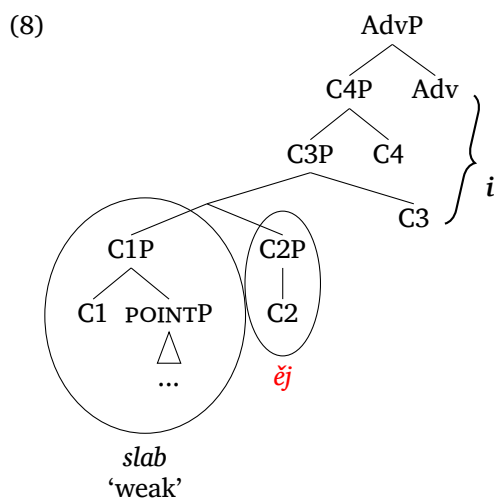
not only C4 (making *-š* disappear) but also C3 (making *-ěj* shrink down to lexicalizing just C2). It turns out that the analysis in (5) cannot be maintained, while the analysis in (6) captures a large range of facts about the morphology of the Czech comparative adverb. Once this is demonstrated, it becomes clear why we posit two different heads underlying *-ěj*: C2 for the comparative adverb, and C2 + C3 for the comparative adjective.

One of the reasons for rejecting the analysis in (5) is that the analysis predicts that the adverbial *-i* will always attach on top of the same base as *-š* (namely to a base that lexicalizes C3P). But this is not the case, as we have already seen on the second row of Table (3). Here we find the adjective *slab-ý* ‘weak,’ which has the comparative *slab-š-í* ‘weaker, ADJ,’ but the comparative adverbial is *slab-ěj-i* ‘weaker, ADV.’ This shows that *-š* and *-i* attach to bases of different sizes. This is only possible if *-ěj* is lexicalising two heads (and not just one).

We show the analysis of the root *slab* ‘weak’ below. According to our analysis, the root lexicalizes a configuration where C1P moves out of C3P, see (7). We assume that the configuration – as depicted in the circle – corresponds to the syntactic side of the lexical entry of the root. We further note that the constituent on the right branch of the root’s structure (containing C2 and C3) is normally lexicalised by *-ěj* (unless overridden by the root).



Now if *-i* has its lowest feature at C3, the base in the comparative adverb only lexicalizes C2P, as in (8).



However, the root *slab* ‘weak’ has no constituent that is identical to the syntactic structure below *-i*: the label-free node dominating C1P and C2P in (8) is not contained in the entry of the root. As a result, C2P must be lexicalised by *-ěj* in the adverb. We shall motivate this derivation in detail as the chapter unfolds, for now we just want to point out one of the reasons why *-ěj* must lexicalise two heads.

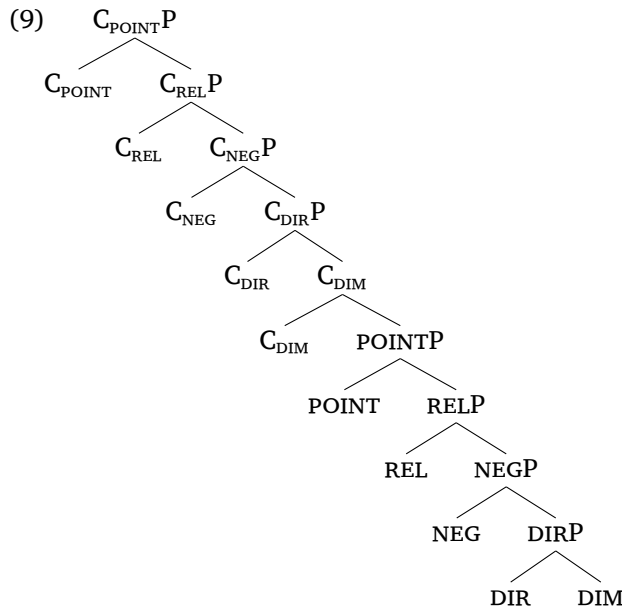
In this chapter, we develop in detail an analysis of the Czech comparative along these lines, and show how it captures an impressive range of 15 different morphological classes of adjectives, and correctly predicts where augments and other morphemes are found. The analysis will crucially rely on (and provide further evidence for) the proposal that *-ěj* lexicalises two heads in the comparative adjective and only one such head in the comparative adverb. If *-ěj* in the adjective indeed lexicalises two heads, then we need four heads in the comparative adjective in total.

Let us now turn to the implications of such findings. In the previous chapter, we argued that positive degree adjectives feature four different heads (with an additional optional head NEG), which jointly form a scale and determine the position of the positive-degree standard that the argument must exceed, i.e., *tall* applies to objects that exceed the contextually-determined standard of comparison.

In this chapter, we observe that the morphological patterns in Czech comparatives necessitate four additional syntactic heads in the comparative (i.e., the same number of heads in the positive). To make sense of this, we propose that the comparative corresponds to very much the same type of scale as the positive, namely one that is constructed on the basis of the positive degree. The comparative scale has the same dimension as the positive-degree scale (the comparative DIM is unspecified and interpreted the same as the positive-degree DIM). However, the comparative scale is different from the

positive-degree scale in that it has a different starting point, e.g., a scale corresponding to *taller than John* starts at John's height. This type of approach has been proposed in Kennedy & Levin (2008) on independent grounds (they use the term derived measure function for what we call the comparative scale). We also note that the comparative scale may have the same or possibly a different direction than the original scale (*more* vs. *less tall*).

As a result, we propose that the four comparative heads are the same heads as in the positive. For ease of reference, and in order to be able to distinguish the positive and the comparative in the representations and the discussions in the text, we mark the comparative heads with a capital C and a subscript that provides their semantics, e.g. C_{DIM} , C_{DIR} , etc. The full adjectival functional sequence (with the optional NEG head included) is given in (9):



Summarising, the four heads of the comparative match those of the positive, both in their number and in the interpretation we assign to them. Both the positive and the comparative involve an independent scale, with a shared dimension DIM. The comparative dimension is interpreted as identical to the dimension of the positive, as no new conceptual content is contributed in comparatives.

The orientation or direction of the comparative scale may be the same or different from the one in the positive. If only C_{DIR} is present, the orientation is the same, and we get a comparative of superiority (i.e., *more A*). If also C_{NEG} is present, we get a comparative of inferiority *less A*.

In Chapt. 1 we already proposed that antonymic pairs of adjectives like *tall* and *short* differ in the orientation of the scale associated with them. As the examples in (10) show, the optional NEG feature may be either absent in

both the positive and the comparative (10a), or present in either the positive (10b) or the comparative (10c), or in both (10d).

(10)

	NEG C _{NEG}	
a. Linnea is taller than Neal.		
b. Neal is shorter than Linnea.	✓	
c. Neal is less tall than Linnea.		✓
d. Linnea is less short than Neal.	✓	✓

As observed in Bobaljik (2012), comparatives of inferiority are never expressed morphologically, a property he describes as Lesslessness.

- (11) Lesslessness
No language has a synthetic comparative of inferiority.

Czech accordingly has no morphological comparatives of inferiority. Since this book restricts itself to morphological comparatives, we shall leave comparatives of inferiority aside here (but see De Clercq & Vanden Wyngaerd 2019a for some discussion).

Comparatives also have a REL and a POINT head, which provide an individual and a point, relative to which the individual's degree on the scale is interpreted. As observed in Chapt. 1, the point of comparison is provided by context, i.e. it is an implicit contextual standard, as in (12a). In the comparative, as proposed in Kennedy & Levin (2008), the standard of comparison expressed in a *than*-phrase provides the starting point of the derived scale (12b).

- (12) a. Linnea is tall
> 'The degree to which Linnea is tall exceeds some contextual standard.'
b. Linnea is taller than Neal.
> 'The degree to which Linnea is tall exceeds the degree to which Neal is tall.'

Since the comparative scale is closed at the bottom, the comparative holds of an argument as long as that argument has a non-zero value on the comparative scale. This is similar to adjectives like *wet*, whose standard is contextually set at some non-zero degree on the scale of wetness.

The rest of the chapter is dedicated to the exploration of the morphological data that lead us to postulate four different heads in the comparative. The chapter is organised as follows. In Section 3.2, we introduce 15 different classes of adjectives in a triplet of forms, that includes the positive-degree adjective, the comparative-degree adjective, and the comparative-degree adverb. In Sections 3.3-3.6, we provide a detailed analysis of these classes including step-by-step derivations. As in the previous chapter, we show that

once the rich structure of comparative is adopted, this is possible to achieve using variable size and shape. Section 3.7 concludes.

3.2 Morphological variation

In the remainder of this chapter, we shall motivate the need for the fine-grained decomposition by the richness of the morphological patterns of marking in Czech. As already mentioned, the comparative marking of Czech adjectives is rather complex, and fifteen different classes of adjectives can be recognised. Fortunately, these fifteen classes can be reduced to two factors (or parameters) of variation.

The first factor concerns the allomorphy of the comparative marker, namely whether it includes *-ěj* or not. The allomorphy is controlled by the morpheme to the left of the comparative marker, which can either be the root or an augment.

In the comparative adjective, this parameter distinguishes adjectives with *-ěj-š* or with *-Ø-š*. For comparative adverbs, the form with *-ěj* is *-ěj-i*, the one without *-ěj* is *-Ø-e*, with an additional distinction between *-i* vs. *-e*. However, the *i/e* distinction is completely reducible to the *-ěj/Ø* distinction. While the distinction between *i/e* must of course be accounted for as well (which we shall achieve by attributing different features to each of these markers), it does not represent an independent parameter of variation and will therefore be ignored in this section.

The second factor of variation in the marking of the comparative concerns the augment. It turns out that adjectives do not combine with the augment in all the forms. The augment can be present in only some forms and absent in others, differently for different adjectives. For instance, some adjectives can have the augment in all forms, other adjectives lack it in the comparative adjective, but have it in the positive adjective and the comparative adverb. The presence/absence of the augment is not governed by the augment itself, but by the root: the same augment can be found in the comparative of some roots, but not others. Interestingly, this parameter of variation is relevant for each of the three augments that we have discussed, i.e., *-ok*, *-n*, and *-k*.

Following these two parameters led Vanden Wyngaerd et al. (2025) to distinguish nine different patterns. However, that analysis was still restricted, because it only considered two augments (*-n* and *-k*), and it did not consider comparative adverbs. Adding the third augment (*-ok*) and the comparative adverbs to the mix results in a total of fifteen different ways of marking the comparative, corresponding to fifteen different classes of adjectives. We argue that a full account of the fifteen patterns that adjectives may display in the comparative is possible using the four different heads on top of the positive (five if we include the optional NEG head). Our aim in this chapter is to show that the rather bewildering variety of data patterns can be accounted

for using exclusively the tools we have laid out in Chapt. 1, in essence root size and shape.

Before we set out to deliver on this goal, it is also worth noting that describing variation in terms of these parameters restricts it into a very narrow niche, where basically two positions in the morphological structure (the augment position and *ěj*) alternate between an overt realisation and no realisation. For example, we do not see adjectives *changing* augments, or using a different allomorph of *-ěj*.

Despite the fact that the variation boils down to the alternations between an overt morpheme and a $\emptyset$, the factors conditioning these alternations are complex. For instance, in order to know whether an augment appears or not, we need to take into consideration three factors: the identity of the root, the degree (positive vs. comparative) and also the specific type of the comparative (adjective vs. an adverb). Given all these conditioning factors, the modelling of the distribution (morpheme vs. $\emptyset$) turns out to be far from trivial, and, as we show, out of the reach for approaches based on context-sensitive rules (as already argued in Caha et al. 2024b).⁴

In the remainder of this section, we present the fifteen classes that our analysis will have to account for. We start out by taking a closer look at each of the two factors of variation in some more detail (Sections 3.2.1 and 3.2.3), and then we put them all together into a big table with all the fifteen patterns (Section 3.2.4). As the discussion unfolds, we address the role of the phonology, in particular palatalisation and vowel length, which will lead us to simplify the data to a certain extent, in that we take out variation that is caused by regular phonological rules (Section 3.2.2).

3.2.1 The difference between *-ějš* and *-š*

This section provides a discussion of the first parameter of variation, where *-ěj* alternates with $\emptyset$. On the surface, we observe that the adjectival comparative in Czech may be marked by either *-ějš* or *-š*, where the peculiarity is that the former contains the latter. This fact is already suggestive of the internal complexity of *-ějš*. We follow the literature and provide the reasons to decompose *-ějš* into two markers, *-ěj* and *-š*. Once this is adopted, the

⁴ In most classes, positive-degree adverbs do not differ from positive-degree adjectives in the presence/absence of the augment. Generally, when a positive-degree adjective includes the augment, so does the adverb, and vice-versa. As such, positive-degree adverbs do not provide much evidence for the complex structures explored here, and they are therefore not given a prominent place in this book. We discuss them in Section 4.2.1 for completeness' sake.

allomorphy of the comparative reduces to explaining the allomorphy of *-ěj* (leading to *-ěj-š*) and *-Ø* (leading to *-Ø-š*).⁵

Before we go into the arguments for decomposing *-ějš*, we show that the distribution of the allomorphs is morphosyntactic in nature, i.e. an arbitrary property of the root (very much like the augments discussed in the previous chapter). This is shown by (13), where (13a) illustrates the allomorph *-ějš*, with *-š* appearing in (13b).

(13)

	POS	CMPR	GLOSS
a.	chab-ý	chab-ějš-í	‘weak’
b.	slab-ý	slab-š-í	‘weak’

The two adjectives in (13) are similar both in their phonology and their meaning. This indicates that the use of one or the other marker is not determined by the phonology or the meaning of the base. The decision which roots require *-š* and which *-ějš* is therefore an idiosyncratic property of the root, i.e. not predictable from any independent property of the root.⁶

We illustrate the arbitrary nature of allomorph selection in more detail in (14). In this table, we put side by side adjectival roots that have similar phonology, but differ in their comparative allomorphs. In the table, the root on the left (e.g. *star* ‘old’ on the first row) always has similar phonology as the root to its right on the same line (e.g. *čir* ‘pure’), but the allomorphs differ.⁷

⁵ This particular form of comparative allomorphy is found not only in Czech (Caha, 2017b; Caha et al., 2019), but also more widely across Slavic (see e.g. Vanden Wyngaerd et al. 2020 and Vyshnevskaya 2022 for a discussion of Slovak and Ukrainian, respectively).

⁶ Křivan (2012) concludes that frequency plays a role, with more frequent roots showing preference for *-š*.

⁷ There are a limited number of roots that allow both allomorphs, like the following (Křivan, 2012):

(i)

	POS	CMPR -š	CMPR -ějš	GLOSS
	bohat-ý	bohat-š-í	bohat-ějš-í	‘rich’
	hust-ý	hust-š-í	hust-ějš-í	‘dense’
	hrub-ý	hrub-š-í	hrub-ějš-í	‘rough’

We shall not try to account for this variation here (see Vyshnevskaya 2022 for similar observations concerning Ukrainian). However, note that the very existence of such variation rules out the possibility that the choice of the allomorph is determined by phonology alone, as the phonology of the root is presumably identical on each row of Table (i).

(14)

	POS	CMPR GLOSS	POS	CMPR GLOSS
a.	star-ý	star-š-í ‘old’	čir-ý	čir-ějš-í ‘pure’
b.	bohat-ý	bohat-š-í ‘rich’	svat-ý	svat-ějš-í ‘saint’
c.	tvrđ-ý	tvrđ-š-í ‘hard’	hrđ-ý	hrđ-ějš-í ‘proud’
d.	drah-ý	draž-š-í ‘expensive’	uboh-ý	ubož-ějš-í ‘poor’

Each of the suffixes triggers particular consonant mutations at the end of the base. We return to these in more detail in Sect. 3.2.2.

As stated above, the (phonological) containment of -š inside -ějš already suggests that the comparative marker -ějš should be decomposed into two independent morphemes, namely -ěj and -š. This decomposition is confirmed by the fact that both of the morphemes that appear in *ěj-š* can occur independently. That -š appears without -ěj is evidenced by the -š-selecting roots discussed above in (14). The -ěj morpheme appears alone in Czech comparative adverbs, as shown in (15). What we see here is that all the -ěj-š adverbs lack -š in the adverb, and only have -ěj before the adverbial -i.⁸

(15)

CMPR ADJ	CMPR ADV	GLOSS
chab-ěj-š-í	chab-ěj-i	‘weak’
rychl-ěj-š-í	rychl-ěj-i	‘fast’
červen-ěj-š-í	červen-ěj-i	‘red’
hloup-ěj-š-í	hloup-ěj-i	‘stupid’

These facts show that -š and -ěj are independent morphemes. As a result, we can describe the allomorphy between -š comparatives and -ějš comparatives as a simple difference in the presence/absence of -ěj.

When we look at comparative adjectives alone, the *ěj/∅* parameter allows us to distinguish two subclasses of comparatives within augmentless adjectives. It turns out that when we look at comparative adverbs, three subclasses emerge. That is because while some adjectives consistently show -ěj, and some consistently show -∅, there is also an intermediate class with -∅ in the comparative adjective and -ěj in the adverb. We shall call these classes Class I-III, in the order in which they appear in Table (16).

⁸ The marker -i only surfaces with comparative adverbs. Positive-degree adverbs show -ě, e.g., *chab-ě* ‘weakly,’ *rychl-e* ‘quickly,’ *hloup-ě* ‘stupidly,’ etc.

It is perhaps worth noting that the palatalisations (consonant mutations) triggered by the positive-degree adverbial -ě are different from those triggered by the comparative -ěj, e.g. *uboh-ý* ‘poor’ has the adverb with a *z* (*uboz-e* ‘poorly’) but a comparative with a *ž* (*ubož-ěj-š-í* ‘poorer’).

(16) Comparative adverbs in augmentless adjectives

POS	ADJ	CMPR	ADJ	CMPR	ADV	GLOSS
chab-ý	chab-ěj-š-í	chab-ěj-i				‘weak’
slab-ý	slab-š-í	slab-ěj-i				‘weak’
drah-ý	draž-š-í	dráž-			-e	‘expensive’

Classes I/II and III further differ in whether their adverbs show an *-i* or *-e*, but this reduces to the presence or absence of *-ěj*: adverbs that have *-ěj* show the adverbial *-i*, roots that lack *-ěj* exhibit an *-e*. The three classes are abstractly depicted in Table (17). In this table, we also have a column for the second parameter (presence/absence) of an augment, but this plays no role for the adjectives under discussion here. The only difference in the three classes is the presence/absence of *-ěj*.⁹

(17) Three classes of augmentless adjectives

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
I	POS ADJ	chab	Ø	—	ý	‘weak’
	CMPR ADJ	chab	Ø	ěj	ší	
	CMPR ADV	chab	Ø	ěj	i	
II	POS ADJ	slab	Ø	—	ý	‘weak’
	CMPR ADJ	slab	Ø	Ø	ší	
	CMPR ADV	slab	Ø	ěj	i	
III	POS ADJ	drah	Ø	—	ý	‘expensive’
	CMPR ADJ	draž	Ø	Ø	ší	
	CMPR ADV	dráž	Ø	Ø	e	

When we organise the data this way, we realise that the context for the *-ěj* ~ *Ø* alternation is quite complex, and rises some challenges for standard accounts of allomorphy. These are traditionally stated in terms of context sensitive rules. The issues arise because of the fact that in order to describe the distribution of the allomorphs, we need to take into account both the left context (the particular root) as well as the right context (whether the form is a comparative adjective or adverb). Moreover, the particular way in which these contexts interact makes it quite difficult to describe the distribution with just a single rule for *-ěj* or *Ø*. This is so regardless of whether we take *-ěj* or *Ø* to be the elsewhere exponent.

⁹ The fourth logically possible class (*-ěj* in the comparative adverb but not in the comparative adjective) is not attested for adjectives with no augments. However, such distribution (even though rare) exists in classes with augments. For instance, the adjective ‘early’ has a comparative adjective with *-ěj* (*dřív-ěj-š-í* ‘earlier, ADJ’), but an adverb without it (*dřív-Ø-e* ‘earlier, ADV’).

Suppose, for instance, that we assumed that $-\check{j}$ was the default lexicalisation of the relevant morphosyntactic position, call it α , as in (18a). Then, to make sure that $-\check{j}$ does not appear with Class III roots, we would need to use the rule in (18b). However, to get the distribution of $-\check{j}$ vs. $\emptyset$ in Class II, we cannot simply add Class II roots in the contextual restriction of (18b). Rather, we need to make a different rule for $\emptyset$ as in (18c).

- (18) a. $\alpha \Leftrightarrow -\check{j}$
 b. $\alpha \Leftrightarrow \emptyset$ in the context of Class III
 c. $\alpha \Leftrightarrow \emptyset$ in the context of Class II and ADJ

Analogous issues arise when we consider $\emptyset$ to be the default allomorph, as in (19a). In such case, $-\check{j}$ would need two separate rules, as in (19b,c). This is because the rule for $-\check{j}$ is insensitive to the ADJ/ADV distinction in Class I, but sensitive to it in Class II.

- (19) a. $\alpha \Leftrightarrow \emptyset$
 b. $\alpha \Leftrightarrow \check{j}$ in the context of Class I
 c. $\alpha \Leftrightarrow \check{j}$ in the context of Class II and ADV

In this chapter, we shall demonstrate that the Nanosyntax approach resolves this conundrum without the need for accidental homophony. That is because in this approach, zeroes are not dedicated markers, but emerge as a consequence of phrasal lexicalisation (the features lexicalised normally as $-\check{j}$ are expressed by other morphemes). On this approach, the patterns of distribution emerge as a result of phrasal lexical items interacting with the invariant lexicalisation procedure described in Ch. 1.

As the discussion unfolds, we will show that analogous issues arise for the distribution of augments. Before we get to this, we shall review some regular phonological processes that affect the base when the markers $-\check{j}$, $-\check{s}$ and $-e$ are added to it. Readers who are not interested in these alternations (and are willing to accept our morphological analysis at face value) may proceed directly to Section 3.2.3.

3.2.2 The effects of phonology

The data in Table (17) represent Czech spelling conventions. These do not reflect certain regular phonological processes, such as degemination. For example, the comparative of *drah-ý* ‘expensive’ is written in the table as *draž-š-í* ‘more expensive.’ Thus, it includes the sequence *žš*, but it is pronounced with a simple (short) [ʃ] due to devoicing and degemination ([draʃi:]). Similarly, analogous comparative forms such as *tich-ý* ~ *tíš-š-í* ‘more silent’ are pronounced as with a single [ʃ] ([ciʃi:]). Some other processes are reflected

in the spelling and we list them (alongside degemination under (20a)) in (20b-d).

- (20) Phonological processes assumed
- Degemination ([ʃʃ] → [ʃ])
 - Different types of palatalisations triggered by the comparative suffixes *-ěj*, *-š* and *-e*
 - Vowel lengthening in comparative adverbs formed by *-e*
 - Vowel shortening in roots followed by *-š*

We describe the processes (20b-d) below in turn.

Palatalisations happen regularly before the various comparative suffixes and correspond to particular mutations of the final consonant (or consonant cluster). We saw an example of palatalisation in the root *drah-* (Class III in Table (17)), which is realised as *draž-* in the comparative. Building on Scheer (2001), Cavarani & Vanden Wyngaerd (2026) indentify three different patterns in Czech palatalisations. These are schematically shown in Table 3.1.

Table 3.1 Three different patterns (excluding the labials)

	PAL ₁	PAL ₂	PAL ₃
/k/	/ts/	/tʃ/	/tʃ/
/x/	/ʃ/	/ʃ/	/ʃ/
/h/	/z/	/ʒ/	/ʒ/
/t/	/c/	/c/	/ts/
/d/	/ʃ/	/ʃ/	/z/
/s/	/s/	/s/	/ʃ/
/z/	/z/	/z/	/ʒ/
/n/	/ɲ/	/ɲ/	/ɲ/
/r/	/r/	/r/	/r/
/l/	/l/	/l/	/l/

The first column shows the base sound, and it is arranged according to the place of articulation (velars above the horizontal line, alveolars below). The column headings (abstractly) identify the nature of the palataliser as either PAL₁, PAL₂ or PAL₃. The columns themselves detail the sound changes with respect to the base sound. Even though each of these patterns has a unique fingerprint, there are instances where palatalisation is not visible in the output, as with /l/ across the board, and as with /s/ and /z/ in the two of the three cases.¹⁰

¹⁰ For reasons we shall not discuss in any detail, Cavarani & Vanden Wyngaerd (2026) do not include labials in their table.

In Table 3.2 we show the data for the palatalisations triggered by the comparative morpheme *-ěj*, which instantiates PAL_2 . The shading follows that of the middle column of Table 3.1. Note that in the orthography (given at the very right in Table (17)), the palatalising effect of the suffix is marked by the *háček*, which sometimes appears directly on the final consonant of the morpheme to the left of *-ěj*, sometimes on the *-ěj* itself. In the orthography, labials are followed by *-ě*. In the pronunciation, orthographic sequences such as *pě* give rise to an extra glide in between the labial and the vowel. These extra consonants are depicted in square brackets in the lowermost section of the table. We shall treat all these alternations as a regular phonological process triggered by the suffix *-ěj*.

Table 3.2 Palatalisations triggered by comparative -ěj

base		PAL ₂	
/k/	divo k -ý ‘wild’	/tʃ/	divo č -ejší
/x/	plach h -ý ‘timid’	/ʃ/	plab š -ejší
/f/	stro h -ý ‘austere’	/ʒ/	stro ž -ejší
/t/	kulat t -ý ‘round’	/c/	kulat č -ejší
/d/	hrd d -ý ‘proud’	/ʒ/	hrd ž -ejší
/s/	lys s -ý ‘bald’	/s/	lys š -ejší
/z/	drz z -ý ‘cheeky’	/z/	drz ž -ejší
/n/	lev n -ý ‘cheap’	/ɲ/	lev ň -ejší
/r/	bujar r -ý ‘merry’	/r̩/	bujar ř -ejší
/l/	rychl l -ý ‘fast’	/l/	rychl l -ejší
/p/	hloup p -ý ‘stupid’	/pj/	hloup p[j] -ejší
/b/	blb b -ý ‘stupid’	/bj/	blb b[j] -ejší
/f/		/fj/	—
/v/	hrav v -ý ‘playful’	/vj/	hrav v[j] -ejší
/m/	lakom m -ý ‘stingy’	/mj/	lakom m[j] -ejší

The comparative marker -š triggers a different set of palatalisations. We provide an overview of its behaviour in Table 3.3. Note that since this is a closed class, the number of adjectives is restricted. Some sound combinations are illustrated by adjectives that have an augment in the positive but lack it in the comparative. Some comparatives are suppletive.

Table 3.3 Palatalisations triggered by comparative -š

base		PAL ₂	
/k/	onak k -ý ‘of that kind’	/tʃ/	ona č -í
/x/	tich h -ý ‘silent’	/ʃ/	ti š -ší
/f/	drah h -ý ‘expensive’	/ʒ/	dra ž -ší
/t/	bohat t -ý ‘rich’	/t/	bohat š -ší
/d/	chud d -ý ‘poor’	/d/	chud š -ší
	snad d -n-ý ‘easy’	/z/	snaz ž -ší
/s/	vy s -ok-ý ‘tall’	/ʃ/	vy š -ší
/z/	ní z -k-ý ‘low’	/ʒ/	ni ž -ší
/n/	ten n -k-ý ‘thin’	/n/	ten š -ší
/r/	star r -ý ‘old’	/r/	star š -ší
/l/	dlouh l -ý ‘far’	/l/	del š -ší
/p/	dobrý ‘good’	/p/	lep š -ší
/b/	slab b -ý ‘weak’	/b/	slab š -ší
/v/	tma v -ý ‘dark’	/v/	tmav š -ší

The status of *-š* in the typology provided by Cavirani & Vanden Wyngaerd (2026) is unclear. The velars show a pattern that is consistent with either PAL₂ or PAL₃. For example, velar *h* palatalises to *ž* in the adjective *draž-ši* ‘more expensive’, which is consistent with it either having a PAL₂ or a PAL₃ palataliser. The fact that sibilant *z* goes to *ž* when followed by *š* (e.g., *niž-ši* ‘lower’, *blíž-ši* ‘closer’, *už-ši* ‘narrower’) could be construed as an argument that *š* contains PAL₃ (see Table 3.1). However, this does not sit well with the fact that *t*, *n*, and *r* fail to show any palatalisation (e.g. *star-ši* ‘older’), which is not consistent with any of the patterns in Table 3.1. Finally, the biggest issue is the behaviour of *d* preceding *-š*, which shows the PAL₃ pattern with *snad-n-ý* ‘easy’ (i.e. *snaz-ši* ‘easier’), but no palatalisation with *chud-ý* ‘poor’ (*chud-š-í* ‘poorer’).

The mixed behaviour following *d* could point to a different morphosyntactic analysis of these two cases. Since *snad-* ‘easy’ is the only root where *d* undergoes palatalisation (compare *tvrđ-ý* ~ *tvrđ-š-í* ‘hard(-er)’, *mlad-ý* ~ *mlad-š-í* ‘young(er)’ or *slad-k-ý* ~ *slad-š-í* ‘sweet(er)’), we will assume that *-š* does not palatalise *d*, and that the *snad-snaz* alternation is a case of suppletion, which we shall account for in morphosyntactic terms, along the line of our treatment of suppletion in Chapt. 1, where we assigned different root sizes to suppletive pairs.

Roots that take the adverbial comparative marker *-e* are few in number, and so it is hard to find sufficient data to conclusively establish the palatalisation pattern. Nevertheless, the attested forms in Table 3.4 are consistent with PAL₃ (in particular, the fact that /z/ goes to /ʒ/ uniquely points to PAL₃). The fact that in *snad-n-ý* ‘easy’ the root-final *d* changes to *z* in the comparative adverb is also consistent with *-e* containing a PAL₃, but note that the *z* also appears in the comparative adjective, before *-š*. This led us to propose that *snaz-* is a suppletive root, and this weakens the evidence this adjective provides for identifying this pattern as PAL₃.

Table 3.4 Palatalisations triggered by comparative adverb marker *-e*

	base	PAL ₃
/k/		
/x/		
/fi/	dra h -ý ‘expensive’	/ʒ/ drá ž -e
/t/		
snad d -n-ý ‘easy’	/z/	sná z -e
/s/	vys s -ok-ý ‘tall’	/ʃ/ vý š -e
/z/	blíz z -k-ý ‘close’	/ʒ/ blí ž -e
/n/		
/r/	š ir -ok-ý ‘wide’	/ř/ š ř -e
/l/	dlou h ý ‘long’	/l/ dél-e

Another factor that is amenable to a phonological analysis concerns certain alternations in the vowel length of the root, in particular the fact that with the comparative adverbial suffix *-e*, the vowel in the root is invariably long, even if it is short in the positive. This is illustrated in (21), where the adjectives above the line have a short vowel in the positive, which becomes long in the comparative adverb. The ones below the line have either a suppletive root (which makes a comparison with vowel length in the positive moot), or a long vowel in the positive adjective.

(21) Standard Czech comparative adverbs with *-e*

POS	ADJ	CMPR	ADJ	CMPR	ADV	GLOSS
	drah-ý		draž-š-í		draž-e	‘expensive’
	šir-ok-ý		šir-š-í		šír-e	‘wide’
	vys-ok-ý		vyš-š-í		výš-e	‘tall’
	snad-n-ý		snaz-š-í		snáz-e	‘easy’
	brz-k-ý		dřív-ěj-š-í		dřív-e	‘early’
	dobr-ý		lep-š-í		lép-e	‘good’
	špat-n-ý		hor-š-í		hůř-e	‘bad’
	dlouh-ý		del-š-í		dél-e	‘long’
	blíz-k-ý		blíž-š-í		blíž-e	‘close’

Caha et al. (2024a) argue that this lengthening in the root vowel is triggered by a nonmelodic component in the adverbial suffix, which they represent as μ , and which is realised as a long vowel in the root. Accordingly, we shall represent such long vowels in the comparative adverb as short in our analysis, which seeks to eliminate the role of regular phonology.

The converse process (vowel shortening) happens in comparatives marked by *-š*, i.e., all adjectives shorten their vowels when they are followed by *-š*.

(22) Vowel shortening in comparative adjectives

POS	ADJ	CMPR	ADJ	GLOSS
	blíz-k-ý		blíž-š-í	‘close’
	úz-k-ý		už-š-í	‘narrow’
	níz-k-ý		niž-š-í	‘low’
	krát-k-ý		krat-š-í	‘short’
	říd-k-ý		řid-š-í	‘thin’

In (23), we repeat the processes discussed up to now.

(23) Phonological processes assumed

- Degemination ([ʃʃ] → [ʃ])
- Different types of palatalisations triggered by the comparative suffixes *-ěj*, *-š* and *-e*
- Vowel lengthening in comparative adverbs formed by *-e*

d. Vowel shortening in roots followed by -š

In some cases, these processes create multiple possibilities for a morphosyntactic analysis of particular cases. Consider, for example, the examples in (24), which show that in the comparative of some adjectives with the augment -*k*, the augment alternates with -*č* in the comparative.

(24) Cases of comparative adjectives with multiple analyses

POS	ADJ	CMPR	ADJ	GLOSS
	hez-k-ý		hez-č-í	'pretty'
	ten-k-ý		ten-č-í	'thin'
	sliz-k-ý		sliz-č-í	'slimy'
	prud-k-ý		prud-č-í	'steep'

The pronunciation of the affricate in forms like *hezčí* 'prettier' or *tenci* 'thinner' is as /tʃ/. Standard Czech spelling reflects this, as it has no -š in the comparative. Likewise, the traditional descriptions of Czech take the comparative to have zero marking in these cases (Dokulil et al., 1986; Karlík et al., 1995; Osolsobě, 2016). The palatalisation of *k* to *č* is, under this analysis, triggered by the soft agreement marker -*í*, i.e., the underlying comparative is *hez-k-í*, which goes to *hez-č-í*. However, an alternative analysis is possible, according to which there is an underlying comparative -š in these forms, which palatalises the preceding *k*, but is left unpronounced as a consequence of degemination (/tʃʃ → tʃ/).

- (25) a. underlying: *root-k-š* /kʃ/
 b. palatalisation: *root-č-š* /tʃʃ/
 c. degemination: *root-č* /tʃ/

In what follows, we shall adopt the latter analysis, but nothing much hinges on the matter, and the general approach that we shall propose can accommodate either approach.

3.2.3 The behaviour of augments

With the phonology clarified, we now return to the second factor of variation, which concerns the augment. In Czech, each of the three augments found in the positive need not be found in all the comparative forms, but only in a subset of them. The table in (26) illustrates this for the augment -*n*.¹¹

¹¹ The numbering of the classes starts at number IV, given that we have already introduced three classes of adjectives without any augments, recall Table (17). The last pattern of distribution, class VII, is only attested with the (weakly) suppletive root 'easy,' recall the discussion in Section 3.2.2. We use non-suppletive roots whenever possible and mark

(26) Four classes of adjectives with the augment *n*

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
IV	POS ADJ	jem	n	—	ý	‘fine’
	CMPR ADJ	jem	n	ěj	ší	
	CMPR ADV	jem	n	ěj	i	
V	POS ADJ	žádouc	Ø	—	í	‘desirable’
	CMPR ADJ	žádouc	n	ěj	ší	
	CMPR ADV	žádouc	n	ěj	i	
VI	POS ADJ	pozd	n	—	í	‘late’
	CMPR ADJ	pozd	Ø	ěj	ší	
	CMPR ADV	pozd	Ø	ěj	i	
VII	POS ADJ	snad	n	—	ý	‘easy’
	CMPR ADJ	snaz	Ø	Ø	ší	
	CMPR ADV	snáz	Ø	Ø	e	

What we see here is that *-n* can be present in all forms, as in Class IV (the adjective ‘smooth’). However, *-n* can also be missing in the positive (Class V) or in the comparative (Classes VI and VII). Classes VI (‘late’) and VII (‘easy’) show the same distribution of the augment, but they are distinguished by the other parameter (presence/absence of *-ěj*).

It is worth noting that after *-n*, we always find *-ěj*, but when *-n* is missing, the root determines whether we find *-ěj* or Ø. This seems to be a locality effect such that the root only matters for the allomorphy *-ěj/Ø* when the augment does not intervene. In our account, where zeroes are the result of phrasal lexicalisation, this effect can be easily derived, since the root will only be able to lexicalise the features of *-ěj* when it first lexicalises the features of *-n*.

The interesting challenge with this class is the very distribution of the augment *-n*. As pointed out in Caha et al. (2024b), we basically find all logically possible patterns, as depicted in Table (27).

(27) Four different root classes

CLASS	POS	CMPR
IV	n	n
V	Ø	n
VI-VII	n	Ø
I-III	Ø	Ø

suppletive roots with a grey shade in the table. Since we give preference to non-suppletive roots, there is more suppletion than shown. E.g., the root for ‘good’ (*dobr-ý—lep-š-í—lep-e*) belongs in Class III, but we do not list it since a non-suppletive root exists. The adjective ‘bad’ (*špat-n-ý—hor-š-í—hůř-e*) is also in Class VII, just like *snad-n-ý* ‘easy.’

Namely, in Class IV, we always find *-n*, see the top row in (27). Class V has *-n* only in the comparative, while Classes VI and VII have *-n* only in the positive. Finally, the classes without any augments (these are Classes I-III) have no *-n* whatsoever, as depicted on the bottom row of (27). Caha et al. (2024b) argue that such a distribution is impossible to capture using context sensitive rules.

To see the issue, assume again that there is a morphosyntactic position, call it β , which the augment is the realisation of. If zero is the default realisation, as in (28a), we need three different rules specifying where *-n* appears, see (28b-d). As far as we are aware, there is no way of unifying these rules into a single non-disjunctive statement that correctly predicts where *-n* is found.

- (28) a. $\beta \Leftrightarrow -\emptyset$
 b. $\beta \Leftrightarrow n$ in the context of Class IV
 c. $\beta \Leftrightarrow n$ in the context of Class V and CMPR
 d. $\beta \Leftrightarrow n$ in the context of Class VI and POS

Moreover, it is unclear whether the rule in (28d) can be stated at all, since, in an account where comparatives contain the positive, the positive feature (whatever feature that may be) would be also contained in the comparative, and activate the rule also there, i.e., beyond its intended domain. The same issues arise if *-n* were chosen as the default realisation, in which case the contexts for the zero marker would have to be stipulated in very much the same way as in (28b-d).

It is our goal in this chapter to show that the Nanosyntax model adopted here is able to model such distributions alongside with all the other classes yet to be discussed. The core difference is that in Nanosyntax, zero markers arise as a result of phrasal lexicalisation, and there is therefore no need to orchestrate a competition between two markers, a zero and an overt marker; the zero does not really exist in Nanosyntax.

Let us next turn to the adjectives that have the augment *-k*. These provide five additional classes, see Table (29). As in (27), dark shade is used when the pattern is only attested for suppletive adjectives. Suppletion is obvious in the case of the adjective ‘early,’ which shows a completely different form in the positive and in the comparative. We also treat as weakly suppletive the adverb root in Class X (‘narrow’), which shows an irregular palatalisation; recall from 3.2.2 that *-ěj* does not seem to palatalise *z* (e.g., *drz-ý* ~ *drz-ěj-š-í* ‘rude(r),’ which contrasts with *úz-k-ý* ~ *úz-ěj-i* ‘narrow ~ more narrowly’).

(29) Five classes of adjectives with the augment *k*

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
VIII	POS ADJ	hez	k	—	ý	‘pretty’
	CMPR ADJ	hez	k	Ø	ší	
	CMPR ADV	hez	k	ěj	i	
IX	POS ADJ	slad	k	—	ý	‘sweet’
	CMPR ADJ	slad	Ø	Ø	ší	
	CMPR ADV	slad	k	ěj	i	
X	POS ADJ	úz	k	—	ý	‘narrow’
	CMPR ADJ	už	Ø	Ø	ší	
	CMPR ADV	úž	Ø	ěj	i	
XI	POS ADJ	blíz	k	—	ý	‘close’
	CMPR ADJ	bliž	Ø	Ø	ší	
	CMPR ADV	blíž	Ø	Ø	e	
XII	POS ADJ	brz	k	—	ý	‘early’
	CMPR ADJ	dřív	Ø	ěj	ší	
	CMPR ADV	dřív	Ø	Ø	e	

The distribution of *-k* is considerably more complex than the distribution of *-n*. Merely describing it in plain words is a challenge, let alone modelling it. Its presence in the positive is sensitive to the kind of root, since not all roots have it (e.g., the augmentless Classes I-III don’t). However, *-k* is not only sensitive to the morpheme to its left, but simultaneously also to the degree (the morpheme to its right). Thus, some roots always have *-k* in the comparative (Class VIII), but other roots lack it in one of the comparative forms (Classes IX-XII). On top of that, it is also relevant whether the comparative form is an adjective or an adverb. Even though most classes lack the augment in both comparative forms (Classes X-XII), Class IX shows a mixed pattern, where *-k* is present in the comparative adverb, but absent in the comparative adjective (Class IX). Which pattern an adjective exhibits is determined by the root.

Let us now turn to the second parameter (presence vs. absence of *-ěj*). When *-k* is present in the comparative, it is always followed by *-ěj* in the adverb, but by *-Ø* in the adjective. (This is similar to augmentless roots of Class II, which show the same pattern.) When *-k* is absent in the comparative, roots may choose to be followed by *-ěj* in the comparative adverb and *Ø* in the comparative adjective (Class X), but they may also have *Ø* in both forms (Class XI). Finally, Class XII shows *-ěj* in the adjective and *Ø* in the adverb.

The complexity of comparative-adverb formation in Czech is significant, and the predictability of the adverbial form is rather small; specifically, it seems impossible to predict what the comparative adverb will be even if we know both the positive and the comparative of the adjective. Consider,

for instance, Classes IX–XI. All three classes have the positive in *root-k-ý*, and a comparative *root-š-í*. However, each of these classes has a different comparative adverb.

This is in contrast to most Slavic languages, where the comparative adverbial marker is simply attached on top of the comparative adjective minus the agreement marker. We show this general pattern in (30), using Ukrainian as an exemplary example (Vyshnevskya, 2025c).

(30) Ukrainian adjectives and adverbs (Vyshnevskya, 2025c)

AUG	POS ADJ	CMPR ADJ	CMPR ADV	GLOSS
∅	tepl-yj molod-yj	tepl-i-š-yj molod- š-yj	tepl-i-š-e molod- š-e	‘warm’ ‘young’
OK	žorst-ok-yj šyr-ok-yj	žorst-ok-i-š-yj šyr- š-yj	žorst-ok-i-š-e šyr- š-e	‘cruel’ ‘wide’
N	rozum-n-yj	rozum-n-i-š-yj	rozum-n-i-š-e	‘smart’
K	čit-k-yj švyd-k-yj	čit-k-i-š-yj švyd- š-yj	čit-k-i-š-e švyd- š-e	‘clear’ ‘fast’

Ukrainian shows many similarities with Czech in the formation of the comparative adjective, such as the allomorphy in the comparative marker (*-i-š* vs *-š*), which is sensitive to the type of root, as shown on the first two lines of (30). Ukrainian has the same augments as Czech, which (like in Czech) may be preserved or dropped in the comparative adjective. The formation of the adverb, however, is subject to an extremely simple descriptive rule in Ukrainian: take the comparative adjective, remove the agreement marker, and replace it by the adverb marker *-e*.

Let us now turn back to Czech and complete the overview of adjective classes by zooming in on the adjectives with *-ok*. They fall into three classes depicted below. Class XIV shows an irregular lengthening of the base in the comparative adverb, compare the regular Class II *hrub-š-í* ~ *hrub-ěj-i* ‘rougher’ ADJ ~ ADV with the irregular *hlub-š-í* ~ *hloub-ěj-i* ‘deeper’ ADJ ~ ADV).

(31) Four classes of adjectives with the augment *ok*

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
XIII	POS ADJ	div	ok	—	ý	‘wild’
	CMPR ADJ	div	ok	ěj	ší	
	CMPR ADV	div	ok	ěj	i	
XIV	POS ADJ	hlub	ok	—	ý	‘deep’
	CMPR ADJ	hlub	Ø	Ø	ší	
	CMPR ADV	hloub	Ø	ěj	i	
XV	POS ADJ	vys	ok	—	ý	‘tall’
	CMPR ADJ	vyš	Ø	Ø	ší	
	CMPR ADV	výš	Ø	Ø	e	

The augment *-ok* shows only two patterns: it is either always present (Class XIII), or absent in both comparative forms (Classes XIV and XV). The last two classes are then distinguished by the pattern of *-ěj*: completely absent in Class XV but present in the adverb of Class XIV.

3.2.4 A summary and a preview

To conclude this section, we recapitulate all the fifteen classes in Table 3.5.

Table 3.5 Fifteen patterns for the positive and the comparative

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
I	POS ADJ	chab	Ø	—	ý	‘weak’
	CMPR ADJ	chab	Ø	ěj	ší	
	CMPR ADV	chab	Ø	ěj	i	
II	POS ADJ	slab	Ø	—	ý	‘weak’
	CMPR ADJ	slab	Ø	Ø	ší	
	CMPR ADV	slab	Ø	ěj	i	
III	POS ADJ	drah	Ø	—	ý	‘expensive’
	CMPR ADJ	draž	Ø	Ø	ší	
	CMPR ADV	dráž	Ø	Ø	e	
IV	POS ADJ	jem	n	—	ý	‘fine’
	CMPR ADJ	jem	n	ěj	ší	
	CMPR ADV	jem	n	ěj	i	
V	POS ADJ	žádouc	Ø	—	í	‘desirable’
	CMPR ADJ	žádouc	n	ěj	ší	
	CMPR ADV	žádouc	n	ěj	i	
VI	POS ADJ	pozd	n	—	í	‘late’
	CMPR ADJ	pozd	Ø	ěj	ší	
	CMPR ADV	pozd	Ø	ěj	i	
VII	POS ADJ	snad	n	—	ý	‘easy’
	CMPR ADJ	snaz	Ø	Ø	ší	
	CMPR ADV	snáz	Ø	Ø	e	
VIII	POS ADJ	hez	k	—	ý	‘pretty’
	CMPR ADJ	hez	k	Ø	ší	
	CMPR ADV	hez	k	ěj	i	
IX	POS ADJ	slad	k	—	ý	‘sweet’
	CMPR ADJ	slad	Ø	Ø	ší	
	CMPR ADV	slad	k	ěj	i	
X	POS ADJ	úz	k	—	ý	‘narrow’
	CMPR ADJ	už	Ø	Ø	ší	
	CMPR ADV	úž	Ø	ěj	i	
XI	POS ADJ	blíz	k	—	ý	‘close’
	CMPR ADJ	bliž	Ø	Ø	ší	
	CMPR ADV	blíž	Ø	Ø	e	
XII	POS ADJ	brz	k	—	ý	‘early’
	CMPR ADJ	dřív	Ø	ěj	ší	
	CMPR ADV	dřív	Ø	Ø	e	
XIII	POS ADJ	div	ok	—	ý	‘wild’
	CMPR ADJ	div	ok	ěj	ší	
	CMPR ADV	div	ok	ěj	i	
XIV	POS ADJ	vys	ok	—	ý	‘tall’
	CMPR ADJ	vyš	Ø	Ø	ší	
	CMPR ADV	výš	Ø	Ø	e	
XV	POS ADJ	hlub	ok	—	ý	‘deep’
	CMPR ADJ	hlub	Ø	Ø	ší	
	CMPR ADV	hloub	Ø	ěj	i	

Not all of these classes are equally productive. The *-ok* adjectives, for example, amount to less than a handful. As a result, some classes have only one member (e.g., Classes XIII and XV, but also Classes VI, X, and XII). We nevertheless continue to refer to the distribution patterns as classes, because they represent a structural and attested possibility of the underlying system, which any empirically and descriptively adequate theory needs to account for.

An interesting property of the complete paradigm, which we would like to highlight, is that for each augment, we find classes that represent a challenge for a common pattern where the comparative is built on top of the positive. Grano & Davis (2018, 133) propose a candidate universal to that effect, see (32).

(32) *Candidate Universal*

Universally, the comparative form of a gradable adjective is derived from or identical to its positive form.

As Grano & Davis (2018, 133) note, ‘it should [...] be impossible to find a language in which both the positive and comparative forms are independently derived from a common base’ (their pattern D; Grano & Davis 2018, 134). But this is exactly what we find with each of the three augments that we have distinguished. In fact, we find this exact pattern in no less than seven out of our fifteen classes (eight if we include the suppletive Class XII), which all drop the augment in the comparative.

In the remainder of this chapter, we will show how this interesting and typologically rare property arises, along with the other puzzles regarding the distribution of augments and zeroes highlighted in this section.

3.3 Augmentless roots (Classes I-III)

In this section, we analyse the three classes of adjectives that have no augment. These classes therefore only differ in the presence/absence of the comparative *-ěj*. We give them in (33).

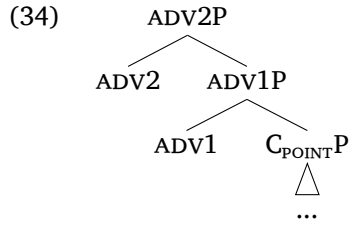
(33) *-ěj* in augmentless adjectives

CLASS	POS	ADJ	CMPR	ADJ	CMPR	ADV	GLOSS
I	chab-ý	chab- ěj -š-í	chab- ěj -i				‘weak’
II	slab-ý	slab-š-í	slab- ěj -i				‘weak’
III	drah-ý	draž-š-í	draž-			-e	‘expensive’

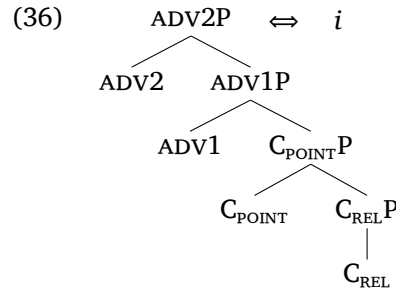
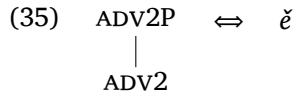
On the top row, we have a class that consistently has *-ěj* (Class I). On the bottom row, we have a class that never has *-ěj* (Class III). The shading brings out the fact Class II is an ‘intermediate’ class, whose comparative adjective

patterns with the $\emptyset$ class, while its adverb looks like that in the $-\check{e}j$ class. In this connection, it's worth recalling two things. The first one is that such a distribution is hard to capture with context-sensitive rules (recall (18)). The second thing is that such intermediate classes naturally lead on to a solution in terms of root shape (recall Section 1.6.3). It is our goal to develop such an account in this section.

Before we set out to explore this goal, we must put in place a proposal regarding adverbs. In particular, we shall assume that adverbs are derived from the adjective by means of two functional heads added on top of the highest projection of the comparative ($C_{\text{POINT}}P$). We show this below.



At the practical level, we need these two heads in order to differentiate adverbs with $-i$ and adverbs in $-\check{e}$. Recall that this distinction is dependent on whether the adjective has an $-\check{e}j$ in the comparative or not. If it does have $-\check{e}j$, the adverb has the $-i$ ending, when there is no $-\check{e}j$, the adverbs have an $-\check{e}$ that attaches directly to the root (since $-\check{e}j$ is missing). To account for this, we shall propose that the adverbial $-\check{e}$ only lexicalises the topmost adverbial projection ADV2, which means that only items reaching all the way to ADV1 will be able to use this ending. Assuming that no affix lexicalises ADV1P, only a restricted set of roots (which will be specified as lexicalising ADV1P) will be able to combine with it. This way, the restriction of $-\check{e}$ to a selected few roots follows.



The adverbial $-i$, on the other hand, lexicalizes both adverbial heads, and it therefore serves as a general marker of adverbs. Recall now that $-i$ does not appear in the positive, and that it makes $-\check{s}$ disappear. Both of these properties can be captured by the proposal that $-i$ lexicalises the two topmost comparative features, i.e., all the way down to C_{REL} , as in (36). By doing so, it

makes *-š* superfluous, since it lexicalises the highest comparative projection C_{POINT} , where *-š* resides.

While this proposal is mainly motivated by the synchronic analysis, it is supported by diachronic considerations. Thus, Lunt (2001, p.78) notes that in Old Church Slavonic (a Slavic language spoken around the 10th century), comparative adverbs were identical to the neuter-singular form of a comparative adjective. The particular form was ambiguous between an adverb and the nominative and accusative singular, as we highlight in the declension table below.¹²

(37) ‘Newer’ in Old Church Slavonic (Aitzetmüller 1978, 132)

	MASC	NEUT	FEM
NOM	nov-ěj-ъ	nov-ěj- -e	nov-ěj-ъš-i
ACC	nov-ěj-ъš-ъ	nov-ěj- -e	nov-ěj-ъš-о
GEN	nov-ěj-ъš-a	nov-ěj-ъš-a	nov-ěj-ъš-ę
LOC	nov-ěj-ъš-i	nov-ěj-ъš-i	nov-ěj-ъš-i
DAT	nov-ěj-ъš-u	nov-ěj-ъš-u	nov-ěj-ъš-i
INS	nov-ěj-ъš-emъ	nov-ěj-ъš-emъ	nov-ěj-ъš-ejо

Two things are worth pointing out. The first thing is that most of the forms in the table have the familiar bipartite structure of the comparative marking, with *-ěj* preceding *-ъš* (the latter followed by agreement markers). However, the neuter singular forms of this particular declension lack *-š* in NOM/ACC, and this is precisely the form that the adverbs developed from. When these adverbial forms are placed in the context of the full declension, we understand why it is tempting to see the absence of *-š* in the NOM/ACC of neuter not as something deep regarding the structure or meaning of the forms, but rather a morphological effect where the neuter singular ending lexicalises the ϕ features along with the higher comparative projections, making *-š* superfluous. This is precisely as we propose for the current comparative-adverb entry *-i* in Czech, recall (36).

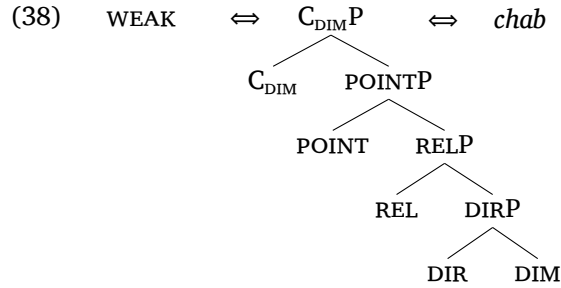
The second thing the paradigm shows is that adverbs can be thought of as regular adjectives (probably modifying/predicated of an event), and carrying neuter singular NOM/ACC ϕ -features. If that is so, it does make sense to think of the adverb as having multiple ‘adverbial’ features on top of it, as in Nanosyntax, each ϕ -feature is an independent head. In sum, these facts provide independent reasons to think that multiple features are to be found in an adverb on top of the comparative structure (singular, neuter, NOM/ACC).

With these issues clarified, let us turn to the analysis of Classes I-III.

¹² Apparently, also in Latin, the accusative form of the neuter singular paradigm was used as the comparative adverb (Greenough et al. 1903, §218).

3.3.1 Class I

We propose that Class I roots, like *chab* ‘weak’, are of size $C_{DIM}P$, as shown in (38).

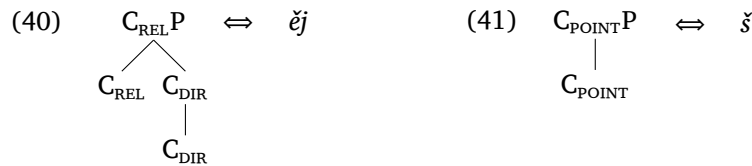


This root will always lexicalize features up to C_{DIM} , and require endings for all the other features. We show the division of labour among the root and the suffixes in (39).

(39) Class I augmentless adjective (*chab* ‘weak’)

	DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ	chab									
CMPR ADJ	chab					ěj	š			
CMPR ADV	chab					ěj		i		

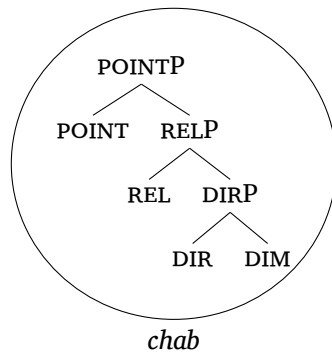
What we see here is that in the comparative adjective, the suffix *-ěj* lexicalizes C_{DIR} and C_{REL} , and *-š* is needed to lexicalise C_{POINT} . We show the presumed entries in (40) and (41).



Turning now to the comparative adverb, we assume that *-i* is the only entry capable of lexicalising ADV1. At the same time, the lowest feature of *-i* is proposed to be C_{REL} , recall (36). This means that in order to lexicalise ADV1, the base must shrink down to C_{DIR} , as shown in the table. This correctly predicts that *-i* makes *-š* superfluous, but *-ěj* must still appear to the left of *-i*.

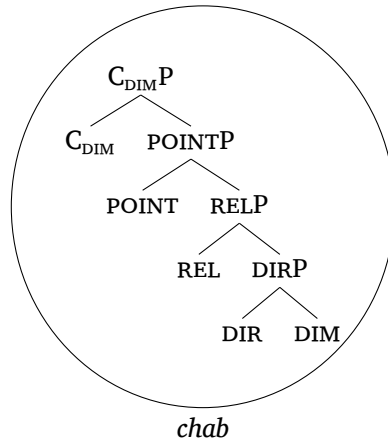
Let us now highlight the crucial steps of the derivation. The positive is straightforwardly lexicalised by the root, which contains $POINTP$ as a sub-constituent, see (42).

(42)



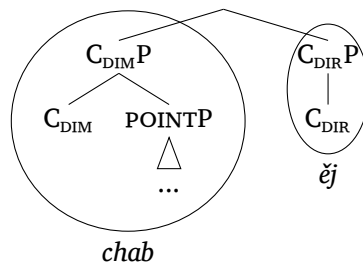
In the comparative adjective, roots like *chab* combine with *-ěj-š*. Since they can lexicalise a constituent as big as $C_{DIM}P$, they lexicalise this without any movement, see (43).

(43)



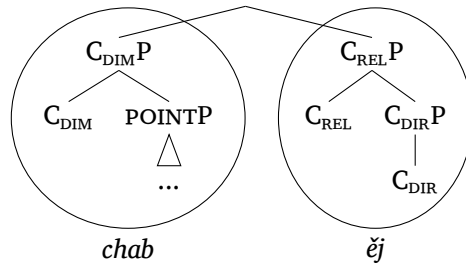
At the merger of C_{DIR} , the root (lexicalising $C_{DIM}P$) moves, by evacuate closest, and $C_{DIR}P$ is lexicalised as *-ěj*. In the tree, we simplify $POINTP$ to a triangle.

(44)



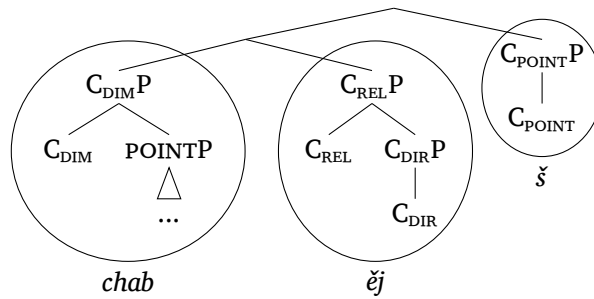
When C_{REL} is merged, evacuate closest applies again, resulting in (45).

(45)



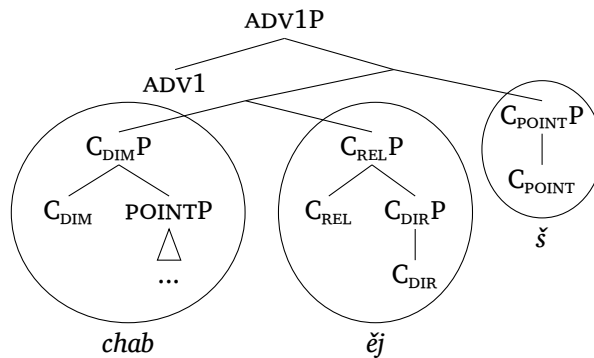
Then, at the merger of C_{POINT} , the evacuation of the root will not be successful, so that pied-piping is applied, resulting in the structure in (46), where $C_{\text{POINT}}P$ is lexicalised as $\check{s}$.

(46)

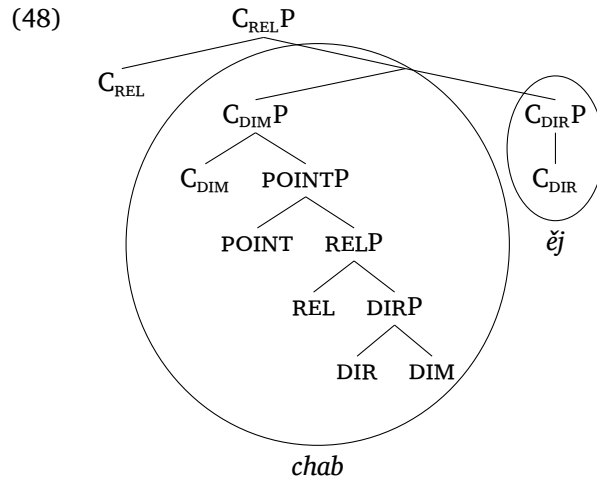


The comparative adverb involves a backtracking derivation (see Section 1.8). Specifically, when ADV1 is merged, as in (47), no evacuation movement succeeds.

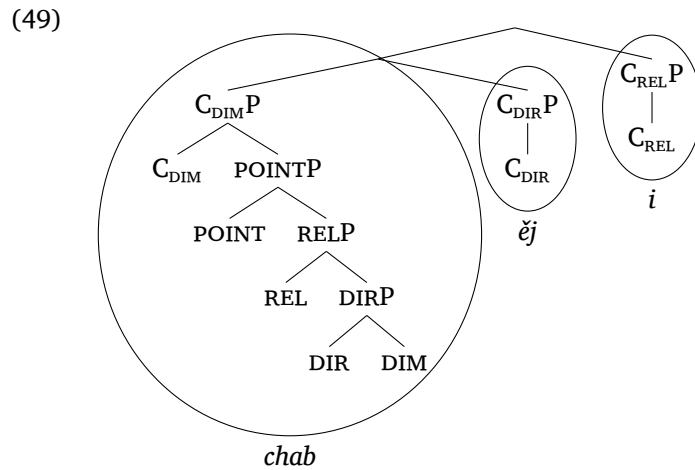
(47)



The derivation will thus need to backtrack and open a new derivational option. This happens when the derivation goes back to the C_{REL} cycle.

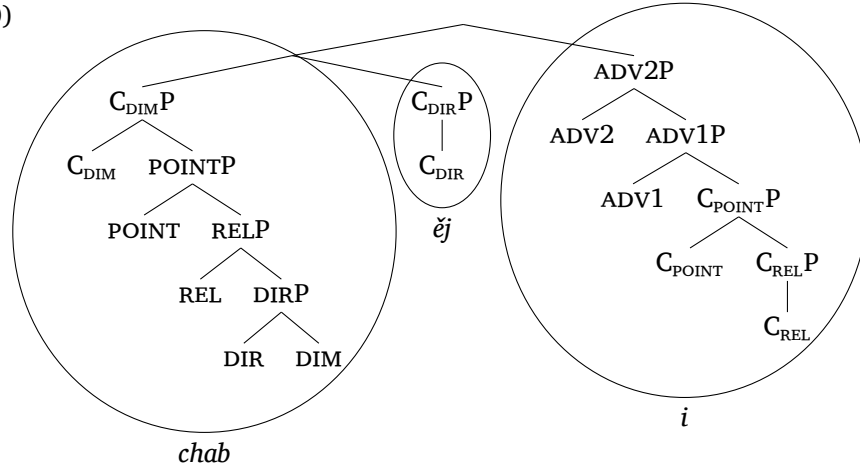


In the first pass at this cycle, we evacuated $C_{DIM}P$ (the closest labelled constituent), leading to (45). But since that did not lead to a viable output for the adverb, we now pied-pipe the node above $C_{DIM}P$, creating a remnant $C_{REL}P$ on the right, which is lexicalised as *-i*.



The remainder of the derivation goes through the usual steps of pied-piping evacuation movement, through which the suffix *-i* eventually grows to size ADV2P:

(50)



The derivation of the adverb concludes the demonstration of how the forms in Class I are derived. We recapitulate below in the form of the lexicalisation table.

(51) Class I augmentless adjective (*chab* ‘weak’)

	DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ			chab							
CMR ADJ			chab			ěj		š		
CMR ADV			chab			ěj			i	

3.3.2 Class II

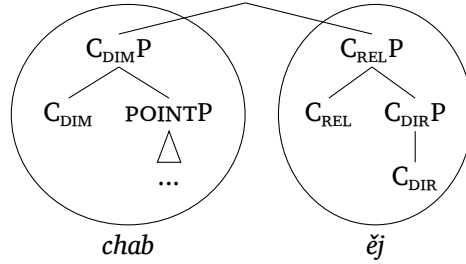
Class II adjectives have no augment in the positive, and they lack *-ěj* in the comparative adjective, e.g., *slab-ý* ~ *slab-ø-š-i* ‘weak ~ weak(er)’. However, *-ěj* surfaces in the comparative adverb *slab-ěj-i* ‘more weakly.’

Roots that show this behaviour have to be bigger in size than the regular roots taking *-ěj* in the adjective. This bigger size makes them able to lexicalise the features that *-ěj* lexicalises, causing its disappearance. In the case of the Class II roots, we can deduce its size from the fact that the root is directly followed by *-š* in the comparative adjective. Since *-š* lexicalises C_{POINT}, roots to the left of *-š* must be of size C_{REL}P. This means that Class II roots have the same size of the constituent that lexicalises the string *root +ěj* in the Class I roots. We could say that $root_{II} = root_I +ěj$. This is clearly so in terms of size, but, more importantly, also in terms of shape: what we propose is that the shape of the lexical entry for the Class II root *slab-* ‘weak’ corresponds

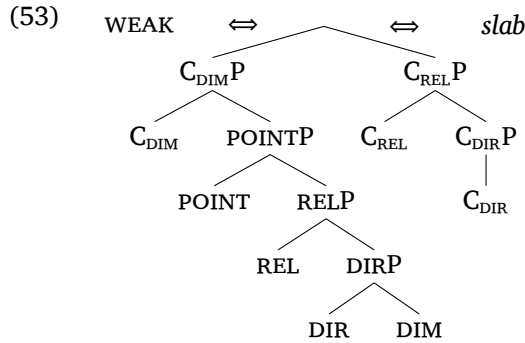
exactly to the shape of the regular root plus the suffix *-ěj* that taken together lexicalise a constituent of size $C_{REL}P$.

The $C_{REL}P$ stage of the Class I roots was given in (45) above, and repeated here as (52).

(52)



The lexical entry for Class II roots that we propose here is in (53), and literally corresponds to the stage in (52), i.e., one that is lexicalised as *root* + *ěj* in Class I.



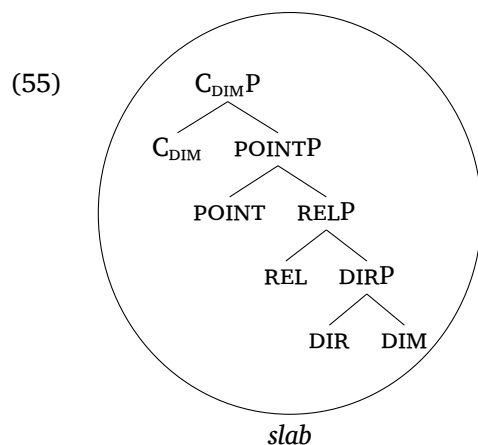
What is interesting about this proposal is that this entry not only allows us to capture the absence of *-ěj* in the comparative adjective, but also the fact that these adjectives regrow *-ěj* in the comparative adverb, i.e., we get *slab-ý—slab-Ø-š-í—slab-ěj-i*.

The lexicalisation table for this class is provided in (54). What we show on the comparative line is that the root, due to its shape, literally overrides *-ěj* in this form due to the fact that lexicalisation always prefers to target the highest node possible.

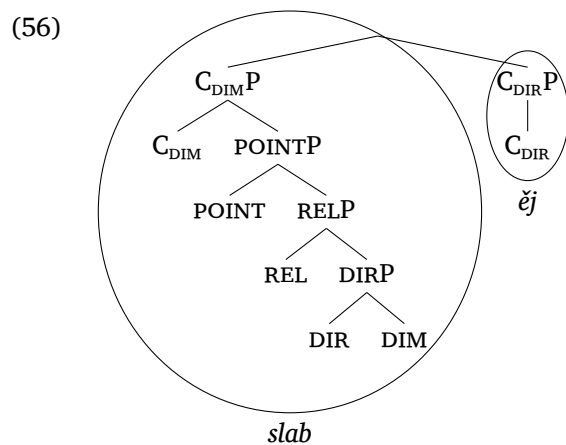
(54) Class II augmentless adjective (*slab* ‘weak’)

	DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ				slab						
CMPR ADJ				slab			ěj	š		
CMPR ADV				slab		ěj			i	

The derivation will follow the same steps as for Class I. First, $C_{\text{DIM}}P$ is constructed and lexicalised by the root.

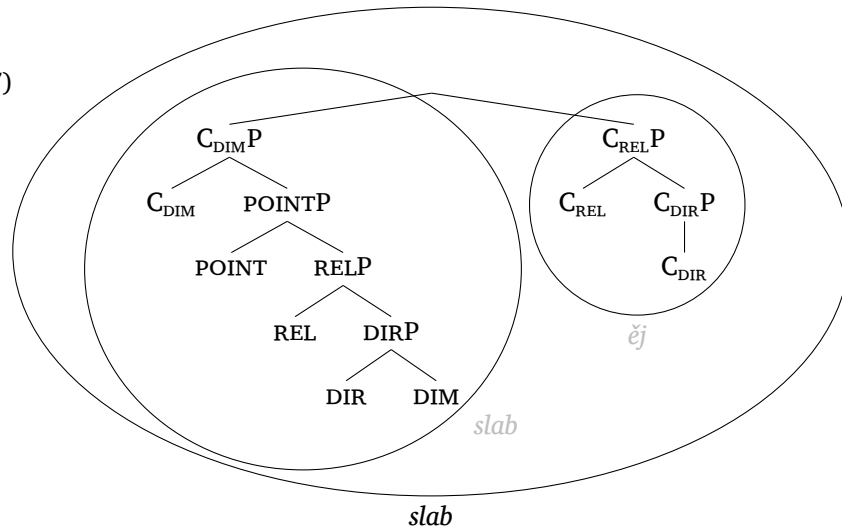


At the merger of C_{DIR} , the root (lexicalising $C_{\text{DIM}}P$) moves, by evacuate closest, and $C_{\text{DIR}}P$ is lexicalised as *-ěj*.



The same process is repeated at the cycle where C_{REL} is merged, resulting in (57).

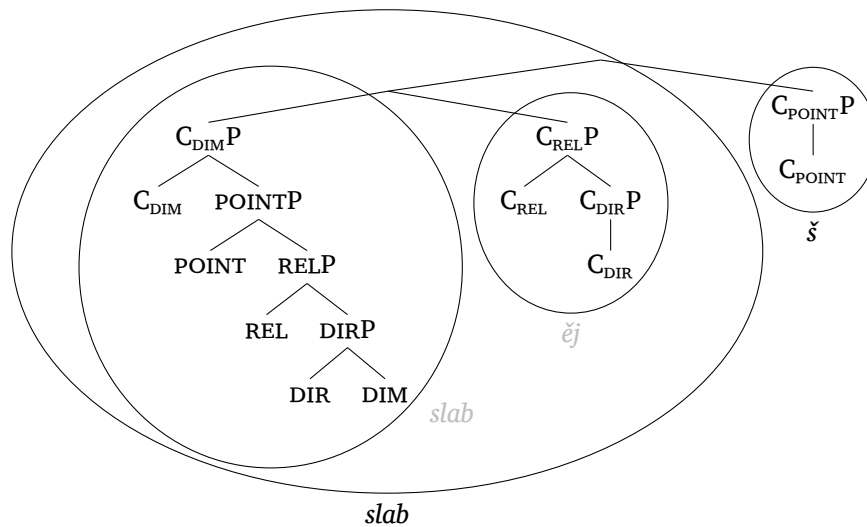
(57)



Now while the process that leads to the tree in (57) is the same as in Class I, the output is different in Class II. What we see here is that instead of lexicalising $C_{\text{DIM}}P$ and $C_{\text{REL}}P$ separately (as was the case in Class I), the root can lexicalise the whole constituent, which is why it has no *-ěj* in the comparative adjective: lexicalisation at a higher node (whenever possible) takes precedence over lexicalising lower nodes.

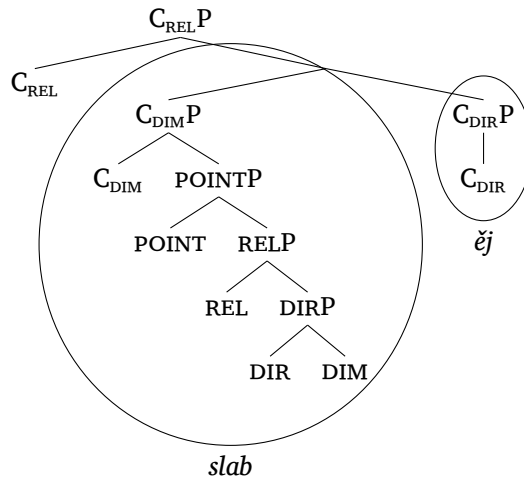
The comparative adjective itself needs one more feature, C_{POINT} , which lexicalises as *-š* in the usual manner: that is, first $C_{\text{DIM}}P$ is evacuated by evacuate closest, but since there is no match, a pied-piping movement produces the tree in (58).

(58)



The comparative adverb *slaběji* ‘more weakly’ will again require lexicalisation of ADV1 by *-i*, forcing backtracking to the cycle of C_{REL} . When the derivation is forced to backtrack to this size, *-ěj* resurfaces, see (59). The reason for this is the constituent structure of the lexical entry in (53): there simply is no identity between the unlabelled node in the structure (59) and any subconstituent in the entry (53).

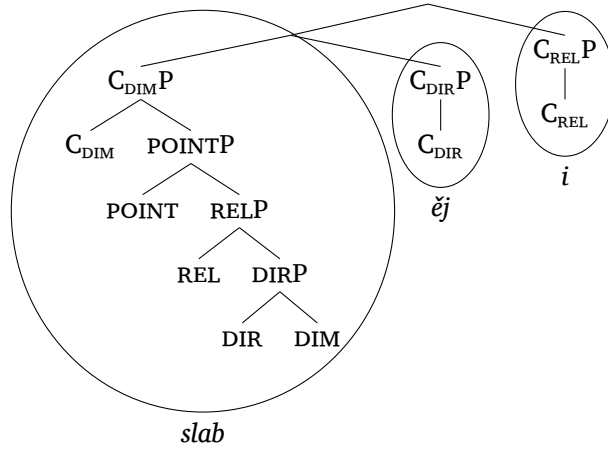
(59)



In the first pass through the cycle in (59), we evacuated the root (creating the structure that matched the entire lexical entry, which led to the overriding of the suffix by the root). But now, we need to pied-pipe and evacuate the

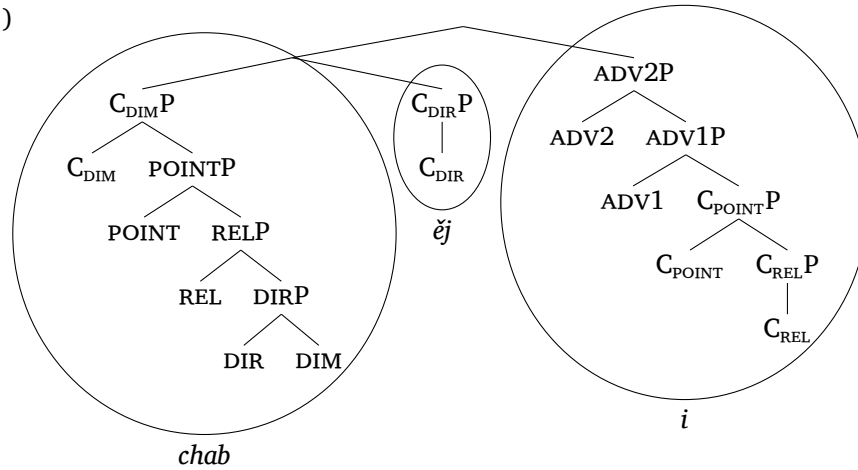
complement of C_{REL} . This will allow the remnant $C_{REL}P$ to be lexicalised as *-i*.

(60)



The rest of the derivation proceeds in the usual fashion, through pied-piping movements that allow the suffix to reach its full size:

(61)



The lexicalisation table summarising the derivations is repeated below.

(62) Class II augmentless adjective (*slab* 'weak')

	DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ			slab							
CMPR ADJ			slab				ěj	š		
CMPR ADV			slab				ěj		i	

As mentioned, the most interesting property of this class is that it follows exactly the same steps as the derivation in Class I. The only difference between the two classes is that the stage given in (57) is stored as a single lexical entry in Class II, while it needs to be lexicalised by two entries in Class I, compare (45). This suggests a strategy where children may not necessarily construct lexical items for each subclass independently of the other classes; rather, they may, at least in some cases, assume the same derivation across different (sub-)classes, and just store different stages of one and the same derivation.

This observation will carry over to other classes: what we shall find is that for a given augment class, the various subclasses generally follow one and the same sequence of steps, but lexicalise different stages of it, with the result that the augments and the innermost comparative marker *-ěj* are absent in various forms. While small deviations from this general pattern will emerge, the overall picture strongly suggests a shared derivation within different augment classes.

If correct, this may shed light on the broader issue of how children arrive at particular lexical entries, some of them with a relatively complex syntactic structure. The results suggested by this research indicate that the child may simply follow one and the same derivation across multiple classes, and associate the sound to one of the stages, as required by the empirical evidence.

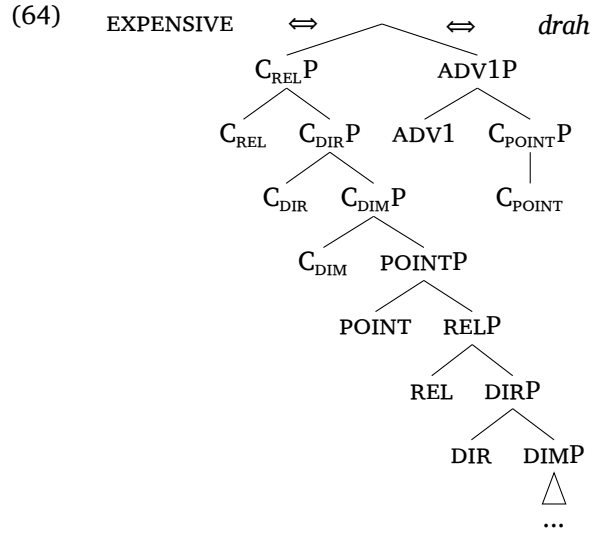
3.3.3 Class III

The last subclass of augmentless adjectives has no *-ěj* in the comparative adjective or adverb, e.g. *drahý—draž-Ø-ší—dráž-Ø-e*. As a consequence of having no *-ěj* in the adverb, it also shows the *-e* allomorph of the adverbial suffix (recall that these two properties go hand in hand). As shown in the lexicalisation table below, we analyse the allomorphy between the adverbial *-i/-e* as a difference in what features they lexicalise. When roots are unable to lexicalise ADV1, *-i* must be used, starting at C_{REL} and going through ADV1 to ADV2. But when a root is able to lexicalise ADV1, *-ě* is used as the realisation of ADV2. Recall that this proposal makes sense of the fact why *-ě* always attaches to roots, as no dedicated affix has its foot at ADV1 in our proposal.

(63) Class III augmentless adjective (*drah* ‘expensive’)

	DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ			drah							
CMPR ADJ			drah				ěj	š		
CMPR ADV			dráž				ěj	š		e

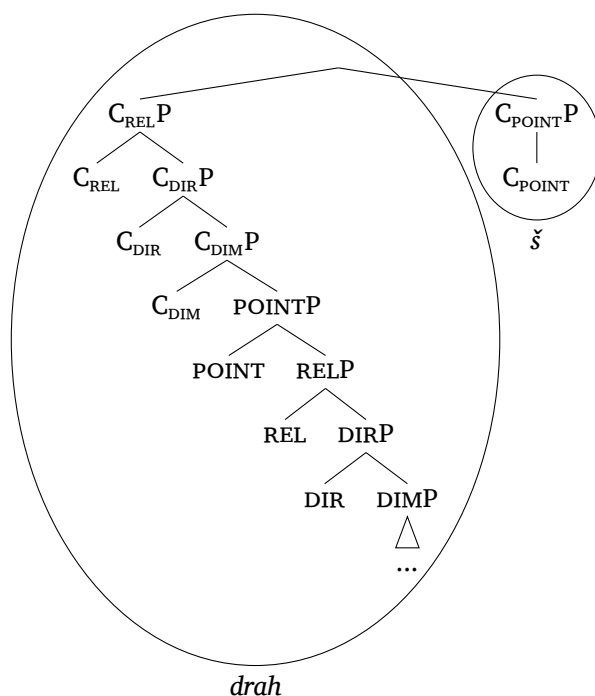
There are at least two ways in which such a lexicalisation table can arise. One way is to assume that the root has a CLB of the size $C_{REL}P$, and ADV1 is placed on the right branch along with C_{POINT} .



This takes care of the following two properties: (1) that we never see *-ěj* with this root, and (2) that *-š* appears in the comparative adjective, despite the root being bigger in size than $C_{POINT}P$.

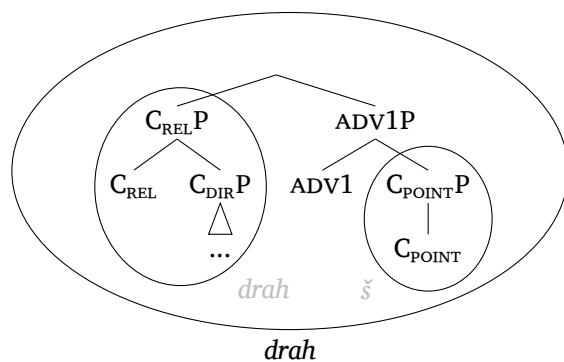
If this was the root entry, the derivation of the positive involves a direct (movementless) lexicalisation of $POINTP$ by the root, as it is a subconstituent of (the left branch of) (64). The comparative continues with direct lexicalisation up to the cycle of C_{REL} , after which an evacuation movement will apply:

(65)



When ADV1 is merged, we evacuate the root, and thus create an exact match for (64), which overrides the earlier lexicalisation as *drah* + *š*.

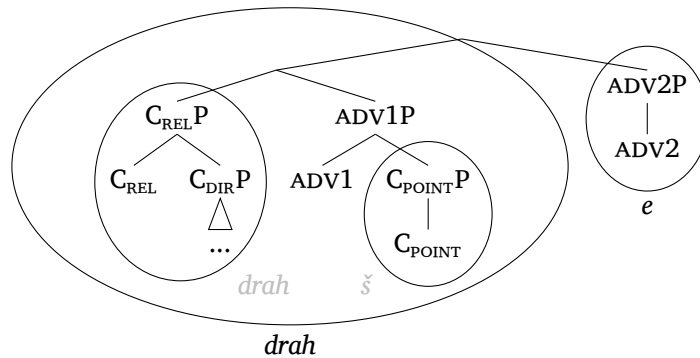
(66)



In the final cycle of the derivation, ADV2 is merged, and, after pied-piping, eventually lexicalised as *-e*.¹³

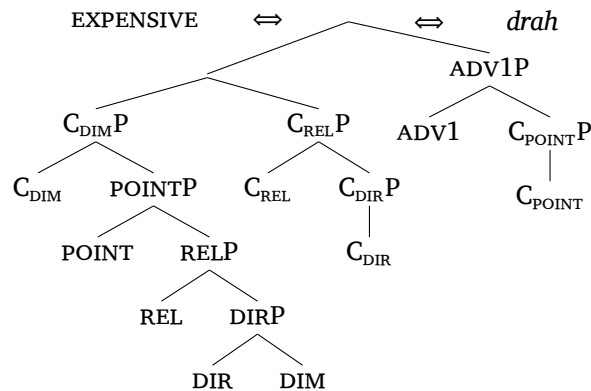
¹³ Recall from 3.2.2 that the adverbial *-e* triggers the lengthening of the root and a palatalisation, so the resulting form is *dráž-e* ‘more expensively.’

(67)



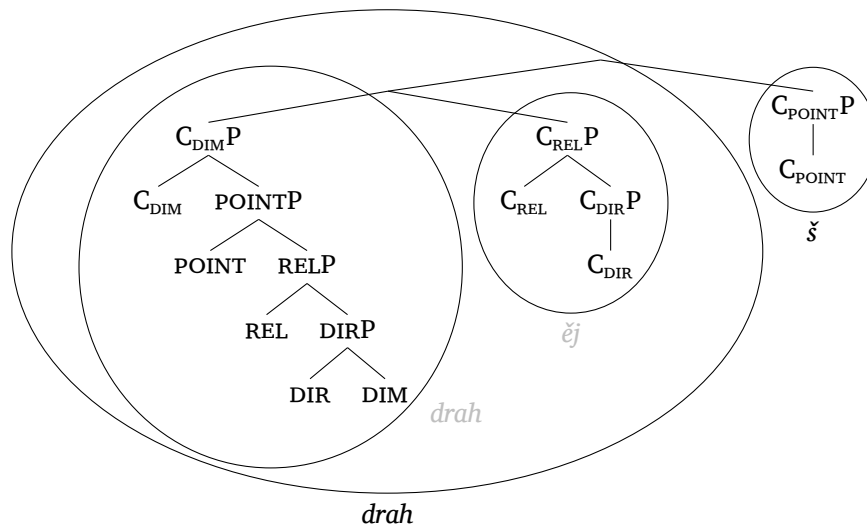
The second possible way to structure the lexical entry of the root would be to assume that it follows the same derivational track as all adjectives without augments. On this approach, a constituent of the size $C_{DIM}P$ would move out of $C_{REL}P$, which is what the entry for *slab* in (53) looks like. However, since such roots would also lexicalise ADV1, there would be an additional right branch to lexicalise ADV1 above C_{POINT} (analogous to the right branch in the first version of the entry in (64)). The entry is given in (68).

(68)



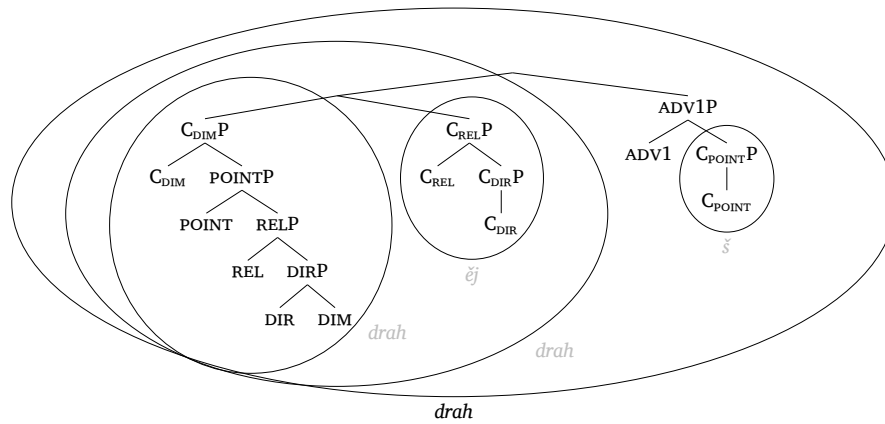
In this case, the derivation would proceed exactly as with Class II *slab* until the comparative adjective, see (69) (and compare (58)).

(69)



When ADV1 is merged to such a structure, the first pied-piping step gives the structure in (70), which is successfully lexicalised by the the root as given in (68).

(70)



The step depicted in (70) is present in all derivations of augmentless adjectives (i.e., in all Classes I-III). In Classes I and II, there is no match, and the derivation therefore has to continue in a different direction. However, in Class III, we get a match, and the derivations therefore diverge. Classes I and II have to backtrack, but the derivation in (70) can merge ADV2 to (70). This ultimately leads to the correct result *dráž-e* ‘more expensively’ after the whole complement of ADV2 is evacuated out of ADV2P due to pied-piping, analogous to (67).

3.3.4 Summary of augmentless classes

The following table summarises our results across the three classes.

(71) Lexicalisation in Classes I-III

	DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ	chab									
CMPR ADJ	chab					ěj	š			
CMPR ADV	chab					ěj			i	
POS ADJ	slab									
CMPR ADJ	slab					ěj	š			
CMPR ADV	slab					ěj			i	
POS ADJ	drah									
CMPR ADJ	drah					ěj	š			
CMPR ADV	dráž					ěj	š			e

The main thing we have achieved is to capture the distribution of *-ěj*: recall from (18) that this was difficult using linear context sensitive rules. The main idea behind our analysis is that the lexical entry of the Class II root has a complex left branch, and so even though it can lexicalise all the features of *-ěj* in the comparative adjective, it cannot do so in the comparative adverb. We further argued that this type of lexical entry allows for Classes I and II to have exactly the same derivation, and differ only in which particular step of that derivation is linked to the phonology of the root. We also suggested a way how this view may be extended to Class III roots, which may also be analysed as lexicalising a step in a shared derivation. However, since that step fails to lexicalise in the other classes, Class III then follows its own course.

3.4 Adjectives with *-n* (Classes IV-VII)

In this section, we turn to adjectives with the augment *-n*. The roots that take *-n* fall into four distinct subclasses. Our goal in this section is to show that three of these subclasses can be captured by assuming the exact same steps in the derivation for all three, with surface differences due to the fact that the roots are linked to different stages of that derivation. The fourth subclass shares all the steps for the comparative adjective, and diverges only for the adverb (just like Class III).

We begin with a preview of this analysis, starting with Class IV. This is the most regular class in that it has the augment throughout, i.e. both in the

positive and comparative forms.¹⁴ Classes V and VI and VII lack the augment in one form or the other. We provide an overview of the facts in (72). As argued in detail in Caha et al. (2024b), this distribution is impossible to capture with contextual rules, unless one assumes widespread homophony or disjunctive rules (recall the discussion surrounding (28)).

(72) Recapitulating the facts: the augment *-n*

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
IV	POS ADJ	jem	n	—	ý	‘fine’
	CMPR ADJ	jem	n	ěj	ší	
	CMPR ADV	jem	n	ěj	i	
V	POS ADJ	žádouc	Ø	—	í	‘desirable’
	CMPR ADJ	žádouc	n	ěj	ší	
	CMPR ADV	žádouc	n	ěj	i	
VI	POS ADJ	pozď	n	—	í	‘late’
	CMPR ADJ	pozď	Ø	ěj	ší	
	CMPR ADV	pozď	Ø	ěj	i	
VII	POS ADJ	snad	n	—	ý	‘easy’
	CMPR ADJ	snaz	Ø	Ø	ší	
	CMPR ADV	snáz	Ø	Ø	e	

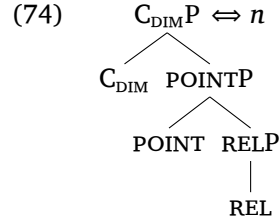
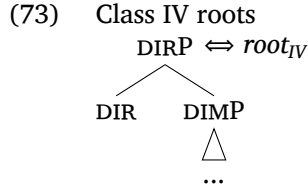
In this section, we first provide a bird’s eye perspective on our overall analysis and then we discuss these classes one by one. In Table 3.6, we provide a lexicalisation table that provides the gist of our solution.

¹⁴ In what follows, we shall use the term ‘regular’ to denote such classes where the augment is found throughout the paradigm. We do not imply anything about the relative frequency of such roots compared with the ones that lose the augment. They are regular in the sense that their derivation is usually trivial, and do not require lexical entries with complex left branches.

Table 3.6 Lexicalisation table for the *-n* roots

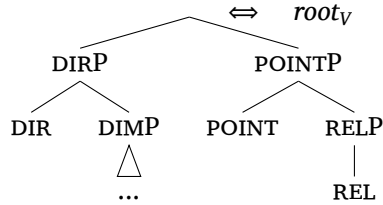
		DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2	GLOSS
IV	POS ADJ	jem		n								‘fine’
	CMPR ADJ	jem		n				ěj		š		
	CMPR ADV	jem		n				ěj			i	
V	POS ADJ	žádouc		n								‘desirable’
	CMPR ADJ	žádouc		n				ěj		š		
	CMPR ADV	žádouc		n				ěj			i	
VI	POS ADJ	pozd		n								‘late’
	CMPR ADJ	pozd		n				ěj		š		
	CMPR ADV	pozd		n				ěj			i	
VII	POS ADJ	snad		n								‘easy’
	CMPR ADJ	snaz		n				ěj		š		
	CMPR ADV	snaz		n				ěj		š	e	

In the top section of the table containing the analysis of Class IV, we assume that Class IV roots are linked to a DIRP-sized constituent, see (73). The lexical entry for the augment changes slightly from what we have assumed in Chapt. 2, in that we take it to lexicalise not only REL and POINT, but also one feature of the comparative sequence, C_{DIM}, as shown in (74). This is needed to capture Class V roots, where *-n* is absent in the positive, but present in the comparative.



The other subclasses of *-n* roots all have a lexical entry with a CLB, which are constructed in a way to mirror agglutinative derivations, where the DIRP-sized root cyclically moves past the individual projections lexicalised by *-n*, and then pied-pipes *-n* across other functional heads. For instance, the root for Class V lexicalizes the stage of the derivation we see in the positive, where the DIRP-sized root moves past the two projections REL and POINT, see (75). This is the same structure as in Class IV, the difference being that the projections on the right branch would be lexicalised by *-n* in Class IV, but are lexicalised along with the root in Class V.

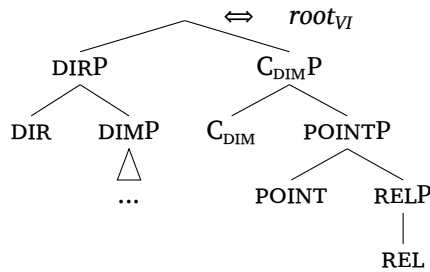
(75) Class V roots



The entry (75) also explains the reappearance of the augment in the comparative, where the DIRP constituent moves past the lowest comparative feature C_{DIM} . At that point, *-n* will be needed again to lexicalise this feature (along with the other features on the right branch).

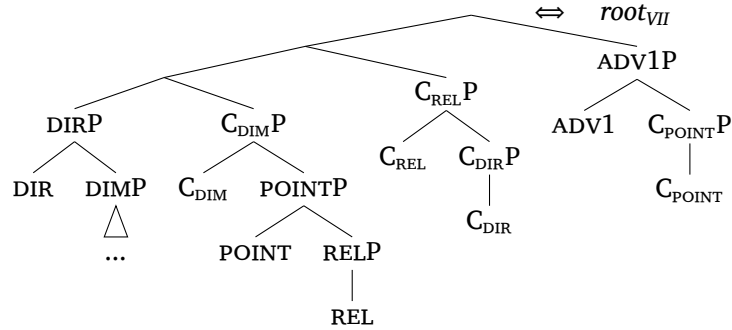
Similarly, Class VI roots literally lexicalise a stage of the derivation where a constituent of the size DIRP (regular Class IV root) moves to the spec of $C_{\text{DIM}}P$. The Class VI root thus literally links its phonology that corresponds to the structure where a Class IV root like *jem-* ‘fine’ sits in the spec of the *-n*-augment:

(76) Class VI roots



The Class VII roots in (77) add the *-ěj* suffix on top of this ($C_{\text{REL}} + C_{\text{DIR}}$), reflecting the fact that *-ěj* is found in the comparative of (76) but not in that of (77). Roots like (76) and (77) will have literally the same structure in the comparative adjective, but lexicalised differently (*root* + *ěj* in Class V, plain *root* in Class VI).

(77) Class VII roots

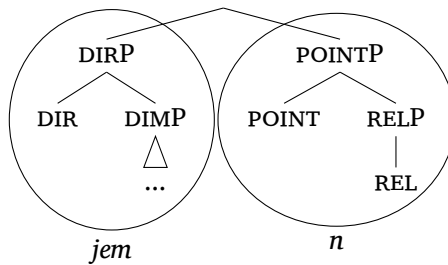


We shall now demonstrate all these derivations in detail.

3.4.1 Class IV

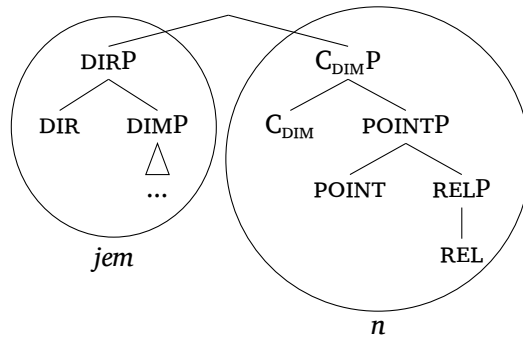
In (78), we show the structure of the positive *jem-n-y* ‘fine’ (repeated from (50) in Chapt. 2).

(78)



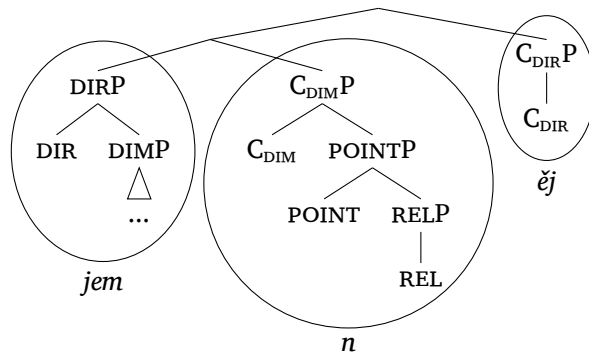
The first step in the derivation of the comparative is the merger of C_{DIM} , followed by the raising of the closest labelled node containing the bottom of the fseq, $DIRP$, and the lexicalisation of the remnant $C_{DIM}P$ by the augment *n*:

(79)



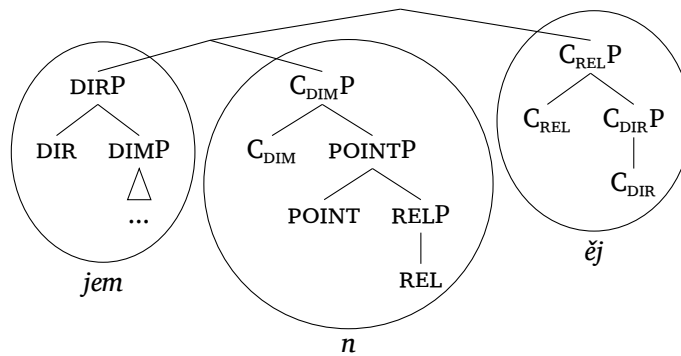
Next C_{DIR} is merged, and again $DIRP$ is evacuated, but this will not be successful, as the augment n cannot grow beyond $C_{DIM}P$ size. We therefore pied-pipe, creating a remnant constituent $C_{DIR}P$, which is lexicalised by *-ěj*.

(80)



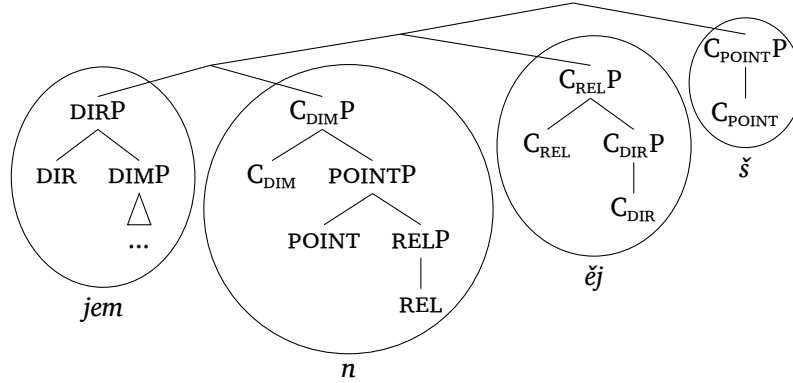
C_{REL} is merged next, and after an initial unsuccessful evacuation movement of the root, pied piping moves the node immediately dominating $DIRP$, resulting in (81), where *-ěj* lexicalises $C_{REL}P$.

(81)



In the final step of the comparative adjective, C_{POINT} is merged, which can be lexicalised by $-\check{s}$ after pied-piping the sister node of C_{POINT} , deriving the correct form *jem-n-ěj-š-í* ‘finer’:

(82)



For the derivation of the comparative adverb *jem-n-ěj-i* ‘more finely’, we will need a backtracking derivation, which is essentially identical to the one we discussed for the Class I root *chab-ěj-i* ‘more weakly’. As we saw there, backtracking is triggered by the need to lexicalise the ADV1 feature, which only the morpheme $-i$ can do. Since $-i$ has a foot (its lowest feature) in C_{REL} , the derivation needs to go back to that cycle and take the next option in the Lexicalisation Algorithm there. Essentially, what needs to happen is that the tree we constructed in (80) must undergo complement movement from below C_{REL} all the way above ADV2. We shall not go through every step of this process, as we have done this previously (recall the discussion in Sect. 3.3.1), and merely show the final step of the derivation.

(83)

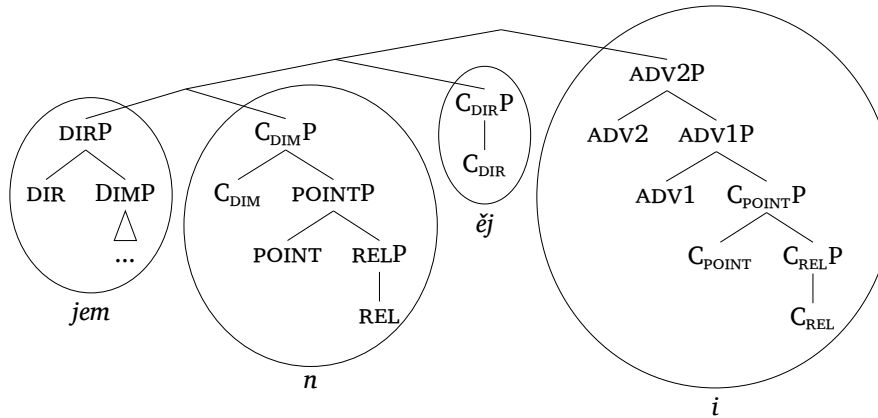


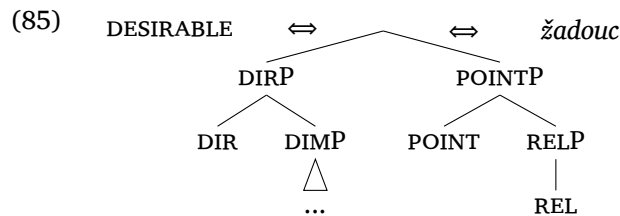
Table (84) recapitulates the lexicalisation table we have just derived for Class IV roots.

(84) Lexicalisation table for Class IV (*jem-n-ý* ‘fine’)

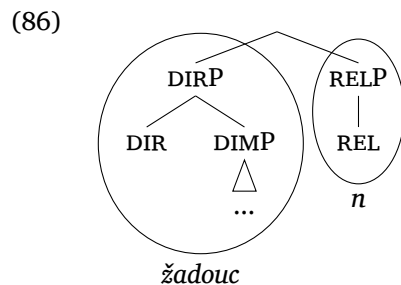
	DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ	jem		n							
CMPR ADJ	jem		n			ěj	š			
CMPR ADV	jem		n			ěj		i		

3.4.2 Class V

The Class V roots show a rather surprising aspect of the distribution of the augment *-n*, which is that it does not occur in the positive but shows up in the comparative. We derive this fact by assuming a lexical entry with a CLB, as in (85):

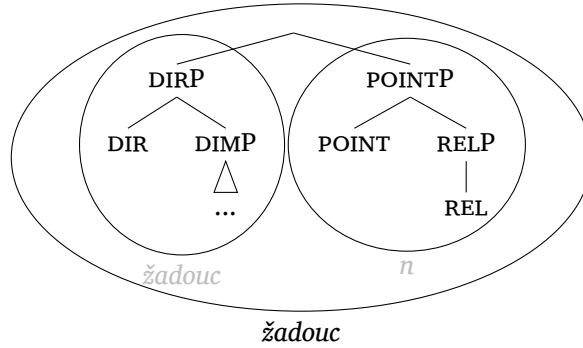


In this case, we need to devote some special attention to the derivation of the positive, since we did not discuss this class in Chapt. 2 above, as its special nature can only be understood in the context of the comparative. As with all the *-n* roots, the derivational path of the positive is that of the regular Class IV *-n* roots. The derivation merges DIRP in the usual manner, which is lexicalised by the root, since DIRP is a subconstituent of (85). When REL is merged, evacuation movement will apply, resulting in (86), where RELP is lexicalised by the augment. Since (86) as a whole is not a subconstituent of (85), the root cannot lexicalise the whole structure, and the augment appears.



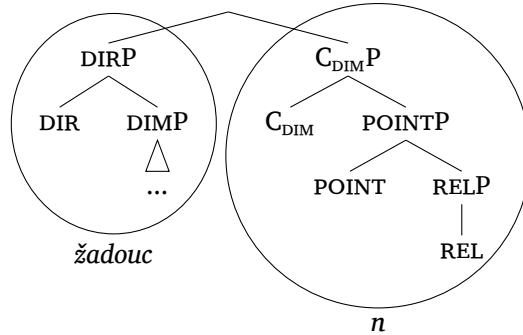
Things change, however, when POINT is merged. Evacuation of the root results in (87), where POINTP is lexicalised by the augment, but where, in addition, the whole structure is an exact match for lexical entry of the root as given in (85), resulting in the overriding of *žadouc* + *n* by the augmentless root *žadouc* in the positive.

(87)



In subsequent steps of the derivation, leading to the comparative forms, this configuration will be destroyed and the augment will reappear. We already see this happening when the C_{DIM} head is merged, which will trigger evacuation movement of the root.

(88)

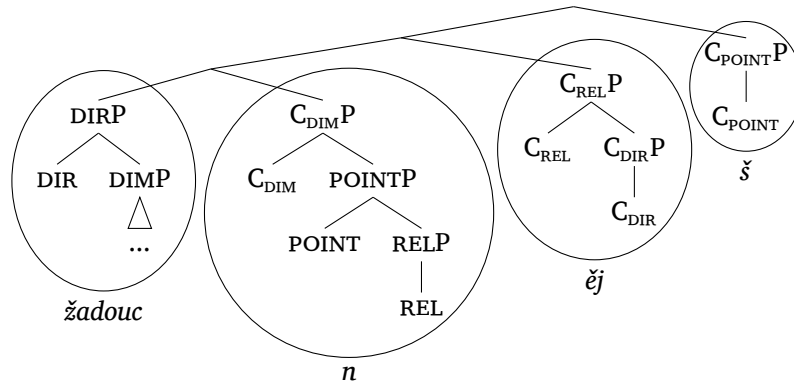


While both the left and right branches of this tree find a lexicalisation in the root and the augment, respectively, the entire structure is no longer a match for (85). The effect of this is that the augment will reappear.

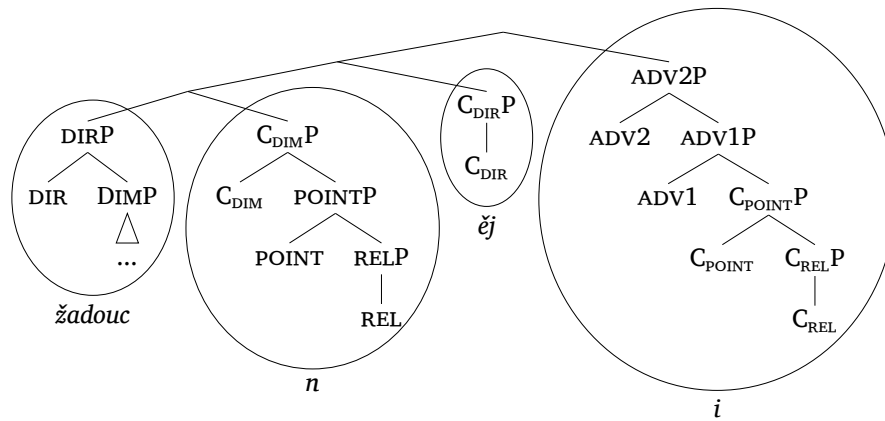
The subsequent steps in the derivation of both the comparative adjective and adverb unfold exactly as for the regular *-n* roots, and so we shall not go through them in detail here, but refer the reader to Sect. 3.4.1 above, where the Class IV roots are discussed.

The final structure of the comparative adjective is shown in (89), and that of the comparative adverb (derived after a backtracking derivation) in (90).

(89)



(90)



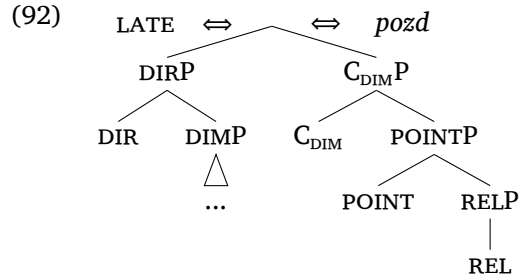
We conclude this section by showing the lexicalisation table that we have just successfully derived.

(91) Lexicalisation table for Class V (*žádouc-í* ‘desirable’)

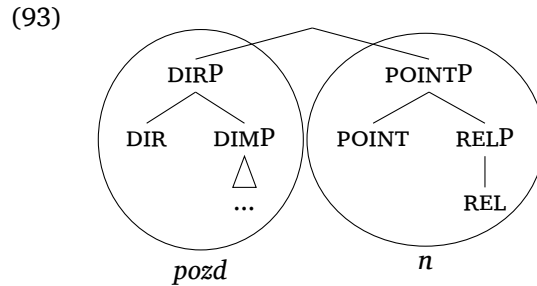
		DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ	žádouc				n						
CMPR ADJ	žádouc				n			ěj	š		
CMPR ADV	žádouc				n		ěj			i	

3.4.3 Class VI

Class VI roots lose the augment in the comparative forms, and consequently require an entry with a CLB. We therefore start out by recalling the lexical entry of the roots in this class.



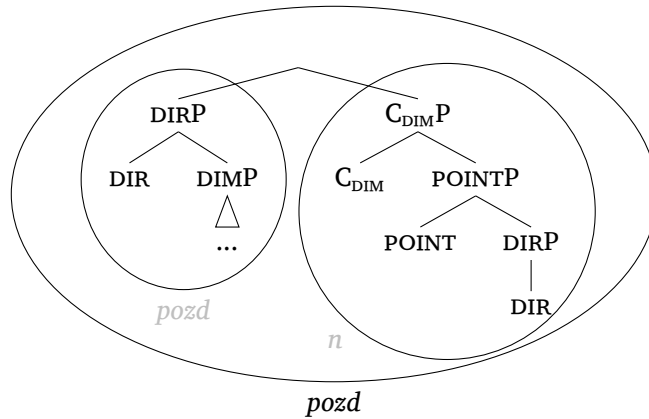
We start the derivation again from the shape of the positive, shown in (93).



It is important to see that, while DIRP is a subpart of the lexical entry of *pozd*- in (92), the root node of (93) is not, and so the root, albeit of size $C_{\text{DIM}}P$, cannot lexicalise the positive, and the augment is needed.

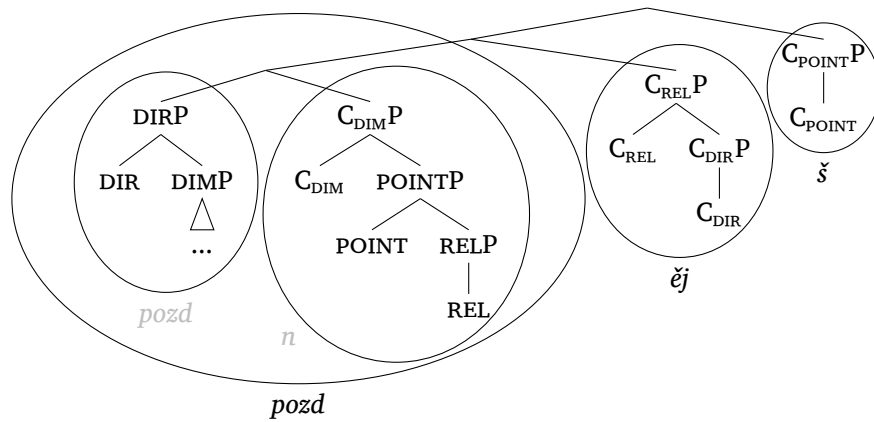
The formation of the comparative starts off with the merger of $C_{\text{DIM}}P$. Since $C_{\text{DIM}}P$ cannot be lexicalised directly, evacuation movement applies, resulting in (94).

(94)



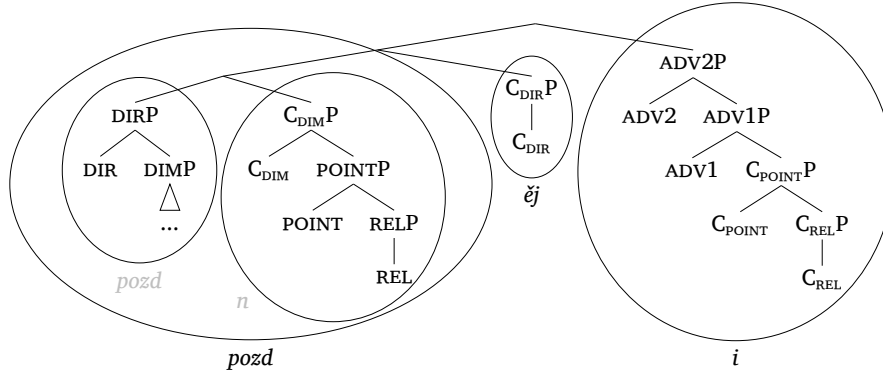
Now at this point $C_{DIM}P$ is lexicalisable by the augment, but in addition, the root node of (94) is also an exact match for the lexical entry (92), so that the root overrides the combination *pozď* + *n*, triggering the disappearance of the augment at the $C_{DIM}P$ level. The augment will not reappear at later stages either, since the merger of C_{DIR} and its lexicalisation by *-ěj* will trigger pied piping movement of the node dominating $C_{DIM}P$, leaving the structure lexicalised by the root intact. The derivation proceeds exactly as with the regular Class IV *-n* roots, with a structural representation of the comparative adjective that is exactly identical to that of the regular root *jem-n-ěj-š-í* 'finer', except that the lexical entry for the root is an exact match for the stage in the derivation where regular roots have *root* + *n*, i.e. the $C_{DIM}P$ stage, leading to the correct form *pozď-ěj-š-í* 'later' in (95). (This is the same structure as for Class IV, compare (82)).

(95)



The same logic holds (*mutatis mutandis*) for the derivation of the comparative adverb, see (96). This is again the same structure as in Class IV (recall (96)), just with the root lexicalising DIRP + *n*.

(96)



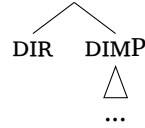
As a summary, we repeat the lexicalisation table just derived.

(97) Lexicalisation table for Class VI (*pozď-n-í* ‘late’)

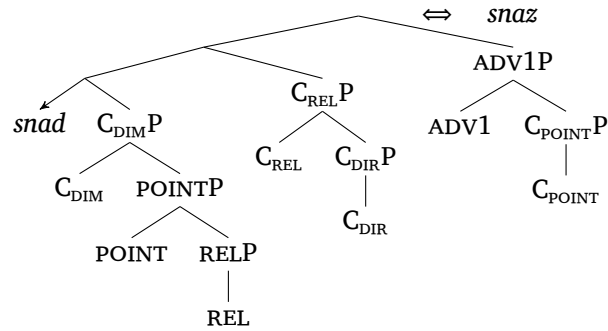
	DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ	pozď	n								
CMPR ADJ	pozď		n			ěj		š		
CMPR ADV	pozď		n			ěj			i	

3.4.4 Class VII

Class VII roots, like the Class VI roots, have the *-n* augment in the positive, but lose it in the comparative. They differ from the Class VI roots in that they have no *-ěj* in the comparative adjective or adverb (where the latter entails that they have an adverb in *-e*). These properties mean that they are bigger in size than the Class VI roots, but also that they require a lexical entry with a CLB. In (77) we have given the lexical entry of a Class VII root in a generic form. In the derivation of this class in this section, we shall be using the concrete root *snad-*. For this root, we argued in Sect. 3.2.2 that it has a suppletive root *snaz-* in the comparative. This means that we need to assign two different lexical entries to these roots, with one containing a pointer to the other (see the discussion about how to capture suppletion in Section 1.6.4). The lexical entry for the *snad-* root is given in (98), and the one for the *snaz-* root in (99).

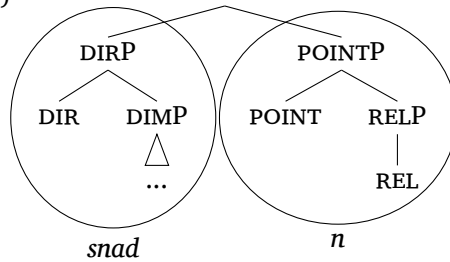
(98) EASY $\Leftrightarrow$ DIRP $\Leftrightarrow$ *snad*

(99)



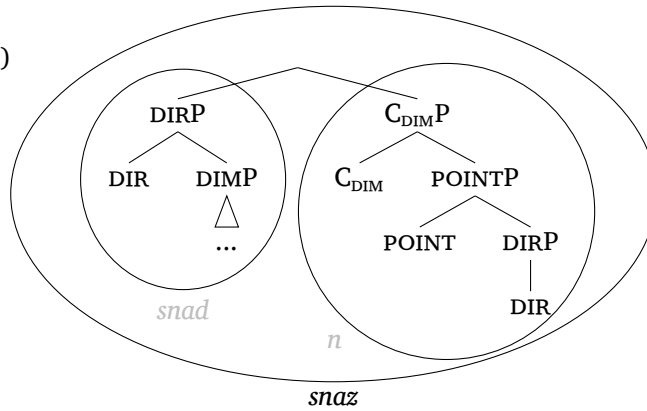
The derivation of the positive proceeds as with the regular *-n* roots. DIRP will find a match in (98), while the entire structure does not match any constituent in (99), so that the augment is needed to lexicalise the rest of the features of the positive.

(100)



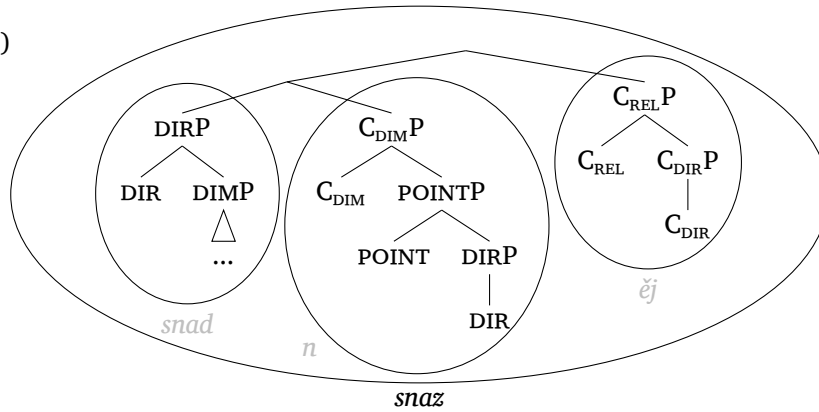
The comparative at first follows the same derivational path as that of the Class VI roots (modulo the fact that in this case there is a suppletive root). This means that at the C_{DIM} cycle, the root is evacuated, and the resulting constituent is lexicalised by the suppletive root *snaz-*, since it is an exact match for a subconstituent of (99) above.

(101)



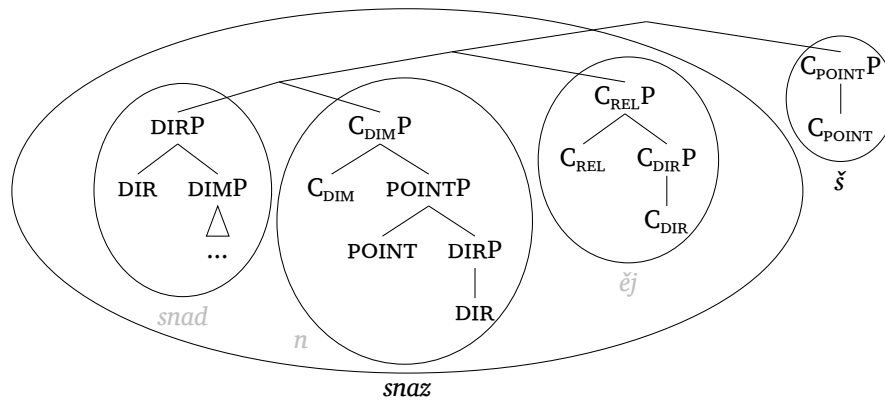
This has the effect that the augment disappears in the comparative, but in addition, there is no *-ěj* marker either. This is because in the subsequent derivation, which unfolds exactly as with the regular *-n* adjectives, a match is created at the $C_{REL}P$ cycle with the lexical entry (99), causing the root to eat up the *-ěj* suffix, as shown in (102).

(102)



This match will be preserved in the next cycle as well, where C_{POINT} is merged, and its sister node is evacuated through pied-piping movement. The full structure of the comparative form *snaz-š-í* 'easier' (disregarding the agreement) is shown in (103).

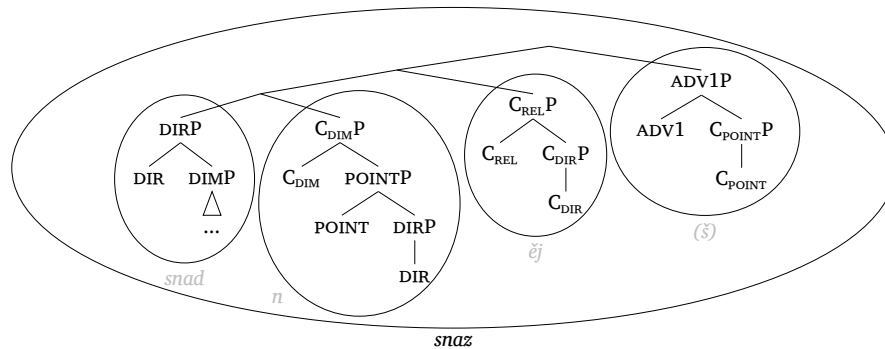
(103)



Note that this tree is not in its entirety a subtree of the lexical tree for the root *snaz*- (99). This will ensure that the suffix *-š* appears in the comparative adjective (though it disappears again in the adverb, as we shall see presently).

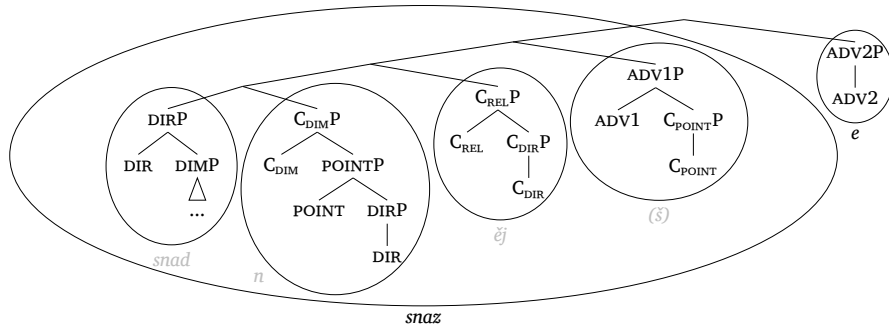
The derivation of the comparative adverb involves the merger of the ADV1 head. Evacuation movements will be unsuccessful at first, until the node dominating $C_{REL}P$ is evacuated. This will create a structure that is an exact match for the root *snaz*- (recall (99)), so that the suffix *-š* will be overridden by the root.

(104)



In the final step of the derivation of the adverb, ADV2 is merged, and after a series of unsuccessful evacuation movements, the maximal pied-piping evacuation movement will produce (105), where ADV2P is lexicalised by *-e*.

(105)



To conclude, we include a summary of the lexicalisations derived in this section.

(106) Lexicalisation table for Class VII (*snad-n-ý* ‘easy’)

		DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2
POS ADJ	snad		n								
CMPR ADJ	snaz			n			ěj		š		
CMPR ADV	snaz			n			ěj		š		e

3.4.5 Summary

By going through the derivations, this section showed how peculiar augment distributions, which seem impossible to capture using context-sensitive rules, fall out from a Nanosyntax account. The crucial feature of this approach is that there are no zeroes whose contextual restrictions must be stated. Rather, the distribution of *-n* and zeroes falls out from the idea that all *-n* adjectives share a single derivation, and the roots of the various subtypes are linked to one of its stages. Once this perspective is adopted, the apparently bizarre and difficult-to-state surface patterns are accounted for, reflecting an underlyingly simple and unifying system.

The lexical items that we use correspond to syntactic structures that differ along various parameters. Specifically, we argue that these entries differ not only in their size (how many features they have), but also in their internal organisation (what is the structure of those features). We call this ‘the shape.’ When we try to be more nuanced, what seems to differentiate these entries is whether they contain an (evacuation) movement or not; in case there is one, it matters how big the moving constituent is, and how high it moves. All of these choices lead to different surface patterns.

It is curious to note that all of these parameters along which entries vary are precisely the same parameters that were used in Cinque's (2005) approach to differentiate between various types of orders within the NP and beyond, as we mentioned briefly in Section 1.2.1. By doing so, we have formulated an account where we are using regular syntactic parameters (like movement height and type) to do the job that used to be done by purely linear statements. Instead of: realise the augment as *-n* next to this root if its a comparative adverb, but not when it's a comparative adjective, we have formulated a theory where such restrictions fall out from independently established syntactic parameters. This seems intellectually more appealing, theoretically less arbitrary and, most importantly, empirically accurate.

3.5 Adjectives with *-k* (Classes VIII-XII)

In this section, we set out to explore the five classes of adjectives with the augment *-k*. We repeat the list of the classes in (107) (originally (29)).

(107) Five classes of adjectives with the augment *k*

	CLASS		ROOT	AUG	ĚJ	ENDING	gloss
VIII	POS	ADJ	hez	k	—	ý	'pretty'
	CMPR	ADJ	hez	k	Ø	ší	
	CMPR	ADV	hez	k	ěj	i	
IX	POS	ADJ	slad	k	—	ý	'sweet'
	CMPR	ADJ	slad	Ø	Ø	ší	
	CMPR	ADV	slad	k	ěj	i	
X	POS	ADJ	úz	k	—	ý	'narrow'
	CMPR	ADJ	už	Ø	Ø	ší	
	CMPR	ADV	úž	Ø	ěj	i	
XI	POS	ADJ	blíz	k	—	ý	'close'
	CMPR	ADJ	bliž	Ø	Ø	ší	
	CMPR	ADV	blíž	Ø	Ø	e	
XII	POS	ADJ	brz	k	—	ý	'early'
	CMPR	ADJ	dřív	Ø	ěj	ší	
	CMPR	ADV	dřív	Ø	Ø	e	

The large number of classes arises due to the fact that both *-k* and *-ěj* exhibit complex distribution patterns that are reminiscent of the ones we discussed in the previous sections, with an intermediate class that behaves like one class in one respect and like another class in another respect. Such intermediate classes, as we have repeatedly pointed out, are difficult to model with

context-sensitive rules, but fall out from the way Nanosyntax derivations proceed.

Let us first focus on the distribution of *-k* itself, which we illustrate in Table (108).

(108) The distribution of *-k* drop and retention

CLASS	POS	ADJ	CMPR	ADJ	CMPR	ADV	GLOSS
VIII		hez- k -ý		hez- k -š-í		hez- k -ěj-i	‘pretty’
IX		slad- k -ý		slad- -š-í		slad- k -ěj-i	‘sweet’
X		úz- k -ý		už- -š-í		úz- -ej-i	‘narrow’

On the first row, we have roots that always include *-k* in their comparative forms, on the bottom row, we have a root that never has *-k* in the comparative. But there is also an intermediate class, which lacks *-k* in the comparative adjective, but includes it in the adverb.

The same type of pattern with an intermediate class arises for the distribution of *-ěj*, as shown in (109).

(109) The absence of *-ěj*

CLASS	POS	ADJ	CMPR	ADJ	CMPR	ADV	GLOSS
X		úz- k -ý		už- -š-í		úz- ej -i	‘narrow’
XI		blíz- k -ý		bliž- -š-í		blíž- -e	‘close’
XII		brz- k -ý		dřív- ěj -š-í		dřív- -e	‘early’

All of the adjectives in (109) belong to the type that loses *-k* in both the comparative adjective and adverb. This type is further divided into three subclasses, depending on where *-ěj* shows up: whether in the comparative adverb (top line of (109)), in the comparative adjective (bottom line of (109)), or not at all (middle line of (109)). Each of these patterns may be by now familiar, yet what makes *-k* adjectives unique is their stacking for subsequent positions of exponence. This empirical picture will require a solution where tree shape becomes relevant not only for roots, but also for the stem marker *-k*.

Before we set out to explore the derivations, we want to point out another issue, which is predictability. What we mean by this is that in some cases, knowing one form of a paradigm allows speakers to predict the remaining forms. In the case of the *-k* class adjectives, this is not the case, and each one form only has a rather small predictive power regarding other forms within that paradigm. This holds to some extent for all the classes, but it is most pronounced for the *-k* class in particular.

To show what we mean by this, we have depicted the complex patterning of *-k* adjectives visually in the diagram in Fig. 3.1. What we do here is take the positive as the starting point of the graph and see whether knowing the

positive helps us determining what the other forms are. It turns out that this form does not allow us to uniquely determine the remaining forms in the paradigm, as the trifurcation shows: some adjectives preserve *-k* in the comparative, while others drop it, in which case they can have two different comparatives (either *-š-í* or *-ěj-š-í*). Even knowing two out of the three forms does not always help: adjectives that have a *-š-í* comparative may still exhibit three different adverbial forms (depending on what the root is). Within this area, the predictability is thus very low: for some classes, to correctly acquire these forms, speakers must encounter all three of them.

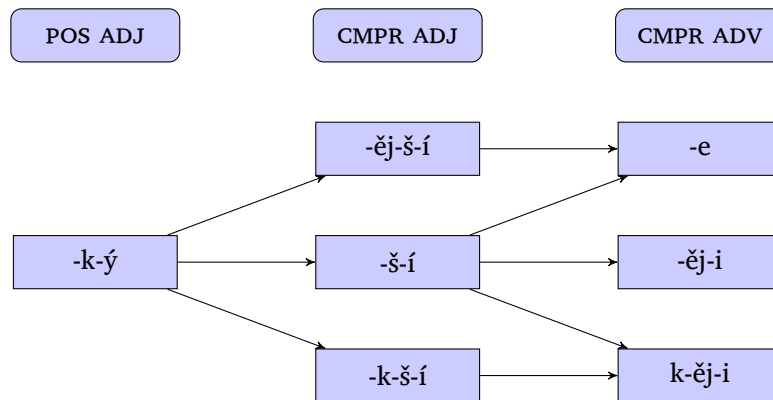


Fig. 3.1 Low predictability of forms for roots with the augment *-k*

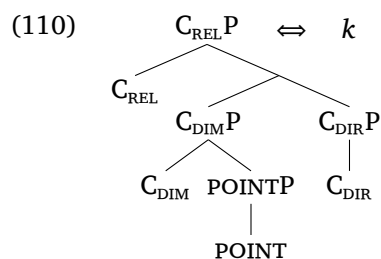
We shall now show how these patterns are derived in Nanosyntax, using again the idea that derivations within a particular augment class tend to proceed along the same lines, with variation restricted to what particular stage we associate the phonology of the root. To give a sense of the direction in which we are headed, we start by presenting an overall lexicalisation table in 3.7. Subsequent sections within this chapter are going to discuss these lexicalisations in detail and provide a step-by-step walk through each derivation.

Table 3.7 Lexicalisation table for the *-k* roots

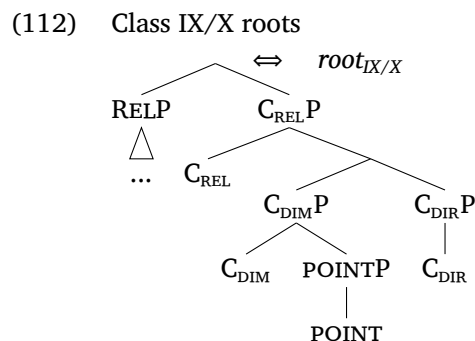
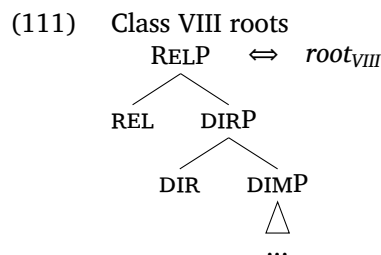
		DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2	GLOSS
VIII	POS ADJ	hez			k							‘pretty’
	CMPR ADJ	hez			k			ěj	š			
	CMPR ADV	hez			k		ěj			i		
IX	POS ADJ	slad			k							‘sweet’
	CMPR ADJ	slad			k			ěj	š			
	CMPR ADV	slad			k		ěj			i		
X	POS ADJ	úz			k							‘narrow’
	CMPR ADJ	úz			k			ěj	š			
	CMPR ADV	úž			k		ěj			i		
XI	POS ADJ	blíz			k							‘close’
	CMPR ADJ	blíz			k			ěj	š			
	CMPR ADV	blíz			k			ěj	š		e	
XII	POS ADJ	brz			k							‘early’
	CMPR ADJ	dřív			k			ěj	š			
	CMPR ADV	dřív			k			ěj	š		e	

Let us begin unpacking this table from the top section (Class VIII). In the comparative adjective, we see that *-k* is directly followed by *-š*, suggesting that *-k* lexicalises all the heads from POINT to C_{REL}. In the adverb of this class, the suffix *-i* has a foot in C_{REL}, which triggers backtracking, as in the other classes we have discussed above. If *-k* had an uncomplicated lexical entry, we could expect that it will simply shrink down to C_{DIR}. The surprising thing is that when the base shrinks to this size, *-k* alone is not enough; instead *-ěj* appears in the adverb as the lexicalisation of C_{DIR}.

We have seen similar behaviour pertaining to roots of Class II, which had a comparative in *-Ø-š-í* and an adverb in *-ěj-i* (e.g. *slab-š-í* ‘weaker’, *slab-ěj-i* ‘more weakly’). We accounted for this by attributing to the root a lexical entry with a CLB. The CLB contained features up to C_{DIM}, and the right branch had the features C_{REL} and C_{DIR}: [C_{REL} [C_{DIR}]], recall (53). In the adverb, the derivation would destroy the configuration that allowed the lexicalisation of the left branch along with C_{DIR}, causing the reappearance of *-ěj* in the adverb. We recall this account of Class II because we shall likewise assume a lexical entry with a CLB for the augment *-k*, which also contains features up to C_{DIM}, and with the ‘adverbial’ C_{DIR} on the right branch, as in (110). The effect of assuming this structure inside the lexical entry of *-k* is precisely to model the effect that when the derivation backtracks below the adverbial *-i*, *-ěj* will have to resurface. By proposing this entry, we are introducing here for the first time a suffix with a complex left branch (so far, we only used roots with CLBs).

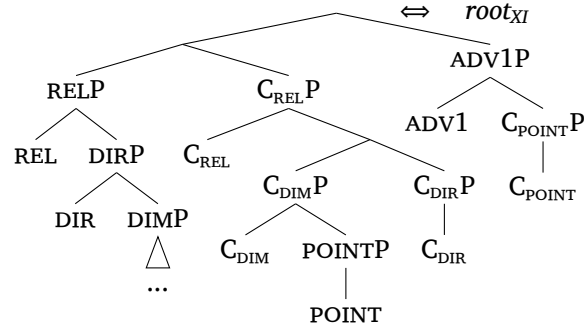


We provide the lexical entries for the $-k$ roots below. The Class VIII roots have a simple lexical entry that corresponds to size RELP (111), as explained in Chapt. 2.



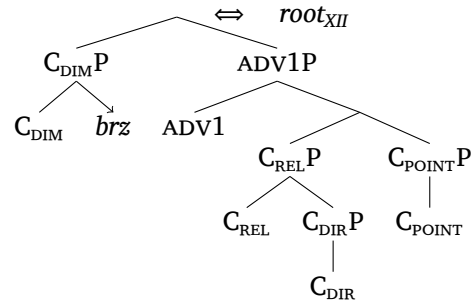
Roots of Class XI correspond to a stage in the derivation when a constituent corresponding to Class IX root moves across ADV1 during evacuation movements, see (113).

(113) Class XI roots



Finally, Class XII (which contains a single suppletive member) will have a RELP sized root for the positive (*brz-*), and the suppletive comparative root *dřív-* ‘earlier’ will have an entry as in (114). As we will see, the lexical item pointed to will need to be adjusted slightly, but we keep that discussion for Section 3.5.5.

(114) Class XII roots



We now provide a detailed discussion of the how the derivations proceed.

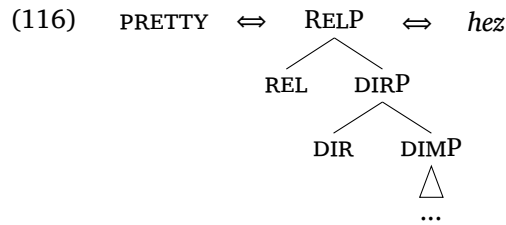
3.5.1 Class VIII

Class VIII roots feature the augment *-k* throughout. As we noted above, the analytical challenge they pose is the resurrection of *-ěj* in the comparative adverb. We start out by giving some representative examples in Table (115).

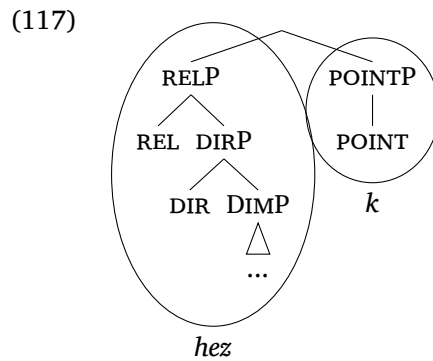
(115) A list of Class VIII adjectives¹⁵

POS	CMPR	ADJ	CMPR	ADV	GLOSS
hez- k -ý	hez- č -š-í	hez- č -ej-i			'pretty'
heb- k -ý	heb- č -š-í	heb- č -ej-i			'smooth'
sliz- k -ý	sliz- č -š-í	sliz- č -ej-i			'slimy'
křeh- k -ý	křeh- č -š-í	křeh- č -ej-i			'fragile'
kluz- k -ý	kluz- č -š-í	kluz- č -ej-i			'slippery'
břit- k -ý	břit- č -š-í	břit- č -ej-i			'sharp'
hoř- k -ý	hoř- č -š-í	hoř- č -ej-i			'bitter'
trp- k -ý	trp- č -š-í	trp- č -ej-i			'bitter'
syp- k -ý	syp- č -š-í	syp- č -ej-i			'loose'
vlh- k -ý	vlh- č -š-í	vlh- č -ej-i			'wet'
leh- k -ý	leh- č -š-í	leh- č -ej-i			'easy'
měk- k -ý	měk- č -š-í	měk- č -ej-i			'soft'
ten- k -ý	ten- č -š-í	ten- č -ej-i			'thin'

The lexical entry for the sample root we shall be using in the derivation is given in (116)

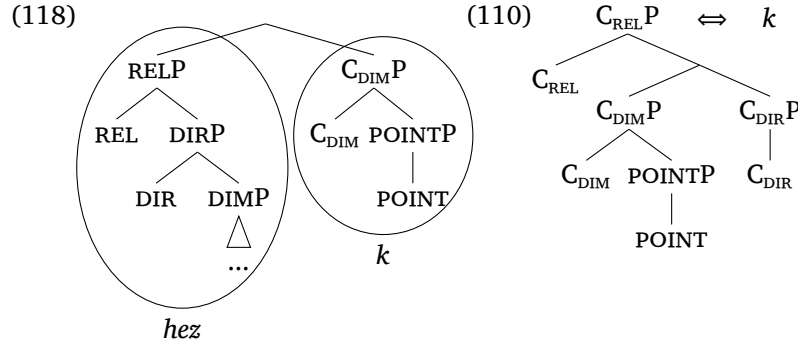


We start the derivation at the point where we left it as (49) in Chapt. 2, i.e. the structure of the positive adjective, repeated here as (117).

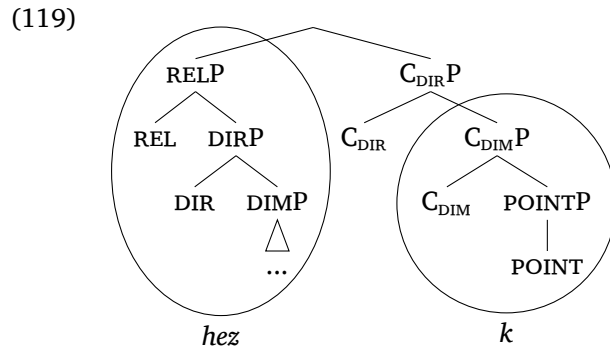


¹⁵ We note that *č-š* in the comparative undergoes degemination, and the orthography reflects this, so all the comparative adjectives are written with just *č* (instead of *čš*).

When C_{DIM} is merged, first the root is evacuated, creating (118), which contains a constituent $C_{DIM}P$ on the right branch that is a subconstituent of (110) (repeated for convenience below), so that we get *hez-k*:

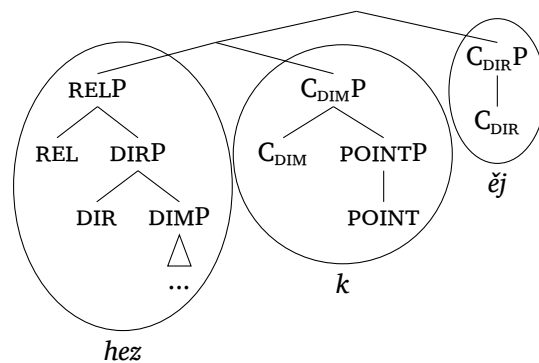


At the next step, C_{DIR} is merged, and evacuation movement of the root will create (119):



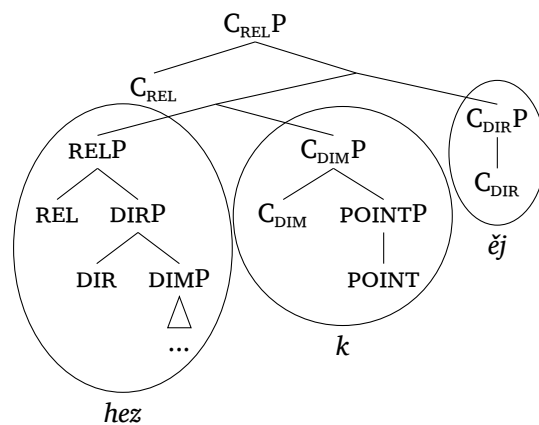
Now although the lexical entry for *-k* does contain the feature C_{DIR} , *-k* cannot lexicalise the constituent $C_{DIR}P$ of (119), since its constituency does not match that of (119) (recall (110) above). We therefore need to undo the movement and pied-pipe, which results in (120), where the comparative marker *-ěj* lexicalises $C_{DIR}P$:

(120)



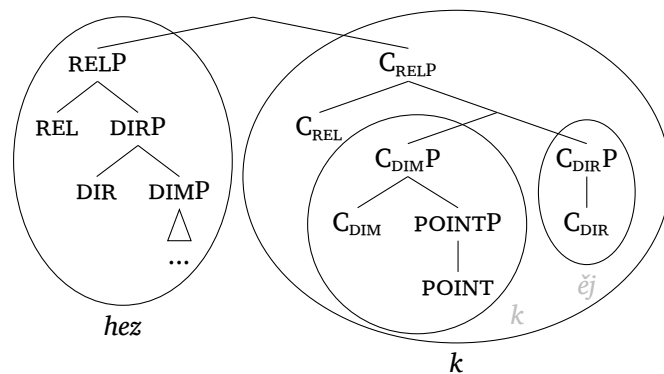
In the next step, we merge C_{REL} :

(121)



Since there is no direct lexicalisation, we apply evacuation movement of the root. In doing so, we create a constituent on the right branch that is an exact match for the lexical entry of *-k*, and *-k* will override the earlier lexicalisation of *-k* and *-ěj*.

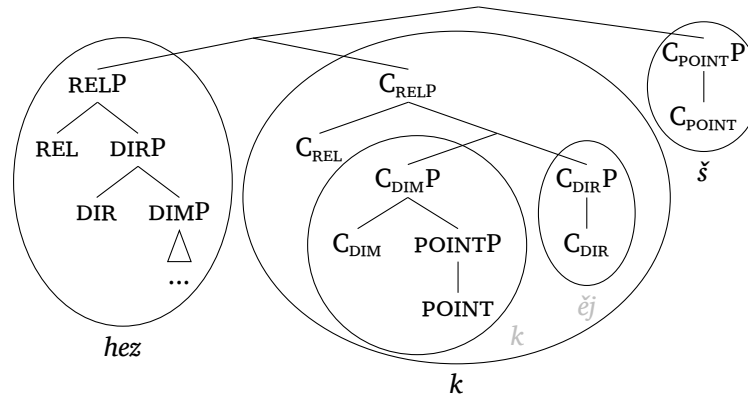
(122)



In this manner, the augment lexicalises a contiguous sequence of four features, leading to the disappearance of *-ěj* in the comparative adjective. The same result could have been achieved with a simple lexical entry for *-k* (which would not contain a CLB). However, the more complex nature of *-k* will prove to be essential in the derivation of the comparative adverb, where *-ěj* reappears. This will be the consequence of a backtracking derivation that goes back to (121), and applies pied-piping movement instead of the evacuation of the root.

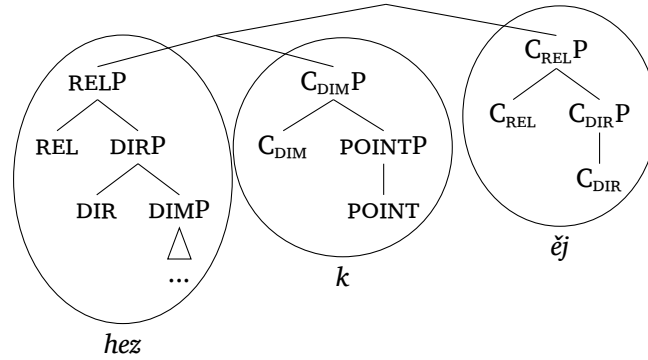
Before moving on to the adverb, there is one more step in the derivation of the adjective, which involves the merger of C_{POINT} , and a series of unsuccessful evacuation movements that we skip here, until the sister of C_{POINT} is moved to its left. At that point, $C_{\text{POINT}}P$ is lexicalised by *-š*.

(123)



The derivation of the adverb *hez-k-ěj-i* ‘more prettily’ involves the merger of ADV_1 , which will trigger backtracking, because no morpheme has a foot in ADV_1 , and the root itself does not contain ADV_1 either. Since the suffix *-i* of the adverb has a foot in C_{REL} , this involves backtracking to the cycle where C_{REL} was merged, i.e. (121) above. In our first pass, we evacuated $DIRP$ lexicalised by the root, which means that we now have to pied pipe the node above it. This creates the tree in (124).

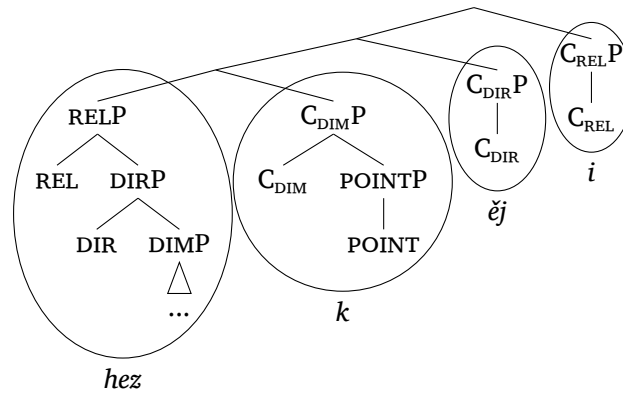
(124)



What we see is that this movement destroys the configuration that was an exact match for *-k*, and led to *-k* overriding *-ěj* at the level of $C_{REL}P$. As a result, *-ěj* reappears, and the feature C_{REL} is now lexicalised by *-ěj*.

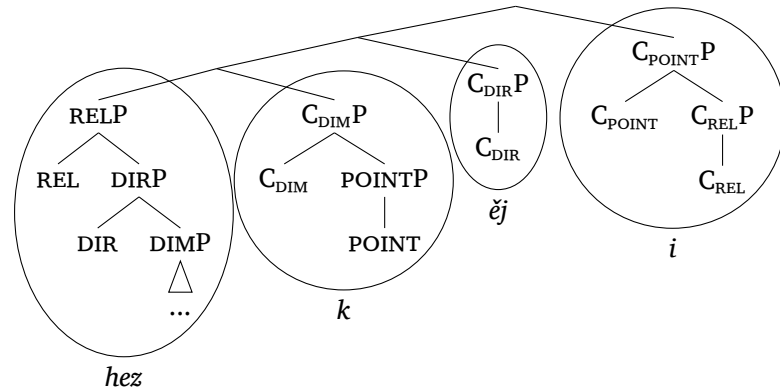
The next cycle sees the merger of C_{POINT} , which will be lexicalised by *-š*. However, when ADV1 is next merged, we run into the same problem as before, namely that the morpheme *-i* is the only one that can lexicalise ADV1, but it has a foot in C_{REL} , but there is no such constituent in (124), nor in the subsequent stages of the derivation based on it. This means we have to backtrack once again to the cycle of C_{REL} (recall (121)), and apply the next option in the Lexicalisation Algorithm, which pied-pipes the sister of C_{REL} . This now allows the lexicalisation of $C_{REL}P$ by *-i*.

(125)



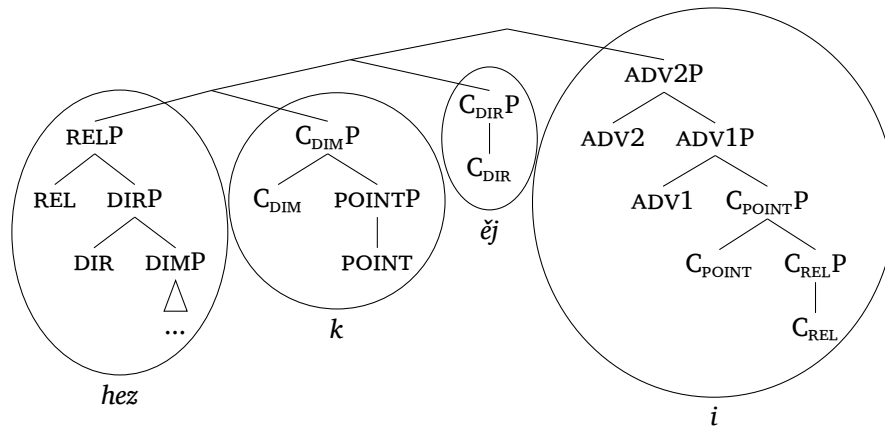
In the next stage, C_{POINT} is merged, and evacuation movements will apply until C_{POINT} is successfully lexicalised by *-i*.

(126)



The subsequent stages in the derivation are the same as for the other comparative adverbs ending in *-i*. After each merge step, evacuation movements apply until *-i* can lexicalise the new head, eventually leading to (127):

(127)



This concludes the derivation of the Class VIII adjectives. Even though they are regular in having *-k* throughout, they required a special lexical entry for the augment, so as to be able to account for the fact that *-ěj* is not present in the comparative adjective (suggesting *-k* lexicalises the features of *-ěj*), but appears in the comparative adverb following *-k* (suggesting that *-k* does not lexicalise the features of *-ěj* in this case). As we have seen, the complex lexical entry that we have assigned to *-k* derives exactly this constellation of facts.

3.5.2 Class IX

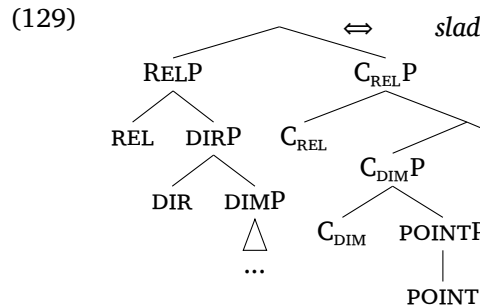
The class IX adjectives have *-k* in the positive, lose it in the comparative adjective, and get it back in the comparative adverb. Similar to the regular Class VIII adjectives, there is no *-ěj* in the comparative adjective, but it regrows in the comparative adverb, resulting in a pattern *slad-k-ý—slad-∅-∅-ší—slad-č-ěj-i* ‘sweet’.

In (128) we list some representative examples of this class.¹⁶

(128) A list of Class IX adjectives

POS	CMPR	ADJ	CMPR	ADV	GLOSS
slad- k-ý	slad-	-š-í	slad- č-ěj-i		‘sweet’
hlad- k-ý	hlad-	-š-í	hlad- č-ěj-i		‘smooth’
prud- k-ý	prud-	-š-í	prud- č-ěj-i		‘steep’
krot- k-ý	krot-	-š-í	krot- č-ěj-i		‘tame’
vrat- k-ý	vrat-	-š-í	vrat- č-ěj-i		‘unstable’
říd- k-ý	říd-	-š-í	říd- č-ěj-i		‘thin’
krát- k-ý	krat-	-š-í	krat- č-ěj-i		‘short’

The lexical entry for the adjectival root *slad-* is shown in (129).

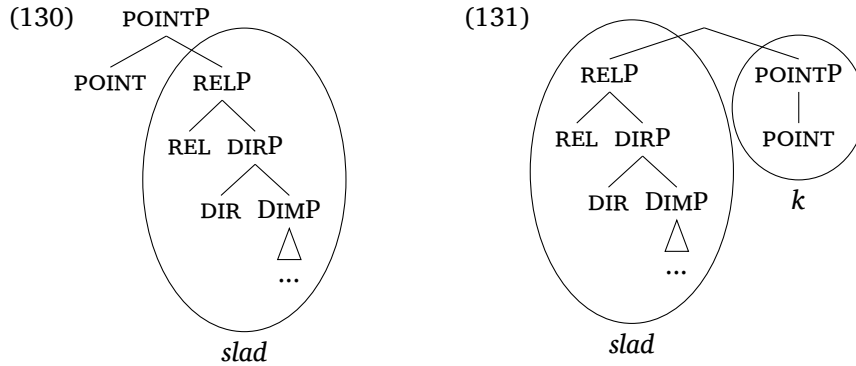


This is a complex structure, with two CLBs inside it. However, both the left and the right branch of the top node correspond to objects that we have encountered before. The left branch (REL P) is the size of the regular Class VIII roots of the previous section (this constituent is a part of the lexical entry of all the *-k* roots, as we shall see). The right branch of the tree (C_REL P) corresponds to the lexical entry of *-k* (recall (110) above). The entire structure therefore represents the stage in the derivation that is depicted in (122) for the regular Class VIII adjectives, i.e. the stage where we get *root + k* (just before adding C_POINT, to be lexicalised as *-š*). At precisely that point, the earlier lexicalisation *slad-k* is overridden by the root *slad-*, leading to the disappear-

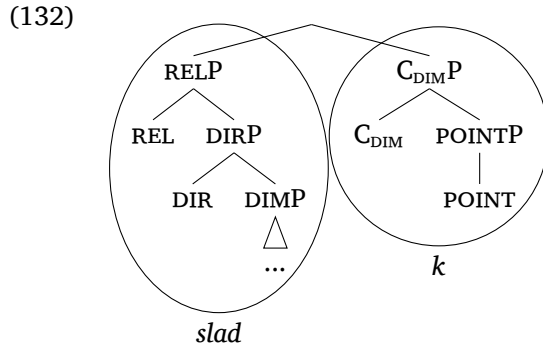
¹⁶ There seems to be a clear commonality here in that the roots end in *t/d*. Note, however, that *břit-k-ý* ‘sharp’ also ends in *t*, but goes into class VIII.

ance of the augment. The addition of *-š* will leave this structure intact, but the backtracking needed for adverbial *-i* will destroy it, leading to the reappearance of both *-k* and *-ěj*, as we shall see below (following exactly the same steps as we did for Class VIII).

We start the derivation with the positive, i.e. at the cycle where POINT is merged (130). Since there is no constituent matching (130) contained in (129), evacuation movement is triggered (131). In spite of the fact that the complex lexical entry in (129) contains POINT, it is not in a configuration that matches that of (131), so that the root cannot lexicalise the entire structure. Instead, the right branch of (131) finds a match in the entry for *-k* (110), so that POINTP lexicalises as *-k*, i.e. we get the augment in the positive, as required.

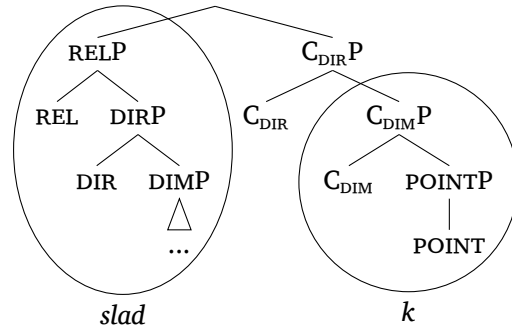


The derivation now proceeds exactly as in the Class VIII adjectives, the difference being that at each step we try to match the structure against the complex lexical entry of the root *slad-* in (129). This is not going to be successful until the cycle of C_{REL} ; we represent the intermediate stages that lead there in succession below.



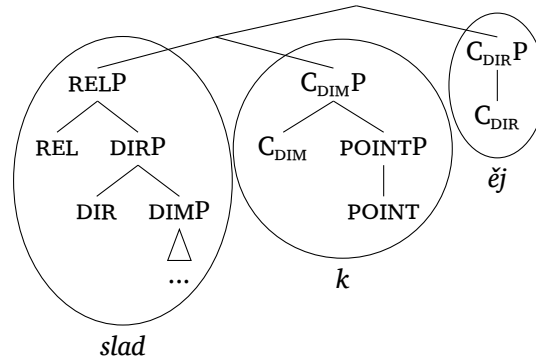
C_{DIR} is merged, and evacuation movement of the root creates (133):

(133)



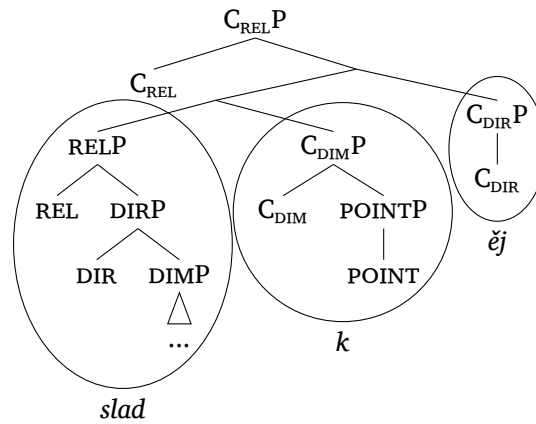
Since *-k* cannot lexicalise the constituent $C_{\text{DIR}}P$ of (133) (recall (110)), we undo the movement of the root and pied-pipe, which results in (134), where the comparative marker *-ěj* lexicalises $C_{\text{DIR}}P$:

(134)



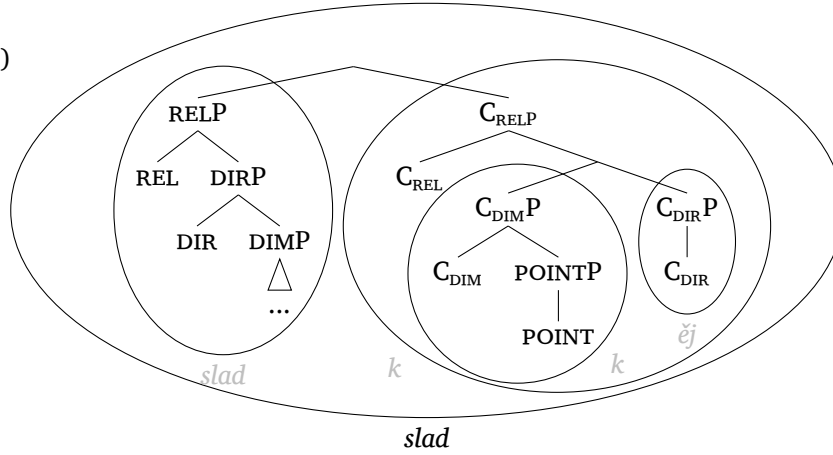
In the next step, we merge C_{REL} :

(135)



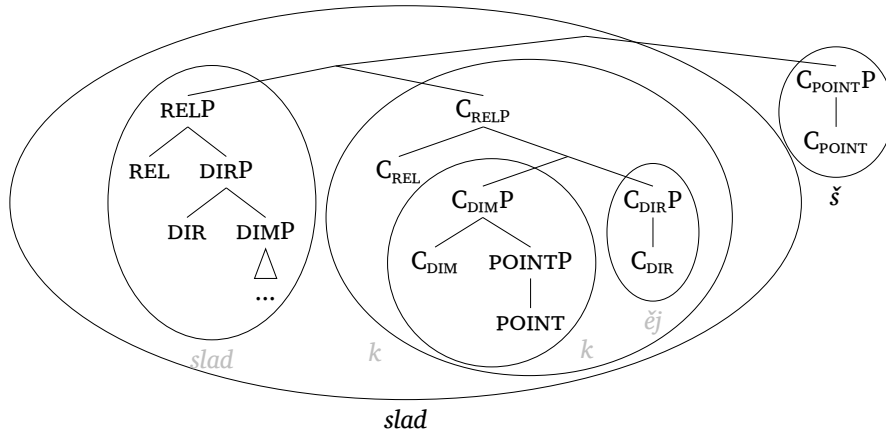
Applying evacuation movement of the root creates a constituent on the right branch that is an exact match for the lexical entry of *-k*, and *-k* will override the earlier lexicalisation of *-k* and *-ěj*. In addition, the entire structure is now an exact match for the root (recall (129)), so that it will in turn override *slad + k*

(136)



At the merger of C_{POINT} , a series of unsuccessful evacuation movements eventually leads to the evacuation of the sister of C_{POINT} , and its lexicalisation by *-š* (137). Observe that the moved constituent sees its internal structure preserved. As a result, its lexicalisation by the root is unaffected, which leads to the augmentless comparative form *sladši* ‘sweeter’.

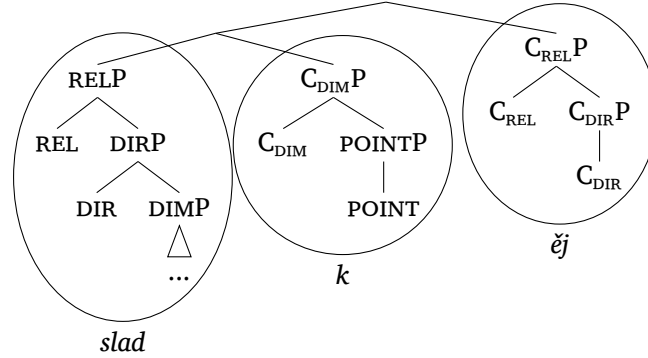
(137)



As with the regular Class VIII adjectives, the derivation of the adverb *slad-k-ěj-i* ‘more sweetly’ involves a backtracking derivation, which is triggered by the need to lexicalise ADV1. Backtracking goes back to the cycle of C_{REL} ,

i.e. (135) above. In our first pass, we evacuated the root, which means that we now have pied pipe the node above it (138).

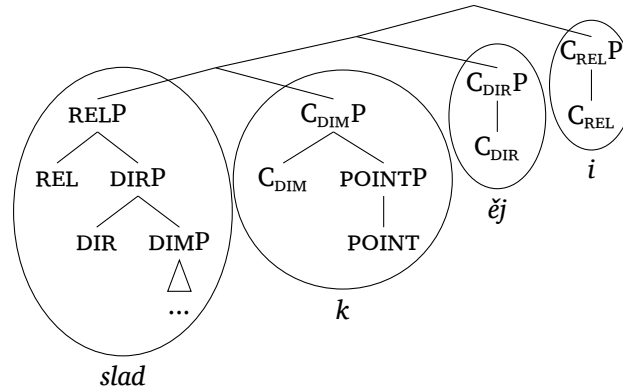
(138)



This movement sees both -k and -ěj reappear. Because the tree in (138) no longer matches the lexical entry for the root (129), earlier lexicalisations are reverted to. This is, first, the lexicalisation of RELP by the root on the leftmost branch. The remainder of the tree can also no longer in its entirety be a match for the lexical entry of -k, since the earlier configuration where the features C_{DIR} and C_{REL} were lexicalised by -k (given in (110)) is no longer present in (138). These features are now lexicalised by -ěj, and C_{DIMP} by -k. This step in the derivation is the crucial one triggering the rebirth of both -k and -ěj in the adverb.

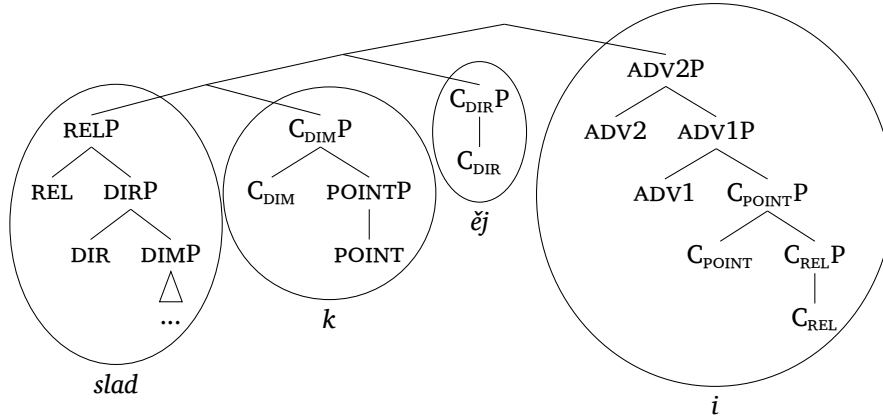
In the next cycle we merge C_{POINT} , which will initially be lexicalised by -š, but this will again result in a failure to lexicalise ADV1, so that backtracking is triggered once again. This reverts to the cycle of C_{REL} (shown in (121)). We apply the final pied-piping step and move the sister of C_{REL} , allowing the lexicalisation of $C_{REL}P$ by -i.

(139)



The subsequent stages in the derivation are the same as for the Class VIII comparative adverbs ending in *-i*. After each merge step, evacuation movements apply until *-i* can lexicalise the new head, eventually leading to (140):

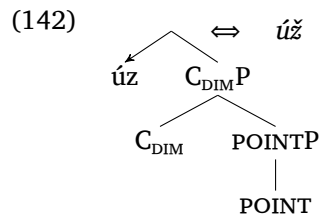
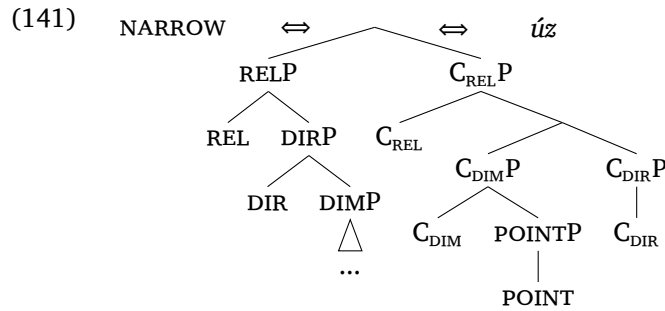
(140)



We have reached the end of the derivation of the Class IX adjectives. Their complex lexical entry represents a stage in the derivation of the comparative adjective, which is why we see neither the augment *-k* nor *-ěj* there, although we do see the augment in the positive, and both the augment and *-ěj* in the comparative adverb.

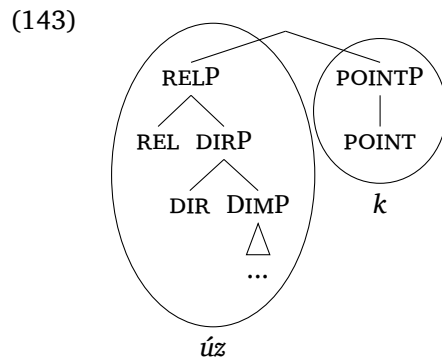
3.5.3 Class X

In Class X, there is just a single adjective: *úzký* ‘narrow.’ This adjective is like the Class IX adjectives except that it has no *-k* in the comparative adverb, i.e. we get the triplet *úz-k-ý—už-Ø-Ø-š-í—úž-Ø-ej-i*. As we observed in Sect. 3.2.2 above, the suffix *-ěj* does not normally trigger palatalisation of *z*, e.g. *drz-ý* ‘rude’ has a comparative adverb *drz-ej-i*, not **drž-ej-i*. Thus, taking the irregular shape of the root in *úž-ej-i* to be non-phonological, we assume that this root is suppletive, and that there are consequently two different lexical entries, with one pointing to the other. Their lexical entries are shown in (141) and (142).



The lexical entry in (141) is the same as that for the class IX roots of the previous section. It will ensure (i) that the positive has *-k*, and (ii) that the comparative adjective lacks both *-k* and *-ěj*. The lexical entry in (142) makes sure that the projections lexicalised by *-k* in the comparative adverb are lexicalised as a part of the suppletive root.

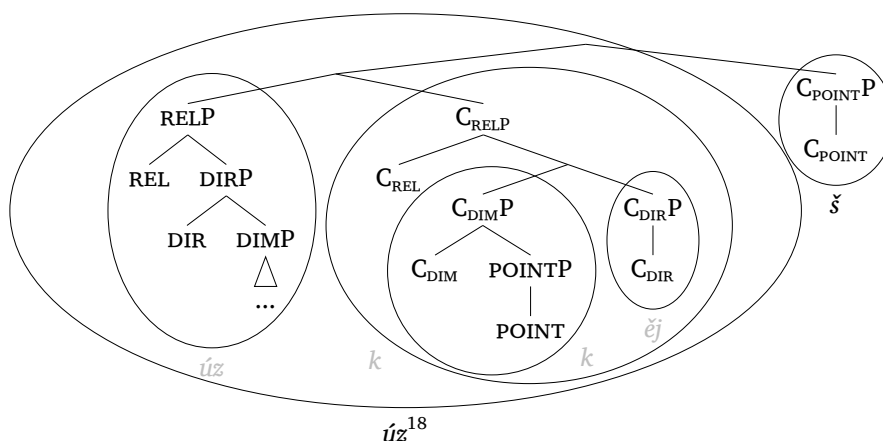
The steps in the derivation of the adjective are exactly the same as in Class IX, so we shall not repeat them, but directly show the final result. For the positive, this is given in (143).



The final stage in the derivation of the comparative adjective is given in (144):¹⁷

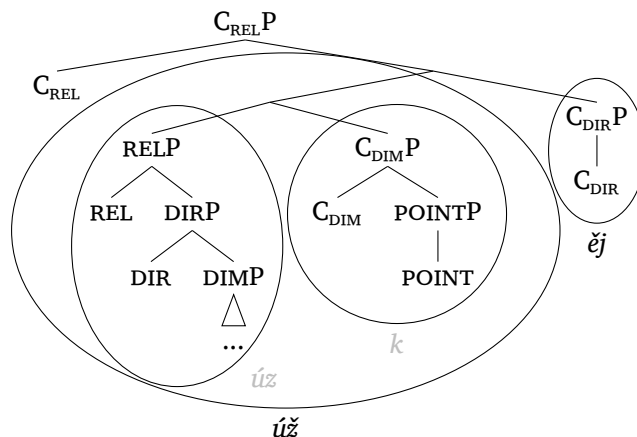
¹⁷ While it is correct to state that the steps in the derivation are the same as for *slad*, the lexicalisation is not always the same, due to the availability of the suppletive root *úž*- in (142). At the stage represented in (132) above, this suppletive root will override the lexicalisation *úž-k* (see (145) below). However, this lexicalisation will not surface, since it will be destroyed at the stage where C_{REL} is merged.

(144)



The respect in which this class differs from the previous one is in the shape of the comparative adverb, which lacks *-k*. Its derivation proceeds identically, but not its lexicalisation. The crucial point is again the one where C_{REL} is merged (see (135) above):

(145)



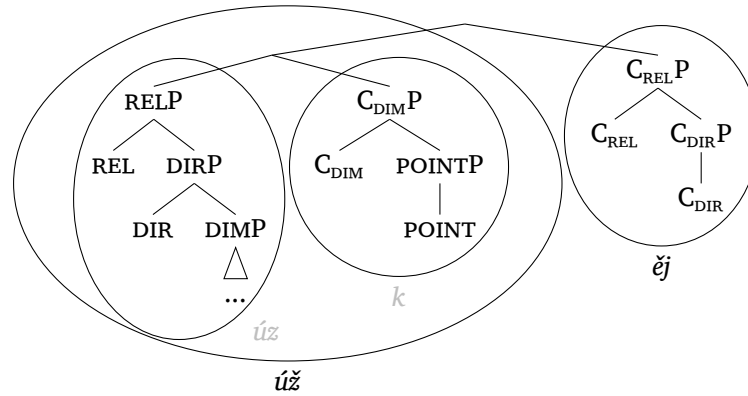
Observe how the suppletive root *úž-* appears in this structure. This is because, at the stage where C_{DIM} is merged, evacuation movement creates a configuration that matches (142), so that the earlier lexicalisation *úž-k-* is overridden (recall also footnote 17).

In the derivation of the adjective, we evacuated the root *REL P* out of $C_{REL} P$, which destroyed the configuration in which the suppletive root applies, so that the nonsuppletive root *úž-* appeared in the adjective. But since now we

¹⁸ Recall that roots with long vowels shorten before *-š*, and also palatalisations triggered by *-š* are different, so the surface form *už-š-í* is correctly derived from the underlying form of the root in this case.

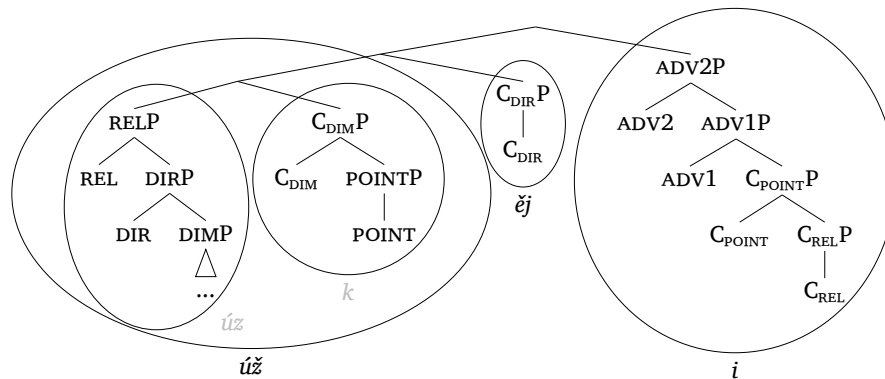
have backtracked to this stage, and we are making our second pass at the C_{REL} cycle, we need to pied-pipe, and this will preserve the configuration of the suppletive root:

(146)



This is how in this class the augment fails to surface in the comparative adverb. The rest of the derivation proceeds exactly as in the previous class, and we shall therefore not go through every step of it. The final stage in the derivation of the adverb is shown in (147).

(147)

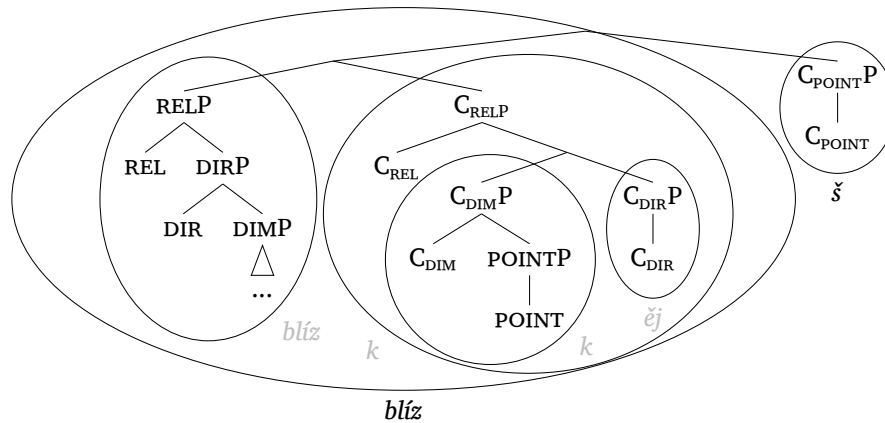


3.5.4 Class XI

The Class XI adjectives have *-k* in the positive, neither *-k* nor *-ěj* in the comparative adjective, and the *-e* marker in the comparative adverb. This gives

The derivation of the comparative adjective follows the same steps as with the regular *-k* adjectives, except that the lexicalisation *blíz* will override the earlier suffixal lexicalisations. The full representation is given in (151).

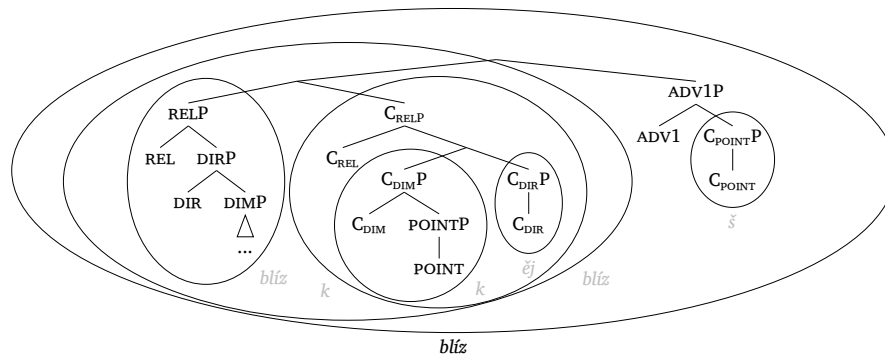
(151)



Observe that in (151) the C_{POINT} head is lexicalised by *-š*, even though the lexical entry of the root (149) contains this feature. This is because only the left branch of the lexical entry is matched in (151), not its right branch, causing it to be lexicalised by *-š*.

A complete match does arise, however, after the adverbial head $ADV1$ is merged and pied-piping evacuation movement takes place. Even though most adjectives with the augment *-k* have to backtrack from the stage depicted in (152), it is still one of the steps of their derivation (even if unsuccessful). So even in this minor class the root corresponds to one of the steps in a derivation shared across all the previous classes.

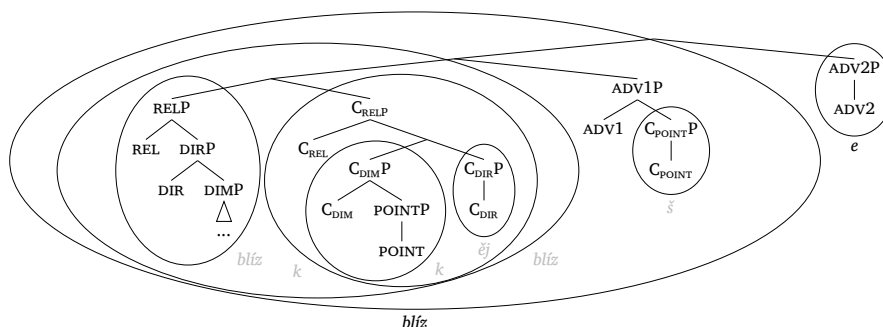
(152)



Due to a successful match in (152), the derivation of the adverb in this class diverges from other adjectives with the augment *-k*. (This is because none of the previous classes allows for the root to match $ADV1P$.) The final step

in this class thus involves the merger of ADV2, followed by several steps of pied-piping evacuation movements, until the entire complement of ADV2 is evacuated:

(153)



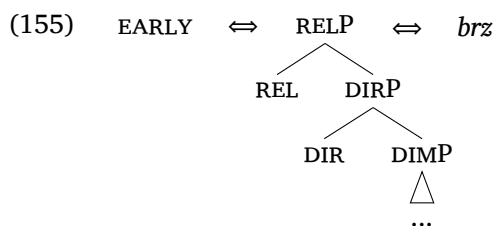
3.5.5 Class XII

Class XII is a single-membered class containing the suppletive adjective ‘early.’ This adjective has two roots *brz-* (for the positive) and *dřív-* (for both the comparative adjective and adverb). The full pattern is *brz-k-ý—dřív-ø-ěj-ší—dřív-ø-ø-e* ‘early’.

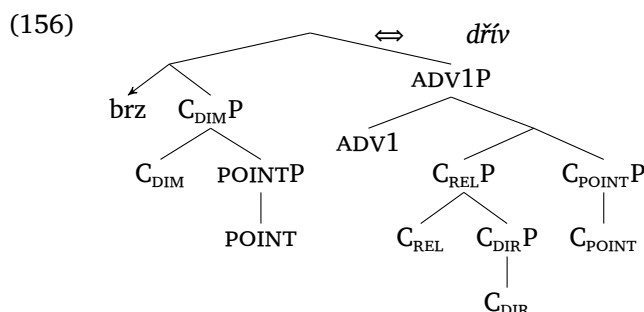
Given that there are no other adjectives in this class, it may be unclear whether we should in fact treat *brz-k-ý* as a decomposable adjective, or whether speakers analyse it as a non-divisible root *brzk-*. In the latter case, it would be an augmentless adjective with a suppletive comparative. We treat it as decomposable here because of the existence of forms with the bare root *brz-*, such as the positive degree adverb in (154a) or the diminutive adjective in (154b), where the diminutive seems to come in between the root and the augment; this is exactly as discussed in the previous chapter in Section 2.3.

- (154) a. *brz-y* ‘early’ ADV
 b. *brz-ič-k-ý* ‘early’ DIM ADJ

Based on this, we shall treat *brz-* as a root of size RELP, which, therefore, needs an augment in the positive.

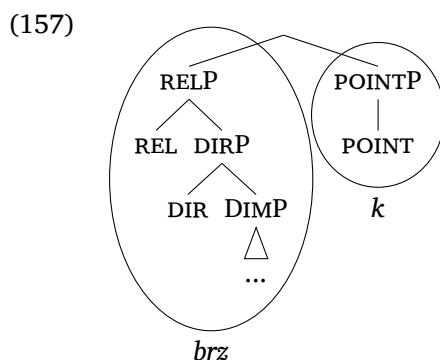


Now what we need for the other root, *dřív-*, is that it overrides this root at the level of $C_{DIM}P$, which is the first comparative feature added. It can't override it earlier, else we would get *dřív* in the positive. It can't override it later, since *dřív-* combines with *-ěj*, which has its foot at C_{DIR} , the second comparative feature. Thus, when C_{DIR} is merged, *dřív-* must already be in place. One way we could conceptualize this proposal would be to propose an entry as in (156).²⁰



The logic behind the entry is that DIRP and POINT should not form a constituent, else the positive would already be lexicalised as *dřív-*, contrary to fact. So in (156), the idea is that the root *brz*, lexicalising RELP, must first move to the Spec of $C_{DIM}P$, and only then is it overridden by *dřív*. However, this proposal ultimately fails, and a slightly more elaborate solution will have to be put in place. Let us first show what is wrong with (156).

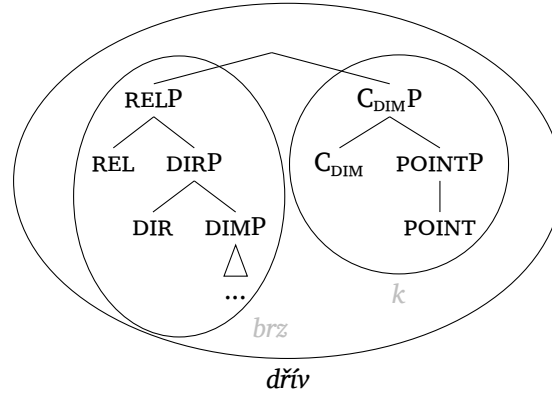
The derivation starts by producing the positive in (157). DIRP is lexicalised by *brz* and POINTP by *-k*. There is no match with the suppletive root (156), as intended.



In the next step, C_{DIM} is merged, and after DIRP is evacuated, we get the structure in (158). As this stage in the derivation, *brz-k-* is overridden by *dřív-*.

²⁰ Recall from Section 1.6.4 that suppletion can also be achieved without a pointer. We use a pointer here for reasons that become clear as the discussion unfolds.

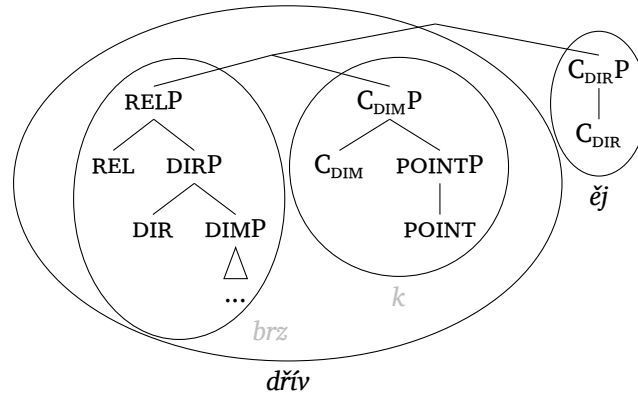
(158)



This happens because there is a match with the left branch of (156).

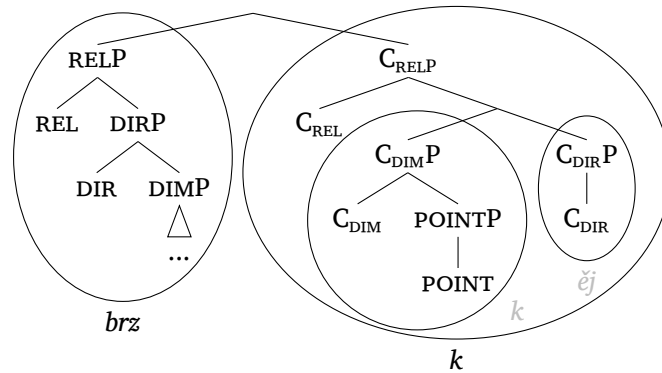
Next C_{DIR} is merged. Since evacuating just RELP does not help, we pied-pipe $C_{DIM}P$ along, yielding (159). We can see that here, $C_{DIM}P$ is lexicalised by $-ěj$.

(159)



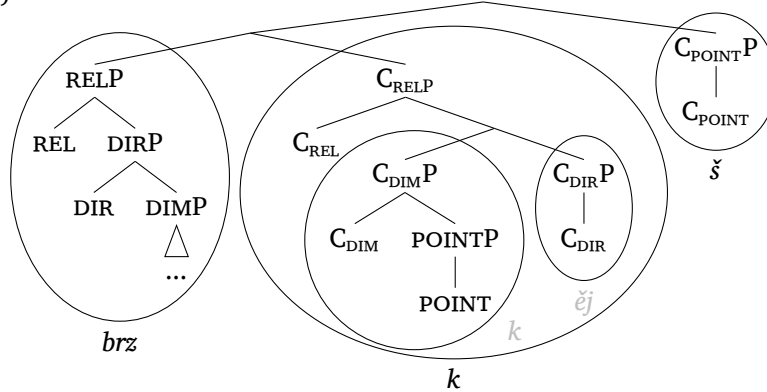
When we now merge C_{REL} , evacuation movement of RELP is triggered, yielding (160). Here, unfortunately, we get a match: RELP is lexicalised as $brz-$, and the right branch as $-k$. In other words, the subextraction of RELP destroys the configuration of the suppletive root, causing it to revert to $brz-$, while C_{REL} will be lexicalised by $-k$:

(160)



Continuing this derivation, we will incorrectly end up with the comparative adjective **brz-k-š-í* (with the expected phonological realisation as **brzčí*), as shown in (161):

(161)



What we need to prevent this unwanted result is to come up with a different lexical entry for *dřív* than the one in (156). The entry, recall, must override *brz-k* at the level of $C_{DIM}P$. But, at the same time, it must do so without moving RELP out of $C_{DIM}P$ (for when this happens, the result is wrong, as we just saw in (161)).

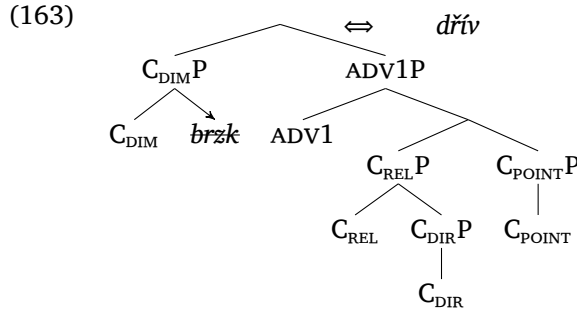
It seems to be the case that the only way to achieve this is to store the positive degree as a separate lexical entry (an idiom) that has no sound or meaning associated to it, see (162).

(162) — $\Leftrightarrow$ $\begin{array}{c} \swarrow \quad \searrow \\ brz \quad k \end{array}$ $\Leftrightarrow$ —

Since no sound or meaning is linked to this structure, the entry changes nothing about the sound or meaning arrived at at the lower nodes, but simply stores the result of the derivation of the positive in an unchanged form. We

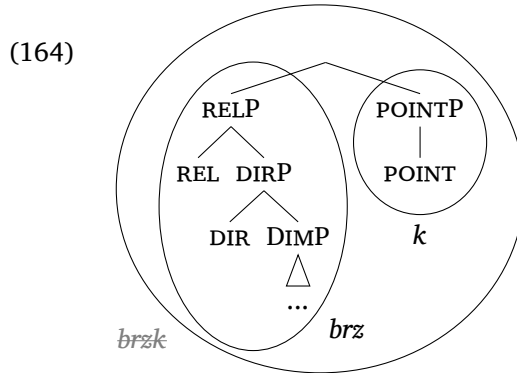
will call this entry ~~*brzk*~~ with a strikethrough just to have a way of talking about it; still, the strikethrough indicates that nothing about the phonology derived at the daughter nodes changes, just like idioms do not change the phonology of the pieces contained inside them.

And then we need *dřív*- to point to the entry ~~*brzk*~~, with C_{DIM} projecting its label over it. The projection of $C_{DIM}P$ acts as the closest labelled constituent for the purpose of extraction, and thus prevents the RELP from evacuating out of $C_{DIM}P$, which was the problem in (161).



Also, it is necessary that suppletion here is implemented via a pointer; if ~~*brzk*~~ did not exist, and we instead placed the structure corresponding to ~~*brzk*~~ at the relevant place, *dřív* would contain the full structure of the positive, and it would be able to pronounce that node. Here, it is unable to do so; recall from Section 1.6.4 that a lexical item like (163) cannot apply just to override ~~*brzk*~~; it must apply at a node higher than the actual pointer.

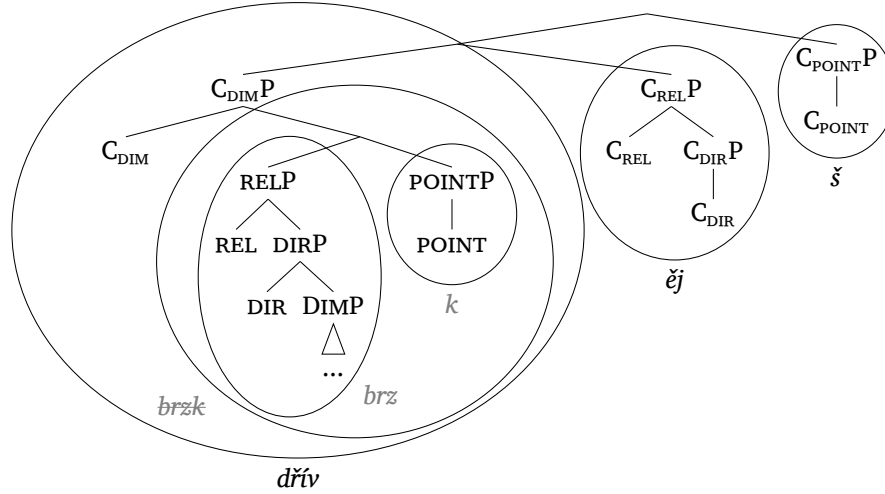
With the entries updated, we now get the following derivation. In the positive, nothing much changes, and we get *brz-k-ý* with the structure in (164). The one crucial difference is, however, that the output of the derivation is matched by the entry in (162), even though that changes nothing about the phonology or meaning.



In the comparative, C_{DIM} is merged, which will allow the suppletive root to override ~~*brzk*~~-, see the left branch in (165). The $C_{DIM}P$ thus derived will not

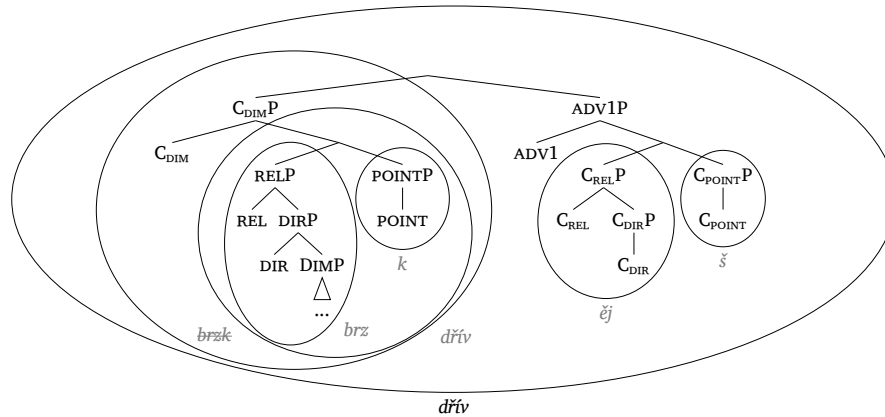
undergo any subextractions, because $C_{DIM}P$ projects over RELP, and represents the closest labelled constituent for extractions. This means that this lexicalisation will be preserved in the rest of the derivation, as $C_{DIM}P$ moves across the suffixes -ěj and -š. The derivation of the comparative will proceed as with Class I roots which lexicalise $C_{DIM}P$, recall Section 3.3.1. Following this derivation will give rise to the comparative adjective, as in (165).

(165)



The markers -ěj and -š are not overridden by *dřív* in (165) since their structures do not match the lexical tree in (163). Only when ADV1 is merged on top of (165), and $C_{DIM}P$ is evacuated, only then will there be a match between such structure and the entry (163). so that we do not see these suffixes surface in the comparative adverb.

(166)



The final step would then be merging ADV2, and after evacuating the whole complement, inserting the final *-ě* as the lexicalisation of ADV2P.

3.5.6 Summary

What this section showed is that the framework adopted here has sufficient power to capture all the five different classes of adjectives that exhibit the augment *-k* in the positive degree. These classes differ in whether they preserve or drop *-k* in the comparative. When they drop it, they can show comparatives both in *-š-í* and *-ěj-š-í*. When the comparative is in *-š-í*, there is again a trifurcation as to what the comparative adverb is: some adjectives regrow both *-k* and *-ěj*, some only *-ěj*, and a third class exhibits neither the augment or *-ěj*.

These patterns are genuinely problematic for linear rules of contextual allomorphy, leading to multiple rules sensitive across multiple positions (to the root, to the degree, and to the adjective/adverb distinction). We believe there is no way of stating such rules for the distribution of *-k* that do not end up postulating multiple *-k* markers.

In the type of approach we pursue here, what children learn are mappings between syntactic structures that are shared across the five different classes and the phonology of the root and/or the suffix. Once these associations are established, the complex patterns of distribution fall out from this set up.

3.6 Adjectives with *-ok* (Classes XIII-XV)

In this section, we complete the derivations by considering the augment *-ok*. We provide an overview of the facts in Table (167), repeated from (31).

(167) Four classes of adjectives with the augment *ok*

CLASS		ROOT	AUG	ĚJ	ENDING	gloss
XIII	POS ADJ	div	ok	—	ý	‘wild’
	CMPR ADJ	div	ok	ěj	ší	
	CMPR ADV	div	ok	ěj	i	
XIV	POS ADJ	hlub	ok	—	ý	‘deep’
	CMPR ADJ	hlub	∅	∅	ší	
	CMPR ADV	hloub	∅	ěj	i	
XV	POS ADJ	vys	ok	—	ý	‘tall’
	CMPR ADJ	vyš	∅	∅	ší	
	CMPR ADV	výš	∅	∅	e	

In Class XIII, the augment *-ok* remains throughout the paradigm, and it is followed by *-ěj* both in the adjective and in the adverb. Class XIV, which is a single-member class, has no *-ok* in the comparative, and shows *-ěj* in the comparative adverb. The root has an irregular length here and we thus treat it as (weakly) suppletive. Finally Class XV lacks both *-ok* and *-ěj* in its comparative forms.

The lexical entry for the Class XIII roots in (168) is the same as in Chapt. 2, where we discussed the internal structure of the positive. The lexical entry for *-ok* in (169) is an update from Chapt. 2, since we now take *-ok* to also lexicalise one feature of the comparative (C_{DIM}). In the positive, this feature is not present, so this update does not affect the lexicalisation of the positive, since (169) will be a match for a constituent that lacks C_{DIM} because of the Condition on Matching.

(168) Class XIII roots
 $\text{DIMP} \Leftrightarrow \text{root}_{\text{XIII}}$
 $\triangle$
 ...

(169)

```

      CDIMP  $\Leftrightarrow$  ok
      /  \
    CDIM  POINTP
          /  \
        POINT RELP
              /  \
            REL  DIRP
                |
              DIR
  
```

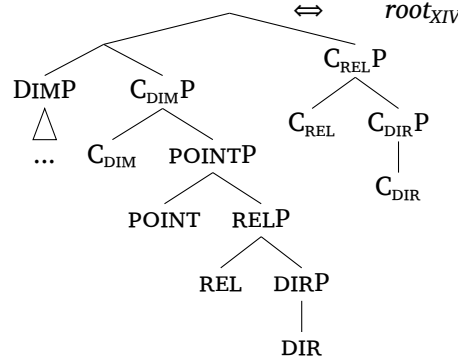
These entries (along with the others assumed so far) lead to the lexicalisation table in (170).

(170) Lexicalisation table for Class XIII

		DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ	div			ok							
CMPR ADJ	div			ok			ěj		š		
CMPR ADV	div			ok		ěj				i	

The next class, Class XIV, is characterised by the fact that the augment disappears in the comparative. This will require a lexical entry for the root with a special shape (a CLB), as shown in (171).

(171) Class XIV roots



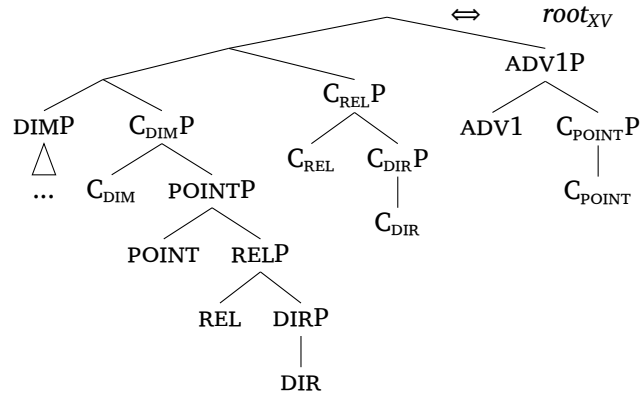
As with the other classes, lexical entries with CLBs correspond to a stage in the derivation that is found with regular forms. Specifically, Class XIV roots correspond to the structure of the derivation that produces *root* + *ok* + *ěj* in the regular Class XIII adjectives, where the *root* lexicalises just the DIMP branch, *-ok* lexicalises the $C_{DIM}P$ branch, and *-ěj* lexicalises the $C_{REL}P$ branch. The root (171) thus leads to a lexicalisation, shown in the upper section of Table (172), when this root overrides the regular morphemes *-ok* and *-ěj*. This is shown by extending the grey colour of the root across these suffixes. When we go through the derivations below, we shall see more details of how this works concretely.

(172) Lexicalisation table for the positive and the comparative

	DIM	DIR	REL	POINT	C_{DIM}	C_{DIR}	C_{REL}	C_{POINT}	ADV1	ADV2
POS ADJ	hlub		ok							
CMPR ADJ	hlub		ok			ěj		š		
CMPR ADV	hloub		ok			ěj		i		
POS ADJ	šir		ok							
CMPR ADJ	šir		ok			ěj		š		
CMPR ADV	šir		ok			ěj		(š)		e

Similarly, Class XV roots, illustrated in (173), lexicalise a stage in the derivation of regular roots that corresponds to the point where *root* + *ok* + *ěj* evacuates out of ADV1P (173). This root leads to the lexicalisation shown in the lower section of Table (172), where it also overrides the C_{POINT} marker *-š*.

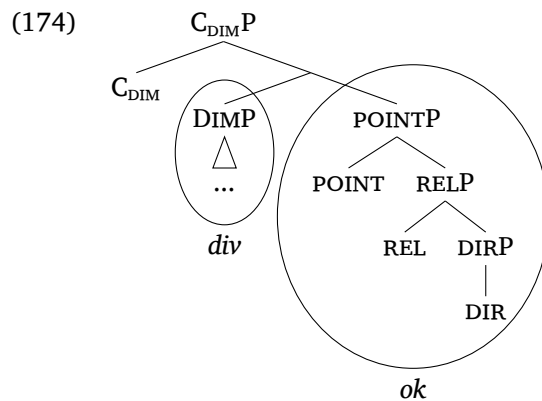
(173) Class XV roots



As in the previous classes, lexical entries with CLBs are a source of surface irregularity in paradigms: at the underlying level, the derivation follows the same steps both in the regular and irregular classes, but with each root class associated to a different stage of it. By showing how the final three classes work, we thus demonstrate that the complex patterns of distribution of the augments and other morphemes can be accounted for in these terms almost without exception.

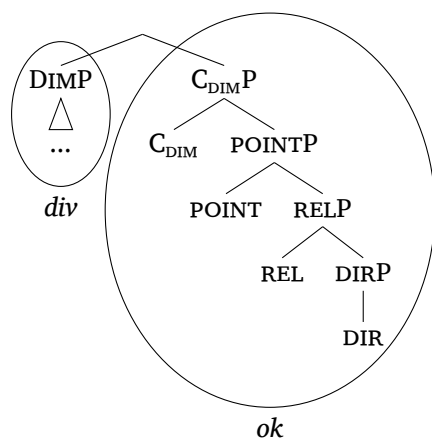
3.6.1 Class XIII

We start the derivation of the Class XIII adjective *div-ok-ý* ‘wild’ at the point where the positive is finished (tree diagram (51) in Chapt. 2), and add the C_{DIM} head to start deriving the comparative:



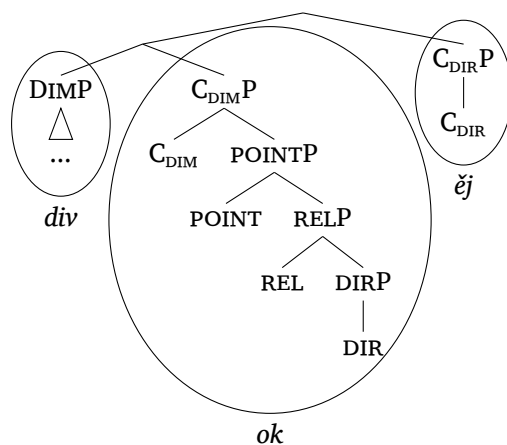
Since there is no match, evacuation movement will target DIMP, resulting in the lexicalisation of C_{DIM} by -ok:

(175)

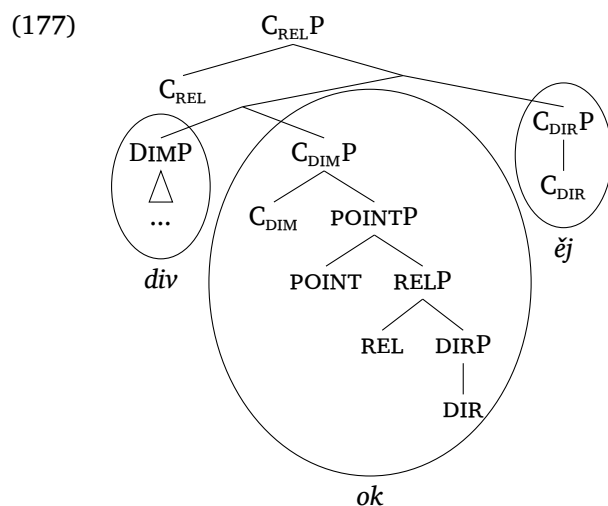


The next head that gets merged is C_{DIR} . Movement will initially target DIMP, but that will be unsuccessful, so that the node dominating it will be moved, creating a remnant constituent $C_{DIR}P$ that can be lexicalised by $-ěj$:

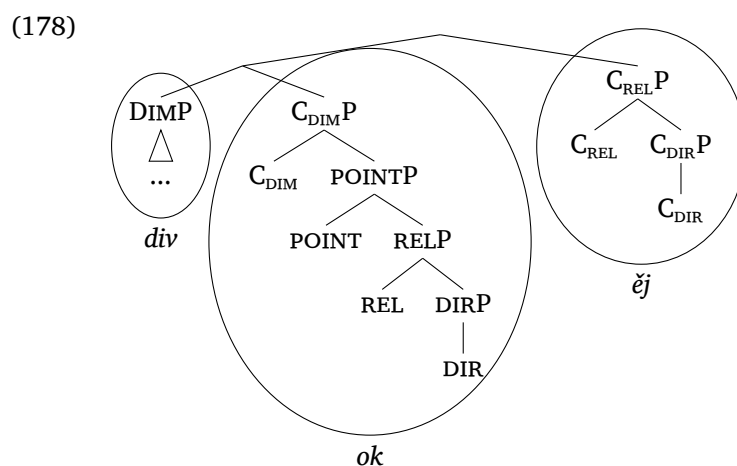
(176)



We next merge C_{REL} , (177), which does not find a match.

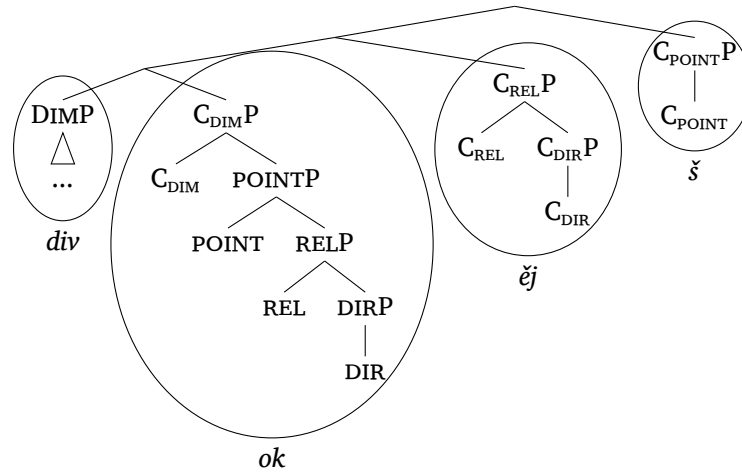


Movement of DIMP will be unsuccessful, and so we move the dominating node. That creates the proper configuration for C_{REL} to be lexicalised by -ěj.



Finally, we merge C_{POINT} , which will not find a match. In this case, only the movement of the entire complement of C_{POINT} will yield a successful lexicalisation:

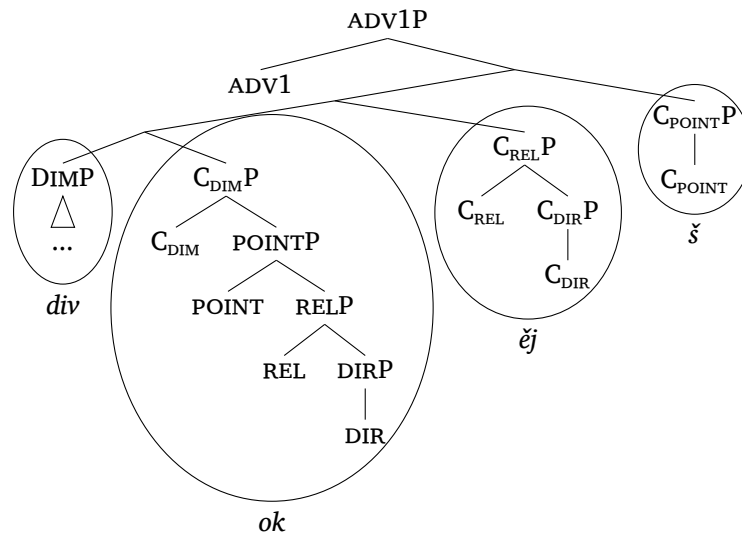
(179)



This completes the derivation of the comparative adjective (barring the agreement, which we leave aside here).

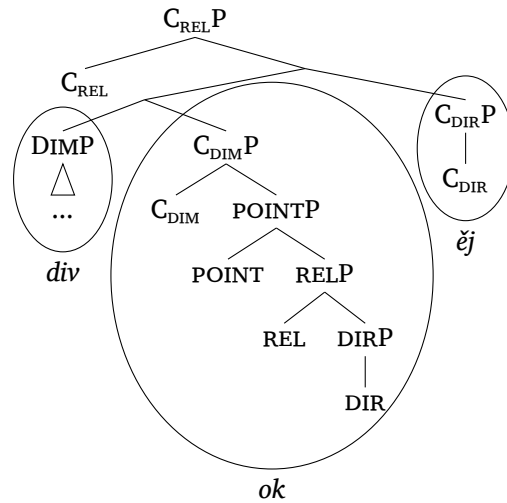
We next turn to the comparative adverb. The derivation begins by merging ADV1, but nothing can lexicalise this head.

(180)



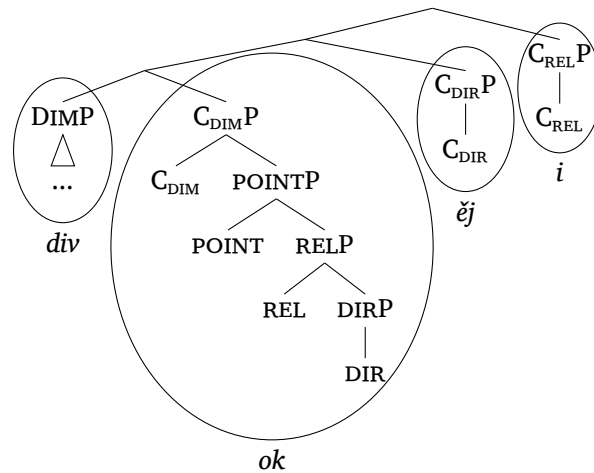
What is more, none of the movements that we go through will be successful either. The only morpheme that can lexicalise ADV1 is *i*, and as we can recall from (36), it has a tree with a foot in C_{REL} . No evacuation movement that we apply to (180) is going to create such a constituent. What we need to do therefore is backtrack to the cycle of $C_{REL}P$, shown in (177) above, and repeated here as (181):

(181)



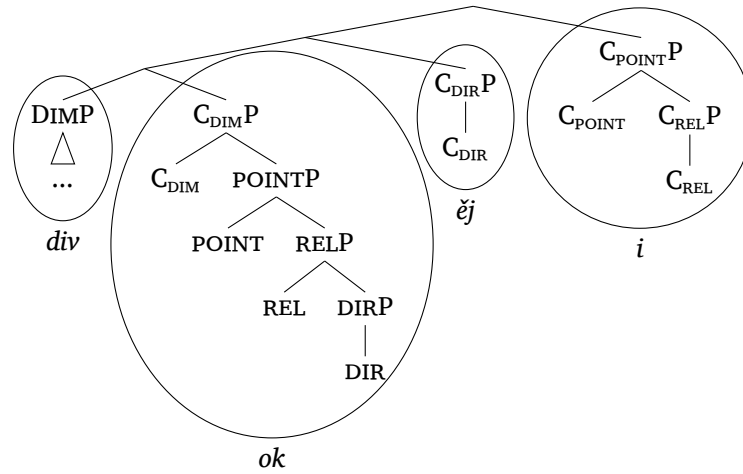
In the previous pass at this cycle, the movement of the node dominating DIMP was successful in lexicalising C_{REL} . Since this is our second pass, we need to apply the next movement option, which targets next dominating node, which is the sister of C_{REL} . This will create the constituent with C_{REL} at the bottom, which we need for *i*, and therefore ultimately for the lexicalisation of ADV1.

(182)



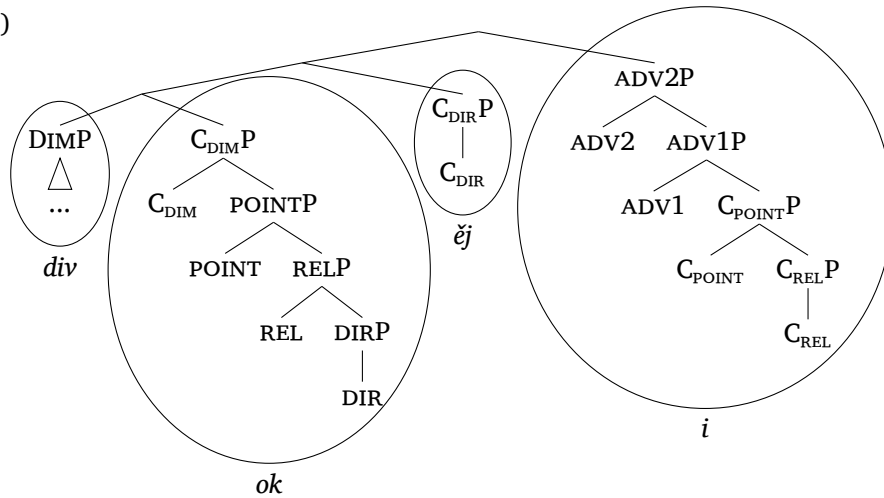
We next merge C_{POINT} , and, after unsuccessful movement of DIMP and the dominating node, move the next dominating node (the one that contains *div-ok-ěj*). This will lead to a successful lexicalisation by *i*:

(183)



When ADV1 is merged, the same procedure is repeated: movement of the node dominating *div-ok-ěj* will be needed to lexicalise the feature, and the same happens when ADV2 is merged. This eventually leads to the comparative adverb *div-oč-ěj-i*:

(184)

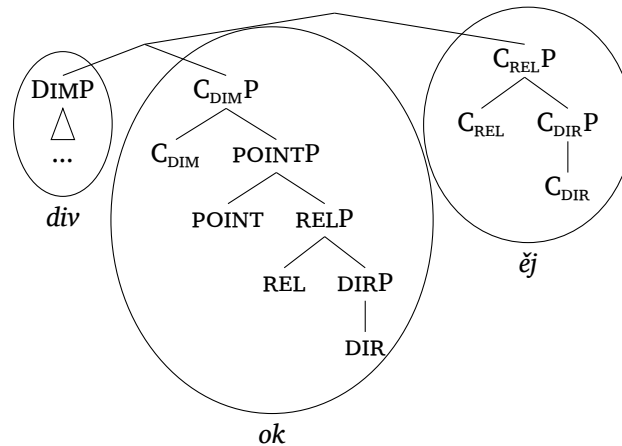


3.6.2 Class XIV

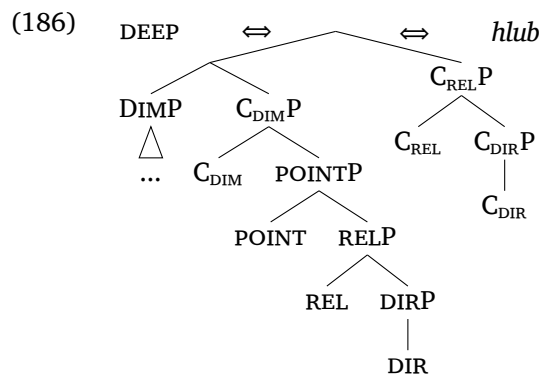
We now turn to Class XIV, which is characterised by the disappearance of the *-ok*-augment in the comparative forms. What we propose is that the lexical entry of these roots has the shape of the derivation that lexicalises as *div-ok-*

ěj. This stage of the Class I roots was given in (178) above, and it is repeated here as (185).

(185)

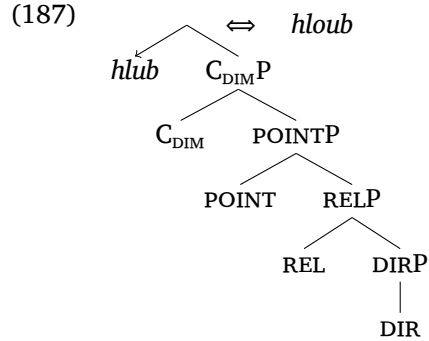


The lexical entry for a Class XIV root like *hlub-* has exactly this shape:

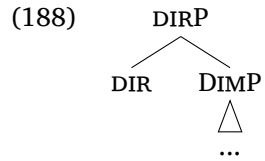


It would be possible to derive the pattern of augment/-ěj distribution using the entry in (186) alone, but, as a matter of fact, the adverb 'more deeply' has a suppletive root *hloub-*.²¹ This form requires another lexical entry, this time with a pointer. The shape of the suppletive entry is below, and it corresponds to a stage in the derivation where DIMP moves above -ok.

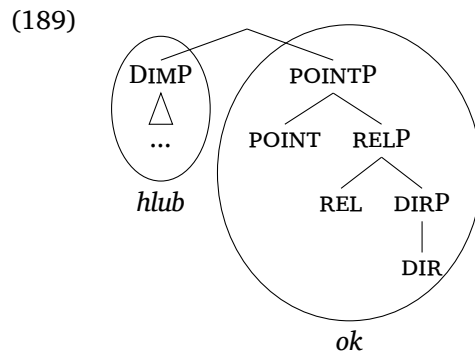
²¹ The root lengthening in the -ěj adverb is not a necessity, e.g. *hrub-ý* 'rough' shows no alternation in the root: *hrub-ěj-š-i* 'rougher', *hrub-ěj-i* 'more roughly' (and not **hroub-ěj-i*).



We first show how it happens that the augment appears in the positive, i.e. at the stage of the lexicalisation of *POINTP*, even with the bigger-sized lexical entries in (186) and (187). The first point to note is that the root initially cannot grow bigger than *DIMP*, like all the other roots that take *-ok* in the positive (Classes XIII–XV). This is because the lexical entry for *hlub-* in (186) contains *DIMP* as a CLB. When *DIR* is merged above *DIMP* (188), the result is not a subconstituent of (186) (nor (187), for that matter), so that the root can no longer lexicalise it, and evacuation movement will be triggered, just as with the Class XIII roots.



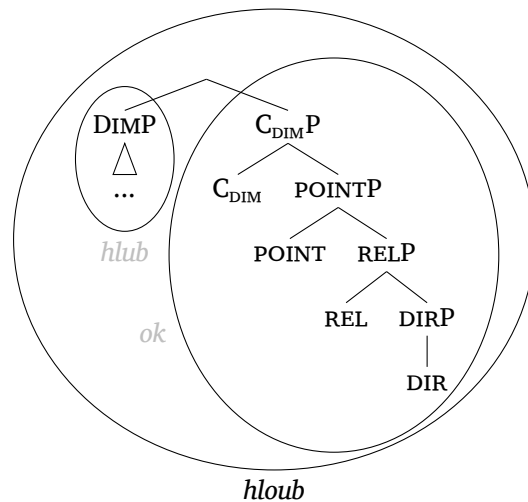
Skipping several intermediate steps on the way to the positive (since they are identical to those already illustrated in Chapt. 2), we arrive at the structure of the positive in (189). This structure is an exact copy of (51) in Chapt. 2, which represents the positive of the *-ok* adjective *div-ok-ý* ‘wild’, except that (189) has the root *hlub-*.



The important point is that this structure is not a subtree of (186), nor of (187), so that the root will not override the combination *hlub-ok*, ensuring that the augment appears in the positive.

The derivation of the comparative adjective in this class will go through exactly the same steps as that of a Class XIII root, leading to the same lexicalisations, except at two points in the derivation, where the combination of *root+suffix* will be overridden by the lexical entry with the CLB. The first point is at the merger of C_{DIM} (see (175) above, repeated here as (190), but with the root *hlub-* substituting for *div-*). At this point, the top node is a match for the lexical entry of the suppletive root *hloub-* (187), so that it overrides *hlub+ok*. This will lead to the absence of the augment in the comparative, as we shall see below. In (190) we mark the overriding by greying out the morphemes that get overridden by the lexicalisation of the larger constituent by the root. The lexicalisation table in (172) also shows the special shape of the Class II root by greying out the suffixes in the lexicalisation bar of the root.

(190)

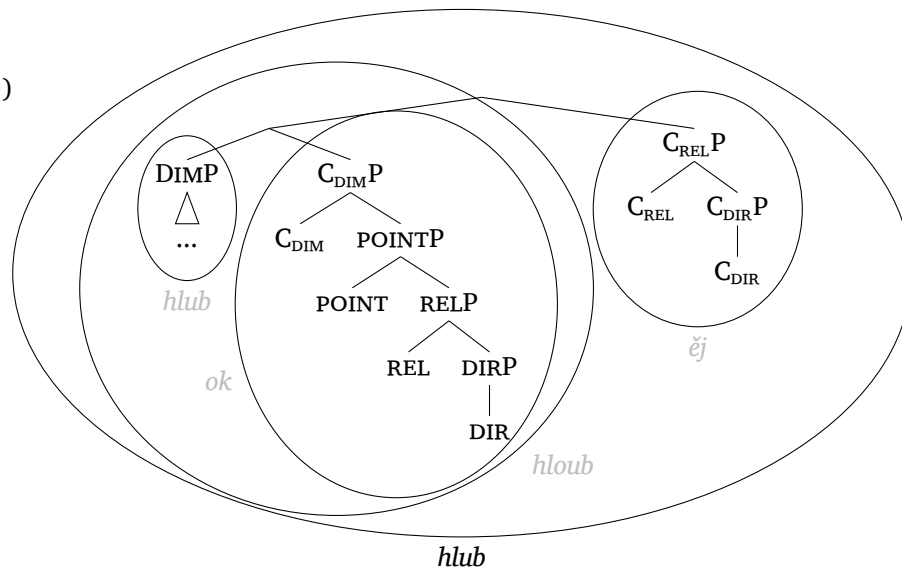


This stage will be relevant when we discuss the derivation of the comparative adverb below. In the comparative adjective, however, the suppletive root represents only an intermediate stage in the derivation, which will not survive to the end, because it will be overridden by the root *hlub-* later on, as we shall discuss next.

The second point in the derivation where a bigger-sized root will override an earlier lexicalisation of *root+suffix* is at the merger of C_{REL} , i.e. the stage depicted in (185) above. At this stage, the entire $C_{REL}P$ (lexicalised as *hloub-ěj*, where *hloub-* itself has overridden *hlub-ok*) will be overridden by the root *hlub-*. This is what triggers the disappearance of both the augment and the comparative marker -ěj in the comparative adjective. The tree in (191)

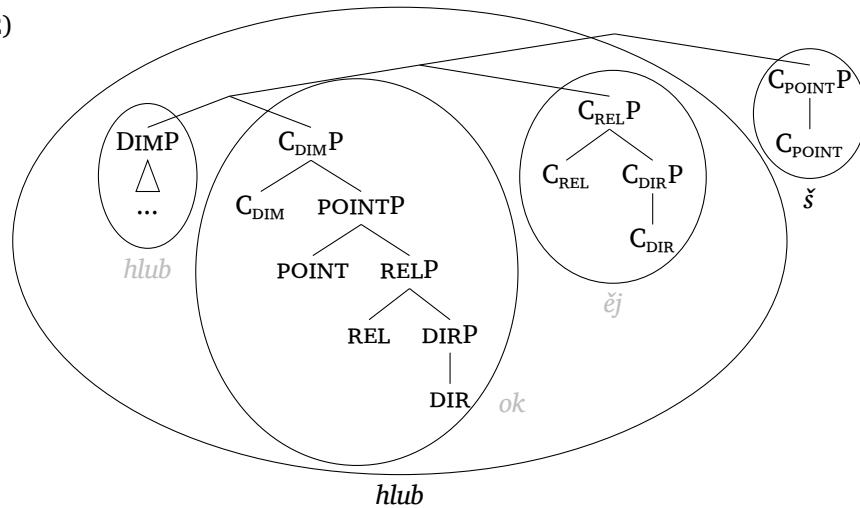
represents this visually, with the lexicalisations of lower nodes than the top node marked in grey, and that of the top node as *hlub* in black.

(191)



Since the derivation of $C_{REL}P$ in Class XIV proceeds in exactly the same fashion as in the Class XIII adjectives, we shall not go through each stage that leads to (191), but refer the reader to the previous section for these steps. The derivation of the comparative adjective involves one additional step in the derivation after (191), and this step is also identical to the same step in the Class I adjectives (see (179) above). It involves the merger of C_{POINT} , and the movement of its complement (which is pied-piped after a series of unsuccessful evacuation movements), followed by the lexicalisation of $C_{POINT}P$ by *-š* (so as not to overburden the representation, the intermediate lexicalisation by the suppletive root *hloub-* is omitted from (192); we return to it when deriving the comparative adverb below):

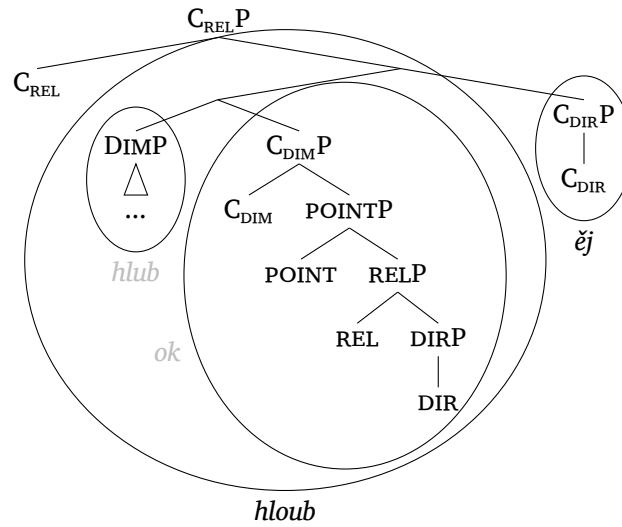
(192)



Importantly, this movement leaves the constituent intact that was lexicalised by the root in the previous state, so that we get an augmentless comparative adjective (*hlub-š-i* ‘deeper’).

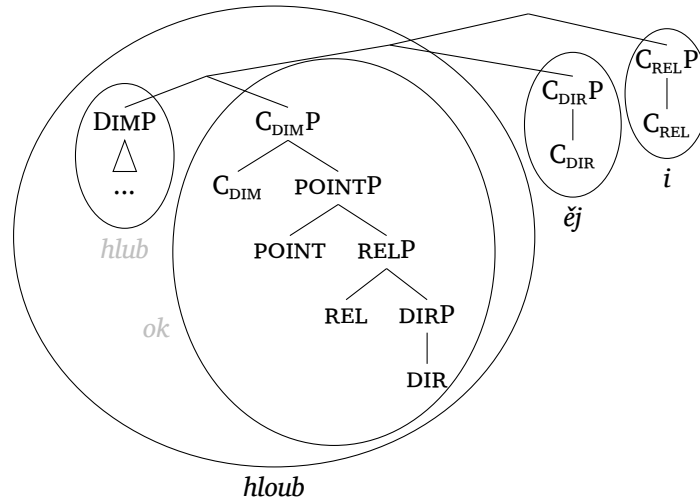
The comparative adverb is likewise augmentless, but it sees the rebirth of the comparative marker *-ěj*: *hloub-ěj-i* ‘more deeply’. Again, we take the derivation of Class XIII as a reference point. There, the adverbial marker *-i* triggered backtracking, because it is the only morpheme that can lexicalise ADV1, and it has a foot in C_{REL} . Backtracking therefore goes back to the cycle of C_{REL} . Note that we are now back at a stage in the derivation where the suppletive root *hloub* lexicalises a constituent of size $C_{DIM}P$. This time, however, the derivation unfolds differently, and the suppletive root will not be overridden.

(193)



In the first pass at this cycle, the movement of the node dominating DIMP was successful in lexicalising C_{REL} as *-ěj*. In the backtracking pass, the next movement option applies, which targets the sister of C_{REL} , creating (194). As we saw in Class I, this creates the constituent with C_{REL} at the bottom, which lexicalises as *-i*.

(194)

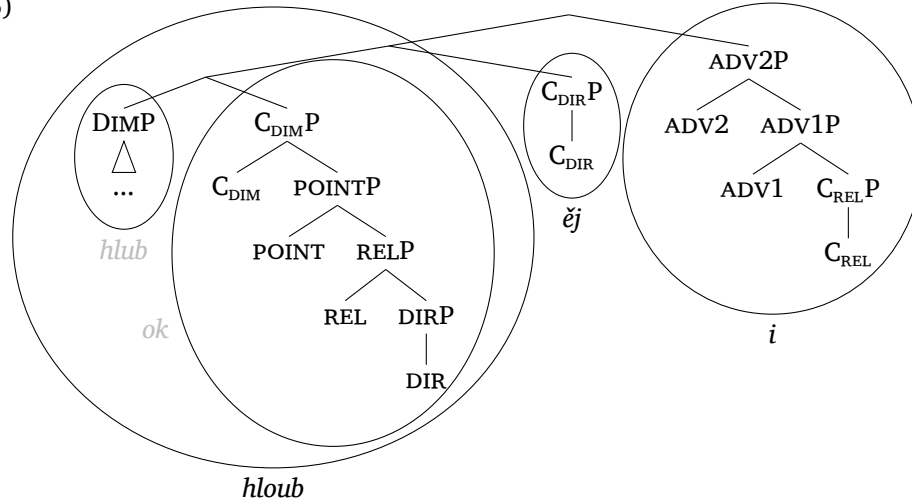


The crucial property of (194), however, is the fact that in the backtracking derivation, there is no longer a match with the the lexical entry of the root *hlub-* in (186). In particular, the constituent [C_{REL} [C_{DIR}]], lexicalised by the root in the comparative adjective, no longer exists in (194), triggering the reappearance of *-ěj* in the comparative adverb. At the same time, the unla-

belled constituent dominating DIMP continues to be a match for (186), so that the suppletive root can surface, while also suppressing the appearance of -ok.

The merger of ADV1 and ADV2 does not change this configuration. The steps in the derivation involving these heads are exactly identical to those in the Class I adjectives, leading to the comparative adverbial form *hloub-ěj-i* ‘more deeply’, shown in (195).

(195)



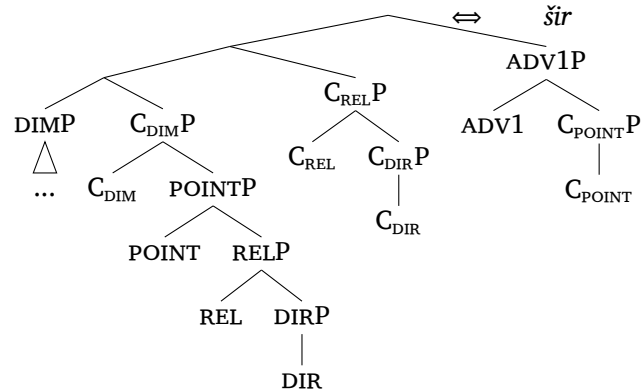
This completes the derivation of the two comparative forms of the Class XIV roots.²²

3.6.3 Class XV

Class XV roots take -ok in the positive, and lose it in the comparative (like the Class XIV roots). They differ from them in that the comparative adverb does not see the rebirth of -ěj. In addition, we see a different allomorph make its appearance in the adverb: -e instead of -i. In (196), we show the lexical entry for the Class XIV root *šir-*.

²² We are aware that this gives rise to a full ABA derivation by the root: DIMP in the positive realized as *hlub-*, then DIMP + ok is realized as *hloub-*, and finally DIMP + ok + ěj is again lexicalised as *hloub-*. This type of derivation has been argued to be necessary in Caha (2025) and Kasenov (2025b).

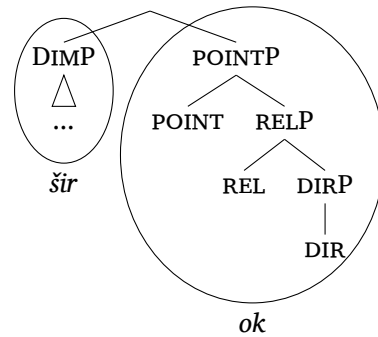
(196)



These roots are larger in size than the Class XIV roots, reaching up to ADV1P. No backtracking will be needed in the derivation, since this was triggered by the need to lexicalise ADV1, but in Class XV the root takes care of this. We also explain the appearance of *-e*, which provides a lexicalisation for the feature ADV2 alone.

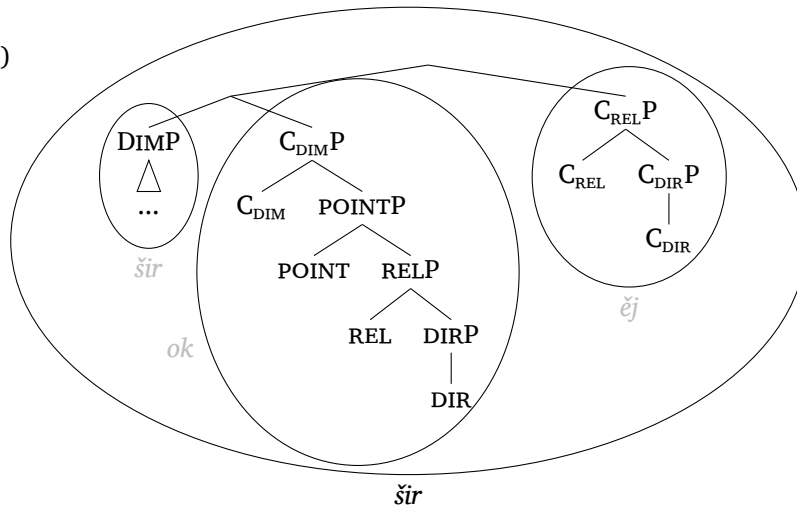
As before, the root will not be able to lexicalise the positive, since there is no constituent in (196) that corresponds to all and only the features of the positive. The augment *-ok* will therefore make its appearance:

(197)



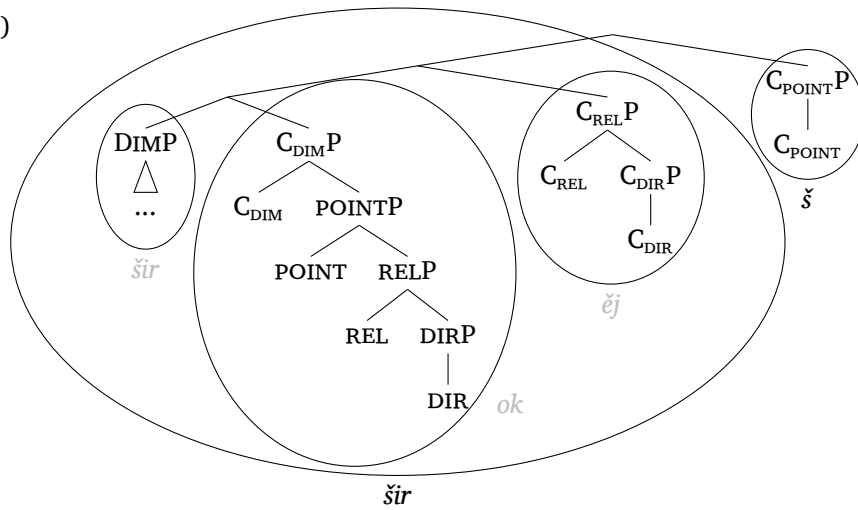
The derivation again unfolds as in Class XIII, with the root lexicalising a constituent of size $C_{REL}P$ on the way to deriving the comparative adjective:

(198)



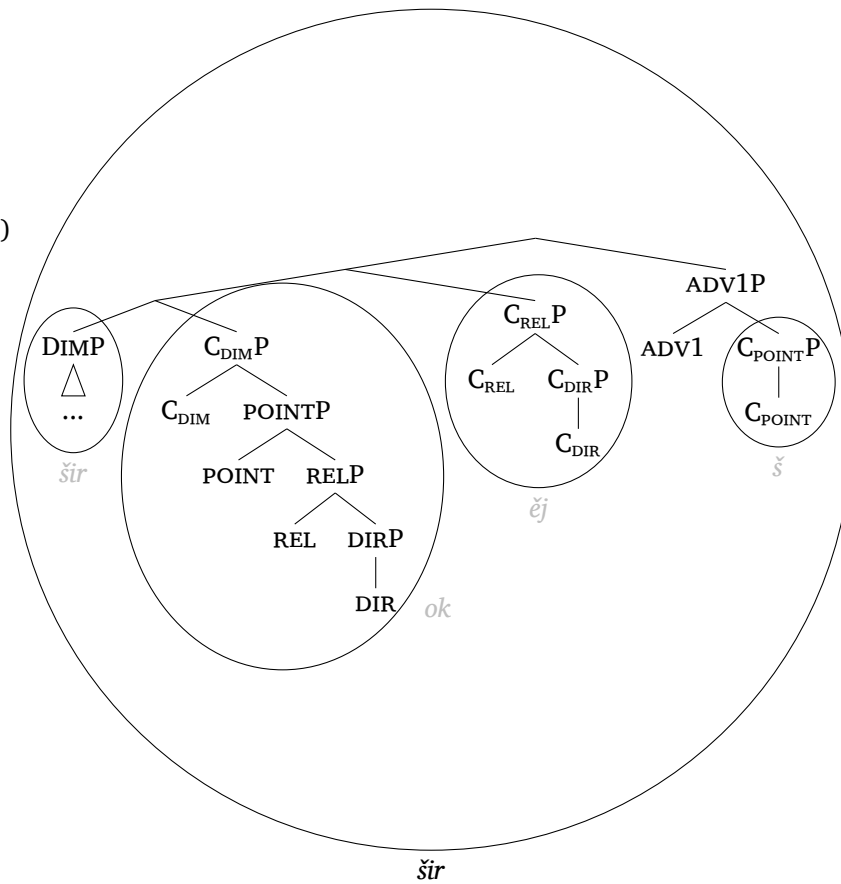
In the next cycle, C_{POINT} is merged, and this will trigger the same evacuation movement steps as in Class XIII, eventually resulting in the lexicalisation of C_{POINT} as -š, and the correct comparative adjective form *šir-ě-í* 'wider'.

(199)

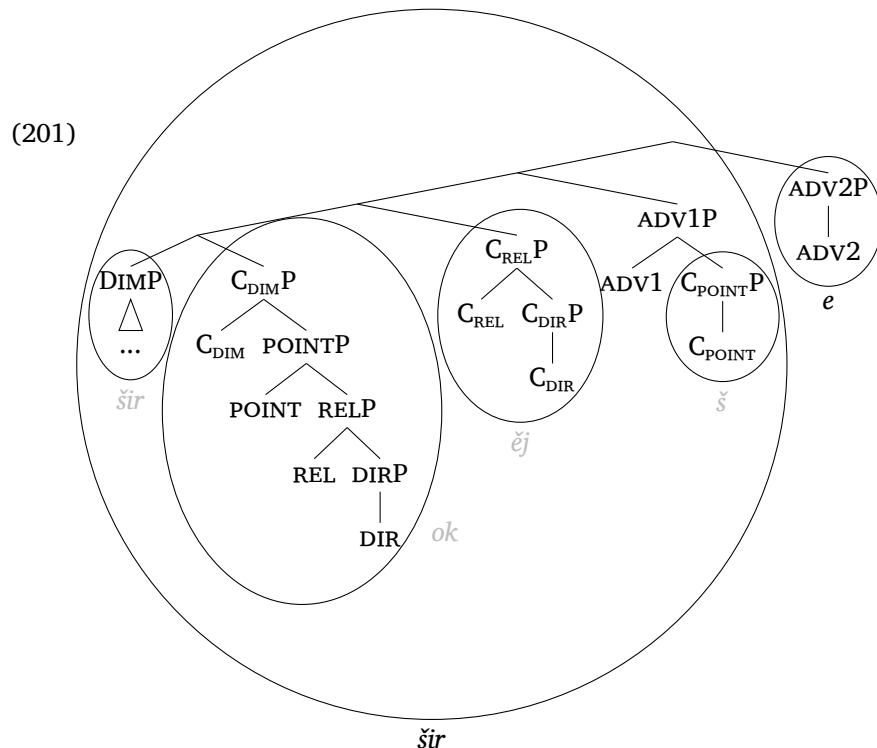


In the derivation of the comparative adverb, the next feature to be merged is ADV1. Since this cannot lexicalise directly, evacuation movements will be triggered, which initially target DIMP, and then subsequent nodes dominating it, until the sister node of $C_{\text{POINT}}P$ moves, resulting in (200). This tree is now an exact match for the lexical entry of the Class XV root in (196) above.

(200)



The final step in the derivation of the adverb involves the merger of ADV2. Again, direct lexicalisation is not possible, which triggers a series of evacuation movements. Eventually, this leads to the evacuation of the sister node of ADV2 and its lexicalisation by *-e*. The PAL₃ and lengthening factors that are a part of this morpheme result in the attested form *šīře* ‘more widely’, with a long vowel in the root and a palatalised *r*.



3.6.4 Summary

In Table 3.8, we show a lexicalisation table for the fifteen patterns. Since the table represents the morphosyntactic analysis, we ignore the effects of the phonology, in particular the palatalisations. Thus, we assign identical shapes to morphemes that we take to be identical, like the various instances of *-ok* and *-k*, which may in fact also be realised as *-oč* and *-č* if they are followed by a palataliser. Similarly, the table shows the root *drah-* (Class III) identically throughout, even though the final *h* of the root is palatalised to *ž* in the comparative, but since this is the result of regular phonological processes, we keep the root identical across the degrees as far as its lexical representation is concerned. We also remove the agreement endings of the adjectives, since we have nothing meaningful to say about them.

Table 3.8 Lexicalisation table for the positive and the comparative

		DIM	DIR	REL	POINT	C _{DIM}	C _{DIR}	C _{REL}	C _{POINT}	ADV1	ADV2	GLOSS
I	POS ADJ			chab								'weak'
	CMPR ADJ			chab				ěj	š			
	CMPR ADV			chab			ěj			i		
II	POS ADJ			slab								'weak'
	CMPR ADJ			slab				ěj	š			
	CMPR ADV			slab			ěj			i		
III	POS ADJ			drah								'expensive'
	CMPR ADJ			drah				ěj	š			
	CMPR ADV			dráž			ěj		š		e	
IV	POS ADJ	jem		n								'fine'
	CMPR ADJ	jem		n				ěj	š			
	CMPR ADV	jem		n			ěj			i		
V	POS ADJ	žádouc		n								'desirable'
	CMPR ADJ	žádouc		n				ěj	š			
	CMPR ADV	žádouc		n			ěj			i		
VI	POS ADJ	pozd		n								'late'
	CMPR ADJ	pozd		n				ěj	š			
	CMPR ADV	pozd		n			ěj			i		
VII	POS ADJ	snad		n								'easy'
	CMPR ADJ	snaz		n				ěj	š			
	CMPR ADV	snaz		n			ěj		š		e	
VIII	POS ADJ	hez		k								'pretty'
	CMPR ADJ	hez		k				ěj	š			
	CMPR ADV	hez		k			ěj			i		
IX	POS ADJ	slad		k								'sweet'
	CMPR ADJ	slad		k				ěj	š			
	CMPR ADV	slad		k			ěj			i		
X	POS ADJ	úz		k								'narrow'
	CMPR ADJ	úz		k				ěj	š			
	CMPR ADV	úž		k			ěj			i		
XI	POS ADJ	blíz		k								'close'
	CMPR ADJ	blíz		k				ěj	š			
	CMPR ADV	blíz		k			ěj		š		e	
XII	POS ADJ	brz		k								'early'
	CMPR ADJ	dřív		k				ěj	š			
	CMPR ADV	dřív		k			ěj		š		e	
XIII	POS ADJ	div		ok								'wild'
	CMPR ADJ	div		ok				ěj	š			
	CMPR ADV	div		ok			ěj			i		
XIV	POS ADJ	hlub		ok								'deep'
	CMPR ADJ	hlub		ok				ěj	š			
	CMPR ADV	hloub		ok			ěj			i		
XV	POS ADJ	šir		ok								'wide'
	CMPR ADJ	šir		ok				ěj	š			
	CMPR ADV	šir		ok			ěj		š		e	

The lexicalisation table has greyed out suffixes in the lexicalisation bar of many of the roots (e.g. Class II). This is a way of visually marking the special shape of the lexical entries of these roots, which makes the relevant suffixes disappear.

3.7 Conclusions

In this chapter, we focussed on the variation in the comparative forms of adjectives and adverbs in Czech. Specifically, we focussed on two parameters: the presence/absence of the augment, and the presence/absence of *-ěj*. We showed that the distribution of these markers is independent of the phonology, and sensitive to both the nature of the root, (the augment, if any) and the particular form (adjective/adverb). We showed that the two parameters combined provide for 15 different classes, and demonstrated that the complexity of the distribution makes it difficult to even describe the facts using context-sensitive statements. For instance, it turned out impossible to come up with a single non-disjunctive statement for morphemes such as *-ěj* along the lines of ‘realise position α as *-ěj* when there is a particular root to its left and the form is an adjective/adverb.’ Such statements had to be multiplied, which amounts to accidental homophony.

Instead, we showed that the presence/absence of these markers follows from the model of phrasal lexicalisation used in this book. We showed that each augment class follows a particular derivation, and individual roots link various stages of this derivation (possibly including movements) to the surface form. These entries, even though restricted in their shape by the rules of syntax, provide exactly the rich and apparently contradictory marker distributions as observed in the data. In the closing chapter, we reflect over the abstract properties of the mechanism that delivers these results.

Chapter 4

Conclusions and future research

Abstract This chapter concludes the book's investigation into the morphological structure of positive and comparative degrees in Czech. It summarises the key proposal that both degrees are built from four syntactic heads, with variation across fifteen classes reducible to two parameters: the size of the moving constituent and the height to which it moves. This system eliminates the need for context-sensitive rules or arbitrary class features, aligning morphology with phrasal syntax. The chapter then outlines four avenues for future research: the formation of positive-degree adverbs, the distinction between two agreement classes (-ý vs. -í), the potential overlap between positive and comparative scales, and the morphology of superlatives. It offers preliminary suggestions for each, including the postulation of an additional head (X) to account for adverb formation and agreement patterns, while leaving a full analysis for further study.

4.1 Modelling morphological parameters

As stated at the outset, this book followed two main goals. The first one was to provide morphological evidence for a particular structure of the positive and the comparative degree. On our account, the positive has four heads. The lowest two features (DIM and DIR) are dedicated to the construction of a scale, with an optional NEG head for negative adjectives. The two upper features (REL and POINT) are dedicated to the introduction of the standard on the scale, which the argument must exceed.

The same four heads, we argued, are also found in the comparative. The comparative takes the positive scale as an input, and constructs a comparative scale on its basis (the 'derived measure function' of Kennedy & Levin 2008). To do so, syntax reuses the same four features. The comparative scale has the same dimension as the one in the positive (i.e., C_{DIM} is anaphoric), but it starts at a different point, namely one corresponding to the comparative

standard provided by the *than*-phrase (which sets the value of the POINT projection in the positive degree). The comparative scale can be oriented in the same direction as the positive scale (featuring just C_{DIR}), deriving a comparison of superiority. It can also be oriented in the opposite direction (featuring also C_{NEG}), deriving a comparative of inferiority. The comparative scale also introduces a standard, using the same two features as in the positive to do so. This standard aligns with the starting point of the comparative scale, i.e., the argument of the comparative must have a non-zero value on the comparative scale. This is similar to adjectives like *wet*, where the standard is also set to zero, and thus, the argument of the adjective counts as *wet* as soon as it has a non-zero degree of wetness. Given that the comparative scale starts at the point introduced by the comparative standard in the *than*-phrase, the argument of the comparative must simply exceed the value of that standard; see Kennedy & Levin (2008).

Our main evidence for this type of architecture was drawn from Czech morphology, even though in Chapter 1, we also found evidence for multiple degree projections in English, Latin and Gulf Arabic (see also Vanden Wyngaerd et al. 2020 for Slovak, Wiland 2023 for Polish, Vyshnevskaya 2025a for Ukrainian, Kasenov 2025a for Russian). Our evidence was presented separately for the positive (Chapter 2) and the comparative (Chapter 3).

For the positive degree, we noticed that the diminutive marker in Czech reveals an asymmetry among the augments such that some of them are found below, and some above the diminutive, which requires at least three projections (one for the root, one for the low augments, and a third one for the high augment). Additional structural asymmetry was established on the basis of the fact that one of the augments only formed positive (but not negative) adjectives.

Regarding the comparative, we presented evidence suggesting that its formation necessitates the postulation of four additional heads. These heads allow us to capture classes where the comparative adds three morphemes on top of the positive, as well as a bewildering variety of morphological patterns (which amount to a total of fifteen different classes).

At the descriptive level, we showed that these fifteen different classes boil down to two parameters of variation, namely the presence/absence of the augment, and the presence/absence of *-ěj*. We demonstrated that these distributions are impossible to capture using context-sensitive rules, unless unrestricted use of accidental homophony is admitted, requiring, in essence, one such rule per distribution pattern. For instance, we observe that in Classes I, IV, V, VI and XIII, we always find *-ěj*, in Classes II, VIII, XI, X and XIV, we find *-ěj* in the adverb, but not in the adjective, and in Class XII, we only find *-ěj* in the adjective but not the adverb. Finally, in Classes III, VII, XI and XV, there is no *-ěj* at all.

(1) The distribution of *-ěj*

CLASS	CMPR ADJ	CMPR ADV
I, IV, V, VI, XIII	<i>-ěj</i>	<i>-ěj</i>
II, VIII, XI, X, XIV	Ø	<i>-ěj</i>
XIII	<i>-ěj</i>	Ø
III, VII, XI, XV	Ø	Ø

In this book, we showed that once phrasal lexicalisation is adopted, no context-sensitive rules are needed to capture these facts. Instead, we analyse the overt exponent as a standard lexical item specified for a particular projection/projections, and zeroes arise due to phrasal lexicalisation by the root and/or by the augment. The correct distribution of morphemes falls out from this idea. In such a system, there is no need to assume accidental homophony or complicated contextual rules sensitive to roots, degrees and parts of speech.

As a further consequence, despite the existence of fifteen different morphological classes, there is no need in our system to postulate any arbitrary class features on the root (see also Caha 2021, Janků 2022, Don et al. 2023, Vanden Wyngaerd 2026). The selection between a root and an affix (be that a specific augment or a particular comparative allomorph) simply falls out from the lexical items themselves, interacting with an invariant lexicalisation procedure.

Encoding class membership without dedicated class features is a welcome consequence, since empirically, it appears that the set of features available to human languages is restricted (Chomsky 2004, Cinque 2013). Cinque observes that even features that correspond to cultural universals (such as shame or taboo) or features important for survival (edibility) are never grammaticalised. If the fifteen different classes were to be implemented as abstract syntactic features that rules of exponence are sensitive to (as in, e.g., Müller 2004), such features would represent a significant deviation from a restrictive theory of features that populate our linguistic system.

This approach is not only empirically adequate, but also theoretically attractive in that the set of possible lexical items is restricted by syntax. Specifically, the syntax side of the lexical entry always has to correspond to a well-formed syntactic tree, derivable by binary Merge and evacuation movements.

Let us now look in greater detail at the various classes of lexical items that we have proposed in the course of the discussion. What we want to show is that the parameters along which these entries differ are reminiscent of parameters that have already been discovered in phrasal syntax. We begin by a parameter that we shall label ‘height of movement.’ To see the effects of this parameter, consider Table 4.1, where we list four of the lexical entries that we were led to propose for the adjectives of the *-n* class.

Table 4.1 Four root classes with the augment -n

<i>jem-</i> (Ch. 3, item (73))	<i>chlad-</i> (Ch. 2, item (95))	<i>žadouc-</i> (Ch. 3, item (75))	<i>pozd-</i> (Ch. 3, item (76))

What we see here is that the classes all share a constituent of the size DIRP (with *-n* starting immediately above, i.e., at RELP). Moving from left to right in Table 4.1, this constituent (DIRP) is first on its own, and then moves higher up in the structure, across REL, POINT, and finally across C_{DIM} . In essence, these entries portray the steps of a single derivation where a constituent of the size DIRP moves cyclically higher and higher in the structure. That is why we call this the ‘height of movement’ parameter.

Each of these entries belongs to a root that takes the augment *-n*, but exhibits a slightly different pattern of augment distribution, with augments missing at the particular stage targeted by lexicalisation (i.e., RELP, POINTP, C_{DIM} going left to right). The entry on the left thus shows a root that has the augment *-n* at all levels above DIRP (i.e., in all the forms that we encountered). We show this pattern in the first column in Table 4.2.

Table 4.2 Movement height determines where augments drop

	‘smooth’	‘cool’	‘desirable’	‘late’
DIM	<i>jem-ṇ-ouč-k-ý</i>	<i>chlad- -ouč-k-ý</i>	<i>žadouc-ṇ-ouč-k-ý</i>	<i>pozd-ṇ-ouč-k-ý</i>
POS	<i>jem-n-ý</i>	<i>chlad-n-ý</i>	<i>žadouc- -í</i>	<i>pozd-n-í</i>
CMPR	<i>jem-n-ěj-š-í</i>	<i>chlad-n-ěj-š-í</i>	<i>žadouc-n-ěj-š-í</i>	<i>pozd- -ěj-š-í</i>

When DIRP extracts out of RELP and no further (as in the second column of Table 4.1), we have a missing augment in the diminutive, see the second column in Table 4.2. This is because DMV takes RELP as a base (recall the discussion in 1.7 how the foot of a suffix determines the size of the base).

When DIRP evacuates out of POINTP and no further (as in the third column of Table 4.1), the augment is missing in the positive and no other form, as in the third column in Table 4.2. Finally, when the root evacuates out of C_{DIM} , the augment drops in the comparative (but remains in all other forms).

This way, we achieve what the contextual rules struggle with. On the one hand, the presence of the DIRP constituent links the roots to a particular augment: all the relevant entries contain DIRP either as the whole entry, or

as a left branch. At the same time, we have a way of capturing the fact that the augment sometimes drops (as defined by the absolute size of the entry, expressed as the features on the right branch).

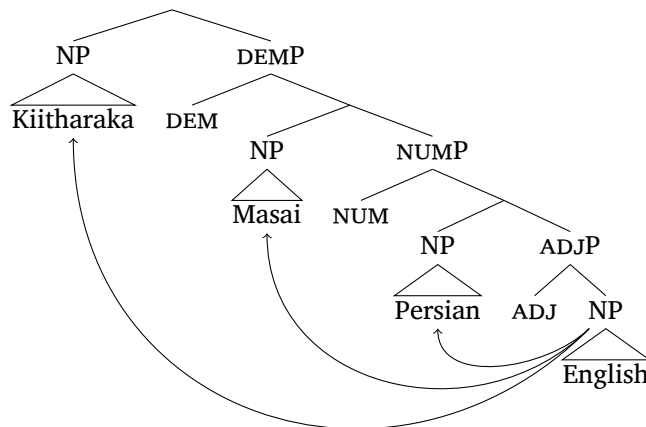
The way these entries encode the differences among the various roots is reminiscent of the ‘movement-height’ parameter identified by Cinque (2005) in the study of phrasal orders (see also Cinque 2009). Cinque (2005) applies this logic to the height of NP movement in a fixed sequence of modifiers, the modifiers being: demonstrative > numeral > adjective. In this order, the NP can either stay in situ, as in English, or move up, which can be either partial movement (as in Persian and Masai), or a total movement, as in Kiitharaka.

(2)

Ordering the NP wrt. the modifiers: the height of movement				
English	DEM	NUM	ADJ	NP
Persian	DEM	NUM	NP	ADJ
Masai	DEM	NP	NUM	ADJ
Kiitharaka	NP	DEM	NUM	ADJ

The tree in (3) shows this graphically, with languages differing in how far the NP moves.

(3)



This variation is reminiscent of the one in Table 4.1, where the DIRP is also pictured as moving to various heights in a fixed sequence of projections. The point we want to make is simply that the format in which class differences are encoded in Table 4.1 finds parallels in other areas of grammar, and does not seem to be a single-purpose *ad hoc* device that can be added to or removed from the theory at will: it is already a part of our theory for independent reasons.

In our account, the ‘height-of-movement’ parameter controls where the n -augment is found, but it does not control the type of the augment. This is controlled by another parameter, which has to do with the size of the constituent that is moving, i.e., with the size of the left branch. For instance, fixing the absolute size of the constituent to $C_{DIM}P$, we have, in the course of the discussion, proposed the entries in Table 4.3 (Since some of the items contained pointers, we have in the table indicated the size of the constituent pointed to). The lexical entries all are of the same size, containing features up to $C_{DIM}P$, which means that they have no augment when preceding $-ěj$ (which starts at the next feature higher up, i.e., at C_{DIR}). However, they differ in the size of the left branch that evacuates out of $C_{DIM}P$, as depicted in the trees. (The variation in the size of the left branch also influences the right branch: when the moving left branch is small, it strands some features behind, and these are found on the right branch.)

Table 4.3 Four root classes of the size $C_{DIM}P$

<i>hloub-</i> (Ch. 3, item (187))	<i>pozd-</i> (Ch. 3, item (76))	<i>úž</i> (Ch. 3, item (142))

The parameter along which the entries in Table 4.3 differ is the size of the left branch. Thus, even though these roots have no augment in the comparative adverb, yielding the forms in (4a), they each have a different augment in the positive, see (4b).

(4)

		deep	late	narrow
a.	CMPR ADVS	hloub- -ěj-i	pozd- -ěj-i	úž- -ej-i
b.	POS ADJ	hlub- ok -ý	pozd- n -í	úz- k -ý

The augments found in the positive lexicalise the different right branches when the left branch sits one notch lower, i.e., just above POINT. We depict this in Table 4.4.

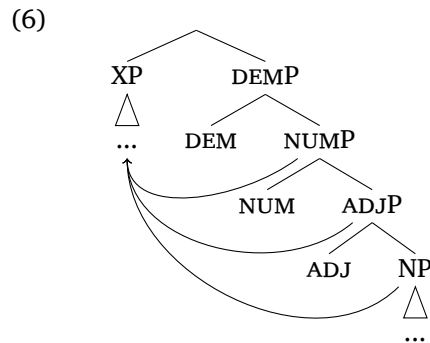
Table 4.4 Four root classes of the size $C_{\text{DIM}P}$ lexicalising the positive

<i>hloub-</i>	<i>pozď-</i>	<i>úž</i>

This parameter also finds an analogue in Cinque's (2005) work. Thus, Cinque observes that in some languages, only the NP moves across the modifiers, leaving them in a fixed order (leading to the order NP > DEM > NUM > ADJ), as in (5a). However, sometimes the NP may pied-pipe the adjective, leading to an order starting with the constituent [ADJ N], and preceding DEM > NUM, as in (5b). When the full complement of DEM moves, we get [NUM [ADJ N]] preceding DEM, as in (5c).

- (5) Variable size of the moving constituent
- [NP] DEM NUM ADJ
 - [ADJ NP] DEM NUM
 - [NUM [ADJ NP]] DEM

We show this arboreally in (6); what distinguishes the orders in (5) is the size of the moving constituent in (6). This is the same type of variation as when constituents of different sizes end up on the left branch above C_{DIM} .



In sum, it appears that the lexical items we are proposing vary across two dimensions, just like phrasal movement. The first parameter is the size of

the left branch (if any). The second parameter is how high the left branch moves in the hierarchy of projections. The first parameter controls the type of augment roots combine with, the second parameter controls the actual distribution of the chosen augment (where it is found, and where it is absent).

The result is therefore a system where complex variation in morphological class is reduced to lexical items that vary along two parameters: the size of the left branch, and how high it moves. These are two parameters that were identified independently of the domain at hand, namely, as parameters that define (and constrain) syntactic variation in the ordering of the noun and its modifiers (and beyond). If correct, this result supports the recent trends that attempt to unify morphological and syntactic variation within a single set of assumptions, ultimately unifying our theory of morphology with syntax. These attempts have been the driving force behind much of the work in Nanosyntax and beyond, pursuing the idea that underlying the human capacity for language is just a single type of generative system, namely syntax.

4.2 Directions for future research

There remain, as always, areas where additional research is needed. In this final section, we address four such areas. In Section 4.2.1, we investigate positive-degree adverbs. In Section 4.2.2, we provide a suggestion how the distinction between different agreement classes of adjectives can be captured. In Section 4.2.3, we consider the extent to which the comparative and positive scales use the same ingredients, or whether they are slightly distinct. In Section 4.2.4, we discuss the superlative degree of adjectives and adverbs.

4.2.1 Positive-degree adverbs

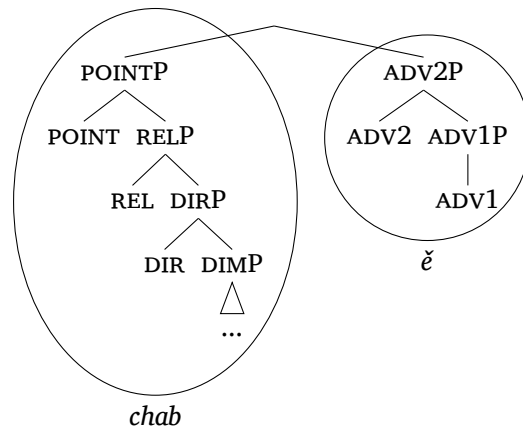
One of the areas that we have not touched upon are positive-degree adverbs. In most classes (eleven out of fifteen), these are simply derived by adding a palatalising *-ě* to the positive degree (i.e., to the same form to which the agreement marker *-ý* attaches). We give several examples in (7) (we return to the missing classes V-VII below).

- (7) Positive degree adverbs are mostly formed by
- ě*

CLASS	POS ADJ	POS ADV	gloss
I	chab-ý	chab-ě	'weak'
II	slab-ý	slab-ě	'weak'
III	drah-ý	draz-e	'expensive'
IV	jem-n-ý	jem-n-ě	'smooth'
VIII	sliz-k-ý	sliz-c-e	'slimy'
IX	slad-k-ý	slad-c-e	'sweet'
X	blíz-k-ý	blíz-c-e	'closely'
XI	úz-k-ý	úz-c-e	'narrow'
XII	úz-k-ý	úz-c-e	'narrow'
XIII	div-ok-ý	div-oc-e	'wild'
XIV	vys-ok-ý	vys-oc-e	'tall'
XV	hlub-ok-ý	hlub-oc-e	'deep'

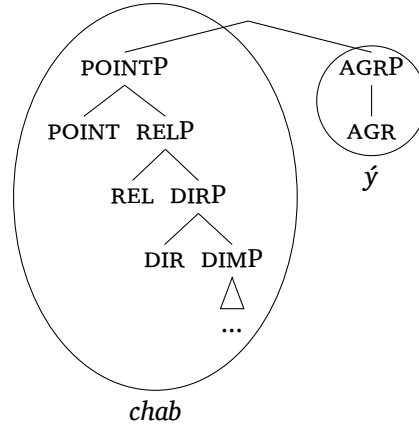
The simplest account of such forms would be one where the adverb is formed by simply placing ADV1 and ADV2 on top of the positive POINTP, and have *-ě* lexicalise these projections, as shown in (8) for Class I root *chab-* 'weak.'

- (8) A hypothetical structure of a positive-degree adverb



The fact that the adverbial *-ě* attaches to the same base as the agreement marker *-ý* could then be understood as shown in (9), where *-ý* also starts immediately above POINTP:

(9) A hypothetical structure of an agreeing positive-degree adjective



The issue with this proposal is that if the adverbial marker *-ě* simply lexicalised ADV1 and ADV2, as in (8), then we would expect *-ě* to be a generic adverb marker, appearing both in the positive and in the comparative adverb. In the comparative, it would take precedence over *-i*, since it requires no backtracking. As a result, we would expect a regular system of adverbial formation across the positive and the comparative degrees, where a single marker (*-ě*) uniformly derives both positive and comparative adverbs. We have seen a glimpse of such a system in Ukrainian, recall (30) from Chapter 2. In Table 4.5, we update the table to include positive-degree adverbs.

Table 4.5 Ukrainian adjectives and adverbs (Vyshnevskya, 2025a)

AUG	POS ADJ	POS ADV	CMPR ADJ	CMPR ADV	GLOSS
∅	vesel-yj molod-yj	vesel-o molod-o	vesel-i-š-yj molod- š-yj	vesel-i-š-e molod- š-e	‘merry’ ‘young’
OK	žorst-ok-yj šyr-ok-yj	žorst-ok-o šyr-ok-o	žorst-ok-i-š-yj šyr- š-yj	žorst-ok-i-š-e šyr- š-e	‘cruel’ ‘wide’
N	mits-n-yj	mits-n-o	mits-n-i-š-yj	mits-n-i-š-e	‘strong’
K	solod-k-yj švyd-k-yj	solod-k-o švyd-k-o	solod-k-i-š-yj švyd- š-yj	solod-k-i-š-e švyd- š-e	‘sweet’ ‘fast’

In the table, there are apparently two different adverbial markers, the positive-adverb marker *-o* and the comparative-adverb marker *-e*. However, these are two different phonological guises of a single morpheme, an underlying *-o* changing to *-e* after *-š* (Vyshnevskya 2025a, p. 146). The relevant point for us is that this marker always attaches to the relevant adjectival

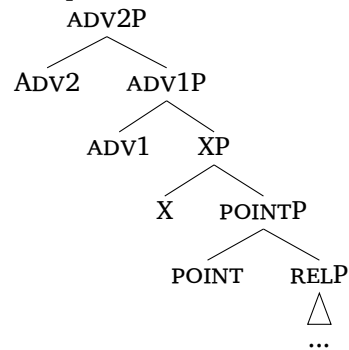
form minus agreement, without requiring any change in the morphological base (i.e., no backtracking).

Given that Czech does not behave like this, we have postulated in the comparative the adverbial marker *-i*, which triggers backtracking and restructures the base. If we are to maintain this proposal for comparative adverbs, the entry for the Czech positive-degree adverb cannot be as simple as suggested above, namely as lexicalising just ADV1 and ADV2. If the entry for *-ě* were that simple, it would apply also in the comparative, following *-š*, as in Ukrainian.

The lexical entry of the Czech positive-degree adverb *-ě* thus represents the first issue that we shall leave open for future research, offering here a few speculations.

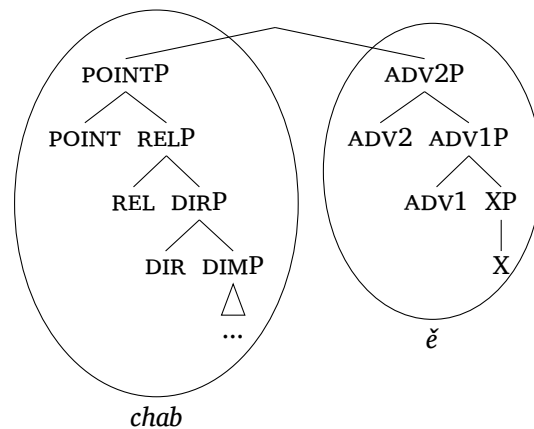
One logical option would be to say that the positive degree adverb has an additional projection, call it X, in between POINTP and ADV1P, as in (10).

- (10) The position of X in the functional sequence



Once the existence of X is admitted, the updated structure of the adverb would look as in (11).

- (11) An updated structure of a positive-degree adverb



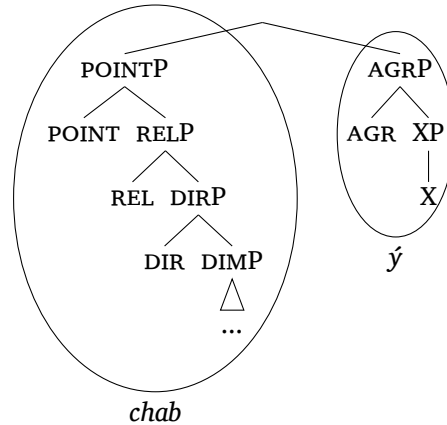
Crucially, if X is missing in the comparative sequence (i.e., there is no X in between C_{POINT} and ADV1), then -ě will be disqualified as a comparative adverbial marker: it does not match a structure where X is missing. That way, the results we have derived in the area of comparative adverbs would be preserved in the face of the positive-degree adverbs.

Notice as well that in (11), nothing changes about the lexicalisation of the POINTP, since X is lexicalised by the adverbial ending. Therefore, our account of the lexicalisation of POINTP remains unaffected. If we further maintain the idea that POINTP is the base to which the first comparative feature C_{DIM} attaches, all results presented in Chapters 2 and 3 are maintained. The only thing that is achieved by the postulation of X is that the ending -ě, found in the positive degree adverbs, will not be applicable in the comparative adverb.

The next question is: what could X be? In Vanden Wyngaerd et al. (2020), we suggested that the functional structure of the positive degree includes (on top of POINTP) and additional head that we have labelled UP (after Neeleman et al. 2004, see also Wiland 2023). The function of this head is to take the denotation of POINTP – a scale with a standard placed on the scale – and explicitly encode the fact that the value of the adjective’s argument is UP from this standard. If we identified X with this head, the proposal would explain why X is not embedded under C_{DIM} . That is because the comparative scale is starting at the POINT (determined by the *than*-phrase), not from a place that is UP from the standard. For Vanden Wyngaerd et al. (2020), this was a part of a larger claim that the positive is not fully embedded as a whole in the comparative, only a part of it is (that part corresponding to POINTP).

Under this view, we would expect that X (= UP) would not only be present in the positive-degree adverb (*chab-ě* ‘weakly’), but also in the positive adjective (*chab-ý* ‘weak’). If that was so, we would be lead to propose that this head is lexicalised in the adjective along with the agreement marker, as in (12).

(12) An updated structure of a positive-degree adjective



The postulation of the new head X may shed some light on certain irregularities in both the adjectival forms and the adverbial forms that we have been ignoring up to now. In order to see this, consider the exhaustive list of classes where positive-degree adverbs are derived by *-ě*. We show these in (13). Compared to Table (7), two new classes appear: V and VI.¹

(13) Positive degree adverbs with the marker *-ě*

CLASS	POS ADJ	POS ADV	gloss
I	chab-ý	chab-ě	‘weak’
II	slab-ý	slab-ě	‘weak’
III	drah-ý	draz-e	‘expensive’
IV	jem- n -ý	jem- n -ě	‘smooth’
V	žádouc- -í	žádouc- n -ě	‘desirable’
VI	pozd- n - í	pozd- -ě	‘late’
VIII	sliz-k-ý	sliz-c-e	‘slimy’
IX	slad-k-ý	slad-c-e	‘sweet’
X	blíz-k-ý	blíz-c-e	‘closely’
XI	úz-k-ý	úz-c-e	‘narrow’
XII	úz-k-ý	úz-c-e	‘narrow’
XIII	div-ok-ý	div-oc-e	‘wild’
XIV	vys-ok-ý	vys-oc-e	‘tall’
XV	hlub-ok-ý	hlub-oc-e	‘deep’

What is remarkable about these two classes is that their positive-degree adjective (minus agreement) is different from the adverb (minus *-ě*). Specifically, they either grow or lose their augment as we go from the adjective to the adverb. This is different from the Class IV root *jem-n-ý* ‘smooth,’ which shows a uniform pattern.

Curiously, in addition to the irregularity linked to the augment, another property that puts the misbehaving classes V and VI together against the other classes is that they have the agreement marker *-í*, unlike all the other classes, which have an *-ý*. We mark this in blue. In the next section, we look more closely at the different agreement markers *-í* and *-ý*. We will show that their distribution is not purely phonological, and we suggest an account that links the presence of the *-í* inflection to the fact that these classes have a different form in the adjective and in the adverb.

¹ There are still two classes missing in the table, namely Class VII and IX. These are both single-member classes and have a different adverb marker than *-ě*. In Class VII, *snad-n-ý* ‘easy’ has the adverb *snad-n-o*; in Class XI, *brz-k-ý* ‘early’ has the adverb *brz-y* ‘early’ (*brz-o* ‘early’ is also a possible adverb). Given that these are single-member classes, we will not tackle them here; in the worst case, these adverbs could be treated as non-decomposable (suppletive) forms lexicalising the full adverbial structure.

4.2.2 Two different agreement markers

In Czech, adjectives belong to two different agreement classes, where in the masculine singular some adjectives have the (hard, nonpalatalising) ending *-ý*, while others have the (soft, palatalising) ending *-í*. The forms differ also in other forms, as illustrated in (14) for masculine and feminine singular. We use here the positive adjective ‘good’ and the suppletive comparative ‘better.’

- (14) Masculine and feminine singular of the *-ý* class and the *-í* class

	good	better	good	better
	MASC.SG	MASC.SG	FEM.SG	FEM.SG
NOM	dobr-ý	lepš-í	dobr-á	lepš-í
ACC	dobr-ý	lepš-í	dobr-ou	lepš-í
GEN	dobr-é-ho	lepš-í-ho	dobr-é	lepš-í
LOC	dobr-é-m	lepš-í-m	dobr-é	lepš-í
DAT	dobr-é-mu	lepš-í-mu	dobr-é	lepš-í
INS	dobr-ý-m	lepš-í-m	dobr-ou	lepš-í

It can be observed that while quite similar in the masculine gender, the two declensions are quite different in the feminine forms. Also, the *-í* inflection makes no distinction between the masculine and the feminine in the nominative. This property generalises across all three genders, where *-í* adjectives have an invariant form in the nominative singular, while *-ý* adjectives exhibit three different forms, as in (15). It thus appears to be the case that the two declensions are not phonologically derived one from the other, but they require a set of their own exponents.

- (15) Nominative singular of the *-ý* class and the *-í* class

	good	better
MASC.SG	dobr-ý	lepš-í
FEM.SG	dobr-á	lepš-í
NEUT.SG	dobr-é	lepš-í

The distribution of the ‘hard’ vs. ‘soft’ endings is partly correlated with phonology; *-ý* does not appear after ‘soft’ consonants *ž, š, č, ř, ď, ě, ň*, while *-í* does not appear after ‘hard’ consonants *h, ch, k, r, d, t, n*. More specifically, when *-í* appears after stems ending in such consonants, it triggers PAL₂ palatalisations as described in Section 3.2.2.

Crucially, even though the distinction between the *-í* agreement and the *-ý* agreement is frequently correlated with phonology, there is a set of consonants where both endings are found. Thus, it seems impossible to fully reduce this distinction to phonology. For instance, *z*-final adjectives may

belong either to the *-í* agreement class or the *-ý* agreement class. We show this in (16).

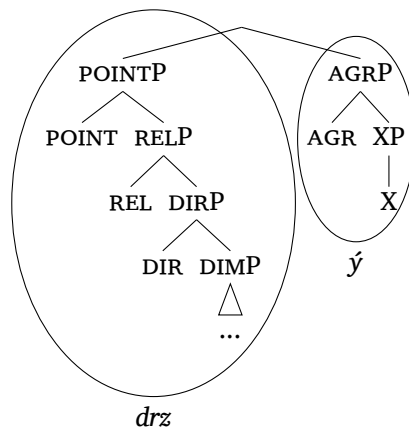
(16) Two z-final adjectives

	pure	rude
MASC.SG	ryz- <i>í</i>	drz- <i>ý</i>
FEM.SG	ryz- <i>í</i>	drz- <i>á</i>
NEUT.SG	ryz- <i>í</i>	drz- <i>é</i>

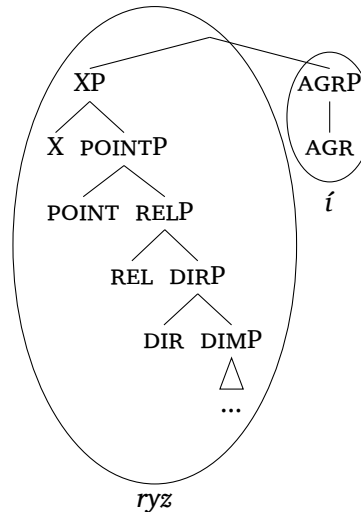
The first column contains the masculine, feminine, and neuter forms of the *-í* adjective ‘pure’ (*ryz-í*), which has the ‘soft’ declension and it is syncretic across the three genders. The second column contains the same three forms of the *-ý* adjective ‘rude’ (*drz-ý*), which patterns with the hard adjectives and shows a three-way distinction.

Given this fact, we suggest that the distinction between ‘soft’ and ‘hard’ declensions should be encoded morphologically. Given our earlier proposal that *-ý* lexicalises the head X plus agreement, as in (12), repeated in (17) for convenience, it seems advantageous to suggest that *-í* lexicalises just the agreement heads (without X), as in (18). In such cases, X has to be expressed by the base.

(17) *-ý* adjective



(18) *-í* adjective



There are two points to be noted related to this account of the hard/soft declension. The first point is related to the comparative. Recall that the comparative always has the soft declension marker *-í*. This is consistent with the conclusion reached in the previous section that X must be missing in the comparative; recall that this is needed in order for *-ě* to be ruled out as the comparative-adverb marker. So, the first point is that the account in (18)

not only provides a way to differentiate the two agreement markers, it is also consistent with our account of adverbs.

The second point is that the proposal in (18) allows us to capture the fact that only soft declension adjectives have irregular adverb forms in the positive degree. To show that, consider first Table (19).

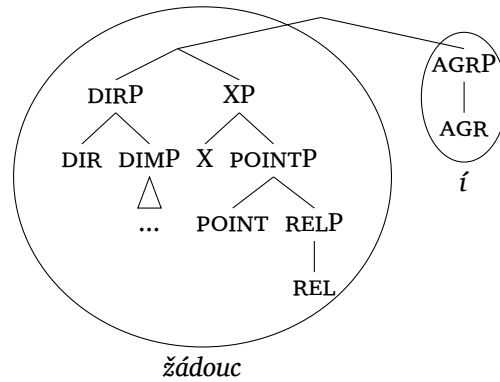
(19) Two *c*-final adjectives

	desirable	hill-less
NEUT. SG	žádouc-í	bezkopc-ě
MASC. SG	žádouc-í	bezkopc-ý
FEM. SG	žádouc-í	bezkopc-á

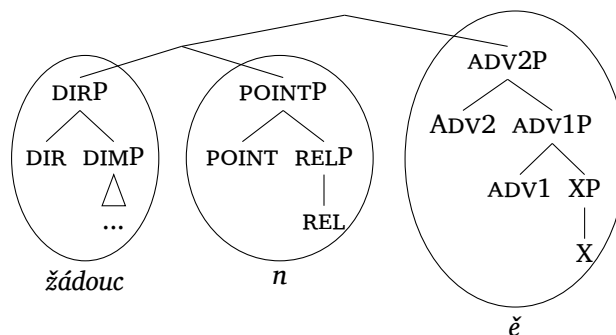
The table shows that for *c*-final adjectives, class membership is also impossible to predict on the basis of phonology alone. This is because adjectives such as *žádouc-í* ‘desirable’ show a soft agreement marker, while the adjective *bezkopc-ý* ‘hill-less’ belongs in the *-ý* paradigm.

Given its soft declension pattern, we hypothesise that in the adjective *žádouc-í* ‘desirable,’ the base must lexicalise *X*. However, recall from Table 4.1 (and also from Ch. 3, item (75)) that *žádouc-* has a complex left branch of the size DIRP. Maintaining this proposal, and adding the idea that *žádouc-* must also lexicalise *X*, we are led to posit the structure as in (20) for the full positive degree of this particular adjective:

(20) The structure of the adjective *žádouc-í* ‘desirable’



Suppose now that an adverb is to be formed from such an adjective. Since the adverbial marker *-ě* has to lexicalise *X*, recall (11), the derivation will have to backtrack below *X* in the positive-degree adverb, yielding the structure in (21).

(21) The structure of the adverb *žádouc-n-ě* ‘desirably’

The crucial point is that the full lexical entry for the root (with the structure as in (20)) no longer matches the unlabelled complex left branch in (21), and the augment *-n* surfaces again. As a result, the adverb has an augment while the adjective has none.

This difference between the adjective and adverb is just a consequence of the general setup where the ‘hard’ agreement markers have the same ‘foot’ (the same lowest feature) as the adverb marker *-ě*, namely *X*. Having the same foot means that the *-ý* agreement and the adverb *-ě* will always combine with the same constituent (POINTP). However, the soft agreement markers have a distinct foot (AGRP), which means that the base has to lexicalise *XP*. As a result we allow for a difference between the form required by *-ě* and *-í*, but there can be no difference between the forms required by *-ě* and *-ý*. This is precisely what the data in Table (13) indicate.

We leave it open how to capture the full range of the positive-degree adverbs; our main point was to indicate a direction in which the investigation may proceed to both capture the core facts while maintaining our previous results. The main fact we attempted to capture is the fact that positive adverbs are (overwhelmingly) derived by the marker *-ě*, attaching to the form of the positive degree, but, at the same time, we do not see this *-ě* in the comparative. We tentatively suggested a solution postulating the head *X*, residing above *POINT* in the structure of the positive, and showed that once this head is adopted, an account of positive-degree adverbs becomes possible without adversely affecting our approach to comparative morphology developed in Chapter 3. We also showed that this head may be used to differentiate ‘soft’ and ‘hard’ agreement, where ‘hard’ agreement lexicalises *X*. This proposal further captures the correlation between agreement class and adverb formation regarding possible changes in the base with soft agreement markers, but no changes with hard agreement markers.

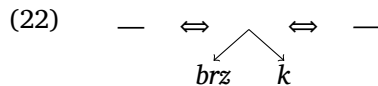
4.2.3 The overlap between the positive and the comparative scale

Another issue left open for future research are the ramifications of the parallel architecture of the positive and the comparative scale. Both are objects with a particular dimension and direction, and in both cases, the argument of the adjective must exceed a standard imposed on that scale.

At the same time, there are also differences between the scales. The positive-degree scale is associated to a dimension that is explicitly expressed by the root. For example, an adjective such as *intelligent* uses its encyclopaedic content to encode the dimension. The comparative scale, as in *more intelligent* is not encoded lexically; rather, *more* uses the context to determine the dimension of the scale it marks.²

We thus treated the scales as partly different, using labels such as DIM for the positive degree and labels such as C_{DIM} for comparatives. At the technical level, this bifurcation had consequences for the matching procedure. For instance, in the positive degree, the augment *-k* lexicalises POINT, while in the comparative, *-š* lexicalises C_{POINT}, without any competition between them. The lack of competition follows from the distinction in the labels: had the labels been radically the same, *-š* would have been expected to win in the positive over *-k*, and we would have to update our account.

We shall not attempt to do so here, though we note that if we decided to do so, we could treat *-š* as the elsewhere exponent of POINT, and *-k* as an exponent that is used in the context of about 30 roots as discussed in Chapter 2, Table 2.2. One way to implement this selection would be to encode it by a pointer, using lexical entries for each of the *-k* adjectives like that for *brz-k-ý* ‘early,’ recall (162) in Chapter 3, repeated in (22) for convenience (see De Clercq & Vanden Wyngaerd 2019b for a case study where such lexical items are used).



The effect of the entry would be to apply at a higher node than both the root and the competing entries *-š* and *-k*. The competition between *-š* and *-k*

² The contextual dependence of the scale conveyed by *more* is also found in noun phrases. As Schwarzschild (2002) notes, phrases such as *more snow* may be interpreted along multiple dimensions as long as the dimension is monotonic on the part-whole relation of the NP. Within this constraint, more than one construal may be available for *more* depending on the context. Thus, *more snow* may be judged on volume, but also on depth. Schwarzschild notes that when there is 10 inches of snow in Rhode Island, and 2 inches in Florida, one may say that Rhode Island has more snow than Florida (judged on depth). However, since Rhode Island is much smaller in terms of land mass, one could also say that there is *more snow* in Florida, judged on volume. In other words, the dimension associated with *more* is context-dependent not only in adjectives, but also in NPs.

would normally be won by *-š*, but due to the higher-level match, *-š* would get overridden by *-k*. This is not a matter of competition, since lexicalisation at a higher node always overrides matches at lower nodes.

While we do not advocate for such an approach here (and maintain the partial difference between the positive and the comparative scales), an approach where the ingredients are fully identical could shed some light on forms where an apparent comparative morphology appears in positive degree adjectives, such as those in (23). Here the rule seems to be that the adjectives remove final vowels of the adverb (if any) and add *-ejší*.

(23) Positive degree adjectives formed by *-ejší*

ADVERB	GLOSS	ADJECTIVE	GLOSS
dole	down	dol-ejší	lower
nahoře ³	up	hoř-ejší	upper
někdy	sometimes	někd-ejší	former
nyní	now	nyn-ejší	current
tam	there	tam-ejší	local
tehdy	then	tehd-ejší	of that time
včera	yesterday	včer-ejší	yesterday's
vedle	beside	vedl-ejší	the one beside
vně	outside	vn-ejší	outside
zde	here	zd-ejší	local
zítra	tomorrow	zítř-ejší	tomorrow's

The fact that *-ejší* may derive positive-degree adjectives could indicate that the ingredients of the comparative scale are the exact same ingredients used in the making of the positive-degree scale.

However, there are also some differences between the *-ejší* seen in the positive and in the comparative, namely palatalisations. Specifically, the positive-degree *-ejší* fails to palatalise *d*-final bases (e.g., *někd-ejš-í* 'former') and *r*-final bases (e.g., *včer-ejš-í* 'yesterday's'). Rather than trying to reach a final conclusion here, we leave this issue for future research.

4.2.4 Superlatives

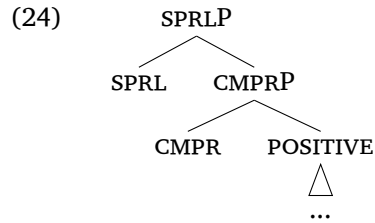
In Czech, superlatives are formed by prefixing *nej-* to the comparative form. We illustrate this in Table 4.6 for all the fifteen classes, both adjectives and adverbs. This table differs from the overview table in Chapter 3, Table 3.5 only by the presence of the superlative *nej-*; all other regularities as well as irregularities are preserved.

³ The locative adverb *na-hoř-e* literally 'on-top-LOC' loses the preposition *na* 'on' in the adjectival form.

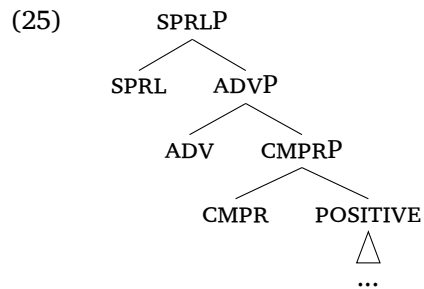
Table 4.6 Fifteen patterns for the positive, comparative and superlative

CLASS	SPRL	ROOT	AUG	ĚJ	ENDING	gloss
I	POS ADJ		chab	Ø	—	ý ‘weak’
	CMPR/SPRL ADJ	(nej)	chab	Ø	ěj	ší
	CMPR/SPRL ADV	(nej)	chab	Ø	ěj	i
II	POS ADJ		slab	Ø	—	ý ‘weak’
	CMPR/SPRL ADJ	(nej)	slab	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	slab	Ø	ěj	i
III	POS ADJ		drah	Ø	—	ý ‘expensive’
	CMPR/SPRL ADJ	(nej)	draž	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	draž	Ø	Ø	e
IV	POS ADJ		jem	n	—	ý ‘fine’
	CMPR/SPRL ADJ	(nej)	jem	n	ěj	ší
	CMPR/SPRL ADV	(nej)	jem	n	ěj	i
V	POS ADJ		žádouc	Ø	—	í ‘desirable’
	CMPR/SPRL ADJ	(nej)	žádouc	n	ěj	ší
	CMPR/SPRL ADV	(nej)	žádouc	n	ěj	i
VI	POS ADJ		pozd	n	—	í ‘late’
	CMPR/SPRL ADJ	(nej)	pozd	Ø	ěj	ší
	CMPR/SPRL ADV	(nej)	pozd	Ø	ěj	i
VII	POS ADJ		snad	n	—	ý ‘easy’
	CMPR/SPRL ADJ	(nej)	snaz	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	snáz	Ø	Ø	e
VIII	POS ADJ		hez	k	—	ý ‘pretty’
	CMPR/SPRL ADJ	(nej)	hez	k	Ø	ší
	CMPR/SPRL ADV	(nej)	hez	k	ěj	i
IX	POS ADJ		slad	k	—	ý ‘sweet’
	CMPR/SPRL ADJ	(nej)	slad	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	slad	k	ěj	i
X	POS ADJ		úz	k	—	ý ‘narrow’
	CMPR/SPRL ADJ	(nej)	už	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	úž	Ø	ěj	i
XI	POS ADJ		blíz	k	—	ý ‘close’
	CMPR/SPRL ADJ	(nej)	bliž	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	blíž	Ø	Ø	e
XII	POS ADJ		brz	k	—	ý ‘early’
	CMPR/SPRL ADJ	(nej)	dřív	Ø	ěj	ší
	CMPR/SPRL ADV	(nej)	dřív	Ø	Ø	e
XIII	POS ADJ		div	ok	—	ý ‘wild’
	CMPR/SPRL ADJ	(nej)	div	ok	ěj	ší
	CMPR/SPRL ADV		div	ok	ěj	i
XIV	POS ADJ		vys	ok	—	ý ‘tall’
	CMPR/SPRL ADJ	(nej)	vyš	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	výš	Ø	Ø	e
XV	POS ADJ		hlub	ok	—	ý ‘deep’
	CMPR/SPRL ADJ	(nej)	hlub	Ø	Ø	ší
	CMPR/SPRL ADV	(nej)	hloub	Ø	ěj	i

This type of situation naturally falls out from the view explored in Bobaljik (2012), and sketched in Section 1.2.1 to the effect that the comparative is contained in the superlative, as in (24).



The data found in Czech (as given in Table 4.6) further require us to place the superlative head (expressed by the marker *nej-*) above the adverb markers, else we would not be able to achieve a joint lexicalisation of the adverbial heads and the comparative heads. Recall that we proposed this to capture the fact that the adverbial *-i* always makes the comparative *-š* disappear (a continuation of the neuter forms from Old Church Slavonic, recall (37)). We show this proposal in (25), simplifying the number of comparative and adverbial heads to the bare minimum.



We leave it open for future research how the scope between the superlative and the adverbial is derived, whether by some kind of transformation (prefix moving up), or whether this is a base-generated structure in Czech (and possibly beyond). The area of prefixation is currently under development and we submit our data as one piece of the puzzle that the ultimate solution must accommodate.

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